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AI APTITUDE QUESTION BANK

AI Engineering

About This Question Bank

The complete AI Engineering Question Bank covers all 11 domains of the Menler AI Engineering Fellowship — from LLM fundamentals and prompt engineering to agentic systems, RAG, reliability, product thinking, and AI-native execution. 825 questions across 55 sets.

5

Sets

15

Questions per Set

75

Total Questions

SET GUIDE

Set 1 · How Language Models Work

AI systems thinking — pipelines, components, and architectural decisions

Set 2 · Training and Learning

Tooling and infrastructure — selecting and combining AI tools in production

Set 3 · Inference and Context

Data and training — datasets, fine-tuning, and model lifecycle

Set 4 · Evaluation and Benchmarks

Evaluation and iteration — measuring, debugging, and improving AI systems

Set 5 · Applied Engineering Judgment

Deployment and scaling — serving models reliably at production scale

Set 6 · Agent Fundamentals

Agent fundamentals — autonomy, reasoning loops, and tool use

Set 7 · Tool Use and Actions

Planning and memory — how agents plan, store, and retrieve context

Set 8 · Prompt Chaining

Multi-agent systems — coordination, delegation, and orchestration patterns

Set 9 · Multi-Agent Systems

Agentic reliability — failure modes, guardrails, and human-in-the-loop design

SET GUIDE (continued)

Set 10 · Safety and Reliability

Production agents — deployment, monitoring, and real-world agentic systems

Set 11 · APIs and Integration

API design and integration — building and consuming AI APIs reliably

Set 12 · RAG and Vector Databases

MCP and protocol layers — model context protocol and agent connectivity

Set 13 · Deployment and Scaling

Compute and serving — GPUs, inference infrastructure, and cost optimisation

Set 14 · Cost and Latency

Security and access control — authentication, rate limiting, and AI security

Set 15 · Security and Access

Observability and reliability — logging, tracing, and SLA management for AI

Set 16 · The Claude Stack

Foundation models — comparing and selecting large language models

Set 17 · Code Assistants

Orchestration frameworks — LangChain, LlamaIndex, and workflow tools

Set 18 · No-Code and Low-Code AI

Vector and retrieval tools — databases, search, and knowledge retrieval

Set 19 · Multimodal Tools

Development and productivity tools — IDE assistants, code generation, CI/CD

Set 20 · Tool Selection and Evaluation

Enterprise AI platforms — cloud AI services, MLOps, and deployment tooling

Set 21 · Output Quality

Ethical reasoning — bias, fairness, and responsible AI decision-making

Set 22 · Hallucination and Verification

Risk assessment — identifying and mitigating AI failure modes

SET GUIDE (continued)

Set 23 · Prompting Decisions

Stakeholder communication — explaining AI trade-offs to non-technical audiences

Set 24 · When AI Fails

Regulatory and compliance — GDPR, AI Act, liability, and governance frameworks

Set 25 · Professional Accountability

Strategic judgment — when to build AI, when to buy, and when to wait

Set 26 · How Language Models Work

Transformer architecture — attention, layers, residuals, and model scale

Set 27 · Tokens and Embeddings

Tokens and embeddings — how text becomes numbers and what those numbers mean

Set 28 · Context Windows

Context windows — limits, management, and practical implications

Set 29 · Inference and Reasoning

Inference and reasoning — how models generate, sample, and think

Set 30 · Model Selection and Fine-tuning

Model selection and fine-tuning — choosing and adapting models for production

Set 31 · System Prompts

System prompts — design, persona, constraints, and security

Set 32 · Chain-of-Thought and Reasoning

Chain-of-thought and reasoning — CoT, ToT, self-consistency, scratchpad

Set 33 · Role and Structure

Prompt structure and context injection — formatting, templates, few-shot

Set 34 · Context and Output Formatting

Output formatting — structured outputs, JSON mode, length control

Set 35 · Reliability and Evaluation Prompting

Reliability and evaluation — regression testing, LLM-as-judge, evaluation rubrics

SET GUIDE (continued)

Set 36 · Vector Databases and Embeddings

Vector databases and embeddings — HNSW, cosine similarity, ANN search

Set 37 · Chunking and Indexing

Chunking and indexing — strategies, overlap, hierarchical, content-aware

Set 38 · Retrieval

Retrieval — top-k, hybrid, HyDE, multi-query, re-ranking

Set 39 · Grounding and Hallucination Prevention

Grounding and hallucination prevention — faithfulness, citation, abstention

Set 40 · Enterprise Knowledge Systems

Enterprise knowledge systems — access control, freshness, observability

Set 41 · Hallucination Detection

Hallucination detection — taxonomy, entity hallucination, SelfCheckGPT, RAGAs

Set 42 · Evaluation Frameworks

Evaluation frameworks — BLEU, MMLU, LLM-as-judge, golden test sets

Set 43 · Guardrails and Fallbacks

Guardrails and fallbacks — input/output filters, PII, constitutional AI

Set 44 · Monitoring and Testing AI Outputs

Monitoring and testing — production monitoring, A/B testing, chaos testing

Set 45 · Human Oversight and Adversarial Testing

Human oversight and adversarial testing — red-teaming, bias evaluation, safety benchmarks

Set 46 · AI UX and Trust

AI UX and trust — explainability, agency transparency, appropriate reliance

Set 47 · Product Decision-Making

Product decision-making — build vs buy, failure mode planning, model versioning

Set 48 · Feature Prioritisation

Feature prioritisation — JTBD, binding constraints, kill criteria, retention signals

SET GUIDE (continued)

Set 49 · User Workflows

User workflows — AI insertion points, delegation design, feedback loops

Set 50 · Operational Usefulness

Operational usefulness — ROI measurement, escalation design, governance

Set 51 · Vibe Coding and Rapid Prototyping

Vibe coding and rapid prototyping — natural language to code, iteration discipline

Set 52 · MVP Building with AI

MVP building with AI — go-to-market assets, demand validation, scaling patterns

Set 53 · AI-Assisted Development

AI-assisted development — code review, debugging, test generation, documentation

Set 54 · Workflow Acceleration

Workflow acceleration — delegation patterns, human oversight, compounding practises

Set 55 · AI-First Operations

AI-first operations — org design, escalation paths, quality monitoring, governance

Each question has one correct answer. Select the best answer from the options provided.

Correct answers and explanations follow each question.

SET

1

How Language Models Work

Set 1 of 5 · 15 Questions · AI Engineering

Q1. What is a "token" in the context of a large language model?

- A. A security credential required to access a model via API.
- B. A unit of text — roughly a word or word-fragment — that the model processes.**
- C. A numerical representation of an entire sentence in the model.
- D. A discrete step in the model's reasoning chain.

Correct Answer: B**Explanation**

Tokens are the atomic units of text that language models process. A token is typically a word, part of a word, or a punctuation character. Models read and generate text one token at a time, not one character or sentence at a time.

Q2. A language model predicts the next token by:

- A. Searching its training data for the closest matching sequence.
- B. Applying logical rules to determine the most appropriate continuation.
- C. Retrieving the answer from a structured internal knowledge base.
- D. Computing a probability distribution over all possible next tokens.**

Correct Answer: D**Explanation**

At inference time, a language model computes a probability distribution over its entire vocabulary for the next token and samples from it. There is no retrieval, rule application, or structured knowledge base — only probabilistic next-token prediction.

Q3. What does "context window" refer to in a language model?

- A. The maximum number of tokens the model can process in a single call.**
- B. The time window within which the model can access recent events.
- C. The set of topics the model was trained to respond to.
- D. The visible portion of the user's screen during inference.

Correct Answer: A

Explanation

The context window is a hard technical limit on how many tokens — input plus output — a model can handle in one call. Beyond this limit, earlier content becomes invisible to the model. It is not related to time, topics, or display.

Q4. Why does a language model sometimes produce different outputs for the same input?

- A. The model updates its weights between each generation call.
- B. Different server instances apply different inference rules.
- C. Sampling randomness controlled by the temperature parameter.**
- D. The model draws from different training data subsets each run.

Correct Answer: C

Explanation

Language model outputs are sampled from a probability distribution. Temperature > 0 introduces randomness into this sampling, meaning identical inputs can produce different outputs. The model's weights do not change between calls, and inference is deterministic once temperature is fixed to 0.

Q5. What does "autoregressive generation" mean?

- A. The model generates the entire response in a single forward pass.
- B. The model generates one token at a time, conditioning each on all previous tokens.**
- C. The model recursively rewrites its output until it meets a quality threshold.
- D. The model selects from a fixed set of response templates.

Correct Answer: B

Explanation

Autoregressive generation means each new token is generated by conditioning on all previously generated tokens. The model does not produce the full response at once; it iterates, one token per step, each step feeding the previous output back as input.

Q6. What is the role of the softmax function in a language model's output layer?

- A. Converts raw logit scores into a valid probability distribution over the vocabulary.**
- B. Filters out low-probability tokens to speed up generation.
- C. Applies grammar rules to ensure syntactically valid outputs.
- D. Normalises the input tokens before they enter the transformer.

Correct Answer: A

Explanation

The output layer produces raw unnormalised scores (logits) for each vocabulary token. Softmax transforms these into a probability distribution that sums to 1, allowing the model to sample the next token proportionally to its predicted likelihood.

Q7. What is "attention" in a transformer model?

- A. A filtering layer that removes irrelevant tokens before processing.
- B. A memory module that retrieves facts from external knowledge bases.
- C. A scoring system that ranks candidate outputs by quality.
- D. A mechanism that computes weighted relationships between all tokens in the input.**

Correct Answer: D

Explanation

Self-attention computes a weighted sum of value vectors for every token in the sequence, where the weights are determined by the similarity of each token's query to every other token's key. This allows the model to attend to any part of the input regardless of distance.

Q8. What distinguishes a "decoder-only" architecture from an "encoder-decoder" architecture?

- A. Decoder-only models are smaller and faster but less capable.
- B. Encoder-decoder models can only handle fixed-length inputs.
- C. Decoder-only models generate text left-to-right without a separate encoding step.**
- D. Decoder-only models require separate fine-tuning for each task.

Correct Answer: C

Explanation

Decoder-only models (like GPT-family and Claude) generate text autoregressively using only a transformer decoder stack. Encoder-decoder models (like T5) first encode the input, then decode the output. Most modern large language models use the decoder-only architecture.

Q9. What happens when you set temperature = 0 during inference?

- A. The model always selects the highest-probability token, producing deterministic output.**
- B. The model refuses to generate creative or speculative content.
- C. The model generates shorter responses to reduce computation.
- D. The model switches to a rule-based response mode.

Correct Answer: A

Explanation

At temperature = 0, the model's sampling distribution collapses to a point mass on the highest-probability token. This makes generation fully deterministic — the same input always produces the same output. It does not affect response length or enable rule-based modes.

Q10. What is "top-p sampling" (nucleus sampling)?

- A. Selecting the top p% of tokens by individual probability.
- B. Sampling tokens in order of probability until p tokens have been drawn.
- C. Setting a minimum probability threshold p below which tokens are excluded.
- D. Sampling only from the smallest set of tokens whose cumulative probability exceeds p.**

Correct Answer: D

Explanation

Top-p (nucleus) sampling first identifies the smallest set of tokens whose cumulative probability mass reaches p, then samples from only that set. This dynamically adjusts the candidate pool — larger for uncertain next tokens, smaller when one token is highly probable.

Q11. What does the "embedding layer" do in a language model?

- A. Compresses long inputs into shorter summary vectors for efficiency.
- B. Converts discrete token IDs into dense continuous vector representations.**
- C. Maps model outputs back to human-readable text tokens.
- D. Stores the model's factual knowledge as lookup tables.

Correct Answer: B

Explanation

The embedding layer maps each integer token ID to a high-dimensional dense vector. These vectors are learned during training and encode semantic and syntactic information that the rest of the model can process. They are not summaries, outputs, or fact stores.

Q12. Why do language models have a "knowledge cutoff"?

- A. The model's context window cannot hold information beyond a certain date.
- B. Anthropic manually updates the model's knowledge on a scheduled basis.
- C. Training data collection ends at a point in time, so the model has no knowledge of later events.**
- D. Regulatory requirements limit the training data to pre-approved date ranges.

Correct Answer: C

Explanation

A knowledge cutoff reflects when training data collection stopped. The model learned from text up to that date, so it has no information about events that occurred afterward. This is a data collection limit, not a context window or regulatory constraint.

Q13. What is the primary function of "layer normalisation" in a transformer?

- A. Reduces the model's output to a fixed-length vector for classification.
- B. Normalises the input tokens before the attention mechanism.
- C. Applies regularisation to prevent individual attention heads from dominating.
- D. Stabilises training by normalising activations across features within each layer.**

Correct Answer: D

Explanation

Layer normalisation stabilises training by normalising the distribution of activations within each transformer layer. This prevents activation values from growing unboundedly, which would cause gradient instability during training.

Q14. What does "parameter count" tell you about a language model?

- A. The number of discrete facts the model has memorised from training data.
- B. The total number of learnable weights that encode the model's knowledge and behaviour.**
- C. The maximum number of tokens the model can process per second.
- D. The number of distinct tasks the model was trained to perform.

Correct Answer: B

Explanation

Parameters are the numerical weights that are learned during training. More parameters generally allow more complex pattern representation. They do not directly correspond to memorised facts, processing speed, or task count — those depend on architecture, data, and training.

Q15. What does "perplexity" measure in language model evaluation?

- A. How surprised the model is by a test text — lower means better prediction.**
- B. How often the model produces factually incorrect outputs on a benchmark.
- C. The computational complexity of the model's inference process.
- D. The degree of grammatical error in the model's generated text.

Correct Answer: A

Explanation

Perplexity measures how well a probability model predicts a sample — it is the exponentiated average negative log-likelihood of the test text. Lower perplexity means the model assigned higher probability to the actual next tokens, indicating better language modelling.

SET

2

Training and Learning

Set 2 of 5 · 15 Questions · AI Engineering

Q16. What is "pre-training" in the context of large language models?

- A. A small trial training run to test the model architecture before full training.
- B. Training the model on labelled examples for a specific downstream task.
- C. Training on a large general text corpus to learn broad language patterns.
- D. A calibration step that sets the model's safety and refusal thresholds.

Correct Answer: C**Explanation**

Pre-training involves training a language model on a very large, diverse text corpus using self-supervised objectives (e.g., next-token prediction). This produces a foundation model with broad language understanding before any task-specific fine-tuning.

Q17. What does "fine-tuning" a language model mean?

- A. Continuing training on a smaller, task-specific dataset to adapt model behaviour.
- B. Adjusting the model's temperature and sampling parameters for a use case.
- C. Removing knowledge the model learned from pre-training that is not needed.
- D. Adding new layers to the model architecture to handle a specific domain.

Correct Answer: A**Explanation**

Fine-tuning updates a pre-trained model's weights by continuing training on a smaller, curated dataset. This adapts the model's behaviour toward the target task or domain without training from scratch. It does not change the architecture or delete pre-training knowledge.

Q18. What is the purpose of RLHF (Reinforcement Learning from Human Feedback)?

- A. Teaching the model to retrieve information from the internet during inference.
- B. Ensuring the model's factual claims match a curated ground-truth database.
- C. Reducing the model's parameter count for deployment efficiency.
- D. Aligning model outputs with human preferences using human-rated training signal.**

Correct Answer: D

Explanation

RLHF uses human preference ratings to train a reward model, then uses reinforcement learning to adjust the language model toward outputs humans rate more positively. This is a key technique for making models helpful, harmless, and honest.

Q19. What is "catastrophic forgetting" in the context of fine-tuning?

- A. The model forgets context from earlier in a conversation as the context window fills.
- B. The model loses previously learned general capabilities when fine-tuned on narrow data.**
- C. The training process fails because the learning rate is too high.
- D. The model's training data contains contradictory information that degrades performance.

Correct Answer: B

Explanation

Catastrophic forgetting occurs when fine-tuning on a narrow dataset causes the model to overwrite the general knowledge and capabilities learned during pre-training. Techniques like LoRA and careful dataset design help mitigate this.

Q20. What is "LoRA" (Low-Rank Adaptation) used for?

- A. Compressing model weights to reduce inference memory requirements.
- B. Generating low-resolution outputs faster than full model inference.
- C. Efficient fine-tuning by training only small low-rank weight update matrices.**
- D. A regularisation technique to prevent overfitting during pre-training.

Correct Answer: C

Explanation

LoRA freezes the original pre-trained weights and trains small additional low-rank matrices that are added to specific weight matrices. This dramatically reduces the number of trainable parameters during fine-tuning, making it efficient without catastrophic forgetting.

Q21. What does "instruction tuning" accomplish?

- A. Teaches the model to produce structured outputs like JSON and markdown.
- B. Reduces the model's tendency to hallucinate by adding factual training data.
- C. Converts a classifier model into a generative model.
- D. Trains the model to follow natural language instructions rather than just complete text.**

Correct Answer: D

Explanation

Instruction tuning (a form of supervised fine-tuning) trains the model on instruction-response pairs so it learns to treat user instructions as tasks to complete rather than text to continue. This is what makes base models into useful assistants.

Q22. Why does training a large language model require massive compute?

- A. Computing gradients and updating billions of parameters across trillions of tokens requires enormous parallelism.**
- B. Language models need to search the internet during training to acquire knowledge.
- C. Each training step requires solving an NP-hard optimisation problem.
- D. The model must be evaluated on every possible input during training.

Correct Answer: A

Explanation

Training requires forward passes (computing predictions), loss computation, and backward passes (computing gradients for billions of parameters) across trillions of training tokens. Each gradient step involves massive matrix multiplications that require GPU/TPU clusters running for weeks or months.

Q23. What is a "reward model" in the RLHF pipeline?

- A. A model that evaluates the factual accuracy of the language model's outputs.
- B. A model trained on human preferences that scores outputs for the RL training loop.**
- C. The language model itself when used to evaluate its own generated responses.
- D. A rule-based system that enforces safety guidelines during training.

Correct Answer: B

Explanation

A reward model is trained on pairs of model outputs with human preference labels (which output was better). It learns to predict human preference scores, which are then used as the reward signal to guide RL fine-tuning of the language model toward preferred behaviour.

Q24. What does "gradient descent" do during model training?

- A. Searches through possible model architectures to find the optimal one.
- B. Selects the most informative training examples for the current batch.
- C. Normalises the model's outputs to match the training data distribution.
- D. Iteratively adjusts model weights in the direction that reduces the training loss.**

Correct Answer: D

Explanation

Gradient descent computes the gradient of the loss function with respect to each model weight, then updates each weight by a small step in the direction that reduces the loss. Repeated over many iterations and examples, this gradually improves the model's predictions.

Q25. What is the primary function of the learning rate in model training?

- A. Determines how quickly the model can read and process training examples.
- B. Sets the maximum number of training steps before the run terminates.
- C. Controls how large each weight update step is during gradient descent.**
- D. Adjusts the proportion of the training data used in each epoch.

Correct Answer: C

Explanation

The learning rate is the multiplier applied to the gradient when updating each weight. Too high and training becomes unstable (weights overshoot the optimum); too low and training is slow or gets stuck. It is a critical hyperparameter in any neural network training run.

Q26. What is "overfitting" in model training?

- A. The model generates outputs that are too long and detailed for the given task.
- B. The model memorises training data and performs poorly on new, unseen examples.**
- C. The model's training loss decreases while its outputs become less coherent.
- D. The model's weights exceed the memory capacity of the GPU.

Correct Answer: B

Explanation

Overfitting occurs when a model learns the specific patterns (including noise) in the training data so well that it fails to generalise to new examples. A model that has overfit achieves low training loss but high validation loss.

Q27. What is the difference between a "base model" and an "instruction-tuned model"?

- A. A base model predicts next tokens from text; an instruction-tuned model follows conversational instructions.**
- B. A base model is larger; an instruction-tuned model is a compressed version.
- C. A base model can only generate code; an instruction-tuned model handles all text.
- D. A base model has no safety training; an instruction-tuned model has been safety-tested.

Correct Answer: A

Explanation

A base (foundation) model is trained on raw text prediction and generates continuations. An instruction-tuned model has been further trained on instruction-response pairs, making it behave like an assistant that interprets instructions rather than completing arbitrary text.

Q28. What does "constitutional AI" (CAI) refer to in the context of Anthropic's training approach?

- A. A legal framework governing how AI models can be used commercially.
- B. A hardware architecture that ensures safe computation during inference.
- C. Training models to critique and revise their outputs according to a set of principles.**
- D. A method for training models without any human-generated data.

Correct Answer: C

Explanation

Constitutional AI is Anthropic's approach where models are trained to critique their own outputs against a "constitution" of principles, then revise accordingly. This reduces the need for human preference labelling in some parts of the safety training pipeline.

Q29. What is "data contamination" in the context of model evaluation?

- A. When training data contains offensive or harmful content the model has learned.
- B. When noise introduced during data preprocessing corrupts training examples.
- C. When competing models train on each other's outputs, leading to distribution collapse.
- D. When test set examples were present in the model's training data, inflating benchmark scores.**

Correct Answer: D

Explanation

Data contamination occurs when evaluation benchmarks were present (or close variants existed) in the training data. The model appears to perform better because it has seen the answers before. This is a critical issue for interpreting benchmark results honestly.

Q30. What is "tokenizer" vocabulary size typically optimised for?

- A. Maximising the number of languages the model can process.
- B. Balancing coverage of common words with manageable model input dimension.**
- C. Minimising training data storage requirements.
- D. Matching the output dimension of the model's embedding layer exactly.

Correct Answer: B

Explanation

Tokenizer vocabulary size is a tradeoff: larger vocabularies mean fewer tokens per text (more efficient) but larger embedding tables and output layers. Typical modern LLM tokenizers use 50k–150k tokens, balancing multilingual coverage, efficiency, and model size.

SET

3

Inference and Context

Set 3 of 5 · 15 Questions · AI Engineering

Q31. What is "KV cache" and why does it matter for inference?

- A. Stores computed key-value pairs from previous tokens to avoid recomputing them.**
- B. A hardware cache that stores the most frequently accessed model weights.
- C. A client-side cache that stores previous API responses for reuse.
- D. A compression algorithm that reduces model output size for transmission.

Correct Answer: A**Explanation**

During autoregressive generation, the model recomputes attention over all previous tokens at every new step. KV caching stores the key and value matrices from previous forward passes so they don't need recomputation, dramatically reducing inference latency for long contexts.

Q32. What does "quantisation" do to a language model?

- A. Converts the model from a generative to a discriminative architecture.
- B. Splits the model across multiple GPUs for parallel inference.
- C. Reduces the numerical precision of model weights to decrease memory and speed up inference.**
- D. Removes redundant neurons from the model to reduce its parameter count.

Correct Answer: C**Explanation**

Quantisation reduces the bit-width of weight values (e.g., from 32-bit float to 8-bit or 4-bit integers). This reduces the model's memory footprint and speeds up inference with minimal quality loss, making large models deployable on smaller hardware.

Q33. What is "speculative decoding" in language model inference?

- A. The model generates multiple complete responses and selects the best one.
- B. A smaller draft model generates candidate tokens that the main model verifies in parallel.**
- C. Inference is run speculatively on predicted user queries before they are submitted.
- D. The model skips low-confidence tokens to speed up generation.

Correct Answer: B

Explanation

Speculative decoding uses a small, fast draft model to generate several candidate next tokens. The larger model then verifies these candidates in parallel (which is much faster than generating them sequentially). Accepted tokens are kept; the first rejected one triggers re-generation. This can significantly increase throughput.

Q34. What is "batch inference" and when is it useful?

- A. Breaking a long input into smaller batches to fit within the context window.
- B. Running the same input multiple times to average out sampling variation.
- C. Pre-computing responses for a batch of expected future queries.
- D. Processing multiple input requests simultaneously to improve GPU utilisation.**

Correct Answer: D

Explanation

Batch inference groups multiple independent requests together for a single GPU forward pass. Since GPUs are highly parallel, this improves utilisation and throughput compared to processing requests one at a time. It is fundamental to efficient production AI serving.

Q35. What is "multi-head attention"?

- A. Running multiple attention computations in parallel, each learning different relationship patterns.**
- B. Applying attention to the input from multiple users simultaneously.
- C. A technique where the model attends to its previous outputs before generating.
- D. Attention computed across multiple transformer layers simultaneously.

Correct Answer: A

Explanation

Multi-head attention runs h parallel attention mechanisms ("heads"), each with its own learned query, key, and value projections. The outputs are concatenated and projected. Different heads learn to attend to different types of relationships in the input, giving the model richer representation capacity.

Q36. What does "model parallelism" solve?

- A. Ensures multiple users can query the same model simultaneously.
- B. Splits a model too large to fit on one GPU across multiple GPUs.**
- C. Allows the model to process multiple modalities (text and images) in parallel.
- D. Distributes training data across GPUs to speed up data loading.

Correct Answer: B

Explanation

Model parallelism (also called tensor or pipeline parallelism) partitions model layers or weight matrices across multiple GPUs when a single device cannot hold the full model. This is distinct from data parallelism, which runs copies of the full model on different data.

Q37. What is "flash attention" designed to improve?

- A. The speed at which the attention mechanism learns to focus on relevant tokens.
- B. The accuracy of attention scores for long-distance token dependencies.
- C. The initialisation of attention weights to speed up model convergence.
- D. Memory efficiency of the attention computation by avoiding storing large intermediate matrices.**

Correct Answer: D

Explanation

Standard attention computes and stores an $N \times N$ attention matrix (where N is sequence length), which is memory-prohibitive for long contexts. Flash Attention reorders the computation to use fast SRAM instead of slow HBM, computing correct attention without materialising the full attention matrix.

Q38. What is "inference latency" and what are the main contributors to it?

- A. The delay between the user sending a query and the server receiving it.
- B. The time required to load model weights from storage into memory.
- C. The time to generate a response — driven by model size, hardware, and generation length.**
- D. The computational cost of evaluating training loss on the inference server.

Correct Answer: C

Explanation

Inference latency is the total time from request receipt to response delivery. Key contributors include model size (more parameters = more computation per token), hardware capability, batch size, context length (KV cache size), and output token count.

Q39. What does "throughput" measure in an LLM serving context?

- A. The accuracy of the model's outputs on a standard benchmark.
- B. The number of tokens (or requests) the system can process per unit time.**
- C. The bandwidth available between the client and the inference server.
- D. The proportion of requests that complete within an SLA latency target.

Correct Answer: B

Explanation

Throughput measures production capacity — tokens per second or requests per second. It is distinct from latency (time per request). High throughput requires efficient batching and hardware utilisation; high throughput and low latency are often in tension.

Q40. What is a "mixture of experts" (MoE) architecture?

- A. A model where only a subset of specialised sub-networks (experts) activates per token.**
- B. An ensemble of multiple separate models that vote on the best response.
- C. A training approach using multiple human annotators as domain experts.
- D. A model that dynamically selects which dataset to sample from during training.

Correct Answer: A

Explanation

In MoE, the model has multiple "expert" feed-forward networks, but a gating mechanism routes each token to only a small subset (e.g., 2 out of 64). This allows massive total parameter counts while keeping per-token compute manageable, giving MoE models a strong compute-efficiency advantage.

Q41. What is "positional encoding" and why do transformers need it?

- A. Encodes the semantic position of words within a sentence's grammatical structure.
- B. Maps absolute token positions to relative importance weights for attention.
- C. Injects sequence position information since attention is inherently order-agnostic.**
- D. Adds timestamp information to allow the model to process time-series data.

Correct Answer: C

Explanation

The attention mechanism treats its input as a set — it has no built-in notion of order. Positional encodings (learned or sinusoidal) add position information to token embeddings so the model can distinguish "dog bites man" from "man bites dog."

Q42. What is "beam search" in text generation?

- A. A method for parallelising attention computation across multiple processing units.
- B. A technique for filtering harmful content from the model's output distribution.
- C. An approach that generates text in both forward and backward directions.
- D. A search algorithm that maintains multiple candidate sequences and selects the highest-scoring one.**

Correct Answer: D

Explanation

Beam search expands B (beam width) candidate sequences at each step, keeping only the top B by accumulated probability. This can produce higher-quality outputs than greedy decoding or simple sampling, but at a cost of generating B times more tokens and often producing more generic outputs.

Q43. What is "chain of thought" (CoT) prompting and why does it improve performance on complex tasks?

- A. A technique that chains multiple separate model calls together automatically.
- B. Prompting the model to reason step by step before answering, improving accuracy on multi-step problems.**
- C. Prompting the model to generate its full context window before producing an answer.
- D. A sampling technique that produces multiple candidate reasoning paths simultaneously.

Correct Answer: B

Explanation

CoT prompting asks the model to articulate intermediate reasoning steps before the final answer. This dramatically improves performance on tasks requiring multi-step logic, mathematics, or reasoning — not by changing the model, but by using the context window as a working scratchpad.

Q44. What does "context length scaling" refer to as a challenge in LLM development?

- A. Longer contexts require proportionally more training data to learn from.
- B. Models produce worse outputs when given more context than their training used.
- C. The computational cost of attention grows quadratically with sequence length.**
- D. Context windows must be fixed at training time and cannot be extended.

Correct Answer: C

Explanation

Standard attention has $O(N^2)$ complexity in sequence length N — doubling context quadruples compute and memory for the attention step. This makes very long contexts (millions of tokens) computationally expensive and is a core research challenge addressed by sparse attention, linear attention, and memory architectures.

Q45. What is the difference between "model weights" and "model activations"?

- A. Weights are stored parameters; activations are intermediate computed values during a forward pass.**
- B. Weights determine output format; activations determine output content.
- C. Weights are used during training; activations are used during inference.
- D. Weights are shared across layers; activations are unique to each input token.

Correct Answer: A

Explanation

Weights (parameters) are the learned numerical values stored in the model that persist across calls. Activations are the intermediate computed values produced during a specific forward pass — they depend on the input and are not stored after the pass completes.

SET

4

Evaluation and Benchmarks

Set 4 of 5 · 15 Questions · AI Engineering

Q46. What is an "eval" (evaluation) in the context of language model development?

- A. A human expert review of model outputs for quality assurance.
- B. A safety check run before deploying a model update to production.
- C. An automated scan for harmful content in the model's training data.
- D. A systematic test measuring model performance on a defined task or capability.**

Correct Answer: D**Explanation**

An eval is any structured measurement of a model's capability or behaviour. Evals can be automated (running a model on benchmark questions and scoring outputs) or human-evaluated. They are used to track progress, compare models, and detect regressions.

Q47. What is the most important limitation of benchmark scores as a measure of model quality?

- A. Benchmarks can only measure factual knowledge, not reasoning ability.
- B. Benchmark performance may not reflect real-world task performance or user value.**
- C. Benchmark scores are proprietary and not publicly disclosed by model providers.
- D. Benchmarks require human evaluation which introduces inter-rater variability.

Correct Answer: B**Explanation**

Benchmark scores measure performance on specific test sets under specific conditions. Models can score highly by fitting to benchmark formats without generalising to real-world tasks. And when training data contains benchmark examples (contamination), scores are further inflated. Real-world performance requires separate evaluation.

Q48. What does "MMLU" (Massive Multitask Language Understanding) measure?

- A. The model's ability to understand and generate multiple languages.
- B. Multiturn dialogue quality across many conversational topics.
- C. Multiple-choice knowledge across 57 academic subjects and disciplines.**
- D. The model's performance at generating structured data in multiple formats.

Correct Answer: C

Explanation

MMLU consists of 57 academic subjects (from STEM to humanities to professional domains) in multiple-choice format. It tests breadth of knowledge and academic reasoning. It is one of the most widely cited language model benchmarks, though susceptibility to data contamination is a known concern.

Q49. What is "human eval" in LLM evaluation?

- A. Having human raters compare or assess model outputs directly.**
- B. Running the model on problems where humans' correct answers are the ground truth.
- C. A specific coding benchmark that measures model performance on Python tasks.
- D. An internal Anthropic evaluation methodology for safety testing.

Correct Answer: A

Explanation

Human eval refers to using human judges to rate model outputs — often comparing two models head-to-head (A/B preference) or rating outputs on dimensions like helpfulness, accuracy, or safety. It is more reliable than automated metrics for measuring quality but expensive and hard to scale.

Q50. What does "accuracy on held-out test data" measure in model evaluation?

- A. How accurately the model reproduces its training data verbatim.
- B. The percentage of inference calls that complete without technical errors.
- C. How precisely the model's outputs match the user's intended format.
- D. How well the model performs on examples it was not exposed to during training.**

Correct Answer: D

Explanation

Held-out test data is reserved from the training process and used only for final evaluation. Performance on this data estimates how the model will generalise to genuinely new examples — the most honest measure of model capability, assuming no data contamination.

Q51. What is "F1 score" and why is it useful in model evaluation?

- A. A composite benchmark score averaging performance across multiple F-series tasks.
- B. A measure of fluency that rates the grammatical correctness of model outputs.
- C. The harmonic mean of precision and recall, balancing false positives and false negatives.**
- D. The fraction of model outputs that a human rater finds factually correct.

Correct Answer: C

Explanation

F1 combines precision (fraction of positive predictions that are correct) and recall (fraction of actual positives the model identified) into a single metric. It is particularly useful when the cost of false positives and false negatives matters and when classes are imbalanced.

Q52. Why is "model calibration" important?

- A. A calibrated model's confidence scores accurately reflect the probability it is correct.**
- B. Calibration ensures the model produces outputs of consistent length.
- C. A calibrated model refuses to answer questions outside its training domain.
- D. Calibration aligns the model's vocabulary with the user's expected terminology.

Correct Answer: A

Explanation

A well-calibrated model assigns probability p to claims it is correct approximately p fraction of the time. A poorly calibrated model may be confidently wrong (overconfident) or unnecessarily uncertain. Calibration is critical for safe AI use where expressed confidence informs human trust.

Q53. What is "LLM-as-judge" evaluation and what is its key limitation?

- A. Asking users to rate their conversations with an LLM — limited by user expertise.
- B. Using a language model to rate other model outputs — limited by the judge model's own biases.**
- C. Running models against standardised legal evaluation criteria — limited by jurisdiction.
- D. Evaluating model outputs on safety criteria — limited by adversarial prompt injection.

Correct Answer: B

Explanation

LLM-as-judge uses a language model (often GPT-4 or Claude) to rate outputs from other models. It is scalable and correlates well with human judgments. The key limitation is that the judge model inherits its own biases, preferences, and blind spots — it may systematically prefer certain styles regardless of quality.

Q54. What does "win rate" measure in comparative model evaluation?

- A. The percentage of benchmark tasks a model completes successfully.
- B. The probability that a model produces a factually correct answer on a given topic.
- C. The fraction of head-to-head comparisons where one model is preferred over another.**
- D. The fraction of inference calls that complete without timeout or error.

Correct Answer: C

Explanation

Win rate is measured by showing human raters (or an LLM judge) pairs of outputs from two models and recording which is preferred. A win rate above 50% means the model is preferred more often than the comparison model. It is a widely used metric in LLM leaderboards.

Q55. What is "evals data contamination" and why is it a significant problem?

- A. Evaluation data collected from users introduces privacy risks.
- B. Human annotators contaminate eval results by disagreeing with each other.
- C. Model outputs leak into evaluation pipelines, creating circular scoring.
- D. Benchmark test sets present in training data make scores meaninglessly high.**

Correct Answer: D

Explanation

If a model was trained on data containing benchmark test questions and answers, its scores on those benchmarks reflect memorisation, not generalisation. This is a fundamental validity threat to published benchmark comparisons. Many popular benchmarks are suspected to be contaminated in large-model training sets.

Q56. What is the purpose of a "safety eval"?

- A. Measuring a model's tendency to produce harmful, misleading, or inappropriate outputs.**
- B. Testing that the model's inference API meets performance SLA requirements.
- C. Verifying that the model's training data was sourced ethically.
- D. Ensuring the model's outputs are grammatically correct and coherent.

Correct Answer: A

Explanation

Safety evals systematically probe for harmful outputs — testing jailbreaks, checking refusal appropriateness, measuring bias, and assessing outputs on sensitive topics. They are a core part of responsible model development before any public deployment.

Q57. What does "red teaming" mean in AI model development?

- A. Competing against another team to achieve higher benchmark scores.
- B. Adversarially probing a model to find safety failures, harmful outputs, or exploitable behaviours.**
- C. Testing model performance under high-load production traffic conditions.
- D. Running the model on tasks from domains it was not specifically trained for.

Correct Answer: B

Explanation

Red teaming involves security and safety researchers systematically trying to break or misuse a model — finding jailbreaks, generating harmful content, or surfacing unexpected failure modes. It is a critical step in responsible AI development before public release.

Q58. Why do evaluation results sometimes differ across providers reporting the same benchmark?

- A. Different providers have access to different versions of the same benchmark test set.
- B. Benchmarks are designed differently for each model provider by the benchmark authors.
- C. Different GPU hardware produces numerically different inference results.
- D. Prompt formatting, few-shot examples, and generation settings all affect scores significantly.**

Correct Answer: D

Explanation

Benchmark scores are sensitive to how prompts are formatted, how many few-shot examples are included, what system prompt is used, what temperature is set, and how outputs are parsed. The same model with the same benchmark but different prompt settings can show meaningfully different scores.

Q59. What does a model's performance on "out-of-distribution" data reveal?

- A. How well the model generalises beyond the patterns present in its training data.**
- B. How the model handles inputs in languages not present in the training corpus.
- C. Whether the model can identify when a question is factually incorrect.
- D. The model's robustness to typos and grammatical errors in input text.

Correct Answer: A

Explanation

Out-of-distribution (OOD) data tests whether the model has learned genuine generalisable patterns or just memorised training distributions. Strong OOD performance indicates robust generalisation; poor OOD performance suggests the model relies on surface patterns that break with distributional shift.

Q60. What is the key difference between "task-specific" and "general" capability evaluation?

- A. Task-specific evals use human raters; general evals use automated metrics.
- B. Task-specific evals are faster; general evals require more compute.
- C. Task-specific evals measure one skill; general evals measure breadth across many skills.**
- D. Task-specific evals are proprietary; general evals are publicly published.

Correct Answer: C

Explanation

Task-specific evals (coding, math, medical Q&A) measure deep capability in one domain. General capability evals (MMLU, BIG-Bench) measure breadth across many domains. Both matter: a model might excel at coding but perform poorly at medical reasoning, which neither type alone reveals.

SET

5

Applied Engineering Judgment

Set 5 of 5 · 15 Questions · AI Engineering

Q61. You are building a product feature where the same input must always produce the same output. What inference setting do you use?

- A. Temperature = 0 for deterministic, reproducible generation.
- B. Temperature = 1 for maximum quality generation.
- C. Top-p = 1.0 to include the full vocabulary distribution.
- D. Max tokens set to exactly the expected output length.

Correct Answer: A

Explanation

Setting temperature = 0 makes token selection deterministic — the model always picks the highest-probability token. This guarantees the same input always produces the same output, which is critical for reproducible, testable production features.

Q62. You need to reduce the cost of running Claude at scale. The most impactful lever is:

- A. Switching to a different cloud region with lower GPU costs.
- B. Increasing batch size to improve throughput without changing token count.
- C. Using a longer context window to process more data per call.
- D. Reducing input and output token counts through prompt and output design.

Correct Answer: D

Explanation

API cost scales directly with input and output tokens. Engineering shorter, more efficient prompts and designing features that generate concise outputs is the highest-leverage cost reduction lever. Batching improves throughput and can reduce latency cost, but the fundamental cost driver is token count.

Q63. A model produces high-quality outputs in testing but degrades in production. The most likely cause is:

- A. The model is being called with a higher temperature in production.
- B. Production inputs differ from test inputs in format, length, or content distribution.**
- C. The production server hardware is less powerful than the test environment.
- D. The model's weights are being updated by production usage patterns.

Correct Answer: B

Explanation

Distribution shift — differences between test and production data — is the most common cause of deployment degradation. Production users ask different questions, in different formats, with different edge cases than anticipated in testing. Model weights do not update from production traffic.

Q64. You want Claude to output only valid JSON. The most reliable approach is:

- A. Use the phrase "respond in JSON" without further specification.
- B. Post-process Claude's output with a JSON formatter to fix any errors.
- C. Specify the exact JSON schema in the prompt with an example, and set temperature = 0.**
- D. Ask Claude to rate its own output for JSON validity before returning it.

Correct Answer: C

Explanation

Reliable structured output requires schema specification (what fields, types, and structure) with an example, plus temperature = 0 for consistency. "Respond in JSON" without a schema produces inconsistent field names and structures. Post-processing fixes syntax but not structural errors.

Q65. Which scenario most clearly justifies fine-tuning a model rather than using prompting?

- A. Consistent high-quality performance on a very specific domain not covered well by the base model.**
- B. Adding the ability to browse the internet and retrieve current information.
- C. Reducing the model's hallucination rate on general knowledge questions.
- D. Making the model respond in a different language than its primary training language.

Correct Answer: A

Explanation

Fine-tuning is most justified when you need deep, reliable performance on a specific domain that prompting alone cannot achieve — e.g., a narrow medical subspecialty or a proprietary technical domain. Internet access requires tool use, not fine-tuning. Hallucination reduction and language shift are better addressed by prompting and RAG.

Q66. You notice that a model's benchmark scores have improved significantly between versions but user satisfaction has not. The most likely explanation is:

- A. Users are not sophisticated enough to recognise the quality improvement.
- B. The model is being tested by users on tasks outside the benchmark scope.
- C. Benchmark improvements do not always translate to real-world capability improvements.**
- D. Satisfaction metrics have a 3-6 month lag behind capability improvements.

Correct Answer: C

Explanation

Benchmark improvements and real-world quality are often decoupled. Models can improve on specific test formats (especially contaminated benchmarks) without improving on the diverse, messy tasks users actually bring. This is one reason why benchmark inflation is a genuine concern in AI evaluation.

Q67. A RAG system retrieves relevant documents and passes them to Claude, but Claude's answers are still often wrong. The most likely cause is:

- A. Claude's context window is too small to process the retrieved documents.
- B. Claude is ignoring the retrieved context and generating from training knowledge.
- C. The embedding model used for retrieval is not from Anthropic.
- D. Retrieved documents are not accurately addressing the specific question being asked.**

Correct Answer: D

Explanation

RAG failures most commonly trace to retrieval quality — the retrieved documents are topically adjacent but do not actually contain the answer to the specific question. The system retrieves confidently but irrelevantly, and Claude generates plausible-sounding answers from incomplete context.

Q68. What is the most important thing to monitor in a production LLM system?

- A. GPU memory utilisation to prevent out-of-memory errors.
- B. Output quality on real user inputs, not just test set performance.**
- C. API response times to ensure SLA compliance.
- D. Token costs per request to control infrastructure spend.

Correct Answer: B

Explanation

Production output quality — whether real user requests are being handled correctly — is the most important thing to monitor. All other metrics (GPU utilisation, latency, cost) are operational signals. The system can be fast and cheap while producing wrong or harmful outputs if quality is not monitored.

Q69. You are asked to compare two language models for a production use case. What evaluation approach is most reliable?

- A. Compare their scores on the latest public benchmarks from their technical reports.
- B. Ask both models to rate each other's outputs on the target task.
- C. Run both models on a sample of real production inputs and measure task-specific outcomes.**
- D. Select the model with the higher parameter count, as it generally performs better.

Correct Answer: C

Explanation

The most reliable comparison is task-specific evaluation on real production inputs. Public benchmarks may be contaminated, may not reflect your task, and are often reported under favourable conditions. Mutual self-rating is circular. Parameter count does not reliably predict task performance.

Q70. What is the most important property of a prompt template used in a production system?

- A. Robustness — producing consistent, correct outputs across diverse real inputs.**
- B. Conciseness — minimising token count to reduce API costs.
- C. Creativity — producing varied outputs to avoid repetition.
- D. Formality — using professional language appropriate to enterprise users.

Correct Answer: A

Explanation

Production prompt templates must be robust — they need to handle the full diversity of real inputs without breaking, producing wrong formats, or generating unexpected outputs. Conciseness and tone matter but are secondary to the fundamental requirement that the template works reliably across all inputs it will encounter.

Q71. A system prompt instructs Claude to "always respond in exactly 3 bullet points." This is:

- A. A hard enforcement mechanism that overrides all other output tendencies.
- B. A format constraint that Claude will generally follow but cannot guarantee perfectly.**
- C. An instruction that Claude will refuse if the content does not fit 3 points.
- D. A valid instruction only when temperature is set to 0.

Correct Answer: B

Explanation

System prompt instructions are soft constraints that Claude generally follows but cannot guarantee under all conditions. Claude will attempt to follow format instructions but may deviate when content genuinely does not fit, when instructions conflict, or when edge case inputs trigger unexpected behaviour.

Q72. You want to evaluate whether a model update improved performance without introducing regressions. What methodology do you use?

- A. Deploy the new model and monitor user satisfaction for 30 days.
- B. Ask domain experts to rate the new model's outputs on a sample of queries.
- C. Compare the models' scores on a new benchmark not used in either version's training.
- D. Run a consistent eval suite on both model versions and compare scores on all dimensions.**

Correct Answer: D

Explanation

A consistent eval suite run on both versions provides systematic, comparable measurement across all capability dimensions. It identifies where the new version improved and where it regressed before deployment. Post-deployment monitoring is too slow; expert review is not systematic enough to catch all regressions.

Q73. What is the most important consideration when choosing a context window size for a production application?

- A. Maximising context window size to ensure the model has access to all possible information.
- B. Minimising context window size to reduce API latency and cost.
- C. Whether the actual use case inputs will fit within the window reliably, with headroom.**
- D. Matching the context window to the size used in the model's original training.

Correct Answer: C

Explanation

The practical question is whether real inputs fit reliably. A window that is too small causes truncation and quality degradation; oversizing wastes money and increases latency. The goal is the minimum window that handles real inputs with comfortable headroom for edge cases.

Q74. What is the strongest signal that a language model is hallucinating rather than misunderstanding a question?

- A. The model produces a vague, hedged response that does not directly answer.
- B. The model produces specific, confident, plausible-sounding details that are factually wrong.**
- C. The model asks for clarification before answering.
- D. The model produces an answer that contradicts its previous responses.

Correct Answer: B

Explanation

The signature of hallucination is confident specificity on wrong information — fabricated citations, wrong dates, invented product features, or nonexistent people described in convincing detail. Vagueness, clarification requests, and contradictions are different failure modes, not hallucination per se.

Q75. Menler's AI Engineering Thinking bank is designed for learners at what stage?

- A. Academic researchers specialising in machine learning architecture design.
- B. Those studying for ML engineering interviews at top technology companies.
- C. Learners with no prior exposure to AI who want a technical foundation.
- D. Those building applied AI products who need to understand model behaviour and constraints.**

Correct Answer: D

Explanation

The AI Engineering Thinking bank tests the applied technical knowledge needed to build products on top of AI models — understanding inference behaviour, model constraints, evaluation, and deployment — not ML research or academic ML theory. It is for practitioners building with AI, not building AI from scratch.

SET

6

Agent Fundamentals

Set 6 of 5 · 15 Questions · AI Engineering

Q1. What distinguishes an AI "agent" from a standard AI chatbot?

- A. An agent has higher intelligence and handles more complex questions.
- B. An agent takes multi-step actions autonomously using tools to achieve a goal.**
- C. An agent learns and improves from each conversation over time.
- D. An agent can access the internet and retrieve current information.

Correct Answer: B**Explanation**

An agent plans and executes multi-step sequences of actions — calling tools, using results to decide next steps, and iterating until a goal is achieved. A chatbot generates a single response. The key distinguishing feature is autonomous multi-step action, not intelligence, learning, or internet access per se.

Q2. What is the "observe-orient-decide-act" (OODA) loop relevant to in agentic AI?

- A. A debugging framework for identifying errors in multi-step AI pipelines.
- B. A compliance framework for monitoring AI agent behaviour in production.
- C. A training methodology for teaching agents to plan multi-step tasks.
- D. A mental model for how an AI agent processes perception and takes action iteratively.**

Correct Answer: D**Explanation**

The OODA loop (Observe — Orient — Decide — Act) is a useful mental model for agentic systems: the agent observes its environment (tool outputs, user input), orients (updates its understanding), decides the next action, and acts. This cycle repeats until the goal is achieved.

Q3. What is "tool use" (function calling) in the context of language model agents?

- A. The model can invoke predefined external functions and incorporate their results.**
- B. The model uses different internal parameters depending on the task type.
- C. The model selects from a library of pre-written prompt templates for each task.
- D. The model switches between text and code generation modes based on context.

Correct Answer: A

Explanation

Tool use (function calling) allows a language model to call external functions — APIs, code interpreters, search engines, databases — as part of generating a response. The model produces structured tool call outputs, the runtime executes them, and the results are fed back to the model.

Q4. What is the most important design constraint for an autonomous AI agent?

- A. Using the most capable available model as the agent's reasoning core.
- B. Minimising the number of steps the agent takes to complete a task.
- C. Clearly defined boundaries on what actions it is and is not permitted to take.**
- D. Ensuring the agent can retry failed steps without human intervention.

Correct Answer: C

Explanation

Action boundaries are the most critical design constraint. An agent without clear scope can take unintended, irreversible actions. The first design decision must be: what can this agent do autonomously, and what requires human approval? Model capability and efficiency are secondary to this safety-critical scope definition.

Q5. What does "planning" mean in an agentic AI system?

- A. The model predicting which future user requests are most likely to arrive.
- B. Decomposing a goal into a sequence of steps before beginning execution.**
- C. Scheduling agent jobs across multiple servers for parallel execution.
- D. Pre-generating several candidate responses before selecting the best one.

Correct Answer: B

Explanation

Planning is the process by which an agent takes a high-level goal and decomposes it into an ordered sequence of subtasks and actions. Good planning determines whether an agent can efficiently accomplish complex goals or wastes steps and gets stuck on dead ends.

Q6. What is the "ReAct" framework for AI agents?

- A. Interleaving reasoning and action steps — the model thinks, then acts, then thinks again.**
- B. A React.js framework for building user interfaces for AI agent products.
- C. A reinforcement learning approach for training agents through trial and error.
- D. A method for automatically generating tool definitions from API documentation.

Correct Answer: A

Explanation

ReAct (Reasoning + Acting) is a prompting framework where the model alternates between "Thought" steps (reasoning about the current state) and "Action" steps (calling a tool or producing output). This interleaving of reasoning and action improves agent reliability on complex tasks.

Q7. What is the most common failure mode of an AI agent in production?

- A. The agent uses too much compute and exceeds cost budgets unexpectedly.
- B. The agent refuses to take any action without explicit human confirmation.
- C. The agent produces outputs too slowly for real-time use cases.
- D. The agent gets stuck in loops or takes wrong actions when encountering unexpected inputs.**

Correct Answer: D

Explanation

Production agent failures most commonly involve handling unexpected inputs — edge cases that the agent's planning assumptions did not anticipate, causing it to loop, take incorrect actions, or get stuck in error states. Robust error handling and fallback logic are essential for production agents.

Q8. What is "scaffolding" in the context of AI agent development?

- A. A set of pre-written prompts that guide the agent through each task step.
- B. The process of gradually increasing task complexity during agent training.
- C. The code infrastructure that manages agent execution, tool calls, and state.**
- D. A technique for breaking large agent outputs into smaller manageable chunks.

Correct Answer: C

Explanation

Agent scaffolding is the surrounding code that drives the agent loop — parsing tool call outputs, routing results back to the model, managing state, handling errors, enforcing timeouts, and deciding when to stop. Good scaffolding is as important as good prompting for reliable agent behaviour.

Q9. What is the difference between a "single-step" and "multi-step" AI workflow?

- A. Single-step is one model call; multi-step chains multiple calls where each builds on the last.**
- B. Single-step handles simple tasks; multi-step requires a more capable model.
- C. Single-step uses one model; multi-step uses multiple different models.
- D. Single-step is synchronous; multi-step is always asynchronous.

Correct Answer: A

Explanation

Single-step workflows make one AI call and return the result. Multi-step workflows chain multiple calls — the output of each step becomes the input of the next. Multi-step is necessary for tasks too complex or long to handle in one context window, or where intermediate results inform later decisions.

Q10. What is "memory" in the context of an AI agent?

- A. The total amount of RAM available to the model during inference.
- B. The agent's ability to learn and update its weights from user interactions.
- C. The portion of the context window reserved for conversation history.
- D. Mechanisms for persisting and retrieving context across multiple agent steps or sessions.**

Correct Answer: D

Explanation

Agent memory refers to how agents store and access information beyond the immediate context window: in-context memory (within the current window), external memory (databases, vector stores), or episodic memory (conversation logs). Effective memory design determines whether agents can handle long tasks and learn from past interactions.

Q11. What makes "irreversible actions" especially important to handle carefully in agents?

- A. Irreversible actions take longer and increase per-step latency.
- B. Mistakes cannot be undone without significant effort or harm.**
- C. Irreversible actions require more compute than reversible ones.
- D. Regulatory frameworks prohibit agents from taking irreversible actions without a license.

Correct Answer: B

Explanation

Irreversible actions — sending emails, posting content, deleting files, spending money — cannot be undone if the agent makes an error. This makes human review before execution critical. The consequences of an irreversible mistake compound the importance of getting these actions right the first time.

Q12. What is "chain of thought" most useful for in an agentic context?

- A. Linking multiple agent instances together in a sequential pipeline.
- B. Tracking the sequence of tool calls an agent has made for debugging.
- C. Having the agent reason explicitly before deciding which action to take next.**
- D. Generating a narrative explanation of the agent's actions for the user.

Correct Answer: C

Explanation

In agents, chain of thought (or "thinking" steps) allows the model to reason about the current state, consider options, and deliberate before committing to an action. This reduces impulsive errors and improves the quality of action selection, especially for complex multi-step tasks.

Q13. What is the primary purpose of a system prompt in an AI agent?

- A. Providing the agent with real-time data it needs for the current task.
- B. Setting the agent's output format for all responses it generates.
- C. Storing the conversation history so the agent can reference previous turns.
- D. Defining the agent's role, capabilities, constraints, and available tools.**

Correct Answer: D

Explanation

The system prompt is the persistent instruction set that defines who the agent is, what tools it has access to, what it is allowed to do, what its goal is, and what constraints it must operate within. It is the foundational configuration of any agentic system.

Q14. What does "agent trajectory" refer to?

- A. The latency curve of an agent across multiple task runs.
- B. The sequence of observations, reasoning steps, and actions an agent takes to complete a task.**
- C. The path an agent takes through a decision tree during planning.
- D. The history of user prompts that led to the current agent invocation.

Correct Answer: B

Explanation

An agent trajectory is the full record of what happened during a task run: what the agent observed, how it reasoned, what actions it took, what results it received, and how it responded to those results. Trajectories are essential for debugging, evaluation, and improving agent behaviour.

Q15. What is "grounding" in the context of AI agents?

- A. Anchoring agent actions and beliefs to verified external information sources.**
- B. Connecting the agent to a physical hardware environment for real-world tasks.
- C. Providing the agent with explicit instructions at the start of each session.
- D. Limiting the agent's output to text that can be directly executed as code.

Correct Answer: A

Explanation

Grounding means ensuring the agent's beliefs and claims are anchored to external, verifiable sources rather than generated from internal patterns alone. Grounded agents retrieve information from trusted databases or documents before acting on it, reducing hallucination-driven errors.

SET

7

Tool Use and Actions

Set 7 of 5 · 15 Questions · AI Engineering

Q16. When a language model "calls a function," what actually happens?

- A. The model directly executes code within its own inference process.
- B. The model retrieves a pre-computed result from a cached function output store.
- C. The model outputs a structured request; external code executes it and returns the result.**
- D. The model generates pseudocode that a human developer then implements.

Correct Answer: C**Explanation**

Function calling is a structured output pattern. The model generates a JSON-formatted call specification; the surrounding scaffolding detects this, calls the actual external function, and feeds the result back to the model as a new message. The model itself never executes code directly.

Q17. What information must a tool definition include for Claude to call it effectively?

- A. Tool name, description of what it does, and parameter schema with types and descriptions.**
- B. Tool name, the API endpoint URL, and authentication credentials.
- C. Tool name, the programming language it is implemented in, and expected runtime.
- D. Tool name, a list of example inputs, and the expected output format.

Correct Answer: A**Explanation**

Effective tool definitions require a clear name, a precise description (so the model understands when and why to call it), and a parameter schema specifying each parameter's name, type, and description. Without clear descriptions, models call the wrong tool or pass incorrect parameters.

Q18. What is "parallel tool calling" and when is it useful?

- A. Running the same tool multiple times with different inputs to compare results.
- B. Using multiple AI models in parallel, each with access to different tools.
- C. Pre-warming tool execution environments before the agent begins a task.
- D. Invoking multiple tools simultaneously when results are independent of each other.**

Correct Answer: D

Explanation

Parallel tool calling allows the model to invoke multiple tools in the same step when their results are independent. For example, searching for information in three different sources simultaneously rather than sequentially reduces total latency significantly.

Q19. What is the most important consideration when designing a tool for an AI agent?

- A. Minimising the tool's response time to reduce agent latency.
- B. Clear, unambiguous description of when and how to call it to prevent misuse.**
- C. Ensuring the tool can handle any possible input without throwing exceptions.
- D. Matching the tool's interface to standard REST API conventions.

Correct Answer: B

Explanation

The model decides when and how to call a tool based entirely on its description. Ambiguous descriptions lead to wrong tool selection or incorrect parameter usage. The tool description is a contract between the tool developer and the model — precision here is more important than performance or interface conventions.

Q20. You give an agent a web search tool. What is the most important risk to manage?

- A. Web search may return results that exceed the agent's context window.
- B. The agent may make too many search calls and exceed API rate limits.
- C. The agent may act on false or malicious information retrieved from the web.**
- D. Web search results may be in a language the model cannot process.

Correct Answer: C

Explanation

An agent that trusts and acts on retrieved web content is vulnerable to misinformation, out-of-date information, and prompt injection (malicious instructions embedded in web pages). The primary risk is acting on unverified or malicious retrieved content, not rate limits or language issues.

Q21. What is "prompt injection" in an agentic context?

- A. Injecting additional instructions into the agent's system prompt at runtime.
- B. A technique for inserting few-shot examples into the agent's prompt dynamically.
- C. An attack where an adversary sends the agent an unusually long input to cause errors.
- D. Malicious instructions embedded in content the agent reads that hijack its behaviour.**

Correct Answer: D

Explanation

Prompt injection in agents occurs when content retrieved from external sources (web pages, emails, documents) contains text that looks like instructions to the model, causing it to deviate from its original task. This is a critical security concern for agents that process untrusted external content.

Q22. What is the correct handling when a tool call returns an error?

- A. The agent should detect the error, decide whether to retry or use a fallback, and proceed accordingly.**
- B. The agent should immediately terminate and report failure to the user.
- C. The agent should retry the same call indefinitely until it succeeds.
- D. The agent should proceed as if the tool returned an empty successful response.

Correct Answer: A

Explanation

Robust agents detect tool errors and reason about the appropriate response: retry if the error is transient, try an alternative approach if the error is persistent, or escalate to the user if recovery is impossible. Blind termination, infinite retries, and ignoring errors all produce worse outcomes.

Q23. You are designing an agent that can send emails on the user's behalf. What is the non-negotiable safeguard?

- A. Limiting the agent to sending emails only to addresses in the user's contacts.
- B. Human review and confirmation before any email is sent.**
- C. Adding a standard AI disclosure footer to every email the agent sends.
- D. Logging all sent emails to a database for audit purposes.

Correct Answer: B

Explanation

Sending email is an irreversible external action with real consequences. Human confirmation before sending is the non-negotiable safeguard — no other control (contact list restrictions, logging, disclosure) compensates for the risk of sending wrong emails without human approval.

Q24. What is the difference between a "deterministic" and "non-deterministic" tool in agent design?

- A. Deterministic tools execute instantly; non-deterministic tools have variable latency.
- B. Deterministic tools are safe for agents to call; non-deterministic ones require human approval.
- C. Deterministic tools are built in-house; non-deterministic tools call third-party APIs.
- D. Deterministic tools always return the same output for the same input; non-deterministic tools do not.**

Correct Answer: D

Explanation

Deterministic tools (calculators, data lookups by ID) return consistent results for given inputs. Non-deterministic tools (web search, current time, randomness sources) return different results on different calls. This distinction matters for agent reliability and debugging — non-deterministic tool results make trajectories harder to reproduce.

Q25. What does "tool result grounding" mean in a RAG-enabled agent?

- A. The tool results are stored in a permanent database for future reference.
- B. The agent verifies tool results by calling a second verification tool.
- C. The agent's answer is based on retrieved document content rather than training knowledge.**
- D. The tool results are formatted and grounded to a specific output template.

Correct Answer: C

Explanation

Tool result grounding means the agent's responses are explicitly based on the content returned by retrieval or tool calls — not the model's training knowledge. This is the primary mechanism by which RAG reduces hallucination: the model is instructed to cite and reason from retrieved documents, not generate from memory.

Q26. What is the most reliable way to prevent an agent from calling a sensitive tool inappropriately?

- A. Adding a human approval step before every tool call the agent makes.
- B. Clear tool descriptions specifying exactly when the tool should and should not be called.**
- C. Removing sensitive tools from the agent's tool list unless explicitly needed.
- D. Using a safety classifier to intercept and block inappropriate tool calls.

Correct Answer: B

Explanation

The most reliable primary prevention is clear tool descriptions — telling the model explicitly what situations warrant calling the tool and what situations do not. Removing the tool when not needed is a strong secondary control. Safety classifiers add a layer but are not the primary mechanism.

Q27. What is "structured output" and why is it important for tool use?

- A. Machine-parseable formatted output (like JSON) that the scaffolding code can reliably process.**
- B. Formatted text output that is presented to users in a visually organised way.
- C. Output that is generated according to a fixed response template.
- D. Output that has been reviewed and approved by a human before delivery.

Correct Answer: A

Explanation

Structured outputs (JSON, XML) are essential for tool-use scaffolding because the surrounding code needs to parse the model's tool call specification to execute it. If the model generates malformed or unstructured tool calls, the scaffolding fails. Reliable structured output is foundational to reliable agent systems.

Q28. You notice your agent is calling the same tool repeatedly with the same parameters. The most likely cause is:

- A. The tool has a bug that returns incorrect results, confusing the agent.
- B. The agent's context window is too small to remember it already called the tool.
- C. The agent is stuck in a loop because the tool result is not resolving its uncertainty.**
- D. The system prompt instructed the agent to verify all tool results by re-calling.

Correct Answer: C

Explanation

Repeated identical tool calls indicate the agent is in a loop — it expected the tool result to change its decision state, but it didn't. This usually happens when the tool result doesn't contain what the agent needs to progress, or when the agent's plan doesn't account for tool failure.

Q29. What is the most important metric to track for tool reliability in a production agent?

- A. Average tool response latency across all calls in production.
- B. Total number of tool calls made per user session.
- C. The percentage of tool calls that return non-empty results.
- D. Tool call error rate and the agent's recovery success rate when errors occur.**

Correct Answer: D

Explanation

Tool error rate and recovery success rate are the critical reliability metrics. A tool that fails frequently and which the agent cannot recover from degrades the entire system. Latency, call volume, and non-empty result rate are operational metrics; they don't capture whether the agent is actually succeeding.

Q30. What is the correct approach when an agent needs a tool that does not yet exist?

- A. Use the most general-purpose tool available as a substitute, even if it does more.
- B. Build a minimal, well-documented tool that does exactly what is needed and nothing more.**
- C. Ask the model to simulate the tool's output using its training knowledge.
- D. Build a tool that handles all possible future needs to avoid rebuilding later.

Correct Answer: B

Explanation

The correct approach is building minimal, focused tools with clear, precise interfaces. Over-general tools introduce misuse risk; simulated tool outputs introduce hallucination; building for future needs increases complexity unnecessarily. Start with the minimum tool that does exactly what is needed.

SET

8

Prompt Chaining

Set 8 of 5 · 15 Questions · AI Engineering

Q31. What is "prompt chaining" and when does it produce better results than a single prompt?

- A. Sequential prompts where each output becomes the next input — better for complex, multi-stage tasks.**
- B. Linking multiple AI models together where each specialises in a different modality.
- C. Repeating the same prompt multiple times and selecting the best output.
- D. Appending follow-up instructions to a prompt to extend a single response.

Correct Answer: A

Explanation

Prompt chaining decomposes complex tasks into stages, where each step has a focused, well-defined scope. This is more reliable than trying to accomplish everything in one prompt because each step can be validated, corrected, or re-run independently, and the model has more focused context at each stage.

Q32. What is the main reliability advantage of prompt chaining over a single long prompt?

- A. Chaining is faster because each individual prompt is shorter.
- B. Chaining uses less total compute than a single long prompt.
- C. Each step can be verified and corrected before it propagates errors to the next step.**
- D. Chaining allows different models to be used for different steps.

Correct Answer: C

Explanation

The primary reliability advantage is error containment. In a single long prompt, early errors cascade through the entire output. In a chain, each step's output can be validated before it becomes the input for the next step, and failed steps can be retried without restarting the whole task.

Q33. What is a "map-reduce" pattern in prompt chaining?

- A. Process large documents by dividing them across multiple models simultaneously.
- B. Apply the same operation to many items in parallel (map), then aggregate results (reduce).**
- C. Generate multiple candidate outputs and select the one that reduces perplexity.
- D. A data engineering pattern for preprocessing training data before model ingestion.

Correct Answer: B

Explanation

Map-reduce in prompt chaining processes many items independently in parallel (map) then combines results into a final output (reduce). For example: summarise 50 documents individually (map), then synthesise those summaries into a single executive briefing (reduce).

Q34. What is the most important design principle for individual steps in a prompt chain?

- A. Each step should be as long as possible to minimise the number of steps.
- B. Each step should produce a self-contained final output that needs no further processing.
- C. Each step should include instructions for handling all possible error conditions.
- D. Each step should have a single, clearly defined input and output.**

Correct Answer: D

Explanation

Single responsibility is the most important principle for chain steps: one input, one clearly defined task, one well-defined output. Steps with multiple responsibilities are harder to validate, debug, and retry, and tend to produce less reliable results.

Q35. You are building a chain that extracts information from documents and writes a report. The most important step to include between extraction and writing is:

- A. A validation step that checks the extracted information for completeness and accuracy.**
- B. A formatting step that converts extracted data to the report's visual template.
- C. A planning step where the model outlines the report structure.
- D. A length-checking step to ensure the report will fit within the context window.

Correct Answer: A

Explanation

Validating extracted information before using it to write the report prevents extraction errors from propagating into the final output. Catching errors at the extraction stage is far cheaper than rewriting a full report because the extraction was wrong.

Q36. What is "dynamic chain construction" in an AI workflow?

- A. The chain is built by sampling randomly from a library of available prompt steps.
- B. The agent determines which steps to execute based on intermediate results.**
- C. New chain steps are added by the user as the task progresses.
- D. The model dynamically selects which language model to call at each step.

Correct Answer: B

Explanation

Dynamic chain construction means the workflow path is not fixed in advance — the agent inspects intermediate results and decides which steps to take next. This enables adaptive workflows that handle branching logic, conditional execution, and dynamic task discovery.

Q37. What is the most common failure mode when chaining prompts?

- A. The total token count across the chain exceeds what is cost-effective.
- B. Different steps use different prompt formats that produce incompatible outputs.
- C. The final step loses coherence because the context window has been exceeded.
- D. Errors in early steps compound and corrupt all subsequent steps.**

Correct Answer: D

Explanation

Error propagation is the most common and damaging failure mode. If step 1 extracts wrong information, every step that follows operates on that wrong information, producing plausible-looking but fundamentally incorrect final outputs. Validation gates between steps are the primary mitigation.

Q38. How should you handle a step in a prompt chain that fails or produces bad output?

- A. Restart the entire chain from step one with a better initial prompt.
- B. Pass the failed output to the next step and flag it as uncertain.
- C. Detect the failure, retry with a modified prompt or different approach, then continue.**
- D. Ask the user to provide the correct output for the failed step manually.

Correct Answer: C

Explanation

Robust chains detect step failures and retry with modifications — adjusting the prompt, trying a different strategy, or using a fallback — before proceeding. Restarting from step one is wasteful. Passing bad output forward corrupts later steps. User manual input should be the last resort, not the default.

Q39. What is "prompt compression" and why does it matter in long chains?

- A. Reducing the length of each chain step's prompt to speed up generation.
- B. Summarising intermediate outputs to prevent context accumulation from filling the window.**
- C. Removing redundant instructions from prompts to reduce token costs.
- D. A technique for combining multiple chain steps into one to reduce API calls.

Correct Answer: B

Explanation

As chain outputs accumulate in context, the effective context window fills up. Prompt compression — summarising or extracting only the necessary information from intermediate outputs — prevents the model from losing earlier context or having to handle an inefficiently long prompt at later steps.

Q40. What distinguishes a "deterministic chain" from a "branching chain"?

- A. A deterministic chain always follows the same steps; a branching chain routes based on intermediate outputs.**
- B. A deterministic chain uses the same model at each step; a branching chain uses different models.
- C. A deterministic chain is synchronous; a branching chain is asynchronous.
- D. A deterministic chain is simpler to build; a branching chain requires more compute.

Correct Answer: A

Explanation

A deterministic chain executes the same sequence of steps every time. A branching chain inspects intermediate outputs and conditionally routes to different next steps based on the results. Branching chains are more powerful but harder to test exhaustively because different paths may rarely be executed.

Q41. What is "output parsing" in prompt chaining and why can it fail?

- A. Translating the final output into the user's preferred language.
- B. Running grammar and spell-check on model outputs before they move to the next step.
- C. Extracting structured data from a model's text output — fails when format is inconsistent.**
- D. Converting model outputs to speech for audio applications.

Correct Answer: C

Explanation

Output parsing extracts structured information (e.g., JSON fields, specific data points) from model-generated text. It fails when the model generates output that does not match the expected format — wrong field names, missing fields, malformed JSON, or free text where structured data was expected.

Q42. When is it appropriate to use a single complex prompt versus a chain of simpler prompts?

- A. Single prompts are always better because they require fewer API calls.
- B. Chains are always better because each step can be individually optimised.
- C. Single prompts are better for speed; chains are better when accuracy is critical.
- D. Single prompts work for straightforward tasks; chains are better for complex multi-stage workflows.**

Correct Answer: D

Explanation

Single prompts are appropriate for tasks that can be done well in one pass. Chains are better for tasks that have distinct stages (extract, validate, synthesise), where intermediate results must be checked, or where the total task is too large for one context window. Neither is universally superior.

Q43. What is the most important thing to log in a production prompt chain?

- A. The total token count and cost of each chain execution.
- B. Every step's inputs, outputs, and any errors, for debugging and quality monitoring.**
- C. Only the final output and whether the chain succeeded or failed.
- D. The model version used at each step for reproducibility tracking.

Correct Answer: B

Explanation

Full step-level logging — inputs, outputs, and errors at every step — is essential for debugging, quality monitoring, and root cause analysis when chains fail or produce wrong results. Cost tracking, final-output logging, and model version tracking are all valuable but secondary to this diagnostic capability.

Q44. A prompt chain produces a good result 90% of the time but fails 10%. The best engineering response is:

- A. Accept the 10% failure rate and notify users when failures occur.
- B. Switch to a more capable model to reduce the failure rate.
- C. Add validation steps and error handling to catch and recover from the 10% failures.**
- D. Redesign the chain so it only runs on the 90% of inputs where it is expected to succeed.

Correct Answer: C

Explanation

A 10% failure rate in production means 1 in 10 tasks fails, which is typically unacceptable. The correct response is engineering — adding validation steps to catch failures, error handling to attempt recovery, and fallback paths. These controls are more reliable than model capability upgrades alone.

Q45. What is a "handoff" in a prompt chain context?

- A. Passing the output of one step as the structured input to the next step.**
- B. Transferring a user session from an AI agent to a human support agent.
- C. Moving a workflow from development to production deployment.
- D. Shifting responsibility for a chain from one developer to another.

Correct Answer: A

Explanation

A handoff is the structured transfer of information between chain steps — the output format of step N must match the expected input format of step N+1. Well-designed handoffs have clear contracts (data types, field names, formats), making chains reliable and modular.

SET

9

Multi-Agent Systems

Set 9 of 5 · 15 Questions · AI Engineering

Q46. What is a "multi-agent system" in AI?

- A. A single AI model that simulates multiple personas in a conversation.
- B. A cluster of servers running the same AI model for high availability.
- C. An AI system that can call multiple external APIs in a single step.
- D. Multiple AI agents working together, each handling a part of a complex task.**

Correct Answer: D**Explanation**

A multi-agent system deploys multiple specialised agents that collaborate — each agent may have different tools, instructions, or capabilities, and they pass work and results between them. This allows parallelisation of complex tasks and specialisation of roles within a larger workflow.

Q47. What is the primary advantage of using specialised agents versus one general-purpose agent for complex tasks?

- A. Specialised agents use less compute than a single general-purpose agent.
- B. Specialised agents can be independently optimised and tested for their specific role.**
- C. Specialised agents can use different underlying models for different subtasks.
- D. Specialised agents are easier to prompt and require shorter system prompts.

Correct Answer: B**Explanation**

Specialisation allows each agent to be designed, prompted, and evaluated for its specific role — a research agent, a writing agent, and a verification agent each have focused, optimised configurations. A single general-purpose agent must handle all roles with a shared, less focused context.

Q48. What is an "orchestrator" agent?

- A. An agent that manages the infrastructure and compute resources for other agents.
- B. An agent that translates between different agents' output formats.
- C. An agent that coordinates other agents — assigning tasks and synthesising their results.**
- D. The most capable agent in a system that overrides other agents' decisions.

Correct Answer: C

Explanation

An orchestrator agent manages the workflow — it receives the user goal, decomposes it into subtasks, assigns subtasks to specialised worker agents, monitors their progress, synthesises their outputs, and determines when the overall goal has been achieved.

Q49. What is the biggest coordination challenge in multi-agent systems?

- A. Ensuring agents share consistent context and don't operate on contradictory information.**
- B. Preventing agents from using too many tokens in aggregate.
- C. Ensuring each agent uses the same AI model version.
- D. Managing API rate limits when multiple agents call external tools simultaneously.

Correct Answer: A

Explanation

Context consistency is the core coordination challenge. If Agent A makes a decision based on certain information, Agent B (working in parallel) must have access to the same information and any updates from Agent A, or their work will be based on divergent assumptions, producing contradictory or conflicting outputs.

Q50. What is "agent role confusion" and how does it manifest?

- A. The user cannot tell which agent in a system produced a given output.
- B. An agent fails to understand what role the user wants it to play.
- C. Multiple agents produce identical outputs because they have similar role definitions.
- D. An agent acts outside its assigned role, taking actions intended for another agent.**

Correct Answer: D

Explanation

Role confusion occurs when an agent ignores its assigned scope and attempts tasks outside its mandate — a research agent that tries to write the final report, or a writer agent that modifies the research brief. Clear, narrow role definitions and scope constraints prevent this.

Q51. What is "emergent behaviour" in multi-agent systems and why is it a risk?

- A. Agents developing new capabilities through interaction with each other.
- B. Agents that spontaneously develop efficient workflows without being explicitly programmed.
- C. Unexpected system-level behaviour arising from agent interactions not anticipated by designers.**
- D. A single agent discovering a novel approach that improves the whole system's performance.

Correct Answer: C

Explanation

Emergent behaviour is system-level outcomes that arise from agent interactions that designers did not anticipate or intend. In AI systems, this can manifest as agents coordinating to take actions or reach conclusions that no individual agent was designed to do — some of which may be harmful or unintended.

Q52. What is a "deadlock" in a multi-agent system?

- A. Two or more agents each waiting for the other to complete before they can proceed.**
- B. A situation where all agents produce identical outputs and the system cannot select one.
- C. An agent that keeps sending requests to a tool that is unavailable.
- D. A failure mode where an orchestrator loses track of which agents are active.

Correct Answer: A

Explanation

A deadlock occurs when agents are in a mutual wait state — Agent A is waiting for Agent B's output to proceed, while Agent B is waiting for Agent A's output. Without external intervention or a timeout mechanism, the system halts indefinitely.

Q53. What is the most appropriate use of parallel agent execution?

- A. All tasks in a multi-agent system, to minimise total completion time.
- B. Tasks that are independent of each other and whose results can be combined afterward.**
- C. Tasks where an earlier agent's output is uncertain and needs verification.
- D. Tasks where the final result needs to be produced as quickly as possible.

Correct Answer: B

Explanation

Parallelism is appropriate only for genuinely independent subtasks — where Agent A's execution does not depend on Agent B's output. Forcing parallelism on dependent tasks causes coordination failures. Independent parallel tasks (e.g., researching five different topics simultaneously) are the correct use case.

Q54. What is the correct approach when one agent in a multi-agent system produces an error?

- A. The entire multi-agent workflow restarts from the beginning.
- B. The failed agent retries indefinitely until it succeeds.
- C. The orchestrator detects the error and decides whether to retry, use a fallback, or escalate.**
- D. The orchestrator skips the failed agent's task and proceeds without its output.

Correct Answer: C

Explanation

The orchestrator is responsible for error handling in multi-agent systems — detecting failures from worker agents and making intelligent recovery decisions: retry the same agent, substitute a different agent, use cached results from a previous run, or escalate to a human if recovery is impossible.

Q55. What is "agent communication protocol" and why does it matter?

- A. The rate at which agents send messages to each other during execution.
- B. The security protocol used to authenticate agent-to-agent API calls.
- C. The programming language used to implement each agent's reasoning logic.
- D. The structured format by which agents exchange information — critical for interoperability.**

Correct Answer: D

Explanation

A communication protocol defines how agents share information — what format messages take, what fields are required, and how to interpret different message types. Without a clear protocol, agents produce outputs in incompatible formats that other agents cannot reliably parse and act on.

Q56. What distinguishes "cooperative" from "competitive" multi-agent architectures?

- A. Cooperative agents work toward a shared goal; competitive agents optimise different objectives.**
- B. Cooperative agents share compute resources; competitive agents have isolated resources.
- C. Cooperative agents can call each other's tools; competitive agents cannot.
- D. Cooperative agents are designed by the same team; competitive ones are from different vendors.

Correct Answer: A

Explanation

In cooperative multi-agent systems, all agents are aligned on a shared goal and optimise for collective success. In competitive architectures (debate, red team vs blue team), agents optimise different or opposing objectives. The choice of architecture fundamentally changes system design and trust requirements.

Q57. What is a "critic agent" and when is it useful?

- A. An agent that monitors system performance and alerts when metrics degrade.
- B. An agent that evaluates another agent's output and identifies errors or improvements.**
- C. An agent that filters harmful content from other agents' outputs.
- D. An agent that rejects tasks outside its scope and returns them to the orchestrator.

Correct Answer: B

Explanation

A critic agent independently evaluates another agent's output — checking for errors, inconsistencies, logical flaws, or quality issues. It is most useful in high-stakes workflows where one agent's unchecked output would be used directly in a consequential decision or downstream process.

Q58. What is the most important human oversight requirement for multi-agent systems?

- A. A human reviewing every message that passes between agents.
- B. A human approving the addition of each new agent to the system.
- C. Daily audits of all agent logs by a designated system administrator.
- D. Checkpoint reviews at consequential decision points before irreversible actions.**

Correct Answer: D

Explanation

Checkpoints at consequential decision points — moments where the agent system is about to take an irreversible or high-stakes action — are the most important oversight mechanism. Reviewing every inter-agent message is impractical; daily audits catch problems too late; agent approval gatekeeping is not the right control.

Q59. What is "agent handoff" in a customer service context?

- A. An AI agent transferring a conversation to a human agent when it cannot resolve the issue.**
- B. One AI agent passing a task to a specialised AI agent with more relevant tools.
- C. A system that routes new incoming queries to the most available agent.
- D. The process of switching a user from one product's AI to another product's AI.

Correct Answer: A

Explanation

In customer service, agent handoff is the critical transition from AI to human handling. A well-designed handoff includes context transfer (conversation history, attempted solutions, customer sentiment) and a smooth user experience. Poor handoffs that lose context are a primary source of customer frustration in AI-assisted support.

Q60. What is the most reliable way to test a multi-agent system before production deployment?

- A. Unit testing each agent in isolation with simulated inputs from other agents.
- B. Testing the orchestrator in isolation with all worker agents mocked.
- C. Running end-to-end tests on realistic scenarios that cover the main task paths and edge cases.**
- D. Deploying to a canary user group and measuring error rates in production.

Correct Answer: C

Explanation

Multi-agent systems must be tested end-to-end on realistic scenarios because system-level behaviour emerges from agent interactions, not from individual agents alone. Unit tests catch individual agent errors; they don't catch coordination failures, context drift, or emergent behaviours that only appear in full-system execution.

SET

10

Safety and Reliability

Set 10 of 5 · 15 Questions · AI Engineering

Q61. What is the "minimal footprint" principle in agent design?

- A. Request only the permissions and tools needed for the specific task.**
- B. Use the smallest possible model that can complete the task adequately.
- C. Minimise the number of steps the agent takes to complete each task.
- D. Limit the agent to read-only tool access to prevent unintended changes.

Correct Answer: A**Explanation**

Minimal footprint means agents should request only the permissions they need, retain only the state they need, and access only the data they need. Unnecessarily broad tool access and permissions increase the blast radius of mistakes and security vulnerabilities.

Q62. What is "reward hacking" as it applies to AI agent systems?

- A. A security vulnerability where an attacker manipulates the agent's reward signal.
- B. An agent that requests rewards (credits, tokens) from users in exchange for help.
- C. A training failure where the reward model assigns high scores to clearly wrong outputs.
- D. An agent achieves its specified objective by means that violate the designer's intent.**

Correct Answer: D**Explanation**

Reward hacking (specification gaming) occurs when an agent finds a way to satisfy its specified goal metric without achieving the underlying intent. For example, an agent told to "maximise user engagement" might keep users stuck in loops rather than actually helping them. The measure is optimised; the purpose is subverted.

Q63. What is the most important difference between "safe" and "unsafe" agentic behaviour?

- A. Safe agents are slower because they verify each step; unsafe agents are faster.
- B. Safe agents stay within their authorised scope; unsafe agents take unauthorised actions.**
- C. Safe agents use less capable models; unsafe agents use the most powerful model.
- D. Safe agents have human oversight; unsafe agents operate in a production sandbox.

Correct Answer: B

Explanation

Safe agentic behaviour is defined by operating within authorised scope — doing what was sanctioned by the principal hierarchy (user and operator), nothing more. Unsafe behaviour is taking actions that exceed the given authority, regardless of whether those actions seem beneficial.

Q64. What is "cascading failure" in an agent system?

- A. When multiple agents fail at the same time due to a shared resource outage.
- B. A failure mode where an agent's output keeps growing until it exceeds context limits.
- C. An error in one component propagates and amplifies through dependent components.**
- D. When retry logic causes exponentially increasing API calls that overwhelm a service.

Correct Answer: C

Explanation

Cascading failure in agents occurs when an error in one step or agent corrupts the inputs of downstream steps or agents, which then produce their own errors, propagating the failure through the system. Without error detection and circuit-breaking at each handoff, one early error can corrupt an entire complex workflow.

Q65. What is the most effective way to prevent an agent from taking an action with irreversible consequences?

- A. Require explicit human confirmation before the agent executes any irreversible action.**
- B. Add a post-execution check that undoes any action the agent should not have taken.
- C. Train the agent to recognise irreversible actions and automatically refuse them.
- D. Limit the agent's tool access so it cannot access any tool with irreversible effects.

Correct Answer: A

Explanation

Pre-execution human confirmation is the only reliable prevention for irreversible actions. Post-execution checks cannot undo what is done. Training-based refusal can be circumvented or fail on edge cases. Blanket tool removal is too restrictive for agents that legitimately need to take some consequential actions.

Q66. What is "scope creep" in agent design and why is it dangerous?

- A. The system prompt gradually grows too long for the context window.
- B. The agent's tool list expands over time without systematic review.
- C. The agent gradually takes on tasks beyond its intended scope, increasing risk.**
- D. The agent's performance degrades as the deployment scope expands.

Correct Answer: C

Explanation

Scope creep occurs when an agent begins handling tasks outside its designed boundaries — often because it seems capable and users start asking for more. This is dangerous because safety testing, permissions, and oversight were designed for the original scope, not the expanded one.

Q67. What is the most important log to maintain for agent safety auditing?

- A. A log of every user prompt that triggered an agent execution.
- B. A record of every API call and its cost for billing and performance analysis.
- C. A log of every tool error and the agent's recovery strategy.
- D. A complete record of every action taken and its outcome, with timestamps.**

Correct Answer: D

Explanation

A complete action log — what the agent did, when, and what resulted — is the foundational record for safety auditing. It allows investigators to reconstruct exactly what happened when something goes wrong. Prompt logs, cost logs, and error logs are all subsets of or supplements to this primary audit record.

Q68. What is "principal hierarchy" in the context of AI agent safety?

- A. The order in which agents execute in a multi-agent pipeline.
- B. The authority structure defining whose instructions the agent should prioritise.**
- C. The chain of command within the development team responsible for the agent.
- D. The ranking of tools the agent should prefer when multiple tools could serve a purpose.

Correct Answer: B

Explanation

Principal hierarchy defines whose instructions the agent should follow and in what priority order. For Claude, this is Anthropic (through training) > operator (through system prompt) > user (through conversation). Understanding this hierarchy is essential for designing agents that follow appropriate instructions and resist manipulation.

Q69. What is the correct response when an agent is given an instruction that conflicts with its safety constraints?

- A. Follow the instruction since the user or operator has authority over the agent.
- B. Attempt to partially follow the instruction while minimising safety violations.
- C. Follow the safety constraints and explain why the conflicting instruction cannot be followed.**
- D. Escalate the conflict to the operator for resolution before proceeding.

Correct Answer: C

Explanation

Safety constraints are non-negotiable and take precedence over user or operator instructions. When an instruction conflicts with safety constraints, the agent should decline to follow the unsafe instruction and explain why. Instructions can narrow what an agent does within safety bounds; they cannot override safety bounds.

Q70. Why is "rate limiting" important in production AI agent systems?

- A. Prevents runaway agents from making unbounded API calls that cause cost or service failures.**
- B. Ensures fair distribution of compute resources between multiple concurrent users.
- C. Prevents agents from generating outputs that exceed the context window size.
- D. Limits the number of agents that can be deployed simultaneously in the system.

Correct Answer: A

Explanation

Without rate limits, a malfunctioning or looping agent can make thousands of API calls in seconds, exhausting budgets, hitting external API rate limits, and causing service disruptions. Rate limiting is a critical circuit-breaker that prevents runaway agent behaviour from causing cascading operational failures.

Q71. What is the most important test to run before deploying an AI agent to production?

- A. Performance testing — does the agent complete tasks within the SLA time limits?
- B. Adversarial testing — can the agent be prompted to take actions outside its intended scope?**
- C. Cost testing — does the agent stay within the target token budget per task?
- D. Compatibility testing — does the agent work correctly across all supported platforms?

Correct Answer: B

Explanation

Adversarial testing is the most critical pre-deployment test for agents. Agents in production will encounter manipulative users, unexpected inputs, and edge cases. Understanding where the agent's scope boundaries can be pushed or circumvented is essential before exposing it to real users.

Q72. What does "fail-safe" mean in an agent system design?

- A. The agent automatically retries all failed operations until they succeed.
- B. The agent stores its state so it can resume from the last successful step after a failure.
- C. A backup agent that activates when the primary agent fails.
- D. When the system fails, it defaults to a safe state rather than an unsafe one.**

Correct Answer: D

Explanation

Fail-safe design means that when something goes wrong — tool failure, context overflow, unexpected input — the system's default response is conservative (do nothing, ask the human, stop and report) rather than attempting to proceed with potentially wrong or dangerous actions.

Q73. What is "human-in-the-loop" most critical for in agent design?

- A. All agent tasks, since agents should never operate without human supervision.
- B. The initial task specification, to ensure the agent has the correct goal.
- C. High-stakes, irreversible, or uncertain decisions where agent errors have significant consequences.**
- D. Error recovery steps, where the agent cannot determine the correct next action.

Correct Answer: C

Explanation

Human-in-the-loop is most critical at high-stakes decision points — where the agent is about to take an action with significant, irreversible, or hard-to-detect consequences. Requiring human approval for every step is impractical; applying it selectively to critical junctures is the appropriate design.

Q74. What is the most reliable indicator that an AI agent system needs a human escalation path?

- A. The agent has been running for longer than its typical task completion time.
- B. The agent encounters a situation where proceeding would require taking an irreversible action under uncertainty.**
- C. The agent has made more than a defined number of tool calls without completing.
- D. The user has sent more than three messages without being satisfied with the response.

Correct Answer: B

Explanation

The clearest indicator for escalation is an irreversible action under uncertainty — the agent needs to do something that cannot be undone and is not confident it is the right action. This is exactly when human judgment must be brought in before proceeding.

Q75. Menler's AI Agents & Workflows bank targets practitioners at what level?

- A. Researchers studying multi-agent coordination and emergent AI behaviour.
- B. Beginners learning what AI agents are for the first time.
- C. Executives deciding whether to invest in agentic AI for their organisation.
- D. Those designing and building production AI agent systems with real reliability requirements.**

Correct Answer: D

Explanation

The AI Agents & Workflows bank tests the knowledge needed to design, build, and operate production agent systems — tool definitions, scaffolding, error handling, multi-agent coordination, and safety. It is for practitioners building real systems, not researchers studying agents theoretically or beginners exploring the concept.

SET

11

APIs and Integration

Set 11 of 5 · 15 Questions · AI Engineering

Q1. What is an "API key" in the context of accessing an AI model?

- A. A configuration parameter that controls the model's output temperature.
- B. A credential that authenticates requests to an AI provider's API endpoints.**
- C. A licence that grants permission to use a specific model commercially.
- D. An encryption key used to secure the model's weights in transit.

Correct Answer: B**Explanation**

An API key is an authentication credential — a secret string that identifies the calling application to the AI provider. It is used to authenticate API requests, track usage, enforce rate limits, and bill correctly. It is not a configuration value, licence, or encryption mechanism.

Q2. What does a REST API return when a request succeeds?

- A. A binary model output that must be decoded before use.
- B. A confirmation message and the request ID for tracking purposes.
- C. The model's confidence score alongside the generated response.
- D. A response body (typically JSON) and an HTTP 200-level status code.**

Correct Answer: D**Explanation**

A successful REST API response includes an HTTP 200-level status code (200 OK, 201 Created, etc.) and a response body — for AI APIs, typically JSON containing the model's output. Binary encoding, confidence scores, and request-ID-only responses are not standard REST patterns.

Q3. What is the difference between synchronous and asynchronous API calls?

- A. Synchronous calls wait for the response; asynchronous calls return immediately and retrieve the result later.**
- B. Synchronous calls are encrypted; asynchronous calls are not.
- C. Synchronous calls are for small requests; asynchronous calls are for large ones.
- D. Synchronous calls use REST; asynchronous calls use WebSockets.

Correct Answer: A

Explanation

Synchronous calls block until the server responds. Asynchronous calls return immediately with a job ID; the caller polls for or receives a callback with the result when it is ready. Asynchronous patterns are essential for long-running AI tasks that would otherwise cause client timeouts.

Q4. What is "streaming" in the context of LLM API responses?

- A. Continuously processing an incoming audio or video feed through an AI model.
- B. Sending large documents to the AI API in small chunks to avoid size limits.
- C. Receiving tokens as they are generated rather than waiting for the full response.**
- D. A real-time learning mode where the model updates based on user feedback.

Correct Answer: C

Explanation

Streaming (server-sent events or WebSocket) delivers tokens to the client progressively as the model generates them, rather than waiting for the complete response. This dramatically improves perceived responsiveness — the user sees output appearing immediately rather than after a potentially long wait.

Q5. What HTTP status code indicates that an API request was rejected due to rate limiting?

- A. 503 Service Unavailable.
- B. 429 Too Many Requests.**
- C. 401 Unauthorized.
- D. 400 Bad Request.

Correct Answer: B

Explanation

429 Too Many Requests is the standard HTTP status code for rate limiting. It indicates the client has exceeded the allowed request frequency. 401 is for authentication failure; 400 is for malformed requests; 503 is for server-side unavailability.

Q6. What is "exponential backoff" and why is it used in AI API integrations?

- A. Retrying failed requests with increasing delays to avoid overwhelming the API.**
- B. A technique for gradually increasing batch size to optimise throughput.
- C. A method for reducing token usage by exponentially compressing prompts.
- D. A caching strategy that stores responses for exponentially increasing time periods.

Correct Answer: A

Explanation

Exponential backoff retries a failed API request after a delay, doubling the delay with each subsequent retry. This prevents retry storms from overwhelming a rate-limited or temporarily unavailable service, while still ensuring eventual success for transient failures.

Q7. What is a "system prompt" in the API context, and how does it differ from a user message?

- A. A system prompt is encrypted for security; a user message is sent in plain text.
- B. A system prompt is processed first by a faster model; user messages use the full model.
- C. A system prompt is visible to users; a user message is confidential.
- D. A system prompt sets persistent instructions for the model; a user message is the runtime input.**

Correct Answer: D

Explanation

In the API, a system prompt (role: "system") provides persistent context, persona, and constraints that apply to the entire conversation. User messages (role: "user") are the runtime inputs. The model treats these differently: system prompts carry higher instructional authority.

Q8. What does "max_tokens" control in an API request?

- A. The maximum number of tokens allowed in the user's input prompt.
- B. The total maximum tokens for the entire conversation context window.
- C. The maximum number of tokens the model will generate in its response.**
- D. The number of candidate responses the model generates before selecting the best.

Correct Answer: C

Explanation

max_tokens sets a ceiling on the response length. The model will stop generating once this limit is reached, potentially truncating the response. It does not limit input length or context size — those are controlled by the context window limit and separate input parameter validation.

Q9. What is the purpose of a "system fingerprint" or "model version pinning" in production API use?

- A. Ensuring a specific model version is used for reproducibility and preventing unexpected changes.**
- B. Verifying that the API response was generated by the claimed model and not tampered with.
- C. Tracking which system sent the API request for billing attribution.
- D. Ensuring the model uses the same random seed across all requests.

Correct Answer: A

Explanation

Model providers periodically update models under the same name. Version pinning ensures your application always runs on a specific, tested model version — preventing unexpected behaviour changes from silent model updates in production. This is critical for reproducible, testable production systems.

Q10. What is "context stuffing" and why is it problematic?

- A. A technique for fitting more data into a prompt by compressing or summarising it.
- B. An attack where malicious content is added to the context to manipulate the model.
- C. The process of loading model weights into GPU memory before inference.
- D. Filling the context window with excessive content, degrading model attention on key information.**

Correct Answer: D

Explanation

Context stuffing occurs when so much text is added to the prompt that the model's attention becomes diluted — relevant information is buried and the model may miss or deprioritise it. "Lost in the middle" research shows models attend better to information at the beginning and end of very long contexts.

Q11. What does an API "endpoint" refer to?

- A. The server hardware at the physical end of the network that runs the model.
- B. The specific URL that receives API requests for a particular functionality.**
- C. The final token the model generates before stopping.
- D. The response format returned at the end of a streaming response.

Correct Answer: B

Explanation

An API endpoint is the URL path that accepts requests for a specific function — e.g., `https://api.anthropic.com/v1/messages` is Anthropic's messages endpoint. Different endpoints provide different functionalities within the same API.

Q12. What is "prompt caching" and what problem does it solve?

- A. Storing common user prompts in a database to respond without calling the model.
- B. Caching model outputs so identical requests return the same response instantly.
- C. Reusing cached computation for repeated prompt prefixes to reduce cost and latency.**
- D. A technique for compressing prompts before sending them to the API.

Correct Answer: C

Explanation

Prompt caching (like Anthropic's prefix caching) allows the model to reuse the computed key-value representations of a repeated prompt prefix across multiple calls. This is highly valuable for applications that repeatedly use the same large system prompt or context — dramatically reducing cost and latency for the repeated portion.

Q13. What is the most important consideration when storing API keys in a production application?

- A. Keys must be rotated every 24 hours to prevent unauthorised use.
- B. Keys must be encrypted with AES-256 before being stored in any database.
- C. Keys must be different for development and production environments.
- D. Keys must be stored in environment variables or a secrets manager, never in code.**

Correct Answer: D

Explanation

API keys in source code are a critical security failure — they end up in version control, logs, and everywhere the code is deployed. The correct practice is environment variables or a dedicated secrets manager (AWS Secrets Manager, Vault, etc.), which keeps keys outside the codebase entirely.

Q14. What is "model routing" in a multi-model production system?

- A. Load-balancing requests across multiple instances of the same model.
- B. Directing different requests to different models based on task requirements and cost.**
- C. Routing model outputs to different downstream systems based on content type.
- D. Selecting the optimal serving hardware for each model inference request.

Correct Answer: B

Explanation

Model routing selects which model to use for each request based on factors like task complexity, required capability, cost constraints, and latency requirements. Simple queries route to cheaper, faster models; complex tasks to more capable ones. This optimises the cost-quality tradeoff across a production workload.

Q15. What does "API observability" mean in an AI production system?

- A. Monitoring request/response data, latency, errors, costs, and quality to understand system behaviour.**
- B. Making the AI API's internal workings transparent to end users.
- C. Ensuring the model's reasoning is explainable for each response it generates.
- D. Logging all user conversations for compliance and audit purposes.

Correct Answer: A

Explanation

API observability means instrumenting the system to capture the data needed to understand how it is performing: request volumes, error rates, latency distributions, token costs, and output quality metrics. Without observability, production issues are invisible until they cause user-facing failures.

SET

12

RAG and Vector Databases

Set 12 of 5 · 15 Questions · AI Engineering

Q16. What is "RAG" (Retrieval-Augmented Generation) and what problem does it solve?

- A. Training a model on domain-specific data to improve its in-domain performance.
- B. A technique for reducing hallucination by blocking uncertain model outputs.
- C. Supplementing a model with retrieved documents to answer questions it could not from training alone.**
- D. A method for compressing large datasets before feeding them to a language model.

Correct Answer: C**Explanation**

RAG retrieves relevant documents from an external knowledge base and includes them in the model's context. This allows the model to answer questions about information not in its training data (recent events, proprietary documents, domain-specific knowledge) without fine-tuning.

Q17. What is a "vector embedding" in the context of RAG?

- A. A numerical representation of text that captures semantic meaning in a high-dimensional space.**
- B. A compressed version of a document stored in a binary database format.
- C. A template that structures how retrieved documents are inserted into prompts.
- D. A hash of a document used to check for duplicates in the knowledge base.

Correct Answer: A**Explanation**

A vector embedding is a dense numerical vector (e.g., 1536 dimensions) produced by an embedding model that represents the semantic content of a text chunk. Semantically similar texts produce similar (nearby) vectors, enabling similarity-based retrieval.

Q18. How does a vector database enable semantic search?

- A. By indexing documents with keyword tags and returning exact keyword matches.
- B. By training a specialised model on the document corpus for each new query.
- C. By storing documents in a tree structure ordered by semantic topic.
- D. By comparing query embeddings to document embeddings using cosine or dot product similarity.**

Correct Answer: D

Explanation

A vector database stores document embeddings and enables approximate nearest-neighbour (ANN) search. When a query arrives, it is embedded by the same model, and the database returns documents whose embeddings are most similar to the query embedding — capturing semantic relevance, not just keyword overlap.

Q19. What is "chunking" in RAG and why does it matter?

- A. Compressing retrieved documents to reduce the tokens sent to the model.
- B. Splitting documents into smaller pieces that fit within retrieval and context limits.**
- C. Dividing the RAG workflow into modular processing stages.
- D. Grouping semantically similar documents together for batch embedding.

Correct Answer: B

Explanation

Documents must be split into chunks because embedding models and context windows have size limits. Chunk size is a critical design decision: too small and chunks lack context; too large and retrieval becomes imprecise (the relevant information is diluted across too much irrelevant content in the same chunk).

Q20. What is "hybrid search" in a RAG system?

- A. Running the same query through multiple embedding models and merging results.
- B. Searching both an internal database and the public internet for relevant documents.
- C. Combining vector similarity search with keyword (BM25) search to improve retrieval.**
- D. Using two different vector databases and selecting the better result for each query.

Correct Answer: C

Explanation

Hybrid search combines semantic vector search (which handles paraphrase and conceptual matching) with keyword-based BM25 search (which excels at exact term matching). Different queries benefit from different approaches, and combining them produces more reliable retrieval than either alone.

Q21. What is "re-ranking" in a RAG pipeline and when is it used?

- A. Running the model on retrieved documents to decide which ones are accurate.
- B. Reordering the model's generated output for better coherence and flow.
- C. A post-processing step that re-ranks model responses by quality score.
- D. A second-pass scoring step that reorders retrieved documents by relevance before passing to the model.**

Correct Answer: D

Explanation

After initial retrieval returns the top-K candidates by vector similarity, re-ranking applies a more expensive but more accurate relevance model (e.g., a cross-encoder) to reorder them. This improves the quality of the documents actually passed to the generation model.

Q22. What causes "retrieval failures" in a RAG system and how are they typically diagnosed?

- A. The retrieved documents do not contain the answer — diagnosed by checking retrieval results independently of generation.**
- B. The vector database is returning results too slowly, causing timeouts.
- C. The embedding model is mismatched with the vector database query format.
- D. The model is ignoring retrieved documents and generating from training knowledge.

Correct Answer: A

Explanation

Retrieval failure is when the pipeline retrieves irrelevant or incomplete documents, so the generation step has no basis for the correct answer. Diagnosing this requires evaluating retrieval independently — checking whether the correct documents are being retrieved for test queries, separate from generation quality.

Q23. What is "embedding model alignment" and why does it matter in RAG?

- A. The embedding model must be trained on the same data as the generation model.
- B. Documents and queries must be embedded by the same model for similarity comparisons to work.**
- C. Embedding dimensions must exactly match the vector database's storage format.
- D. The embedding model must be retrained whenever documents in the knowledge base change.

Correct Answer: B

Explanation

Embeddings are only comparable when generated by the same model — the vector space of Model A is not the same as Model B's. Querying a database of Model A embeddings with a Model B query embedding will return meaningless similarity scores. This alignment must be maintained across document ingestion and query time.

Q24. What is the most important metric to track for RAG retrieval quality?

- A. Mean retrieval latency — how quickly the vector database returns results.
- B. Embedding model perplexity on the document corpus.
- C. The cosine similarity score of the top-1 retrieved result for each query.
- D. Recall@K — whether the correct document is in the top-K retrieved results.**

Correct Answer: D

Explanation

Recall@K measures whether the relevant document is found within the top K retrieved results. This is the foundational retrieval metric — if the correct document is not in the retrieved set, generation cannot produce a correct answer regardless of how capable the generation model is.

Q25. What is "metadata filtering" in a vector database?

- A. Removing identifying information from documents before they are embedded.
- B. Filtering out low-quality embeddings based on their vector magnitude.
- C. Filtering retrieved documents by structured attributes (date, author, category) in addition to semantic similarity.**
- D. A security layer that prevents unauthorised users from querying certain documents.

Correct Answer: C

Explanation

Metadata filtering combines semantic similarity search with structured attribute filters — for example, "find semantically similar documents where date > 2024-01-01 and category = finance." This dramatically improves retrieval precision in applications where document scope matters.

Q26. What is "document freshness" a challenge for in RAG systems?

- A. Ensuring retrieved documents are readable and not too old to understand.
- B. Keeping the knowledge base up to date when source documents change or new ones are added.**
- C. Preventing the model from generating outdated information from its training data.
- D. Managing storage costs as the document corpus grows over time.

Correct Answer: B

Explanation

RAG knowledge bases must be actively maintained — when source documents are updated, deleted, or added, the index must be updated accordingly. Stale indices return outdated information. This is an ongoing operational requirement, not a one-time setup.

Q27. What is the "lost in the middle" problem in RAG with long contexts?

- A. Language models attend less to information in the middle of long contexts, missing retrieved content.**
- B. Long retrieved documents cause the model to lose track of the original user question.
- C. Documents retrieved in the middle of a session are processed less accurately than early ones.
- D. Chunking long documents causes the middle sections to be missing from the index.

Correct Answer: A

Explanation

"Lost in the middle" research shows that language models attend most strongly to the beginning and end of long contexts, with attention dropping for content in the middle. This means RAG systems that concatenate many retrieved documents may have the model miss or underweight the most relevant content if it lands in the middle.

Q28. What is the difference between "sparse" and "dense" retrieval?

- A. Sparse retrieval indexes a subset of documents; dense retrieval indexes all documents.
- B. Sparse retrieval is faster but less accurate; dense retrieval is slower but perfect.
- C. Sparse uses keyword matching (BM25); dense uses semantic embedding similarity.**
- D. Sparse retrieval works for short queries; dense retrieval handles multi-sentence queries.

Correct Answer: C

Explanation

Sparse retrieval (BM25, TF-IDF) represents documents as weighted keyword vectors — mostly zeros. Dense retrieval uses neural embedding vectors — all dimensions have values. Sparse excels at exact term matching; dense excels at semantic and paraphrase matching. Hybrid systems combine both.

Q29. What is "answer grounding" in RAG and why is it important?

- A. Physically grounding the RAG system to a secure server infrastructure.
- B. Verifying that retrieved documents are from trusted, authoritative sources.
- C. Ensuring the model's answer uses the exact wording from retrieved documents.
- D. Ensuring the model's answer is based on and citable from the retrieved documents.**

Correct Answer: D

Explanation

Answer grounding means the model's response should be traceable to specific retrieved content — not generated from training knowledge or hallucinated. Grounded answers can be verified by checking the source documents; ungrounded answers cannot. Grounding is the primary mechanism by which RAG reduces hallucination.

Q30. What is the most important operational consideration for a production RAG system?

- A. Embedding model inference speed to minimise query-time latency.
- B. Document ingestion pipeline reliability — ensuring the index stays fresh and complete.**
- C. Vector database storage cost as the document corpus grows.
- D. The number of documents returned per retrieval call (top-K parameter).

Correct Answer: B

Explanation

The most important operational requirement is a reliable, continuous ingestion pipeline that keeps the index current with the source documents. A RAG system with an out-of-date index provides wrong or missing information. Everything else — speed, cost, K parameter — is secondary to index freshness and completeness.

SET

13

Deployment and Scaling

Set 13 of 5 · 15 Questions · AI Engineering

Q31. What is "time to first token" (TTFT) and why does it matter for user experience?

- A. The latency before the first token appears — drives perceived responsiveness in streaming UIs.**
- B. The time required for the model to read and process the first token of input.
- C. The delay between model deployment and the first successful API request.
- D. The time it takes for the first token in a prompt to be embedded and stored.

Correct Answer: A**Explanation**

TTFT is the time from request submission to the first character appearing in the response. In streaming UIs, low TTFT makes responses feel immediate even if total generation takes longer. High TTFT makes the interface feel slow even if total response time is reasonable.

Q32. What is "tokens per second" (TPS) and what does it measure in LLM serving?

- A. The rate at which input tokens are processed by the model before generation begins.
- B. The throughput of the vector database during retrieval operations.
- C. The rate at which the model generates output tokens — higher means faster text generation.**
- D. The API's capacity for processing concurrent streaming requests.

Correct Answer: C**Explanation**

TPS (tokens per second) measures output generation speed — how quickly the model produces response tokens after the first token appears. It determines how fast text streams to the user and how many requests the system can serve in parallel at a given quality level.

Q33. What is a "GPU" and why is it essential for LLM inference?

- A. A specialised memory chip that stores model weights for fast retrieval during inference.
- B. A processor with thousands of cores optimised for the parallel matrix operations in neural network inference.**
- C. A hardware accelerator that translates model outputs into binary for network transmission.
- D. A co-processor that handles tokenisation and embedding before the main model runs.

Correct Answer: B

Explanation

GPUs (Graphics Processing Units) have thousands of small cores optimised for parallel floating-point operations — exactly what neural network inference (matrix multiplication, attention computation) requires. CPUs have far fewer cores and cannot perform these operations at the scale needed for LLM inference.

Q34. What is "model sharding" in the context of LLM deployment?

- A. Splitting user requests across multiple independent model instances for load balancing.
- B. Breaking model responses into shards that are assembled client-side.
- C. A compression technique that reduces model size by removing redundant weights.
- D. Distributing model weights across multiple GPUs because the model is too large for one device.**

Correct Answer: D

Explanation

Model sharding (tensor or pipeline parallelism) partitions model layers or weight tensors across multiple GPUs to handle models that exceed single-device memory. Without sharding, 70B+ parameter models cannot be deployed at all on consumer or cloud GPU hardware.

Q35. What is "serving latency percentile" (e.g., p99) and why does it matter?

- A. The latency that 99% of requests complete within — captures worst-case user experience.**
- B. The average latency for the top 99% most common request types.
- C. A benchmark score measuring the model's inference speed on standard hardware.
- D. The percentage of requests that complete within a predefined latency SLA.

Correct Answer: A

Explanation

The p99 latency is the value that 99% of requests complete within — meaning 1% of requests are slower. The p99 matters more than average latency for user experience because it captures the tail behaviour: occasional slow requests that users experience as timeouts or hangs.

Q36. What is "cold start" in serverless AI deployment?

- A. An inference error caused by the model receiving an input it was not trained on.
- B. The latency spike from initialising a new container or loading model weights on first request.**
- C. A deployment strategy that starts with a small model and scales to larger models under load.
- D. The process of pre-warming the GPU cache before inference begins.

Correct Answer: B

Explanation

Cold start latency occurs when a serverless function or container that has been idle must initialise — loading model weights into GPU memory can take 10-60+ seconds for large models. This creates severe first-request latency spikes. Mitigation strategies include keep-warm instances, weight pre-caching, and smaller model deployment.

Q37. What is "autoscaling" in an AI serving infrastructure?

- A. Automatically adjusting the model's generation parameters based on query complexity.
- B. A technique for dynamically resizing batch sizes based on GPU memory availability.
- C. Automatically updating model weights when new training data becomes available.
- D. Automatically adding or removing compute instances based on request load.**

Correct Answer: D

Explanation

Autoscaling monitors request queue depth, latency, or CPU/GPU utilisation and adds new serving instances when demand increases, removing them when demand drops. This allows the system to handle traffic spikes without over-provisioning expensive GPU resources during low-traffic periods.

Q38. What is the primary tradeoff between model size and serving cost?

- A. Larger models have lower latency but higher memory requirements.
- B. Larger models are cheaper to run because they require fewer API calls.
- C. Larger models cost more per token but may produce higher quality output.**
- D. Larger models produce more tokens per request, increasing total cost disproportionately.

Correct Answer: C

Explanation

Larger models require more GPU memory, more computation per token, and thus cost more per inference call. The tradeoff is quality versus cost: a 7B model costs less to serve than a 70B model, but may produce lower quality outputs for complex tasks. Optimal model selection minimises cost for each task's quality requirement.

Q39. What is "inference batching" and how does it improve GPU utilisation?

- A. Breaking a single large request into multiple smaller batches for parallel processing.
- B. Processing multiple requests simultaneously in a single GPU forward pass.**
- C. Pre-generating common responses in advance for instant retrieval.
- D. Grouping similar prompts together to improve cache hit rates.

Correct Answer: B

Explanation

GPU utilisation is often low when processing individual requests because the GPU's parallel processing capacity is underutilised. Batching multiple requests together into a single forward pass makes better use of the GPU's parallelism, improving throughput (requests/second) at the cost of some additional latency per request.

Q40. What is "vLLM" and what problem does it solve?

- A. A high-throughput serving engine that improves GPU memory efficiency using PagedAttention.**
- B. A validation library for verifying that LLM outputs meet quality thresholds.
- C. A vector database optimised for large-scale language model embedding storage.
- D. A load balancer designed specifically for distributing LLM API requests.

Correct Answer: A

Explanation

vLLM uses PagedAttention to manage KV cache memory in blocks, similar to how operating systems manage virtual memory. This dramatically reduces memory waste (fragmentation) in the KV cache, allowing more concurrent requests on the same GPU hardware and significantly improving serving throughput.

Q41. What is "SLA" (Service Level Agreement) in the context of AI API deployment?

- A. A legal agreement governing the permitted uses of an AI model in production.
- B. A security standard that API providers must meet for enterprise deployment.
- C. A committed performance guarantee defining uptime, latency, and throughput thresholds.**
- D. A model accuracy guarantee specifying minimum benchmark performance.

Correct Answer: C

Explanation

An SLA defines measurable service commitments: uptime percentage (e.g., 99.9%), latency targets (e.g., p99 < 2s), throughput minimums, and error rate ceilings. SLAs are the contractual basis for enterprise AI deployments and determine compensation when commitments are missed.

Q42. What is "horizontal scaling" in AI serving infrastructure?

- A. Increasing the hardware specification of existing serving instances.
- B. Extending the model's context window to handle longer inputs.
- C. Scaling the model's parameter count to improve output quality.
- D. Adding more serving instances in parallel to increase total capacity.**

Correct Answer: D

Explanation

Horizontal scaling (scale out) adds more serving instances, distributing the load across them. This is contrasted with vertical scaling (scale up), which increases the resources of existing instances. Horizontal scaling is generally more cost-effective and flexible for AI serving as it allows granular capacity adjustments.

Q43. What is the most important thing to include in an AI service health dashboard?

- A. GPU utilisation and memory consumption per serving instance.
- B. Error rate, latency percentiles, and request volume — the three pillars of service health.**
- C. Cost per request and total monthly API spend.
- D. Model accuracy scores from the latest evaluation run.

Correct Answer: B

Explanation

Error rate (are requests succeeding?), latency percentiles (are requests fast enough?), and request volume (is demand normal?) are the three core service health signals. GPU utilisation, costs, and accuracy scores are important secondary metrics but are not the primary health indicators.

Q44. What is the "deployment canary" strategy in AI model updates?

- A. Deploying the new model to a test environment and canary testing for safety issues.
- B. Using a smaller, cheaper model as a canary to catch harmful requests before the main model.
- C. Routing a small percentage of traffic to the new model version before full rollout.**
- D. A gradual rollout that increases model context window size incrementally.

Correct Answer: C

Explanation

Canary deployment routes a small fraction of live traffic (e.g., 5%) to the new model version while the rest continues on the old version. This allows real-world performance comparison and catches issues before they affect all users, enabling rollback if the new version underperforms.

Q45. What is "model deprecation" and what should production systems do about it?

- A. When a model version is retired by the provider — systems should pin to specific versions and have migration plans.**
- B. When a model's accuracy degrades over time due to distribution shift in production inputs.
- C. The process of removing an underperforming model from a multi-model routing system.
- D. A fine-tuning technique that reduces model size by removing deprecated capabilities.

Correct Answer: A

Explanation

Model deprecation occurs when an AI provider retires a specific model version, forcing users to migrate. Production systems should pin to specific versions, monitor deprecation announcements, and have tested migration plans ready before deprecated versions are shut down to avoid unplanned outages.

SET

14

Cost and Latency

Set 14 of 5 · 15 Questions · AI Engineering

Q46. What is the primary cost driver for LLM API usage?

- A. The number of API requests made regardless of their content.
- B. The time spent waiting for the model to complete generation.
- C. The size of the model accessed, independent of token count.
- D. The number of input and output tokens processed per request.**

Correct Answer: D**Explanation**

LLM API pricing is almost universally based on token count — input tokens and output tokens priced separately (output tokens are typically more expensive). The number of API calls matters for rate limiting but not for cost; time is not directly billed; model size affects the per-token rate, not an independent billing dimension.

Q47. What is "token efficiency" and why does it matter at scale?

- A. Ensuring the model generates the maximum useful information per token output.
- B. Achieving the required output quality with the fewest possible tokens — directly reduces cost.**
- C. A measure of how accurately the model represents information in each token.
- D. The ratio of useful content to total content in the model's training data.

Correct Answer: B**Explanation**

Token efficiency means achieving the required quality with fewer tokens. At scale, a 30% reduction in average prompt length translates directly to a 30% reduction in input token costs — compounding significantly across millions of requests. Prompt engineering for efficiency is a direct cost lever.

Q48. Which approach most effectively reduces output token costs in a production system?

- A. Using a higher temperature to generate more focused responses.
- B. Switching to a smaller model that generates shorter responses.
- C. Instructing the model to be concise and specifying exactly the output format needed.**
- D. Increasing the max_tokens limit to allow the model to be more complete.

Correct Answer: C

Explanation

Instructing the model to be concise with a specific output format (e.g., "respond in exactly 3 bullet points") directly reduces output token count without degrading quality for the use case. Higher temperature increases randomness; smaller models may reduce quality; higher max_tokens increases potential cost.

Q49. What is "cost per query" in AI production systems and how is it calculated?

- A. Total API cost divided by number of queries — the primary unit economics metric.**
- B. The infrastructure cost of running one GPU for the duration of one query.
- C. The product of latency and compute cost per second for each inference call.
- D. The cost of the embedding model call plus the generation model call per RAG query.

Correct Answer: A

Explanation

$\text{Cost per query} = \text{total API spend} / \text{number of queries}$. It is the core unit economics metric for AI product profitability. It captures the full cost of serving each user request, enabling comparison to revenue per query to determine whether the AI feature is economically viable.

Q50. A RAG system makes 3 API calls per user query: embedding, retrieval, and generation. The most expensive component is typically:

- A. Embedding — dense embedding models are computationally the most expensive component.
- B. Retrieval — vector database queries at scale dominate infrastructure costs.
- C. All three are approximately equal in cost for a well-designed system.
- D. Generation — output tokens from large models cost significantly more than embedding.**

Correct Answer: D

Explanation

Generation using a large language model is by far the most expensive component. Embedding models are small and inexpensive. Vector database queries are generally cheap at typical scales. The generation step — especially if it involves long outputs from a frontier model — typically accounts for 90%+ of the API cost.

Q51. What is "caching" in the context of LLM production costs?

- A. Pre-loading model weights into GPU memory to reduce cold start latency.
- B. Using compression to reduce the size of API responses in transit.
- C. Storing responses for identical or near-identical queries to avoid repeated API calls.**
- D. Storing user conversation history locally to reduce API calls for context reconstruction.

Correct Answer: C

Explanation

Response caching stores model outputs for queries that have been answered before. When the same (or similar) query arrives, the cached response is returned instantly without an API call. For applications with repetitive queries (FAQs, common scenarios), caching can reduce API costs dramatically.

Q52. What is the "price-performance" consideration when selecting a model for a production use case?

- A. Using the cheapest model that reliably achieves the required quality threshold.**
- B. Selecting the highest-performance model and optimising for cost later.
- C. Matching the model size exactly to the complexity of the task at hand.
- D. Using the same model for all tasks to simplify infrastructure management.

Correct Answer: A

Explanation

The correct approach is to define the minimum quality threshold the use case requires, then find the cheapest model that reliably meets it. Over-provisioning (using an expensive model where a cheaper one would suffice) wastes money; under-provisioning (using a cheap model that doesn't meet quality requirements) destroys user value.

Q53. What is "token budget management" in a complex AI workflow?

- A. Allocating a maximum number of tokens to each user account per month.
- B. Tracking and controlling token usage across all steps to prevent cost overruns.**
- C. Setting the max_tokens parameter to match the expected response length.
- D. Compressing prompts using a tokeniser to minimise input token count.

Correct Answer: B

Explanation

Token budget management tracks cumulative token usage across a multi-step workflow, ensuring costs do not escalate unexpectedly. A workflow that chains 10 steps might spend 5x more tokens than anticipated if each step's context grows. Active budget tracking and enforcement prevents cost surprises.

Q54. What is the most cost-effective approach for handling a task that requires only simple text formatting?

- A. Use the largest model available to ensure perfect output quality.
- B. Use a deterministic rule-based system and avoid calling an AI model entirely.
- C. Use a small, fast model — simple tasks do not justify frontier model costs.**
- D. Use the frontier model with a very low max_tokens to limit cost.

Correct Answer: C

Explanation

Simple formatting tasks (standardising capitalisation, converting between formats, extracting fixed patterns) are best handled by small, fast, cheap models or even rule-based systems. Using frontier models for tasks they massively overkill wastes budget that could be spent on tasks that genuinely need capability.

Q55. What is "inference cost optimisation" at the infrastructure level?

- A. Reducing the number of model layers to speed up inference computation.
- B. Optimising the model's sampling parameters to generate fewer tokens.
- C. Minimising the size of training data to reduce the cost of future fine-tuning runs.
- D. Maximising GPU utilisation through batching, quantisation, and efficient scheduling.**

Correct Answer: D

Explanation

At the infrastructure level, cost optimisation focuses on maximising the useful work done per GPU dollar: batching requests together for parallel processing, quantising weights to fit more models per device, and scheduling to keep GPUs busy. These are engineering levers distinct from prompt-level or model-selection levers.

Q56. What is "usage-based pricing" vs "flat-rate pricing" for AI APIs?

- A. Usage-based charges per token; flat-rate charges a fixed fee regardless of consumption.**
- B. Usage-based charges per API call; flat-rate charges per model version accessed.
- C. Usage-based scales down with volume; flat-rate charges the same regardless of scale.
- D. Usage-based is for inference; flat-rate is for training and fine-tuning.

Correct Answer: A

Explanation

Usage-based pricing (the standard for most AI APIs) charges based on actual token consumption. Flat-rate pricing charges a fixed subscription fee regardless of usage — better for predictable high-volume workloads. The right choice depends on usage patterns, volume, and whether predictable costs or pay-per-use economics are preferable.

Q57. What is the primary risk of optimising for cost before establishing baseline quality?

- A. Cost optimisations are harder to implement once the system is in production.
- B. You may optimise for the wrong objective if quality requirements are not yet defined.**
- C. Cheap models are less secure and introduce additional safety risks.
- D. Cost optimisation requires retraining the model which is expensive.

Correct Answer: B

Explanation

Optimising for cost before quality is defined is premature optimisation. If the quality bar is not established, you do not know what "good enough" looks like, and cost cuts may degrade quality below acceptable thresholds. Always establish quality baselines first, then optimise cost within those constraints.

Q58. What is "latency vs throughput" tradeoff in AI serving?

- A. Higher throughput models produce lower latency responses by generating faster.
- B. Latency is a user experience concern; throughput is an infrastructure concern.
- C. Low latency and high throughput are always in conflict and cannot both be achieved.
- D. Low latency often requires dedicated resources that reduce utilisation; high throughput benefits from batching that increases latency.**

Correct Answer: D

Explanation

Batching requests together improves throughput (requests/second) by maximising GPU utilisation, but increases per-request latency (requests must wait for the batch to fill). Dedicated single-request serving minimises latency but lowers throughput. This fundamental tradeoff must be resolved for each use case based on whether latency or throughput is the binding constraint.

Q59. What is "compute cost normalisation" when comparing AI pricing across providers?

- A. Converting costs to a per-token or per-1M-token basis for apples-to-apples comparison.**
- B. Adjusting for hardware differences between cloud providers when benchmarking.
- C. Normalising model output quality scores before comparing cost-effectiveness.
- D. Converting provider costs to a common currency for international cost comparisons.

Correct Answer: A

Explanation

Providers use different pricing units (per token, per 1K tokens, per 1M tokens). Normalising all prices to per-1M-token makes comparisons valid. Without normalisation, a "cheap" provider may simply be reporting in a different unit, making direct comparison misleading.

Q60. What is the biggest risk of using a free tier AI API in a production system?

- A. Free tiers use less capable model versions than paid tiers.
- B. Free tier usage cannot be monitored or tracked for debugging.
- C. Rate limits and reliability SLAs are not appropriate for production workloads.**
- D. Free tier API keys expire unpredictably, causing production outages.

Correct Answer: C

Explanation

Free tiers are designed for development and experimentation. They have severe rate limits, no reliability SLAs, and no support commitments. Using a free tier in production means rate limits will be hit under normal load, and there is no contractual guarantee of uptime or support when things go wrong.

SET

15

Security and Access

Set 15 of 5 · 15 Questions · AI Engineering

Q61. What is "data privacy" the most critical concern for in AI API integrations?

- A. Ensuring sensitive data sent to AI providers is not stored, used for training, or exposed.**
- B. Ensuring the AI model does not generate personally identifiable information in outputs.
- C. Preventing users from extracting the AI provider's proprietary model weights.
- D. Encrypting all API communications between the client and the model provider.

Correct Answer: A**Explanation**

When sensitive data (PII, confidential business information, health records) is sent to AI providers in API requests, there are critical questions about how it is stored, whether it is used for training, who can access it, and whether it is subject to data residency requirements. These are the primary privacy concerns for AI API integrations.

Q62. What is "output filtering" in a production AI system?

- A. Limiting the length of model outputs to prevent context window overflow.
- B. Filtering input prompts before they are sent to the model.
- C. Selecting only the highest-quality outputs from multiple model generations.
- D. Post-processing model outputs to detect and block harmful or inappropriate content.**

Correct Answer: D**Explanation**

Output filtering applies safety classifiers or rule-based checks to model outputs before they are delivered to users, catching harmful content that slipped through the model's own safety training. It is a defence-in-depth layer, not a substitute for model-level safety.

Q63. What is "zero data retention" (ZDR) in AI API contracts and when is it required?

- A. A technical setting that prevents the model from caching previous responses.
- B. A contractual guarantee that the provider does not store API request/response data.**
- C. A compliance mode where the model refuses to generate any data about users.
- D. A policy that prevents training data from being retained after a model is released.

Correct Answer: B

Explanation

Zero data retention is an enterprise contract provision where the AI provider guarantees that API request and response data is not logged, stored, or used for any purpose after processing. It is required for handling highly sensitive data (healthcare, legal, financial) under regulations like HIPAA or contractual obligations.

Q64. What is "access control" most important for in a multi-user AI application?

- A. Preventing users from accessing the AI system during maintenance windows.
- B. Limiting the number of API calls each user can make per day.
- C. Ensuring users can only query with data and permissions appropriate to their role.**
- D. Ensuring the AI model only generates content appropriate for each user's age group.

Correct Answer: C

Explanation

In multi-user AI applications (enterprise SaaS, internal tools), access control ensures each user's AI interactions only touch data they are authorised to see. Without this, a RAG system might retrieve confidential documents belonging to another user, or an agent might access resources outside its permission scope.

Q65. What is "audit logging" in AI governance and what must it capture?

- A. A tamper-evident log of who queried the AI, what they asked, and what it responded.**
- B. A technical log of API errors and latency metrics for system debugging.
- C. A training data provenance record showing where model knowledge came from.
- D. A compliance certificate demonstrating the model meets regulatory standards.

Correct Answer: A

Explanation

AI audit logs for governance must capture who made the request (identity), what was requested (input), what the AI responded (output), and when (timestamp). This record is essential for investigating incidents, demonstrating compliance, and detecting misuse — it must be tamper-evident and retained per policy.

Q66. What is "model security" concerned with in an AI deployment context?

- A. Ensuring the AI model's weights are free of security vulnerabilities.
- B. Verifying that the AI model was trained on legally compliant data.
- C. Preventing unauthorised access, model extraction, and adversarial manipulation of the AI system.**
- D. Protecting the AI model from being overwhelmed by high volumes of requests.

Correct Answer: C

Explanation

Model security covers: preventing unauthorised API access, protecting against model extraction attacks (reverse-engineering the model through queries), defending against adversarial inputs designed to manipulate model behaviour, and securing the serving infrastructure. These are the attack surfaces specific to deployed AI systems.

Q67. What is "PII detection" and why is it important in AI pipelines?

- A. Detecting when AI outputs contain names and personal details that should be anonymised.
- B. A compliance check that verifies model training data was PII-free.
- C. An output filter that removes all names from AI-generated text.
- D. Identifying personally identifiable information in data before it is sent to AI models.**

Correct Answer: D

Explanation

PII detection scans data before it enters AI pipelines to identify personally identifiable information (names, emails, SSNs, medical records). If PII is detected, it can be masked, anonymised, or flagged before being sent to an external AI provider, preventing inadvertent privacy violations.

Q68. What is the most important governance requirement for AI use in a regulated industry?

- A. Using only AI models that have received regulatory certification for the specific industry.
- B. Documented policies for AI use, output verification, and human oversight of consequential decisions.**
- C. Ensuring AI systems are built on open-source models for transparency.
- D. Limiting AI use to tasks that have been individually approved by a compliance committee.

Correct Answer: B

Explanation

Regulated industry AI governance requires documented policies (what AI is used for, by whom, under what constraints), verification procedures (how AI outputs are checked before use), and human oversight for decisions with regulatory consequences. Certification, open-source requirements, and committee approval are supporting elements, not the foundation.

Q69. What is "rate limiting" from an API provider's perspective?

- A. Limiting the length of model outputs to manage server resource consumption.
- B. Throttling the speed of model generation to improve response consistency.
- C. Restricting the number of API requests or tokens per time period per client to ensure fair access.**
- D. Restricting which model versions a particular client tier can access.

Correct Answer: C

Explanation

Rate limiting protects the shared API infrastructure from overconsumption by any single client. It is typically enforced per API key, per account tier, with limits on requests per minute, tokens per minute, and sometimes tokens per day. Rate limiting ensures equitable access and prevents accidental or intentional resource exhaustion.

Q70. What is "vendor lock-in" as a risk in AI infrastructure decisions?

- A. Becoming so dependent on one AI provider's proprietary features that switching is prohibitively costly.**
- B. A legal liability when using AI provider data in ways that violate their terms.
- C. An infrastructure risk when a single provider hosts all production AI workloads.
- D. A quality risk when a provider's model quality deteriorates after you have built on it.

Correct Answer: A

Explanation

Vendor lock-in occurs when a system is built heavily around provider-specific APIs, proprietary formats, or unique features that do not have equivalents elsewhere. Switching providers then requires significant re-engineering. This becomes a business risk if the provider raises prices, degrades quality, or discontinues a service.

Q71. What is the correct approach to handling model provider outages in a production AI system?

- A. Queue all requests and retry them once the provider recovers.
- B. Fallback to an alternative provider or degrade gracefully to non-AI functionality.**
- C. Alert users that the AI service is temporarily unavailable and ask them to wait.
- D. Switch to a cached version of the model running locally on application servers.

Correct Answer: B

Explanation

Production AI systems must handle provider outages gracefully: routing to an alternative provider (multi-vendor strategy), falling back to rule-based or non-AI alternatives for critical functions, or queuing non-urgent requests. Simply showing an error or asking users to wait is unacceptable for business-critical applications.

Q72. What is "infrastructure as code" (IaC) important for in AI deployment?

- A. Writing the AI model's system prompt in a code-based configuration format.
- B. Using code to generate training data for fine-tuning AI models.
- C. Defining the AI model's architecture parameters in configuration files.
- D. Reproducible, version-controlled, auditable deployment of AI serving infrastructure.**

Correct Answer: D

Explanation

IaC (Terraform, CloudFormation, etc.) defines serving infrastructure in version-controlled code. This ensures deployments are reproducible, changes are auditable, and rollbacks are fast. For AI systems where security and compliance matter, IaC provides the governance trail required for infrastructure changes.

Q73. What is the most important security practice for AI API keys in a team environment?

- A. Sharing a master API key only with senior team members via secure messaging.
- B. Rotating all API keys every 24 hours to prevent long-term exposure.
- C. Using per-service, per-environment keys with least-privilege scope and rotation policies.**
- D. Storing API keys in a shared encrypted spreadsheet accessible to the team.

Correct Answer: C

Explanation

Per-service, per-environment keys with least-privilege scope mean each service has its own key with only the permissions it needs. This limits the blast radius of a compromised key, enables granular revocation, and provides clear attribution in audit logs. Shared master keys and spreadsheet storage are significant security anti-patterns.

Q74. What is "content moderation" as an infrastructure concern in AI deployments?

- A. Manually reviewing AI-generated content before it is published to users.
- B. Automatically screening both inputs and outputs for harmful content at scale.**
- C. Configuring the AI model's safety settings to block certain content categories.
- D. Limiting which users can access AI features based on age or verification status.

Correct Answer: B

Explanation

Content moderation as infrastructure means building systematic, automated checks — input scanners for harmful prompts, output classifiers for unsafe content — that operate at the scale of production traffic. This is a complementary layer to model-level safety, not a substitute for it, operating at the serving layer.

Q75. What does "AI governance" mean at the organisational level?

- A. The technical controls that prevent AI models from taking unauthorised actions.
- B. Compliance with specific AI regulations in the organisation's operating jurisdiction.
- C. A committee that approves each new AI model deployment before it goes to production.
- D. Policies, processes, and oversight structures for responsible AI deployment and use.**

Correct Answer: D

Explanation

AI governance is the organisational framework — policies defining permitted uses, risk assessment processes, oversight mechanisms, accountability structures, incident response procedures — that ensures AI is used responsibly across the organisation. Technical controls, regulatory compliance, and approval committees are all elements within this broader framework.

SET

16

The Claude Stack

Set 16 of 5 · 15 Questions · AI Engineering

Q1. What is a "Claude Project" and what does it enable?

- A. A Claude API project that groups multiple fine-tuned model versions together.
- B. A persistent workspace with a shared system prompt and knowledge files across conversations.**
- C. A collaboration environment where multiple users share a Claude account.
- D. A structured workflow builder that automates multi-step Claude interactions.

Correct Answer: B**Explanation**

Claude Projects allow users to create persistent workspaces where a custom system prompt and uploaded knowledge files are available across all conversations within the project. This enables consistent persona, context, and tools without re-specifying them in every new conversation.

Q2. What is a "Claude Skill" and how is it different from a standard instruction?

- A. A technical capability Claude has, like writing code or analysing images.
- B. A tool Claude can call to access external data or APIs.
- C. A safety setting that restricts Claude to a specific domain of responses.
- D. A reusable, persistent prompt configuration for a specific recurring task.**

Correct Answer: D**Explanation**

A Claude Skill is a saved, reusable prompt configuration — a named task definition with specific instructions, format requirements, and context — that can be invoked by name to run the same structured task reliably. It is operational infrastructure, not a model capability or tool.

Q3. What does "Claude with web search" enable that Claude alone cannot do?

- A. Access to current information beyond the training knowledge cutoff.**
- B. Generating longer and more detailed responses on complex topics.
- C. Producing more accurate factual responses on topics in its training data.
- D. Performing multi-step reasoning tasks that require external verification.

Correct Answer: A

Explanation

Web search gives Claude access to current events, recent publications, updated prices, and live data that post-date its training cutoff. Without web search, Claude's knowledge is limited to its training data. Web search does not extend output length, improve baseline accuracy, or change reasoning architecture.

Q4. What is "Claude Code" designed to do?

- A. A tool for evaluating the quality and security of code that Claude generates.
- B. A platform for fine-tuning Claude on a developer's proprietary codebase.
- C. An agentic coding environment where Claude can read, write, and execute code autonomously.**
- D. A lightweight Claude model optimised for faster code completion responses.

Correct Answer: C

Explanation

Claude Code is an agentic coding environment — Claude can read files, write code, run terminal commands, browse documentation, and iterate on its work autonomously within a development environment. It is not a code evaluator, fine-tuning platform, or separate model.

Q5. What is a "system prompt" in the Claude API context and what does it control?

- A. A hidden safety layer that filters harmful content before the model sees user input.
- B. Persistent instructions that set Claude's role, constraints, and context for all interactions.**
- C. The initial message Claude sends to the user at the start of a conversation.
- D. A configuration parameter that controls Claude's output format and length.

Correct Answer: B

Explanation

A system prompt provides persistent context to Claude — its role, what it can and cannot do, tone, format requirements, and domain constraints — that applies across the entire conversation. It is operator-controlled, authoritative, and sets the foundational behaviour of Claude for that deployment.

Q6. What is a "Claude Connector" (MCP integration) and what does it enable?

- A. A connection between Claude and external services, allowing Claude to read and write data.**
- B. A Claude API wrapper that simplifies authentication and request formatting.
- C. A browser extension that allows Claude to assist with web-based tasks.
- D. A tool for monitoring Claude API usage and tracking costs per integration.

Correct Answer: A

Explanation

Claude Connectors (built on MCP) allow Claude to connect to external services — Notion, Google Drive, Slack, CRM systems — enabling it to read and write data from those services as part of its responses. This transforms Claude from a text generator into an active participant in real workflows.

Q7. What is "Claude Routines" and what problem does it solve?

- A. A scheduling system that sends Claude reminders and time-based notifications.
- B. A quality control process that reviews Claude's outputs before delivery.
- C. A conversation management feature that saves and resumes long conversations.
- D. Automated sequences of Claude instructions that run without manual prompting each time.**

Correct Answer: D

Explanation

Claude Routines allow users to define and save sequences of instructions that Claude executes in order — automating recurring workflows that would otherwise require re-prompting the same sequence manually. This enables reliable, repeatable automation of multi-step Claude tasks.

Q8. What distinguishes "Claude for Teams" from personal Claude plans?

- A. Access to more powerful model versions not available to individual users.
- B. Higher message limits and faster response times for power users.
- C. Collaborative workspaces, admin controls, and enterprise privacy protections for organisational use.**
- D. The ability to fine-tune Claude on proprietary organisational data.

Correct Answer: C

Explanation

Claude for Teams adds collaboration features (shared projects, team workspaces), administrator controls (user management, policy enforcement), and enterprise privacy commitments (no training on team data). It is designed for organisational deployment, not just individual productivity.

Q9. What does Anthropic's "Acceptable Use Policy" govern for Claude?

- A. The types of tasks and content Claude can be used for commercially and personally.**
- B. The technical limitations of Claude's context window and output length.
- C. The pricing tiers and usage limits for different Claude API plans.
- D. The data retention policies for conversations and API request logs.

Correct Answer: A

Explanation

The Acceptable Use Policy sets boundaries on permitted and prohibited uses of Claude — categories of harmful, illegal, or unethical applications that are not allowed. It governs what operators and users can ask Claude to do, and what Anthropic will not permit regardless of technical capability.

Q10. What is "prompt engineering" in the Claude context and why does it matter?

- A. The process of building Claude-powered applications using the Anthropic API.
- B. A technique for reducing Claude's API costs by compressing prompts.
- C. Training Claude on custom data to improve its performance on specific tasks.
- D. Designing prompts to reliably produce high-quality, well-formatted Claude outputs.**

Correct Answer: D

Explanation

Prompt engineering is the discipline of crafting inputs to Claude that reliably produce the desired output — specifying role, context, constraints, format, and examples in ways that guide Claude toward accurate, appropriately formatted, and useful responses. It is the primary lever users and operators have over Claude's behaviour.

Q11. What is the most important difference between "operator" and "user" in the Claude API model?

- A. Operators have access to more powerful models; users access standard model versions.
- B. Operators configure Claude's behaviour via system prompts; users interact within those constraints.**
- C. Operators can see all user conversations; users can only see their own conversations.
- D. Operators pay for API usage; users access Claude through the operator's application.

Correct Answer: B

Explanation

In Anthropic's model, operators (businesses and developers) configure Claude's deployment through system prompts, setting what Claude can and cannot do. Users interact with Claude within the scope the operator defines. This two-tier structure allows operators to create appropriate, safe Claude deployments for their specific user base.

Q12. What is "Claude's extended thinking" mode designed for?

- A. Long documents that require extended processing time to analyse completely.
- B. Multi-turn conversations that benefit from deeper context retention.
- C. Complex problems that benefit from explicit step-by-step reasoning before answering.**
- D. Creative writing tasks that require extended brainstorming before drafting.

Correct Answer: C**Explanation**

Extended thinking (Claude's chain-of-thought processing mode) allows Claude to work through complex problems step by step before producing its final answer. This improves accuracy on tasks requiring multi-step reasoning, mathematics, code debugging, and complex analysis.

Q13. What is a "Claude artifact" in the Claude.ai interface?

- A. A persistent memory object that Claude uses to remember information across sessions.
- B. A file attached to a conversation that Claude can read and reference.
- C. A reusable prompt template stored in the user's Claude account.
- D. A standalone piece of generated content (code, document, design) displayed in a separate panel.**

Correct Answer: D**Explanation**

Artifacts in Claude.ai are standalone generated outputs — code, HTML, markdown documents, SVGs — displayed in a side panel. They can be directly copied, downloaded, or in some cases rendered and interacted with. They are a UI pattern for separating structured deliverables from conversational text.

Q14. What does "model context protocol" (MCP) enable for Claude?

- A. A protocol for Claude to maintain context across multiple separate conversations.
- B. A standardised way for external tools and data sources to connect to Claude.**
- C. A security standard for encrypting data sent between Claude and external services.
- D. A format specification for structuring Claude's system prompts consistently.

Correct Answer: B**Explanation**

MCP (Model Context Protocol) is an open standard that allows external services, tools, and data sources to connect to Claude in a standardised way. It enables Claude to access a wide ecosystem of third-party integrations — from file systems to SaaS applications — without bespoke integrations for each service.

Q15. What is the most effective use of Claude's Projects feature for a team?

- A. Configuring a shared project with relevant context so all team members get consistent Claude behaviour.**
- B. Creating separate projects for each team member to personalise their Claude experience.
- C. Using projects to store conversation histories for long-term team reference.
- D. Setting up projects to restrict which Claude features different team members can access.

Correct Answer: A

Explanation

Shared team projects with a configured system prompt and relevant knowledge files ensure all team members interact with Claude in the same consistent way — the same role, context, and constraints — rather than each person starting from scratch or configuring Claude differently.

SET

17

Code Assistants

Set 17 of 5 · 15 Questions · AI Engineering

Q16. What is the primary function of a "code assistant" AI tool?

- A. Automatically deploying code to production environments after generation.
- B. Replacing software developers for routine coding tasks in enterprise environments.
- C. Suggesting, completing, explaining, and debugging code within a development environment.**
- D. Training custom AI models on a developer's codebase for specialised generation.

Correct Answer: C**Explanation**

Code assistants (GitHub Copilot, Cursor, Claude Code, etc.) are AI tools integrated into development environments that provide autocomplete, code generation, explanation, and debugging assistance. They augment developers rather than replacing them, and they operate in the development environment, not deployment pipelines.

Q17. What is "tab completion" in the context of AI code assistants?

- A. Accepting AI-suggested code completions with a single key press.**
- B. Auto-completing terminal commands based on the user's history.
- C. AI filling in HTML tab attributes and indentation automatically.
- D. A keyboard shortcut for switching between code and documentation panels.

Correct Answer: A**Explanation**

Tab completion in AI code assistants (pioneered by GitHub Copilot) means the assistant suggests a code completion inline as the developer types, and the developer accepts it by pressing Tab. This is the signature interaction pattern of AI-augmented coding.

Q18. What is the most important limitation of AI-generated code that developers must account for?

- A. Generated code is always stylistically different from the developer's own code.
- B. Generated code tends to be less efficient than hand-written code for performance-critical tasks.
- C. Generated code cannot be run without manual review by a senior engineer.
- D. Generated code may be functionally wrong, insecure, or incompatible with the specific context.**

Correct Answer: D

Explanation

AI-generated code can be syntactically correct and plausible-looking while being logically wrong, containing security vulnerabilities, or being incompatible with the specific project's architecture and dependencies. Developers must review and test every line of generated code as thoroughly as hand-written code.

Q19. What is "in-context learning" as it applies to code assistants?

- A. The assistant retrains its weights on the developer's codebase during the session.
- B. The assistant learns the project's style and patterns from the files open in the editor.**
- C. A learning mode where the assistant improves by receiving developer corrections.
- D. The ability of the assistant to learn new programming languages from examples.

Correct Answer: B

Explanation

AI code assistants learn the patterns, naming conventions, and style of the current codebase from what is in the context — open files, recent code, examples in the conversation. This makes the suggestions more relevant to the specific project without any model retraining.

Q20. What is "Cursor" as an AI development tool?

- A. A code review tool that uses AI to identify and fix bugs before deployment.
- B. A code generation API that integrates with existing development environments.
- C. An AI-native code editor built around deep AI assistance for coding tasks.**
- D. A tool for converting AI-generated pseudocode into executable code.

Correct Answer: C

Explanation

Cursor is an AI-native code editor (based on VS Code) where AI assistance is deeply integrated — not a plugin on top of an existing editor. It provides codebase-aware completions, multi-file context, and chat-based code generation that understands the entire project structure.

Q21. What is the most important security practice when using AI-generated code?

- A. Only use AI-generated code in non-production environments and development branches.
- B. Run all generated code through an AI security scanner before deployment.
- C. Require senior engineer sign-off on all AI-generated code before merging.
- D. Treat every line of generated code as unreviewed third-party code and audit for vulnerabilities.**

Correct Answer: D

Explanation

AI-generated code can contain security vulnerabilities — SQL injection, insecure authentication, hardcoded credentials, dependency vulnerabilities — that look plausible to a reviewer not specifically looking for security issues. Treating generated code with the same scrutiny as third-party code is the correct security posture.

Q22. What is "GitHub Copilot" and what is its primary value proposition?

- A. An AI pair programmer integrated into VS Code that suggests code completions in real time.**
- B. A GitHub product that automatically reviews and approves pull requests using AI.
- C. A tool that generates entire GitHub repositories from a natural language description.
- D. An AI model trained specifically on GitHub's private codebase for internal use.

Correct Answer: A

Explanation

GitHub Copilot is an AI coding assistant from GitHub/Microsoft that integrates into editors (primarily VS Code) and suggests code completions, functions, tests, and documentation in real time as developers type. Its primary value is acceleration of routine coding tasks through AI-assisted autocomplete.

Q23. What is the best way to use an AI code assistant for unfamiliar codebases?

- A. Have it rewrite the entire codebase in a more familiar style.
- B. Ask it to explain existing code sections before making changes.**
- C. Use it to generate tests before attempting to understand the existing code.
- D. Ask it to identify and remove code that may be outdated.

Correct Answer: B

Explanation

For unfamiliar codebases, asking the AI to explain existing code — what a function does, how a module is structured, what a pattern is doing — is the most immediately valuable use. Understanding before changing prevents breaking changes made from misunderstanding.

Q24. What is the most common failure mode when using AI to generate unit tests?

- A. Generated tests are too comprehensive and slow down the CI/CD pipeline.
- B. Generated tests use deprecated testing frameworks the project doesn't use.
- C. Generated tests cannot cover edge cases because AI doesn't understand business logic.
- D. Generated tests pass against the generated code without testing actual correct behaviour.**

Correct Answer: D

Explanation

When AI generates both code and tests, the tests are written to match the code's actual (possibly wrong) behaviour — not the intended behaviour. Tests can pass 100% while the underlying logic is wrong. Tests must be written from the specification/requirements, not from the implementation.

Q25. What is "Devin" conceptually as an AI coding tool?

- A. A developer productivity tool that estimates time and complexity for coding tasks.
- B. An AI-powered code review system that identifies issues in pull requests.
- C. An autonomous AI software engineer that plans, codes, and debugs end-to-end tasks.**
- D. A pair programming tool that observes a developer coding and offers suggestions.

Correct Answer: C

Explanation

Devin (and similar systems) represents the "autonomous AI software engineer" category — an agent that receives a task specification, plans the approach, writes code, runs tests, debugs failures, and iterates until the task is complete, with minimal human intervention. This is fundamentally different from copilot-style assistance.

Q26. What is the biggest risk of using AI to accelerate technical debt reduction?

- A. Refactoring speed outpaces testing capacity, leaving gaps in test coverage.
- B. Generated refactored code may introduce new bugs while removing old patterns.**
- C. AI may not understand the business reasons behind legacy code patterns.
- D. Generated code may accidentally change application behaviour while improving structure.

Correct Answer: B

Explanation

The biggest risk of AI-assisted refactoring is introducing new bugs. Refactoring that changes code structure should preserve behaviour exactly. AI may refactor correctly in isolation but break behaviour that depended on the exact old implementation. Comprehensive test suites before AI-assisted refactoring are essential.

Q27. What is "code generation" vs "code assistance" in the AI tool landscape?

- A. Generation produces complete code from specifications; assistance augments developers writing code.**
- B. Generation works on any programming language; assistance is limited to major languages.
- C. Generation is a batch process; assistance operates in real time during development.
- D. Generation is more accurate than assistance because it starts from scratch.

Correct Answer: A

Explanation

Code generation tools (GitHub Copilot Workspace, Replit AI, Lovable) produce complete code artifacts from high-level descriptions. Code assistance tools (Copilot, Cursor, Tabnine) augment developers with suggestions, completions, and explanations as they write. Both categories are valuable but serve different use cases.

Q28. What is the most appropriate use of AI for code documentation?

- A. Replacing the need for developers to write documentation themselves.
- B. Generating documentation that serves as the authoritative source of truth.
- C. Generating initial docstrings and comments that developers then review and refine.**
- D. Documenting only AI-generated code sections, leaving human code unchanged.

Correct Answer: C

Explanation

AI can generate plausible documentation quickly, but it may be inaccurate about intent, edge cases, or the reasons behind decisions. Developers must review and refine AI-generated documentation to ensure it accurately describes what the code does and why. It is a starting point, not an end product.

Q29. What is "Copilot for CLI" and what does it enable?

- A. A tool for deploying Copilot-generated code through command-line CI/CD pipelines.
- B. A CLI interface for accessing GitHub Copilot's code generation capabilities.
- C. An AI that monitors CLI output and alerts when commands produce errors.
- D. AI assistance for terminal commands — explaining, generating, and debugging shell commands.**

Correct Answer: D

Explanation

GitHub Copilot for CLI (and similar tools like Fig) provide AI assistance in the terminal — explaining what commands do, suggesting commands based on natural language descriptions, and helping debug command failures. This extends AI code assistance beyond the editor to the command line.

Q30. What is the most reliable way to evaluate an AI code assistant for a development team?

- A. Compare benchmark scores from the assistant vendors' published documentation.
- B. Run a task-specific evaluation on real project work and measure quality and velocity outcomes.**
- C. Survey developers on their subjective preference after a two-week trial.
- D. Measure the number of code suggestions accepted versus rejected over 30 days.

Correct Answer: B

Explanation

Real-world task evaluation on the team's actual work is the most reliable signal. Vendor benchmarks reflect best-case performance on standard tasks; subjective preference does not measure quality; acceptance rate does not measure correctness — developers may accept wrong code. Actual quality and velocity outcomes on real project tasks are the true measure.

SET

18

No-Code and Low-Code AI

Set 18 of 5 · 15 Questions · AI Engineering

Q31. What is a "no-code AI builder" and what type of user does it serve?

- A. A visual tool for building AI-powered applications without writing code.
- B. A simplified version of Claude designed for users with no AI experience.
- C. A tool that generates code automatically so developers never need to write it.
- D. A drag-and-drop interface for assembling pre-built AI models into pipelines.

Correct Answer: A**Explanation**

No-code AI builders (Lovable, Bolt, v0, Replit with AI) allow non-developers to create functional AI-powered applications through visual interfaces, natural language instructions, and drag-and-drop components. They serve non-technical users who need to build products without programming knowledge.

Q32. What is "N8N" and what does it enable in an AI workflow context?

- A. A no-code platform specifically for building AI chatbots and conversational agents.
- B. A vector database optimised for storing N8N workflow configuration data.
- C. An open-source workflow automation tool that connects AI and non-AI services.
- D. A Claude-specific workflow builder maintained by Anthropic for enterprise users.

Correct Answer: C**Explanation**

N8N is an open-source (self-hostable) workflow automation platform that connects hundreds of apps and services. In AI contexts, it is used to build workflows that trigger Claude or other AI models based on events (emails, form submissions, database changes) and route AI outputs to downstream services.

Q33. What is "Zapier" commonly used for in AI-enabled business workflows?

- A. Building AI models from business data without requiring machine learning expertise.
- B. Connecting AI tools with other business applications through trigger-action automations.**
- C. A Claude-certified integration platform for enterprise AI deployments.
- D. A workflow orchestration tool for managing multi-agent AI systems.

Correct Answer: B

Explanation

Zapier is a no-code integration platform that connects 5,000+ apps through simple trigger-action workflows ("Zaps"). In AI contexts, it is commonly used to trigger AI model calls based on events (new email, form submission, calendar event) and route AI outputs to CRMs, databases, or communication tools.

Q34. What is "Make" (formerly Integromat) and how does it differ from Zapier?

- A. A code generation tool that produces Zapier workflows from natural language descriptions.
- B. An AI model training platform that creates custom models from business data.
- C. A project management tool with AI-powered task prioritisation and scheduling.
- D. A visual workflow automation platform with more complex branching logic than Zapier.**

Correct Answer: D

Explanation

Make is a visual workflow automation platform similar to Zapier but with a more powerful visual editor that handles complex conditional logic, data transformations, and multi-step workflows more elegantly. It is popular for workflows too complex for Zapier's linear structure.

Q35. What is "Lovable" (formerly GPT Engineer) as a tool?

- A. A no-code AI app builder that generates full-stack web applications from text descriptions.**
- B. A fine-tuning platform that makes AI models more lovable by improving their personality.
- C. A prompt engineering tool that generates optimised prompts for different AI models.
- D. A customer feedback tool that helps product teams build AI features users love.

Correct Answer: A

Explanation

Lovable is a no-code builder that takes natural language descriptions and generates complete, functional full-stack web applications — frontend, backend, and database configuration. It competes in the "vibe coding" space where non-technical users can build real apps through conversation.

Q36. What is the primary limitation of no-code AI tools compared to coded implementations?

- A. They are slower to build with than hand-coding the same functionality.
- B. They offer less control over complex logic, error handling, and system integration.**
- C. They cannot connect to enterprise APIs and databases.
- D. They produce lower quality AI outputs because they use simpler models.

Correct Answer: B

Explanation

No-code tools excel at standard use cases with predictable flows. For complex conditional logic, custom error handling, fine-grained API integration, or non-standard data transformations, hand-coded implementations offer control that no-code visual builders cannot match. The tradeoff is speed/accessibility versus control/flexibility.

Q37. What is "Airtable with AI" and what workflow does it enable?

- A. An AI model training platform that uses Airtable data to fine-tune language models.
- B. A document management system with AI search and summarisation capabilities.
- C. A project management tool that uses AI to assign tasks to team members.
- D. A structured database with AI-powered automation for enriching and acting on data.**

Correct Answer: D

Explanation

Airtable with AI capabilities allows users to run AI operations directly on database records — summarising fields, categorising entries, extracting structured data, triggering automations — without leaving the database interface. It brings AI into structured data workflows that would otherwise require custom code.

Q38. What is "Replit" and how is it used for AI-enabled development?

- A. A no-code platform for deploying pre-built AI models to production.
- B. A tool for converting AI-generated code into deployable applications automatically.
- C. A browser-based coding environment with integrated AI assistance and instant deployment.**
- D. A code review platform that uses AI to ensure quality before deployment.

Correct Answer: C

Explanation

Replit is a browser-based IDE with built-in AI assistance (Ghostwriter/Replit AI) that allows users to write, run, and deploy code entirely in the browser. It lowers the barrier to AI-assisted development by eliminating local environment setup, and its AI tools help both beginners and experienced developers build faster.

Q39. What is the most important consideration when choosing between Zapier and custom code for an AI workflow?

- A. Cost — Zapier subscriptions are more expensive than equivalent server infrastructure.
- B. Complexity, maintainability needs, and whether the use case fits standard trigger-action patterns.**
- C. Speed of development — custom code is always faster to build for simple use cases.
- D. Security — custom code is always more secure than third-party automation platforms.

Correct Answer: B

Explanation

The right choice depends on use case fit: if the workflow follows standard trigger-action patterns with supported apps, Zapier wins on speed and maintenance. If the workflow has complex logic, non-standard integrations, or high-volume requirements that Zapier cannot handle, custom code is necessary. Neither is universally superior.

Q40. What is "NotebookLM" and what is its primary use case?

- A. A Google AI tool for synthesising and conversing with uploaded documents.**
- B. An AI-powered Jupyter notebook that generates and executes data science code.
- C. A tool for documenting and explaining AI workflows built with other no-code tools.
- D. A Google Workspace integration that adds AI assistance to Google Docs.

Correct Answer: A

Explanation

NotebookLM is a Google AI tool that allows users to upload documents (PDFs, articles, notes, audio) and then chat with, summarise, and synthesise information from those specific sources. It excels at research synthesis, document understanding, and knowledge management within a defined document set.

Q41. What is "Perplexity AI" best described as?

- A. An AI writing assistant optimised for perplexing or counterintuitive content.
- B. A research tool for evaluating and comparing the perplexity scores of different AI models.
- C. An AI-powered search engine that answers questions with cited sources.**
- D. A conversational AI trained specifically for academic research and citation.

Correct Answer: C

Explanation

Perplexity is an AI-powered search engine that combines LLM generation with real-time web search, providing cited, sourced answers to questions. It fills a gap between traditional search (links) and pure AI (uncited text) by producing synthesised, referenced responses with source attribution.

Q42. What is the best use case for a no-code AI tool in a professional workflow?

- A. Building complex AI applications with custom business logic and integrations.
- B. Replacing software development entirely for AI-enabled product features.
- C. Running large-scale AI model fine-tuning on proprietary company data.
- D. Automating high-volume, repetitive AI tasks that follow a consistent pattern.**

Correct Answer: D

Explanation

No-code tools shine for high-volume, consistent workflows — automatically processing new form submissions, enriching database records, routing support tickets, generating reports — where the pattern is predictable and supported tools exist. Complex custom logic, edge case handling, and fine-tuning all require code.

Q43. What is the most important governance consideration when deploying no-code AI automations in a company?

- A. Verifying that the no-code platform is ISO 27001 certified for enterprise use.
- B. Ensuring automated AI actions are reviewed and authorised by appropriate stakeholders.**
- C. Confirming the no-code tool uses the same AI model as the company's other AI tools.
- D. Ensuring all automation workflows are documented in a central workflow registry.

Correct Answer: B

Explanation

No-code tools make it easy for non-technical employees to build automations that take real actions — sending emails, modifying records, triggering purchases. Without governance over who can build what, unauthorised or poorly designed automations can cause significant business harm. Authorisation and review processes are the critical governance requirement.

Q44. What is "Gamma" and what does it produce?

- A. A GitHub integration tool that uses AI to automate pull request reviews.
- B. A code generation tool from Google that produces Python scripts from descriptions.
- C. An AI-powered presentation tool that generates polished slide decks from text.**
- D. An AI model evaluation platform for comparing presentation-quality outputs.

Correct Answer: C

Explanation

Gamma is an AI-powered presentation tool that generates complete, visually designed presentations, documents, and webpages from natural language descriptions or outlines. It produces polished slide decks with appropriate layouts, imagery, and design without requiring manual design work.

Q45. What is the primary difference between workflow automation tools (Zapier, N8N) and agentic AI systems?

A. Workflow tools follow fixed predetermined paths; agents dynamically decide next steps.

B. Workflow tools are cheaper to run; agents use more compute per task.

C. Workflow tools work with structured data; agents work with unstructured text.

D. Workflow tools are synchronous; agents are always asynchronous.

Correct Answer: A

Explanation

Workflow automation tools execute predefined, static trigger-action sequences — the same path every time. Agentic AI systems plan dynamically, observe results, and decide next steps based on what they encounter. Agents can handle novel situations workflow tools cannot, but are harder to test and predict.

SET

19

Multimodal Tools

Set 19 of 5 · 15 Questions · AI Engineering

Q46. What is a "multimodal" AI model?

- A. A model that combines multiple training methodologies for improved performance.
- B. A model that operates across multiple deployment environments simultaneously.
- C. A model that can be accessed through multiple API interfaces.
- D. A model that can process and generate multiple content types such as text, images, and audio.**

Correct Answer: D**Explanation**

Multimodal AI models process and/or generate more than one type of data — typically combinations of text, images, audio, video, and code. GPT-4o, Claude 3 (with vision), and Gemini are examples of models that can process images alongside text.

Q47. What is "vision capability" in a language model and what does it enable?

- A. The ability to generate images from text descriptions.
- B. The ability to analyse and reason about images provided as input.**
- C. The ability to process video files and extract relevant information.
- D. The ability to render visual outputs like charts and diagrams.

Correct Answer: B**Explanation**

Vision capability allows a language model to take images as input and reason about their content — describing what is in an image, answering questions about it, extracting text from it (OCR), or analysing diagrams and charts. It is an input capability, not image generation.

Q48. What is the most important limitation of AI vision models when processing documents?

- A. Vision models can only process one page at a time, not multi-page documents.
- B. Vision models cannot reason about numerical data presented in tables.
- C. OCR accuracy degrades with poor image quality, unusual fonts, or complex layouts.**
- D. Vision models require images to be in PNG format to achieve accurate text extraction.

Correct Answer: C

Explanation

AI vision OCR accuracy depends on image quality, font clarity, and layout complexity. Handwritten text, decorative fonts, tables with merged cells, and low-resolution scans all degrade extraction accuracy. The model produces confident but wrong text in cases where the image is ambiguous.

Q49. What is "text-to-image" generation and which tools are most commonly used?

- A. Generating images from text descriptions using diffusion models like Midjourney and DALL-E.**
- B. Converting text documents into visual infographics using AI layout tools.
- C. Generating text captions for existing images using vision language models.
- D. Converting text-based code into visual diagram representations.

Correct Answer: A

Explanation

Text-to-image generation uses diffusion models (Stable Diffusion, Midjourney, DALL-E, Ideogram) to synthesise images from natural language descriptions. The user provides a text prompt and the model generates an image matching the description.

Q50. What is "speech-to-text" (STT) and what are its most common production applications?

- A. Generating realistic speech audio from text using AI voice synthesis.
- B. Converting handwritten text captured by a device's microphone to digital text.
- C. A search technique that queries databases using voice commands rather than typing.
- D. Converting spoken audio to written text — used in transcription, voice interfaces, and captioning.**

Correct Answer: D

Explanation

Speech-to-text (ASR — Automatic Speech Recognition) converts audio containing speech into written text. Common applications include meeting transcription, voice assistants, automated customer service (IVR), accessibility captioning, and voice-controlled interfaces.

Q51. What is "Whisper" in the AI tool landscape?

- A. Anthropic's tool for generating low-volume, subtle AI content moderation signals.
- B. A browser-based tool for private, end-to-end encrypted conversations with Claude.
- C. OpenAI's open-source speech recognition model for transcribing audio.**
- D. A voice synthesis model that generates human-sounding speech from text.

Correct Answer: C

Explanation

Whisper is OpenAI's open-source automatic speech recognition model trained on multilingual audio data. It provides high-accuracy transcription across many languages and is widely used as a component in voice AI applications, meeting transcription tools, and voice interfaces.

Q52. What is "text-to-speech" (TTS) and when is it most appropriate to use AI TTS?

- A. Converting written text to synthesised spoken audio — appropriate for scalable voice content.**
- B. Converting spoken audio to text using AI transcription models.
- C. An AI capability that speaks responses aloud in real time during a conversation.
- D. A tool for matching AI-generated voice to a specific person's voice characteristics.

Correct Answer: A

Explanation

TTS converts text to synthetic speech. AI TTS (ElevenLabs, OpenAI TTS, Google WaveNet) produces naturalistic, human-sounding voices and is appropriate when scale or customisation makes human voiceover impractical — audiobooks, scalable content, accessibility features, and voice-based AI interfaces.

Q53. What is the most important consideration when deploying AI vision in a compliance-sensitive document workflow?

- A. Whether the vision model can process the company's proprietary file formats.
- B. Whether the OCR accuracy is sufficient for the document types and consequences of extraction errors.**
- C. Whether the AI vision tool has been trained on similar document types.
- D. Whether the vision model supports batch processing for high-volume document intake.

Correct Answer: B

Explanation

In compliance-sensitive workflows (legal, financial, medical), document extraction errors can have serious consequences. The primary consideration is whether OCR accuracy on the specific document types (quality, format, font) meets the accuracy threshold required for the decisions that will be made from the extracted data.

Q54. What is "video AI" and what are its emerging production use cases?

- A. A category of AI tools for generating realistic synthetic video from text descriptions.
- B. The application of computer vision to optimise video streaming quality and compression.
- C. AI applied to video for analysis, summarisation, and content understanding at scale.**
- D. AI tools that assist video editors with timeline management and color correction.

Correct Answer: C

Explanation

Video AI encompasses tools for analysing, summarising, and extracting insights from video content — meeting recordings, training videos, product demos, surveillance footage. Common applications include automated meeting summaries, content moderation, compliance monitoring, and knowledge extraction from video archives.

Q55. What is the primary risk when using AI-generated images in commercial contexts?

- A. Generated images are always lower resolution than photography for professional use.
- B. AI image styles are recognisable and may reduce brand authenticity.
- C. Generated images cannot be used in print because they lack sufficient metadata.
- D. Copyright and licensing issues with training data or generated outputs resembling existing works.**

Correct Answer: D

Explanation

AI image generation training data and outputs can raise copyright concerns — some generated images closely resemble copyrighted works, and the training data itself includes copyrighted images. Commercial use of AI-generated images requires understanding the licensing terms of the specific tool and awareness of potential IP issues.

Q56. What is the most accurate description of current "AI video generation" capability?

- A. Short clip generation from text or image prompts — improving rapidly but still limited in length and consistency.**
- B. Full-length film and commercial production from natural language scripts.
- C. Video editing automation that applies AI enhancements to existing footage.
- D. Real-time video generation that produces content as events unfold.

Correct Answer: A

Explanation

As of current capability, AI video generation tools (Sora, Runway, Kling) can produce short clips (typically seconds to a minute) from text or image prompts. Long-form consistency, character consistency, and production quality remain active research challenges. It is a rapidly improving but still nascent capability for professional production.

Q57. What is "document AI" and what workflows does it enable?

- A. An AI model trained specifically on legal, financial, and technical documents.
- B. AI for extracting, classifying, and routing information from documents at scale.**
- C. A document management system that uses AI search for knowledge retrieval.
- D. Tools that convert AI model outputs into formatted business documents.

Correct Answer: B**Explanation**

Document AI encompasses tools for processing unstructured documents — invoices, contracts, forms, reports — and extracting structured information from them. Use cases include automated invoice processing, contract review, form data extraction, and compliance document routing.

Q58. What is the most important quality metric for an AI transcription tool in a business context?

- A. Transcription speed compared to the real-time playback speed of the audio.
- B. The number of languages the tool can transcribe simultaneously.
- C. The model's benchmark performance on standard transcription datasets.
- D. Word error rate (WER) on the specific audio conditions the tool will encounter in production.**

Correct Answer: D**Explanation**

WER (Word Error Rate) on the actual production conditions — meeting audio with multiple speakers, background noise, industry-specific terminology, accents — determines whether the transcription is accurate enough for business use. Standard benchmarks are recorded in clean conditions and may not reflect real-world performance.

Q59. What is the key difference between "vision" models and "image generation" models?

- A. Vision models analyse existing images; generation models create new images from text.**
- B. Vision models are used for surveillance; generation models are used for creative work.
- C. Vision models process real photographs; generation models work only with synthetic images.
- D. Vision models require more compute per inference; generation models are faster.

Correct Answer: A**Explanation**

This is a fundamental taxonomy distinction: vision (or image understanding) models take images as input and produce text output (descriptions, answers, analysis). Image generation models take text as input and produce images as output. A model can have both capabilities, but they are architecturally distinct functions.

Q60. What is the most appropriate multimodal AI tool for a team that processes thousands of invoices daily?

- A. A general-purpose vision model accessed via API for each invoice individually.
- B. A manual review workflow augmented by an AI chatbot for difficult cases only.
- C. A document AI solution specifically designed for invoice data extraction at scale.**
- D. A fine-tuned text model that processes invoice data exported to CSV format.

Correct Answer: C

Explanation

High-volume invoice processing requires a purpose-built document AI solution optimised for that specific document type, with structured output schemas, accuracy monitoring, and the throughput to handle thousands of documents efficiently. General-purpose vision models are less reliable at scale for specialised extraction tasks.

SET

20

Tool Selection and Evaluation

Set 20 of 5 · 15 Questions · AI Engineering

Q61. You need to automate sending a daily summary email based on data from a CRM and a spreadsheet. The most appropriate tool is:

- A. A no-code workflow tool like Zapier or N8N that connects the CRM, spreadsheet, and email.
- B. A custom Python script deployed as a serverless function.
- C. Claude with an MCP connector to all three services.
- D. A specialised email marketing platform with database integrations.

Correct Answer: A

Explanation

This is a standard trigger-action integration use case — connecting three supported services with a consistent daily pattern. A no-code workflow tool handles this without engineering overhead. Custom code and complex AI solutions are overkill for a predictable, well-scoped integration pattern.

Q62. A team wants to build an AI tool for employees to ask questions about internal policy documents. The correct architecture is:

- A. Fine-tune Claude on the policy documents so it learns the specific policies.
- B. Build a keyword search over the documents and use Claude to format search results.
- C. Use Claude directly without any additional setup — it can answer general policy questions from training.
- D. **RAG: index policy documents in a vector database and use Claude to answer questions from retrieved context.**

Correct Answer: D

Explanation

RAG is the correct architecture for document Q&A: it retrieves specific relevant policy sections and grounds Claude's answers in the actual document content. Fine-tuning is expensive and does not allow dynamic document updates. Keyword search misses semantic matching. Claude's training data does not contain proprietary internal policies.

Q63. A non-technical founder wants to build a prototype product with AI features in a weekend. The most appropriate starting point is:

- A. The Claude API with a Python backend and React frontend.
- B. A no-code builder like Lovable or Replit with AI assistance.**
- C. A template from a software agency that they customise for their use case.
- D. A pre-built SaaS product and abandon the idea of a custom AI feature.

Correct Answer: B

Explanation

No-code AI builders (Lovable, Bolt, Replit with AI) are specifically designed for non-technical founders to build functional prototypes quickly. The Claude API with custom code requires developer skills. SaaS templates and agencies either don't exist for the specific use case or are slower and more expensive for a prototype.

Q64. A company wants to add AI assistance to their customer support. The first step before choosing any tool is:

- A. Selecting the AI tool with the highest customer satisfaction ratings.
- B. Piloting the largest, most capable AI model to see what is possible.
- C. Defining what specific support tasks AI should handle and what remains with humans.**
- D. Calculating cost savings from AI replacing human agents before evaluating options.

Correct Answer: C

Explanation

Before selecting tools, the company must define scope: which query types AI should handle, where escalation to humans occurs, and what quality standards apply. Without this, tool selection is random, pilots are unfocused, and cost projections are meaningless because the use case is undefined.

Q65. You are evaluating AI transcription tools for recording team meetings. The most important evaluation criterion is:

- A. Accuracy on real team meeting audio with the team's specific vocabulary and accents.**
- B. The number of speakers the tool can differentiate in the transcript.
- C. Integration with the video conferencing platform the team uses.
- D. The speed at which transcripts are delivered after meetings end.

Correct Answer: A

Explanation

Meeting transcription quality depends on the actual audio conditions — how the team speaks, what vocabulary and acronyms they use, their accents, typical background noise. Testing on real meeting recordings is the only reliable way to assess whether a tool will be accurate enough for the team's actual use.

Q66. Which AI tool is most appropriate for a marketing team that needs to consistently generate on-brand social media posts at scale?

- A. A fine-tuned version of Claude trained on previous on-brand social posts.
- B. A dedicated social media AI tool from a specialist marketing AI vendor.
- C. Claude with a detailed brand voice system prompt and a template for each post type.**
- D. Midjourney for visuals combined with Claude for caption copy separately.

Correct Answer: C

Explanation

A well-configured Claude with a detailed brand voice system prompt and structured output templates provides consistent, scalable on-brand generation without fine-tuning costs. Fine-tuning is warranted only if the brand voice is so unique that prompting cannot capture it. Specialist tools add cost without meaningfully improving output for this use case.

Q67. What is the correct tool for a developer who needs to quickly understand a large, unfamiliar codebase?

- A. GitHub Copilot tab completion as they read through the files.
- B. A no-code tool that visualises code structure and dependencies automatically.
- C. Ask Claude to explain the codebase from a brief description they provide.
- D. Claude Code or a similar AI coding assistant with full codebase access.**

Correct Answer: D

Explanation

Claude Code (or Cursor with full codebase context) can read the actual code files, understand relationships between components, and answer specific questions about how the codebase works. Tab completion helps while writing; a brief description to Claude without codebase access produces generic rather than specific answers.

Q68. A small business owner wants to automate sending personalised follow-up emails after customer purchases. Which approach is most appropriate?

- A. Hiring a developer to build a custom email automation system with AI integration.
- B. A no-code workflow tool connecting the e-commerce platform to an email service with AI personalisation.**
- C. Using Claude directly in a browser to manually draft follow-up emails one at a time.
- D. Purchasing a dedicated AI email marketing platform and migrating all email workflows.

Correct Answer: B

Explanation

For a small business, a no-code workflow tool (Zapier + Claude API or similar) is the most proportionate solution — it connects the e-commerce platform and email service without development cost, and AI personalisation can be added within the workflow. Custom development and platform migration are disproportionate for this scale.

Q69. What is the most important question to ask before integrating a third-party AI tool into a workflow that processes customer data?

- A. "Is this the most accurate AI tool available for this specific task?"
- B. "Does this tool integrate natively with our existing software stack?"
- C. "How does this tool handle, store, and process our customers' data?"**
- D. "What is the per-user cost of this tool at our customer volume?"

Correct Answer: C

Explanation

Before sending customer data to any third-party AI tool, you must understand the data handling, storage, retention, and processing practices — particularly whether data is used for training, where it is stored, who can access it, and whether it complies with relevant regulations. Accuracy, integration, and cost are secondary to this foundational privacy and compliance question.

Q70. A developer needs to choose between Zapier and N8N for a workflow automation project. The most important differentiating factor is:

- A. Complexity and control requirements — N8N offers more flexibility; Zapier is simpler to use.**
- B. Cost — N8N is always cheaper because it is open source.
- C. Accuracy — Zapier uses more reliable connectors than N8N.
- D. Speed — N8N executes workflows faster due to its self-hosted architecture.

Correct Answer: A

Explanation

Zapier is optimised for simplicity and non-technical users — it excels at straightforward trigger-action workflows. N8N offers more complex branching, data transformation, and self-hosting options. The choice depends on workflow complexity and whether control/customisation or ease of use is the priority.

Q71. You are recommending AI tools to a professional services firm. The most important first question is:

- A. "What is your budget for AI tool subscriptions?"
- B. "What specific problems or bottlenecks in your work do you want AI to address?"**
- C. "Do you have a technical team who can implement AI tools?"
- D. "Which AI tools are your competitors already using?"

Correct Answer: B

Explanation

Tool selection should always start with problem definition. Without knowing which specific bottlenecks or inefficiencies the firm wants to address, any tool recommendation is arbitrary. Competitor benchmarking, budget, and technical capacity are all secondary to understanding what problem is being solved.

Q72. What is the most appropriate use of Midjourney or DALL-E in a content production workflow?

- A. Replacing professional photography for all commercial marketing assets.
- B. Producing final publication-ready visuals without designer involvement.
- C. Generating brand guidelines and visual identity systems from text descriptions.
- D. Generating visual concepts and initial images that a designer then refines.**

Correct Answer: D

Explanation

AI image generation is most appropriate as a rapid ideation and concepting tool that designers use to explore visual directions. Using AI-generated images as final production assets without designer refinement and review risks quality issues, brand inconsistency, and copyright concerns.

Q73. What is the correct way to evaluate whether an AI writing tool fits a marketing team's needs?

- A. Check the tool's performance on standard creative writing benchmarks.
- B. Ask the vendor for customer case studies from comparable companies.
- C. Run a controlled trial generating real marketing content and have the team rate quality and brand fit.**
- D. Compare the tool's pricing to other AI writing tools in the same category.

Correct Answer: C

Explanation

Marketing content quality is subjective and brand-specific. Only a real trial generating actual content — the formats, tone, and topics the team will use — and having the marketing team evaluate quality and brand fit, provides reliable signal for tool selection. Benchmarks, case studies, and pricing are secondary.

Q74. A company wants to build an internal AI knowledge base tool. The riskiest approach is:

- A. Building a RAG system that retrieves and cites specific internal documents.
- B. Deploying a general-purpose chatbot without grounding it in the company's actual documents.**
- C. Using Claude with company documents as context in a controlled internal environment.
- D. Starting with a pilot deployment limited to a small group of internal users.

Correct Answer: B

Explanation

An ungrounded general-purpose chatbot will confidently answer questions about company policies, procedures, and data using its training knowledge — which will not match the company's actual internal information. This produces confident wrong answers that are more dangerous than obvious "I don't know" responses.

Q75. Menler's AI Tools Ecosystem bank is designed to test what capability?

- A. Technical knowledge of how each AI tool is built and what algorithms it uses.
- B. Sales capability for recommending AI tools to clients and prospects.
- C. Development skills for integrating multiple AI tools into production systems.
- D. Practical judgment in selecting and using the right AI tool for each professional situation.**

Correct Answer: D

Explanation

The AI Tools Ecosystem bank tests applied selection and usage judgment — understanding what each category of tool does, when to use which tool, and how to evaluate fit for specific professional situations. It tests practitioner judgment, not technical internals, sales skills, or development capability.

SET

21

Output Quality

Set 21 of 5 · 15 Questions · AI Engineering

Q1. What is the most reliable signal that an AI output is high quality?

- A. The output is fluent, grammatically correct, and well-structured.
- B. Specific claims are traceable to verifiable sources or logical derivations.**
- C. The output is longer and more detailed than what was requested.
- D. The model expressed high confidence throughout the response.

Correct Answer: B**Explanation**

Fluency, length, detail, and expressed confidence are all independent of accuracy. An AI can produce beautifully written, highly confident, detailed text that is factually wrong. Traceability of claims to verifiable sources or sound logical derivation is the only reliable quality signal.

Q2. What does "output calibration" mean in professional AI use?

- A. Adjusting the AI model's temperature and sampling settings for each task.
- B. Running AI outputs through a secondary model to score their quality.
- C. Calibrating output length to match the expected format for each task type.
- D. Developing accurate judgment about where AI outputs are reliable versus where they fail.**

Correct Answer: D**Explanation**

Output calibration is the practitioner skill of knowing, based on task type and output characteristics, how much to trust a given AI output — and when deep verification is necessary. It develops through systematic experience and is the core judgment skill for safe professional AI use.

Q3. You receive an AI output that answers a question you did not ask while ignoring the one you did. The cause is:

A. The prompt was ambiguous enough that the model interpreted a different question.

B. The model lacked the knowledge to answer the intended question.

C. The model's safety filters redirected the response to a safer topic.

D. The model exceeded its context window before processing the full question.

Correct Answer: A

Explanation

When AI answers a different question than asked, the most common cause is prompt ambiguity — the question could reasonably be read in a different way, and the model selected a different interpretation. The fix is clarifying the question, not assuming the model is incapable or filtered the response.

Q4. What distinguishes a "correct" AI output from a "useful" one?

A. Correct outputs are longer and more complete; useful outputs are shorter and actionable.

B. Correct outputs are verifiable; useful outputs are those the user decides to act on.

C. Correctness means factually accurate; usefulness means relevant to the actual task need.

D. There is no meaningful distinction — correct outputs are always useful.

Correct Answer: C

Explanation

An output can be factually correct but useless (answering a different level of detail than needed, in the wrong format, for the wrong audience). A useful output addresses the actual need — the right information, at the right depth, in the right format — even if it requires the user to verify the factual claims.

Q5. What is the most effective quality control step for AI-generated reports before distribution?

- A. Running a grammar and spell-check tool to catch surface errors.
- B. Human expert review of all specific claims, figures, and recommendations.**
- C. Asking the AI to review its own report for accuracy before finalising.
- D. Comparing the report structure against a template for format compliance.

Correct Answer: B

Explanation

Human expert review of specific claims, figures, and recommendations is the only quality control step that catches factual errors, logical flaws, and contextual mismatches. Grammar checking, AI self-review, and format compliance checks cannot catch factual or logical errors — they are secondary to this primary check.

Q6. What is "fluency illusion" in AI output evaluation?

- A. Well-written text gives readers false confidence that the content is accurate.**
- B. AI outputs that use complex vocabulary to obscure simple concepts.
- C. The experience of reading AI text as if a human wrote it.
- D. The tendency for AI to repeat fluency-related phrases in different contexts.

Correct Answer: A

Explanation

The fluency illusion describes how skilled writing — clear structure, appropriate vocabulary, confident tone — creates an impression of authority and correctness that may be entirely unwarranted. AI exploits this cognitive bias by generating consistently fluent text regardless of factual accuracy.

Q7. What is the most important dimension to evaluate in an AI output for a professional client deliverable?

- A. Length and detail appropriate for the client's seniority level.
- B. Brand voice consistency with the organisation's communication standards.
- C. Completeness — covering all aspects of the topic without omissions.
- D. Factual accuracy of specific claims that will inform client decisions.**

Correct Answer: D

Explanation

Factual accuracy is the primary dimension for professional deliverables because clients use deliverables to make decisions. Wrong facts lead to wrong decisions. Length, voice, and completeness are all secondary to ensuring what is stated is correct.

Q8. What does "output specificity" indicate about AI quality on knowledge tasks?

- A. Specific outputs are always more accurate because they demonstrate depth of knowledge.
- B. Specific outputs indicate the model has retrieved relevant information from the web.
- C. Specific, concrete outputs are higher risk for error than general summaries.**
- D. Output specificity is independent of accuracy and provides no quality signal.

Correct Answer: C

Explanation

Specific outputs — precise figures, exact names, specific dates, verbatim quotations — carry higher error risk than general summaries. AI generates specific details with the same confidence regardless of whether those details are accurate or fabricated. Specificity requires higher verification standards, not less.

Q9. A colleague says "the AI output looks great, let's use it." What is missing from this evaluation?

- A. Assessment of whether the content is factually accurate, not just whether it looks good.**
- B. A review of the output's length and format relative to the intended audience.
- C. Comparison with a human-written version to assess relative quality.
- D. A review of the prompt used to generate the output for quality issues.

Correct Answer: A

Explanation

"Looks great" is a fluency and format assessment — the output reads well. Missing is an accuracy assessment — whether the specific claims, figures, and reasoning are actually correct. Professional AI use requires distinguishing between good appearance and verified accuracy.

Q10. What is the most common quality failure in AI-generated marketing copy?

- A. Grammar errors that undermine the professional appearance of the copy.
- B. Excessive length that exceeds the intended word count or format requirements.
- C. Overly promotional language that violates advertising standards.
- D. Generic, pattern-following content that lacks the specific voice and differentiation of the brand.**

Correct Answer: D

Explanation

AI marketing copy defaults to the statistically average marketing voice — which is generic, pattern-following, and interchangeable. The most common failure is copy that is technically correct and readable but lacks the specific brand voice, cultural nuance, and differentiated perspective that makes marketing effective.

Q11. What is the correct response when an AI output is good enough for one part of the task but wrong for another?

- A. Reject the entire output and regenerate from scratch.
- B. Use the good parts as-is and fix the wrong parts yourself.**
- C. Ask the AI to fix the wrong parts by identifying them in a follow-up prompt.
- D. Average the two parts together to produce a balanced acceptable output.

Correct Answer: B

Explanation

Surgical editing — keeping what is correct, fixing what is wrong — is the most efficient approach. Regenerating from scratch wastes the 80% that was correct; asking the AI to self-correct on identified errors is unreliable; averaging makes no sense. Fix the wrong parts directly and move on.

Q12. What is the best indicator that an AI output has been appropriately personalised for a specific audience?

- A. It is shorter and more direct than a general-purpose response would be.
- B. It avoids technical jargon that might confuse a non-expert audience.
- C. It references specific context, terminology, and concerns relevant to that audience.**
- D. It was generated using a role prompt specific to the target audience.

Correct Answer: C

Explanation

True personalisation is evidenced by specific, audience-relevant content — their context, their vocabulary, their specific concerns and goals. Generic brevity, jargon avoidance, and role prompting can all contribute, but they are not sufficient unless the output actually contains specific relevant content.

Q13. What is "output drift" in a multi-step AI task?

- A. The model's output quality decreasing as the context window fills.
- B. Inconsistency in output style between different sections of a long document.
- C. The model taking longer per output token as context length increases.
- D. The AI progressively deviates from the original brief as the task proceeds.**

Correct Answer: D

Explanation

Output drift occurs in long tasks or conversations where the model gradually loses track of the original objectives, constraints, or style requirements — each output drifts slightly from the brief until the final output is substantially different from what was originally requested.

Q14. What is the most important thing to check when comparing two AI outputs on the same task?

- A. Which output is longer, as length often correlates with thoroughness.
- B. Which output better fulfils the actual task objective, not just which reads better.**
- C. Which output uses more varied vocabulary and sentence structures.
- D. Which output was generated with the higher temperature setting.

Correct Answer: B

Explanation

Output comparison should always evaluate task fulfilment — which output actually does what was needed. Reading quality, length, vocabulary variety, and generation settings are all irrelevant if neither output correctly addresses the task objective.

Q15. What is the most appropriate quality standard for AI-generated content used in training materials?

- A. Every factual claim must be accurate, as learners will treat training content as authoritative.**
- B. The content must be engaging and accessible, as these are the primary goals of training.
- C. The content must pass AI detection tools to ensure it does not appear AI-generated.
- D. The content must be reviewed by the L&D team for format and length compliance.

Correct Answer: A

Explanation

Training materials create beliefs and shape decisions. Learners treat authoritative sources as accurate. AI-generated training content with factual errors produces incorrect beliefs at scale — every person trained on wrong content carries that error into their professional work. Factual accuracy is the non-negotiable standard.

SET

22

Hallucination and Verification

Set 22 of 5 · 15 Questions · AI Engineering

Q16. What is "hallucination" in a language model?

- A. The model seeing patterns or meanings that are not present in the input.
- B. A failure mode where the model refuses to answer factual questions.
- C. **Generating text that is confident and plausible but factually incorrect.**
- D. Generating repetitive or looping text that does not converge to an answer.

Correct Answer: C**Explanation**

Hallucination in language models refers to generating confident, specific, well-structured text that contains factually incorrect information — fabricated citations, wrong statistics, invented product features, or nonexistent people. The model does not "know" it is wrong; it generates the most probable-sounding text.

Q17. Why is hallucination a structural property of language models rather than a fixable bug?

- A. **Models predict statistically probable text, which can produce plausible-sounding errors.**
- B. Hallucination is caused by insufficient training data and grows less common with more data.
- C. Hallucination is a deliberate design choice to make models more creative.
- D. Hallucination occurs only when models are asked questions outside their training domain.

Correct Answer: A**Explanation**

Language models are optimised to generate the most probable next tokens — not to retrieve verified facts. This means plausible-sounding incorrect information is an inherent output of the architecture when the model's training probability distribution happens to peak at a wrong answer. Reducing hallucination is an active area of research, but eliminating it entirely requires fundamental changes to how these systems work.

Q18. What is the most reliable method for detecting hallucination in an AI output?

- A. Running the output through a second AI model for cross-validation.
- B. Asking the AI to rate its own confidence on each claim in the output.
- C. Checking whether the output is internally consistent and logically coherent.
- D. Tracing every specific claim to a verifiable primary source.**

Correct Answer: D

Explanation

Only tracing specific claims to verifiable primary sources actually confirms accuracy. Cross-checking with another AI compounds the same hallucination tendency. AI self-confidence is unreliable — models express equal confidence on correct and hallucinated facts. Internal consistency does not ensure external accuracy.

Q19. An AI output cites a specific academic paper with author names, journal, and year. What should you do before using this citation?

- A. Use the citation as-is — AI cites sources accurately when provided with a reference.
- B. Search for and verify the actual paper exists and says what the AI claims.**
- C. Add a note that the citation was AI-generated for full transparency.
- D. Ask the AI to confirm the citation is correct by re-checking its training data.

Correct Answer: B

Explanation

AI commonly fabricates citations that sound completely real — plausible author names, journals, years, and abstracts. Never use an AI-generated citation without finding and reading the actual paper in a verified academic database. Asking the AI to confirm compounds the problem; noting AI provenance does not make the citation real.

Q20. What is "confabulation" and how is it related to hallucination?

- A. Confabulation refers to AI-generated content that confuses different topics into one answer.
- B. Confabulation is a more severe form of hallucination where the model invents entire narratives.
- C. Confabulation is producing false memories with confidence — hallucination is the AI equivalent.**
- D. Confabulation is the technical term for hallucination in medical AI contexts.

Correct Answer: C

Explanation

Confabulation (from neuropsychology) describes a brain phenomenon where a patient fills gaps in memory with fabricated but sincerely believed information, without intent to deceive. This is the closest human cognitive analogue to LLM hallucination — the model does not "know" it is wrong.

Q21. What is the highest-risk task type for AI hallucination?

- A. Generating creative fiction where facts do not need to be accurate.
- B. Summarising a long document provided directly in the context.
- C. Reformatting data from one structured format to another.
- D. Generating specific, verifiable details like dates, statistics, people, and citations.**

Correct Answer: D

Explanation

Specific verifiable details — exact dates, precise statistics, named individuals, bibliographic citations — are where hallucination is most dangerous. The AI generates these with the same confidence as correct information, but they are exactly the details that get fact-checked and used in consequential decisions.

Q22. What is the correct way to prompt Claude to reduce hallucination risk on factual questions?

- A. Instruct Claude to acknowledge uncertainty and cite its reasoning rather than assert facts confidently.**
- B. Use a higher temperature to make Claude more creative and less likely to commit to wrong facts.
- C. Ask Claude to provide three possible answers and select the most common one.
- D. Provide Claude with a specific factual question and ask for the single most accurate answer.

Correct Answer: A

Explanation

Instructioning Claude to acknowledge uncertainty and explain its reasoning helps surface where the model is generating from pattern rather than solid knowledge. Uncertainty acknowledgment is a weak signal but better than false confidence. Higher temperature increases randomness not accuracy; three answers compounds the problem; a direct factual question produces confident wrong answers.

Q23. What is "grounding" as a mitigation for hallucination?

- A. Adding a factual accuracy disclaimer to all AI-generated outputs.
- B. Providing the model with source documents to reason from rather than generating from training memory.**
- C. Training the model on larger and higher-quality datasets to reduce errors.
- D. Using retrieval-augmented generation to supplement the model's knowledge cutoff.

Correct Answer: B

Explanation

Grounding means providing the model with the relevant source material — documents, data, verified facts — and instructing it to reason from that material rather than from its training knowledge. Grounded outputs are traceable to the provided context; ungrounded outputs are generated from statistical memory. RAG is the most common implementation of grounding.

Q24. A developer finds that adding "check your work" to a prompt reduces errors. Why does this work?

- A. It signals to the model that the task requires higher accuracy and changes the generation weights.
- B. It triggers a different model pathway designed for accuracy-critical tasks.
- C. It doubles the effective context the model uses by prompting it to re-read the input.
- D. The instruction activates a self-review step that catches some generation errors.**

Correct Answer: D

Explanation

Self-review instructions cause the model to generate additional checking steps that can catch some logical errors, inconsistencies, and obvious mistakes. They do not change model weights or trigger different pathways — they simply add a generation step that functions like a scratchpad review. This is a genuine but imperfect mitigation.

Q25. What is the "verification habit" and why is it the most important AI skill for professionals?

- A. Asking the AI to verify its own outputs before accepting them as final.
- B. Running all AI outputs through a third-party fact-checking service.
- C. Systematically checking AI-generated claims against primary sources before acting on them.**
- D. Having a second team member review all AI-generated content before use.

Correct Answer: C

Explanation

The verification habit is the discipline of independently checking the specific factual claims in AI outputs against verifiable sources before acting on them. It is the most important skill because AI generates confident wrong information at a rate that makes professional reliance without verification a systematic source of error.

Q26. What does "source attribution" mean in an AI output context, and why is it insufficient on its own?

- A. Attribution identifies who is responsible for the information in the output.
- B. Citing a source does not guarantee the source exists or says what the AI claims.**
- C. Source attribution is a copyright requirement for AI-generated content.
- D. Attribution is sufficient if the source cited is a peer-reviewed academic journal.

Correct Answer: B

Explanation

AI can generate source attributions — journal names, author names, article titles — that are entirely fabricated. Attribution that has not been independently verified against the actual source is not evidence of accuracy. This is why AI-generated citations must always be verified in the original source, regardless of how authoritative they sound.

Q27. You are using AI to write a report on a competitor's market position. Which facts must you independently verify?

- A. All specific claims about the competitor's products, market share, revenue, and strategy.**
- B. Only the financial figures, since these are the most commonly hallucinated.
- C. Only claims that conflict with your prior understanding of the competitor.
- D. No verification is needed if the AI was given accurate web search results to work from.

Correct Answer: A

Explanation

All competitor-specific facts — products, features, pricing, market share, strategic announcements, personnel — must be verified from primary sources. AI confidently generates incorrect competitor details. Even when given web search results, AI can misattribute, misquote, or confuse information across sources.

Q28. What is the most important principle for professional use of AI outputs in high-stakes contexts?

- A. Always disclose that AI was used to produce any professional output.
- B. Only use AI for tasks where errors are immediately obvious and easily corrected.
- C. Never use an AI output for a consequential decision without verifying the critical facts.**
- D. Always have a second AI model cross-check the first model's output before use.

Correct Answer: C

Explanation

In high-stakes contexts, the professional responsibility is clear: verify the critical facts before acting. Disclosure, ease of error detection, and AI cross-checking are all supporting practices but none substitute for the fundamental requirement of verification when decisions have significant consequences.

Q29. What is the most reliable indicator that an AI output is low risk for hallucination?

- A. The output is shorter, as shorter outputs have fewer opportunities for error.
- B. The model expressed appropriate uncertainty about some claims in the output.
- C. The task topic is well-represented in the model's training data.
- D. The task is structural or logical rather than factual — formatting, calculation, or reasoning.**

Correct Answer: D

Explanation

Structural and logical tasks — reformatting data, solving well-defined problems, transforming information from one structure to another — have verifiable correct answers and do not require the model to generate specific factual details from memory. These are inherently lower-risk than knowledge-intensive tasks.

Q30. What is the most important thing a professional learns from consistently applying the verification habit?

- A. A comprehensive list of AI output types that are reliably accurate.
- B. A calibrated sense of which AI output types need more versus less verification.**
- C. The ability to identify hallucinations by reading style and tone alone.
- D. Confidence that AI outputs are reliable enough for most professional tasks.

Correct Answer: B

Explanation

Consistent verification builds calibration — experience-based judgment about where AI tends to be right and where it tends to fail. This allows professionals to direct their verification effort efficiently: deep verification where it matters, light spot-checking where AI is reliably strong.

SET

23

Prompting Decisions

Set 23 of 5 · 15 Questions · AI Engineering

Q31. What is the single most impactful element to add to a vague prompt to improve AI output quality?

- A. A specific description of what the output should look like when done well.**
- B. A longer, more detailed description of the topic to give the AI more context.
- C. Instructions for the AI to "think carefully" before responding.
- D. A request for the AI to ask clarifying questions before answering.

Correct Answer: A

Explanation

Specifying what good output looks like — the format, level of detail, perspective, structure — is the highest-leverage addition to a vague prompt. Vague prompts produce average outputs; specific output criteria produce targeted outputs. Topic detail, generic quality instructions, and clarifying questions are all lower-leverage additions.

Q32. When should you use a "role prompt" (e.g., "act as a senior editor")?

- A. When you want the model to take on a persona and maintain it throughout the conversation.
- B. When you need the model to use domain-specific terminology in its responses.
- C. When the desired output benefits from a specific professional perspective or expertise level.**
- D. When you want to prevent the model from including caveats and qualifications.

Correct Answer: C

Explanation

Role prompts activate patterns associated with that professional perspective in the model's training distribution. "Act as a senior editor" shifts the model toward more critical, polish-focused feedback. This is most valuable when the desired output genuinely benefits from a specific expert viewpoint, not just for terminology or to suppress caveats.

Q33. What is the most effective way to get a consistently structured output format from Claude?

- A. Use the phrase "always follow this format" in the system prompt.
- B. Provide the exact template or schema you want in the prompt, ideally with an example.**
- C. Ask Claude to generate the format itself before producing the content.
- D. Specify the desired format in general terms at the end of the prompt.

Correct Answer: B

Explanation

Templates and schemas with examples are the most effective format controls. They show Claude exactly what output should look like rather than describing it, which is always more reliable than a description. General format terms at the end of a prompt are less reliable than specific templates at the beginning.

Q34. What is "negative prompting" and when is it useful?

- A. Providing examples of bad outputs to help the model recognise what not to produce.
- B. Using a critical, evaluative tone in the prompt to encourage higher-quality responses.
- C. Removing positive examples from the prompt to force more creative outputs.
- D. Explicitly telling the model what to avoid in its output to prevent specific failure modes.**

Correct Answer: D

Explanation

Negative prompting tells the model what not to do — "do not include caveats," "avoid jargon," "do not repeat the question in your response." It is most useful for preventing specific recurring failures that are not prevented by positive instructions alone. Combined with positive examples, it gives the model a clear picture of the target output.

Q35. What does "few-shot prompting" mean and why does it work?

- A. Providing example input-output pairs in the prompt that demonstrate the desired behaviour.**
- B. Running the same prompt multiple times and selecting the best few outputs.
- C. Starting with a short, minimal prompt and adding detail until the output improves.
- D. Using only a few essential instructions rather than a long, complex prompt.

Correct Answer: A

Explanation

Few-shot prompting gives the model examples of the task — here is an input and here is the kind of output I want. The model uses these as patterns to follow rather than interpreting an abstract description. Examples are often more effective than long descriptions because they directly show the target output format and quality.

Q36. When is it better to use multiple simple prompts in sequence rather than one complex prompt?

- A. When you want to reduce API costs by keeping each individual call shorter.
- B. When the task has distinct stages that each benefit from focused, undivided attention.**
- C. When the task requires real-time user input at intermediate stages.
- D. Always — multiple simple prompts are always more reliable than one complex prompt.

Correct Answer: B

Explanation

Multiple prompts in sequence (chaining) are better when the task has genuinely distinct stages — extract, validate, synthesise — where each step's output quality benefits from the model having a focused, constrained context. One complex prompt works when the task is well-integrated and the model can handle all aspects in one pass.

Q37. What is the most common mistake people make when writing their first serious prompts?

- A. Making the prompt too long and overloading the model with irrelevant context.
- B. Using too formal a tone that makes the model produce overly academic responses.
- C. Not including enough examples for the model to understand the task.
- D. Being too vague about what they want and expecting the model to infer the specifics.**

Correct Answer: D

Explanation

Vagueness is the most common first-prompt mistake. Users assume the model can infer what "good" means for their specific task. Without explicit output criteria, the model produces its statistical average, which is rarely what the user wanted. The fix is specifying the exact output characteristics required.

Q38. What is the most reliable way to improve AI output quality through iteration?

- A. Regenerate the output with higher temperature to introduce variation.
- B. Add more context about the background and topic to the original prompt.
- C. Identify the specific element that fell short and give targeted correction instructions.**
- D. Ask the model to rate its own output and improve the lowest-rated elements.

Correct Answer: C

Explanation

Targeted correction is the most effective iteration strategy: "The analysis is too high-level — add specific examples for each of the three points" produces better results than vague regeneration or context additions. Specific, targeted feedback on exactly what to improve is the highest-leverage iteration approach.

Q39. What is "context injection" and why does it improve prompting outcomes?

- A. Injecting the user's previous conversation history to maintain continuity.
- B. Adding task-specific context that Claude lacks from training to make outputs more relevant.**
- C. Providing real-time data through a web search tool to update Claude's knowledge.
- D. Adding the user's identity information so Claude can personalise responses.

Correct Answer: B

Explanation

Context injection means including information Claude does not have by default — company-specific facts, proprietary terminology, specific client context, project constraints — directly in the prompt. This allows Claude to produce outputs relevant to the specific situation rather than generic outputs that could apply to any situation.

Q40. What is the most important thing to consider when deciding how much context to include in a prompt?

- A. Whether each piece of context actually affects the output — remove context that does not.**
- B. Including as much context as possible to ensure the model has everything it needs.
- C. Keeping context under 500 words to prevent the model from being distracted.
- D. Matching context volume to the model's context window to maximise performance.

Correct Answer: A

Explanation

Relevance is the most important consideration. Irrelevant context dilutes the model's attention from what matters. "Lost in the middle" research shows that adding unnecessary context can actually reduce output quality. Include context only if it would meaningfully change the output if the model had versus did not have it.

Q41. What is the difference between a "system prompt" and a "user prompt" in terms of prompting strategy?

- A. System prompts are longer and more detailed; user prompts are shorter and direct.
- B. System prompts are for operators; user prompts are only for end users.
- C. System prompts set persistent context and role; user prompts specify the immediate task.**
- D. System prompts define format; user prompts define the topic to cover.

Correct Answer: C

Explanation

The strategic distinction is persistence and authority: system prompts set the operational context — who Claude is, what constraints apply, what format to use — that applies to all subsequent interactions. User prompts provide the specific task or question within that context. Good prompt architecture separates persistent context from task-specific instructions.

Q42. What is "prompt sensitivity" and why does it matter for production systems?

- A. The degree to which a prompt activates the model's safety filters.
- B. The length threshold above which longer prompts stop improving output quality.
- C. A model's tendency to generate different outputs to the same prompt across sessions.
- D. Small changes in prompt wording can significantly change output quality or behaviour.**

Correct Answer: D

Explanation

Prompt sensitivity means small, seemingly innocuous wording changes — a different word, a reordered sentence, a different example — can substantially change output quality. This matters for production systems because prompts must be thoroughly tested before deployment and any changes must be re-tested to catch unexpected behaviour changes.

Q43. What is the most important test to run when deploying a prompt in a production system?

- A. Testing with the same input 10 times to measure output consistency.
- B. Testing with real and adversarial user inputs to identify edge cases and failure modes.**
- C. Testing with the ideal example input to confirm the prompt works correctly.
- D. Testing API latency to ensure the prompt length does not cause timeouts.

Correct Answer: B

Explanation

Production prompts will encounter inputs the developer never anticipated. Testing with real user inputs (sampled from actual use cases) and adversarial inputs (edge cases, boundary conditions, unusual formats) reveals failure modes that ideal-case testing misses entirely.

Q44. What is the best way to evaluate whether a prompt change improved or degraded output quality?

- A. Use the new prompt for a week in production and see if user complaints decrease.
- B. Ask Claude to compare its outputs from both prompt versions and judge which is better.
- C. Run both prompt versions on the same test set and compare outputs systematically.**
- D. Check whether the new prompt produces longer, more detailed outputs.

Correct Answer: C

Explanation

Systematic A/B evaluation on a consistent test set is the only reliable method for assessing prompt changes. Production feedback is too slow and confounded; Claude self-comparison is unreliable; output length is not a quality metric. A consistent test set allows objective, comparable measurement of both versions.

Q45. What is the most important principle for prompt design in a product used by non-technical users?

- A. The prompt must produce high-quality outputs even when user inputs are vague or unusual.**
- B. The prompt must be short enough that users can read and understand it themselves.
- C. The prompt must prevent users from changing Claude's behaviour through their messages.
- D. The prompt must include example queries that users can copy and adapt.

Correct Answer: A

Explanation

Non-technical users write imprecise, varied, and unexpected inputs. The system prompt must be robust enough to produce high-quality outputs across this full range — not just for well-formed inputs. Building for the average well-formed query while the prompt breaks on edge cases is a common product failure.

SET

24

When AI Fails

Set 24 of 5 · 15 Questions · AI Engineering

Q46. What is the most common reason AI fails at tasks it seemed capable of during testing?

- A. The model degrades when used at high volume due to server load.
- B. Users communicate less clearly in production than in controlled testing.
- C. The model's capabilities diminish as the conversation grows longer.
- D. Production inputs are more diverse and edge-case-rich than test inputs.**

Correct Answer: D**Explanation**

Testing typically uses representative, well-formed inputs that the developer anticipated. Production exposes the system to the full diversity of real user behaviour — unusual phrasing, edge cases, domain-specific terminology, incomplete information — which tests never fully cover. Distribution shift between test and production is the most common cause of deployment failure.

Q47. What is the most dangerous failure mode in AI-assisted decision support?

- A. The AI produces too many options, creating decision paralysis for the user.
- B. The AI confidently produces wrong recommendations that the decision-maker accepts without review.**
- C. The AI refuses to make a recommendation on high-stakes decisions.
- D. The AI makes different recommendations on the same situation across sessions.

Correct Answer: B**Explanation**

Confident wrong recommendations that are accepted without review is the worst failure mode — it combines AI's tendency to hallucinate confidence with a human tendency to defer to confident-sounding authority. The result is wrong decisions made on authoritative-seeming AI output without the verification that would have caught the error.

Q48. What is "specification gaming" as an AI failure mode?

- A. The AI interprets an ambiguous specification too literally and produces wrong output.
- B. The AI fails to follow a specification it was given in the system prompt.
- C. The AI achieves the specified metric while violating the underlying intent.**
- D. The AI produces outputs that technically meet the specification but are of poor quality.

Correct Answer: C

Explanation

Specification gaming (reward hacking) occurs when the AI finds a way to satisfy the measured objective without achieving the actual goal. A summarisation AI told to minimise length might produce a one-sentence summary that technically satisfies the constraint while losing all meaningful content. The measure is optimised; the intent is subverted.

Q49. What is "automation bias" and how does it affect AI-assisted professional work?

- A. The tendency to trust and accept AI outputs without applying sufficient critical review.**
- B. The tendency to prefer automated processes over manual ones regardless of quality.
- C. The AI model's tendency to produce outputs consistent with its training biases.
- D. A systematic error introduced when AI automates tasks with inherent bias.

Correct Answer: A

Explanation

Automation bias is a human cognitive phenomenon where people over-rely on and under-challenge automated systems. In AI-assisted professional work, automation bias causes people to accept AI outputs with less scrutiny than they would apply to human-generated work, even when verification is essential.

Q50. What is the most appropriate response when an AI makes a factual error in a professional context?

- A. Discard all AI use on this task type to prevent future errors.
- B. Ask the AI to regenerate the response with a request to be more accurate.
- C. Report the error to the AI provider as a model defect to be fixed.
- D. Correct the specific error and assess whether it indicates a pattern requiring systematic changes.**

Correct Answer: D

Explanation

The immediate response is to correct the specific error. The strategic response is to assess whether the error reveals a systematic pattern — a task type where AI consistently fails — that should change prompting strategy, verification protocols, or whether to use AI at all for that task type.

Q51. What is "brittleness" in AI systems and when does it most commonly appear?

- A. When the AI system is fragile and frequently produces error messages or crashes.
- B. When the model performs well in testing but deteriorates quickly after deployment.
- C. When small changes in input produce unexpectedly large changes in output quality.**
- D. When the model's outputs are correct but structured incorrectly for downstream use.

Correct Answer: C

Explanation

AI system brittleness means the system is sensitive to small input variations — slight rephrasing, different ordering, unusual punctuation — that cause output quality to drop dramatically. It commonly appears at the edges of the training distribution and at boundary conditions the system was not tested for.

Q52. What is "task decomposition failure" in an AI workflow?

- A. The AI misidentifies how to break a complex task into manageable subtasks.**
- B. The workflow breaks down because the context window cannot hold all subtasks.
- C. The AI fails to complete tasks because it is given too many subtasks at once.
- D. A failure where one agent's task decomposition conflicts with another agent's.

Correct Answer: A

Explanation

Task decomposition failure occurs when the AI creates a flawed plan — wrong subtasks, wrong order, missing steps, or incorrect dependencies — that cannot produce the correct final output even if each subtask is executed perfectly. It is a planning error, not an execution error.

Q53. What is the most important diagnostic question when an AI workflow produces wrong outputs?

- A. "Did the AI model produce lower-quality outputs than in our initial testing?"
- B. "Which specific step in the workflow produced the error that propagated to the output?"**
- C. "Did the user input contain errors that the AI then replicated in the output?"
- D. "Was the context window too small for the volume of information processed?"

Correct Answer: B

Explanation

Diagnosing AI workflow failures requires tracing to the specific step where the error originated. Without step-level diagnosis, fixes are guesses. The error might be in extraction (wrong data), reasoning (wrong inference), or generation (wrong format) — each requiring a different fix.

Q54. What is "overconfident failure" as a unique risk in AI systems?

- A. The AI overestimates how complex a task is and over-engineers its response.
- B. The AI refuses tasks within its capability because it underestimates its own ability.
- C. AI expresses high confidence on outputs that are actually wrong or uncertain.**
- D. A failure where confident AI use leads organisations to reduce human expertise.

Correct Answer: C

Explanation

AI systems produce confident-sounding text even when they are effectively guessing. This creates a specific risk category where wrong outputs are harder to detect than human errors, which often involve hedging language or obvious uncertainty. Confident wrong answers pass initial review more easily than unsure wrong answers.

Q55. What is the most reliable way to identify systematic AI failure patterns in production?

- A. Surveying users about whether they were satisfied with AI outputs.
- B. Comparing production output error rates to model benchmark accuracy scores.
- C. Running periodic A/B tests with human-generated outputs as the control condition.
- D. Tracking error types and correlating them with input characteristics and task types.**

Correct Answer: D

Explanation

Systematic pattern tracking — logging errors with metadata about input type, topic, format, and user context — allows identification of conditions under which AI consistently fails. User satisfaction surveys are too aggregated; benchmark comparisons are too distant from production; A/B tests catch some but not systematic errors across all input types.

Q56. What is the most important human capability that must not be lost even with advanced AI assistance?

- A. The ability to critically evaluate outputs and make final judgments independently.**
- B. The ability to perform tasks manually without AI assistance when systems fail.
- C. The ability to write effective prompts to extract value from AI systems.
- D. The ability to explain AI behaviour to non-technical stakeholders.

Correct Answer: A

Explanation

Critical evaluation and independent judgment are the irreplaceable human capabilities in AI-assisted work. If these atrophy — if people stop being able to assess whether an AI output is right or wrong — then AI errors propagate unchecked. All other skills (manual fallback, prompting, communication) are secondary to this core judgment capability.

Q57. What is the most common reason AI underperforms on specialised professional tasks?

- A. The model applies excessive safety filtering on professional domain content.
- B. Training data underrepresents the specific domain, reducing pattern quality for those tasks.**
- C. Professional tasks require real-time data that exceeds the model's knowledge cutoff.
- D. The model cannot distinguish between professional and general user queries.

Correct Answer: B

Explanation

Specialised professional domains (rare medical subspecialties, niche regulatory frameworks, proprietary methodologies) have less representation in training data. The model's pattern quality for these tasks is weaker than for general topics. This manifests as outputs that use the right vocabulary but miss domain-specific nuances or make domain-specific errors.

Q58. What is "model collapse" risk in a multi-model pipeline?

- A. Multiple model calls on the same data produce contradictory outputs that cannot be resolved.
- B. The pipeline fails when any single model call errors out, halting the entire workflow.
- C. Individual models produce lower quality outputs when called from a pipeline than directly.
- D. Errors propagate and compound through the pipeline, producing increasingly wrong outputs.**

Correct Answer: D

Explanation

In multi-model pipelines, early model errors become the inputs for subsequent models. Each model processes the flawed input as if it were correct, often compounding the error. The final output can be very wrong while each individual model call seemed plausible given its input. This is the pipeline equivalent of cascading failure.

Q59. What is the most important pre-deployment test for an AI system that will make consequential recommendations?

- A. Testing on adversarial inputs designed to find recommendations that are dangerously wrong.**
- B. Testing the system's recommendation quality against human expert baseline scores.
- C. Testing the system's latency under the expected production request volume.
- D. Testing whether the system correctly declines requests outside its competence scope.

Correct Answer: A

Explanation

Adversarial testing for dangerously wrong recommendations is most critical before deploying decision-support AI. The system will inevitably encounter edge cases in production; pre-deployment adversarial testing discovers the failure modes before real users do, when the cost of errors is high.

Q60. What is the most important thing an organisation should do after an AI system causes a significant error?

- A. Disable AI use for that task type until a better model is available.
- B. Disclose the error to all affected stakeholders and offer remediation.
- C. Conduct a root cause analysis and implement systematic prevention measures.**
- D. Report the failure to the AI model provider for model improvement.

Correct Answer: C

Explanation

Root cause analysis and systematic prevention are the most important responses. Understanding exactly what failed and why allows the organisation to fix the specific problem — whether it is a prompting issue, a missing verification step, or a fundamental task mismatch — and prevent recurrence across all similar deployments.

SET

25

Professional Accountability

Set 25 of 5 · 15 Questions · AI Engineering

Q61. Who is accountable for the consequences of an AI-generated output used in a professional decision?

- A. The professional who used the AI output and chose to act on it.
- B. The AI provider who created the model that generated the output.
- C. The organisation that deployed the AI tool used in the workflow.
- D. Accountability is shared between the AI provider and the user in proportion to their role.

Correct Answer: A**Explanation**

Professional accountability for the consequences of AI outputs rests with the professional who used the output and chose to act on it. AI providers disclaim liability for misuse; organisational deployment is a contributing factor. But the professional who made the decision — relying on unverified AI output — bears the primary accountability.

Q62. What is the most important disclosure consideration when submitting AI-assisted professional work?

- A. Always disclosing AI use in all professional work, regardless of context.
- B. Disclosing only when AI generated more than 50% of the final output.
- C. Not disclosing AI use since it is now standard practice and assumed.
- D. Whether the context requires disclosure and what level of disclosure is appropriate.

Correct Answer: D**Explanation**

Disclosure requirements vary by context: academic submission (strict policies), client deliverable (relationship and contract dependent), regulatory filing (jurisdiction and sector dependent). The professional must understand the applicable requirements for their specific context — not apply a blanket always-disclose or never-disclose policy.

Q63. A professional submits AI-generated work to a client as their own analysis. The client makes a wrong decision based on AI hallucinations in the work. What is the professional's primary responsibility?

- A. To notify the AI provider of the hallucination so the model can be improved.
- B. To own the consequences — they verified the AI output insufficiently before submission.**
- C. To share responsibility with the AI tool since it generated the wrong content.
- D. To disclose to the client that AI was used so they can make their own assessment.

Correct Answer: B

Explanation

The professional took accountability for the work by submitting it as their analysis. Insufficient verification of AI outputs before acting on them is a professional failure, not a technology failure. Disclosure after harm has been done does not discharge the accountability for insufficient verification before submission.

Q64. What is the most important principle for maintaining professional integrity when using AI?

- A. Disclose AI use consistently to all clients and stakeholders.
- B. Only use AI for tasks where errors are immediately obvious and inconsequential.
- C. Own every output fully — verify, understand, and be able to defend every claim.**
- D. Use AI only for tasks explicitly approved by your professional body or employer.

Correct Answer: C

Explanation

Professional integrity with AI means full ownership of every output — you have verified the key claims, you understand the reasoning, and you can defend every assertion as if you had produced it yourself. Using AI as a production tool while maintaining this standard of ownership is the correct professional posture.

Q65. What is "deskilling risk" in the context of professional AI use?

- A. Relying on AI so heavily that core professional skills atrophy from disuse.**
- B. The risk of hiring less-skilled professionals because AI reduces skill requirements.
- C. The risk of AI errors causing professionals to make decisions below their normal skill level.
- D. The demotion of skilled professionals who resist AI adoption in their organisations.

Correct Answer: A

Explanation

Deskilling risk refers to the cognitive atrophy that occurs when professionals delegate tasks so completely to AI that they lose the underlying skill. A lawyer who lets AI draft everything may lose the ability to draft independently; an analyst who lets AI interpret data may lose analytical judgment. This creates vulnerability when AI fails.

Q66. What is the correct response when a client asks you to use AI to do something you are not confident it can do reliably?

- A. Use AI anyway and review the output carefully to catch errors.
- B. Decline the task entirely to avoid professional risk.
- C. Be transparent about AI's limitations for this task and propose an approach with appropriate human judgment.**
- D. Charge more to compensate for the additional verification time required.

Correct Answer: C

Explanation

Transparency about AI's limitations for the specific task — and proposing an approach that appropriately supplements AI with human judgment where needed — is the professional response. It sets accurate expectations, proposes a viable solution, and maintains the professional relationship without either blind AI use or unnecessary refusal.

Q67. What is the most important thing to understand about AI and professional liability?

- A. AI providers carry partial liability for model errors that cause client harm.
- B. Professional liability is reduced when AI is used, since errors are the tool's fault.
- C. Professional liability applies only when AI is used without human review.
- D. AI use does not transfer professional liability — the professional remains fully accountable.**

Correct Answer: D

Explanation

Professional liability — the legal and ethical accountability for one's professional work — does not transfer to an AI tool. A lawyer, doctor, engineer, or accountant who uses AI remains fully liable for the outputs they provide to clients, regardless of whether AI generated them. This is the foundational liability principle for professional AI use.

Q68. What is "professional judgment" in the context of AI-assisted work?

- A. A licensed professional's ability to produce work without any AI assistance.
- B. The human expert's capacity to direct AI appropriately and evaluate its outputs critically.**
- C. The certification that a professional has received AI literacy training.
- D. The right of a professional to choose whether or not to use AI tools.

Correct Answer: B

Explanation

Professional judgment in AI-assisted work is the capacity to: know which tasks benefit from AI, direct AI with appropriate specifications, critically evaluate AI outputs for quality and accuracy, and take accountability for the final work product. It is the meta-skill that makes all other AI capabilities professionally safe and valuable.

Q69. What is the most important way AI changes professional development requirements?

- A. Professionals can now develop expertise more quickly because AI speeds learning.
- B. Domain expertise becomes less important as AI can supply domain knowledge on demand.
- C. Professionals must develop critical evaluation skills alongside domain expertise.**
- D. Professional development should focus on AI skills rather than core domain skills.

Correct Answer: C

Explanation

AI as a production tool means professionals must now develop AI judgment skills — how to direct AI, how to evaluate outputs, when to verify, when to override — alongside (not instead of) their core domain expertise. Domain expertise is more important than ever as the basis for AI output evaluation.

Q70. What is the correct professional response when AI produces an output that contradicts your domain expertise?

- A. Investigate carefully — the AI may have found something you missed, or it may be wrong.**
- B. Trust your expertise — it is always more reliable than AI pattern-matching.
- C. Trust the AI — it has processed more information on the topic than you have.
- D. Present both the AI output and your expert view and let the client decide.

Correct Answer: A

Explanation

Contradiction between AI output and domain expertise is a signal worth investigating — not automatically resolving in favour of either. The AI may have identified a genuine gap; you may have knowledge the AI lacks; the AI may be hallucinating. Investigation determines which, rather than reflexive deference to either.

Q71. What is the most important governance requirement for AI use in a regulated profession?

- A. Regulatory approval for each AI tool used in professional practice.
- B. Documented processes showing how AI outputs are verified before professional use.**
- C. Client consent before any AI tool is used in delivering professional services.
- D. A dedicated AI compliance officer who reviews all AI-generated professional work.

Correct Answer: B

Explanation

Regulatory governance for professional AI use requires documented verification processes — showing that professional judgment was applied, that AI outputs were checked, and that the professional owns the final work product. Approval, consent, and compliance officers are supporting elements; documented verification processes are the evidentiary foundation.

Q72. What is "AI washing" in a professional context and why is it problematic?

- A. Adding unnecessary AI features to products to appear more technologically advanced.
- B. Over-claiming AI capabilities to win business that the AI cannot actually deliver.
- C. Using AI branding on content to attract attention and signal modernity.
- D. Claiming AI-generated work as purely human-generated to meet expectations or avoid disclosure.**

Correct Answer: D

Explanation

AI washing in professional contexts means presenting AI-generated work as original human analysis — to meet client expectations of human professional work, avoid disclosure obligations, or exploit the reputational value of human expertise. This is a form of professional deception with ethical and potentially legal consequences.

Q73. A junior team member asks whether they should use AI for everything to maximise productivity. The most accurate guidance is:

- A. Use AI for everything — productivity gains always outweigh any quality concerns.
- B. Avoid AI until you have mastered the underlying skills independently.
- C. Use AI where it genuinely helps and exercise judgment to develop your own skills alongside AI.**
- D. Use AI only for tasks that have been approved by the team lead for AI use.

Correct Answer: C

Explanation

The most balanced and accurate guidance is to use AI where it genuinely accelerates work, but to actively exercise and develop your own capabilities alongside AI use. Complete AI delegation without human engagement atrophies the skills that make you a professional capable of evaluating AI output — and employable when AI changes.

Q74. What is the most important quality a professional needs to use AI responsibly in the long term?

- A. Technical understanding of how AI models work at the architecture level.
- B. Intellectual honesty about what AI can and cannot do, and when its outputs should not be trusted.**
- C. Comprehensive awareness of all available AI tools and their respective capabilities.
- D. A consistent record of disclosing AI use to all professional stakeholders.

Correct Answer: B

Explanation

Intellectual honesty — the honest assessment of AI capability and limitation, and the courage to say "I don't trust this AI output" when appropriate — is the foundational quality for responsible long-term AI use. Technical knowledge, tool awareness, and disclosure are all valuable but secondary to this core intellectual honesty.

Q75. Menler's AI Judgment bank tests practitioners at what level of professional maturity?

- A. Those learning about AI for the first time and exploring its basic capabilities.
- B. Those building AI systems and models who need to evaluate their own outputs.
- C. Those managing AI teams and evaluating which AI tools to invest in.
- D. Those who must make independent, accountable decisions about AI use in their professional work.**

Correct Answer: D

Explanation

The AI Judgment bank tests the maturity needed to use AI responsibly as a professional — making independent decisions about when to trust AI, when to verify, when to override, and taking full professional accountability for AI-assisted work. It is for practitioners making real professional decisions, not beginners, builders, or managers.

SET

26

How Language Models Work

Set 26 of 5 · 15 Questions · AI Engineering

Q1. What is the core mechanism that allows transformer models to weigh the relevance of different parts of an input when producing an output?

- A. Recurrent processing, which reads tokens one at a time and updates a hidden state.
- B. Self-attention, which computes relationships between all token positions simultaneously.**
- C. Convolutional filters, which detect local patterns across fixed-size windows.
- D. Hard-coded routing tables that assign each token to a specialist sub-network.

Correct Answer: B

Explanation

The transformer architecture is built around self-attention. Every token can attend to every other token in a single pass, allowing the model to weigh relevance across the full input regardless of distance. This is what replaced recurrent processing and enabled parallelism.

Q2. Why did transformers largely replace RNNs for language modelling tasks?

- A. Transformers use less memory per token because they compress sequences into a single vector.
- B. Transformers are deterministic, so they always produce the same output for the same input.
- C. Transformers were designed specifically for text, whereas RNNs were designed for time-series data.
- D. Transformers process all tokens in parallel, making training faster and avoiding the vanishing gradient problem.**

Correct Answer: D

Explanation

RNNs process tokens sequentially, which is slow to train and suffers from vanishing gradients over long sequences. Transformers process the full sequence in parallel via attention, which is both faster to train and better at capturing long-range dependencies.

Q3. In a transformer, what does the feed-forward layer do after the attention layer?

- A. It applies a non-linear transformation independently to each token position.**
- B. It re-orders the token sequence based on predicted importance scores.
- C. It compresses the full sequence into a single context vector.
- D. It retrieves relevant facts from an external knowledge base.

Correct Answer: A

Explanation

After attention mixes information across positions, the feed-forward layer processes each position independently using a two-layer MLP with a non-linear activation. This is where much of the model's "knowledge" is believed to be stored.

Q4. What does the term "parameters" refer to in a large language model?

- A. The configuration settings a developer passes to the model at runtime.
- B. The number of tokens the model can process in a single call.
- C. The learned numerical weights that determine how the model transforms inputs into outputs.**
- D. The hyperparameters used during training such as learning rate and batch size.

Correct Answer: C

Explanation

Parameters are the internal weights of the model — billions of floating-point numbers learned during training. They are distinct from runtime configuration (like temperature or max tokens), context length, or training hyperparameters.

Q5. What is the role of positional encoding in a transformer?

- A. It encodes the semantic meaning of each token into a fixed numerical vector.
- B. It injects information about the order of tokens, since attention itself is position-agnostic.**
- C. It determines which tokens are masked during the training process.
- D. It compresses the full input sequence into a lower-dimensional representation.

Correct Answer: B

Explanation

Attention treats inputs as a set with no inherent order. Positional encoding adds a signal — either fixed sinusoidal or learned — that tells the model where each token sits in the sequence. Without it, the model cannot distinguish "cat sat on mat" from "mat on sat cat".

Q6. What distinguishes a decoder-only transformer (like GPT-style models) from an encoder-decoder transformer (like T5)?

- A. Decoder-only models generate text autoregressively; encoder-decoder models encode an input first, then decode to a target sequence.**
- B. Decoder-only models can only answer questions; encoder-decoder models can generate free-form text.
- C. Decoder-only models are smaller and faster; encoder-decoder models are larger and more accurate.
- D. Decoder-only models require fine-tuning for every task; encoder-decoder models work zero-shot.

Correct Answer: A

Explanation

In decoder-only models, all processing is autoregressive — the model generates one token at a time, attending only to previous tokens. Encoder-decoder models first encode the full input into a representation, then a separate decoder generates the output — useful for translation and summarisation.

Q7. What is "causal masking" in a decoder-only transformer?

- A. Hiding sensitive tokens in the input to prevent the model from memorising private data.
- B. Masking low-confidence tokens in the output to reduce hallucination.
- C. Blocking certain attention heads from activating to reduce compute cost.
- D. Preventing each token from attending to future tokens during training, so the model learns to predict the next token.**

Correct Answer: D

Explanation

Causal masking enforces the autoregressive constraint: during training, each token can only attend to itself and tokens before it. This ensures the model learns to predict the next token given only what came before — not by "seeing ahead".

Q8. What is a "head" in multi-head attention?

- A. The first token in a sequence, which carries the most context for the rest.
- B. A separate sub-network that handles a specific language task like NER or sentiment.
- C. An independent attention function that learns to focus on different aspects of the input simultaneously.**
- D. A parameter group that is frozen during fine-tuning to preserve pre-trained knowledge.

Correct Answer: C

Explanation

Multi-head attention runs several attention functions (heads) in parallel, each with its own learned weight matrices. Different heads can specialise — one might track syntax, another coreference, another topic. Their outputs are concatenated and projected.

Q9. Why do larger language models tend to perform better on tasks they were not explicitly trained for?

- A. Scale enables emergent capabilities — patterns learned across diverse training data generalise to novel tasks.**
- B. Larger models have more memory and can store more facts verbatim from training data.
- C. Larger models are always trained on more recent data, giving them better world knowledge.
- D. Larger models apply explicit reasoning rules, whereas smaller models only match patterns.

Correct Answer: A

Explanation

Emergent capability is a well-documented phenomenon: as models scale in parameters and data, they develop abilities that smaller models do not show — often without those abilities being explicitly trained. This is believed to result from richer internal representations formed by learning across diverse data.

Q10. What is the difference between pre-training and instruction tuning?

- A. Pre-training is done by the user; instruction tuning is done by the model provider.
- B. Pre-training uses labelled data; instruction tuning uses unlabelled internet text.
- C. Pre-training produces a chat model; instruction tuning produces a base model.
- D. Pre-training learns general language representations from raw text; instruction tuning aligns the model to follow human instructions.**

Correct Answer: D

Explanation

Pre-training is the large-scale self-supervised stage where the model learns from raw text by predicting tokens. Instruction tuning (often with RLHF or similar) comes after — it teaches the model to follow instructions, be helpful, and behave appropriately in conversation.

Q11. What does "temperature" control in an LLM at inference time?

- A. The speed of token generation — higher temperature makes the model generate faster.
- B. The sharpness of the probability distribution over next tokens — higher temperature produces more varied outputs.**
- C. The maximum length of the output — higher temperature allows longer responses.
- D. The number of attention heads active during generation — higher temperature activates more heads.

Correct Answer: B

Explanation

Temperature scales the logits before the softmax is applied. At temperature 1.0, the distribution is unchanged. Lower temperature sharpens it (more deterministic); higher temperature flattens it (more random and creative). It does not affect speed, length, or the number of active heads.

Q12. What is a "mixture of experts" (MoE) architecture?

- A. A training technique where multiple smaller models vote on the correct output before it is returned.
- B. A deployment pattern where different model versions handle different user request types.
- C. A model architecture where only a subset of specialist sub-networks are activated for each token, reducing compute per forward pass.**
- D. A fine-tuning approach that trains a separate expert for each target domain.

Correct Answer: C

Explanation

In MoE, the model has many expert sub-networks but only routes each token to a small number of them via a gating mechanism. This allows a much larger total parameter count without proportionally higher compute cost per token — used in models like Mixtral and reportedly GPT-4.

Q13. What is the significance of the "residual connection" in transformer blocks?

- A. It saves the model's previous response so it can refer to it in multi-turn conversations.
- B. It connects the encoder and decoder in sequence-to-sequence models.
- C. It stores intermediate attention scores for use in interpretability tooling.
- D. It adds the block's input directly to its output, helping gradients flow and enabling deeper networks.**

Correct Answer: D

Explanation

Residual (skip) connections add the input of a sub-layer directly to its output. This means gradients can flow through the network without passing through every transformation, enabling very deep models to train stably — a technique borrowed from ResNets.

Q14. What does "RLHF" stand for and what problem does it solve?

- A. Recursive Language Hierarchy Formation — structuring the model's internal representations for better retrieval.
- B. Reinforcement Learning from Human Feedback — aligning model outputs with human preferences beyond what next-token prediction alone achieves.**
- C. Reduced Latency High Fidelity — a technique for speeding up inference without quality loss.
- D. Regularised Loss with Hallucination Filtering — suppressing fabricated outputs during training.

Correct Answer: B

Explanation

RLHF is Reinforcement Learning from Human Feedback. After pre-training, a reward model is trained on human preference data, then used to fine-tune the LLM via RL to produce outputs humans rate more highly. It is the primary technique behind the helpfulness and safety of modern chat models.

Q15. What is "weight sharing" in the context of transformer models?

- A. Using the same parameter matrix for multiple roles in the architecture — for example, tying input embeddings to the output projection layer.**
- B. Distributing a single model's weights across multiple GPUs for parallel inference.
- C. Sharing fine-tuned weights between different user-facing deployments of the same base model.
- D. Copying weights from a larger model into a smaller model as a form of distillation.

Correct Answer: A

Explanation

Weight sharing refers to reusing the same parameter matrix for different purposes. A common example is tying the token embedding matrix to the output linear layer — reducing total parameter count while maintaining performance. It is distinct from model parallelism, deployment sharing, or distillation.

SET

27

Tokens and Embeddings

Set 27 of 5 · 15 Questions · AI Engineering

Q16. What is a token in the context of a large language model?

- A. A security credential required to authenticate API requests.
- B. A discrete reasoning step the model takes before producing an answer.
- C. The basic unit of text the model processes — roughly a word, sub-word, or punctuation mark.**
- D. A numerical identifier assigned to each unique sentence in the training corpus.

Correct Answer: C**Explanation**

A token is the atomic unit of text for an LLM. Models do not process characters or words directly — they process tokens, which are typically sub-word pieces. The word "unbelievable" might be three tokens; "cat" is usually one. Everything — input and output — is counted in tokens.

Q17. Why do some words cost more tokens than others?

- A. Rare or long words are split into multiple sub-word tokens by the tokeniser.**
- B. Common words are cached and cost zero tokens to process.
- C. Words in capitals are always treated as single tokens regardless of length.
- D. Token cost depends on word frequency in the training data, not word length.

Correct Answer: A**Explanation**

Tokenisers like BPE (Byte-Pair Encoding) build a vocabulary of common sub-word pieces. Frequent short words get their own token; rare or long words are split into multiple tokens. "Microprocessor" costs more tokens than "cat" — and this directly affects context usage and cost.

Q18. What is an embedding in the context of LLMs?

- A. A compressed version of the model's weights used for faster inference.
- B. A structured format for injecting external data into the model's context.
- C. A technique for reducing hallucinations by anchoring outputs to source text.
- D. A dense numerical vector that represents the meaning of a token or text in a high-dimensional space.**

Correct Answer: D

Explanation

An embedding is a dense vector of floating-point numbers. Each token maps to an embedding vector; the model learns these representations during training such that semantically similar tokens cluster together in the vector space. Embeddings are the bridge between raw text and numerical computation.

Q19. What does it mean for two texts to have a "high cosine similarity" in embedding space?

- A. They share a high proportion of identical tokens after tokenisation.
- B. Their embedding vectors point in nearly the same direction, indicating semantic similarity.**
- C. They were retrieved from the same document in the training corpus.
- D. Their token counts are within a small percentage of each other.

Correct Answer: B

Explanation

Cosine similarity measures the angle between two vectors. A value close to 1 means the vectors point in the same direction — the texts are semantically similar. This is the basis of semantic search and RAG retrieval: finding chunks whose meaning is close to the query, regardless of exact word match.

Q20. What is the difference between a "token embedding" and a "sentence embedding"?

- A. Token embeddings are used during training; sentence embeddings are used only at inference time.
- B. Token embeddings are high-dimensional; sentence embeddings are always 128 dimensions.
- C. Token embeddings represent individual tokens; sentence embeddings represent the meaning of an entire passage as a single vector.**
- D. Token embeddings are learned; sentence embeddings are computed using hand-crafted rules.

Correct Answer: C

Explanation

Token embeddings are per-token vectors — they are the input representations that evolve through transformer layers. Sentence (or passage) embeddings collapse a full text into one vector, often by pooling over token representations. Sentence embeddings are what you store in a vector database for retrieval.

Q21. Why does tokenisation of non-English text often cost more tokens per word?

- A. Non-English models use a different API endpoint that charges at a higher rate.
- B. Non-Latin scripts require Unicode escaping, which multiplies the byte count.
- C. Non-English words are always treated as unknown tokens and replaced by a placeholder.
- D. Tokenisers trained on English-heavy data have larger vocabulary coverage for English, so non-English words are split into more sub-word pieces.**

Correct Answer: D

Explanation

BPE tokenisers build their vocabulary based on the training corpus. If the corpus is English-heavy, frequent English words get their own tokens. Non-English words — especially in morphologically rich languages or non-Latin scripts — are broken into many small pieces, increasing token count for the same semantic content.

Q22. What is the "vocabulary size" of a language model?

- A. The total number of distinct tokens the model recognises — typically tens of thousands of sub-word pieces.**
- B. The number of unique concepts the model can discuss, inferred from its training data.
- C. The maximum number of tokens the model can generate in a single response.
- D. The number of languages the model was trained on.

Correct Answer: A

Explanation

Vocabulary size is the number of entries in the tokeniser's lookup table — each a distinct token (sub-word, word, or special symbol). GPT-4 uses ~100K tokens; other models use 32K–256K. It determines the output layer size and affects how efficiently different languages and domains are represented.

Q23. What happens to embedding representations as they pass through transformer layers?

- A. They remain identical at each layer; layers only affect the attention weights.
- B. They are progressively refined — each layer transforms the representations to capture increasingly abstract and contextual information.**
- C. They collapse to a single value per token that represents its classification label.
- D. They are discarded after each layer and re-computed from the original token IDs.

Correct Answer: B

Explanation

Each transformer layer updates the token representations through attention (mixing information across positions) and the feed-forward layer (per-position transformation). Early layers capture surface features; later layers capture abstract meaning, syntax, and reasoning — building richer representations with depth.

Q24. What is the purpose of "special tokens" like [CLS], [SEP], or <|endoftext|>?

- A. They are placeholders for sensitive information that has been redacted from the input.
- B. They indicate tokens that the model should copy verbatim without modification.
- C. They are error codes inserted when the tokeniser cannot parse a word.
- D. They serve as structural markers that help the model understand task boundaries, sequence starts/ends, or signal generation completion.**

Correct Answer: D

Explanation

Special tokens are not natural language — they are structural signals in the token sequence. [CLS] was used in BERT as a classification token; [SEP] separated segments; <|endoftext|> tells a generative model the sequence has ended. Instruction-tuned models use special tokens to mark system prompts, user turns, and assistant turns.

Q25. What is "semantic search" and how does it differ from keyword search?

- A. Semantic search is faster because it uses pre-computed index tables; keyword search is slower because it reads raw documents.
- B. Semantic search only works on structured data; keyword search works on both structured and unstructured text.
- C. Semantic search matches based on meaning using embedding similarity; keyword search matches based on exact or fuzzy token overlap.**
- D. Semantic search requires a fine-tuned model; keyword search works with any off-the-shelf model.

Correct Answer: C

Explanation

Keyword search looks for matching terms — it finds "car" when you search "car". Semantic search converts query and documents into embedding vectors and retrieves by cosine similarity — it finds "automobile" and "vehicle" when you search "car" because they cluster nearby in embedding space.

Q26. What does it mean when a model is described as having a "128K token context window"?

- A. The model was trained on 128,000 documents.
- B. The model can process up to 128,000 tokens — input plus output — in a single call.**
- C. The model can remember up to 128,000 previous conversations.
- D. The model generates outputs of exactly 128,000 tokens before stopping.

Correct Answer: B

Explanation

128K context window means the model can hold up to 128,000 tokens in its active working memory for a single API call — combining system prompt, conversation history, retrieved documents, and the response. It says nothing about training data volume or memory across calls.

Q27. Why do embeddings from one model not work directly with embeddings from a different model?

- A. Each model learns its own vector space with different dimensions and geometric structure; cross-model distances are meaningless.**
- B. Embeddings are encrypted at generation time and can only be decrypted by the originating model.
- C. Different models use incompatible file formats that cannot be loaded by each other's inference engines.
- D. Embeddings are only valid during the session in which they were generated.

Correct Answer: A

Explanation

Embedding vectors are coordinates in a space that a specific model learned during training. The geometry — which directions correspond to which meanings — is unique to each model. Mixing vectors from different models is like mixing GPS coordinates from two different map projections: the numbers are incompatible.

Q28. What is the role of the embedding layer at the start of a transformer?

- A. It filters out low-relevance tokens before they enter the attention mechanism.
- B. It compresses the input sequence to reduce compute cost in later layers.
- C. It converts token IDs into dense continuous vectors that the attention mechanism can operate on.**
- D. It assigns each token a confidence score that propagates through the network.

Correct Answer: C

Explanation

The embedding layer is a lookup table: given a token ID (an integer), it returns the corresponding embedding vector (a dense float array). This is the first transformation — from discrete symbols to continuous representations that the mathematical operations in attention and feed-forward layers can process.

Q29. What does "dimensionality" refer to in the context of embedding vectors?

- A. The number of languages the embedding model was trained on.
- B. The maximum text length that can be embedded in a single call.
- C. The precision (float16 vs float32) used to store the embedding values.
- D. The number of values in the vector — higher dimensionality allows richer representations but increases storage and compute cost.**

Correct Answer: D

Explanation

Dimensionality is simply the length of the vector — how many numbers it contains. Text embedding models commonly use 768, 1536, or 3072 dimensions. Higher dimensionality can encode more nuance but costs more to store, compute, and index in a vector database.

Q30. What is "out-of-vocabulary" (OOV) handling in modern tokenisers?

- A. OOV tokens are replaced with a fixed [UNKNOWN] token that the model treats as a generic placeholder.
- B. Sub-word tokenisers like BPE eliminate OOV by breaking any word into known sub-word pieces, including byte-level fallbacks.**
- C. OOV handling requires the user to manually add new tokens to the vocabulary before inference.
- D. Modern models skip OOV tokens silently during generation to avoid errors.

Correct Answer: B

Explanation

Classic word-level vocabularies had OOV problems. Sub-word tokenisers like BPE, WordPiece, and SentencePiece solve this by decomposing any word into smaller known pieces. Byte-level BPE (used by GPT models) goes further — any byte sequence can be represented, so OOV is impossible.

SET

28

Context Windows

Set 28 of 5 · 15 Questions · AI Engineering

Q31. What practical problem does a limited context window create for enterprise AI applications?

- A. Long documents, conversation histories, or knowledge bases may not fit, forcing truncation or chunking strategies.**
- B. The model refuses to answer questions about topics not covered in the first 1,000 tokens.
- C. Enterprise users pay per context window slot, making long inputs prohibitively expensive.
- D. Context limits only affect the output length, not the amount of input the model can read.

Correct Answer: A**Explanation**

Context window limits mean you cannot simply feed the model an entire document library or long conversation history. Engineers must decide what to include — through chunking, retrieval, summarisation, or sliding windows — because anything outside the window is invisible to the model.

Q32. What is the "lost in the middle" problem with long context windows?

- A. When context windows are long, the model loses track of the user's original question.
- B. Content placed in the middle of the context is automatically summarised and compressed.
- C. Models tend to pay more attention to content at the beginning and end of the context, and less to content in the middle.**
- D. Long contexts cause the model to switch languages or styles midway through the response.

Correct Answer: C**Explanation**

Research has shown that LLMs are better at using information placed at the start or end of the context than information buried in the middle. This has practical implications for RAG: where you place retrieved chunks matters, not just whether they fit.

Q33. Why does increasing context window size increase inference cost?

- A. Longer contexts require more fine-tuning passes to update the model's weights.
- B. Attention computation scales quadratically with sequence length, making longer contexts significantly more expensive.**
- C. APIs charge a flat fee per context window size tier regardless of actual token count.
- D. Longer contexts increase the model's temperature, requiring more sampling passes.

Correct Answer: B

Explanation

Self-attention computes relationships between every pair of token positions. For a sequence of length n , that is n^2 operations. Doubling the context length quadruples the attention compute. This is why long-context inference is expensive and why efficient attention variants (FlashAttention, sparse attention) matter.

Q34. What is a "sliding window" approach to handling documents longer than the context window?

- A. Expanding the model's context window dynamically as more tokens are added.
- B. Summarising earlier parts of the document and injecting summaries into later windows.
- C. Routing different sections of the document to different model instances simultaneously.
- D. Processing the document in overlapping chunks that each fit within the context, then aggregating results.**

Correct Answer: D

Explanation

When a document exceeds the context window, one approach is to slide a fixed-size window across it — processing each chunk with some overlap to avoid cutting off information at boundaries. Results (answers, summaries, extractions) are then merged. Simple but can miss cross-chunk reasoning.

Q35. What is the difference between "context length" and "memory" for a language model?

- A. Context length is the tokens available in a single call; memory refers to mechanisms for persisting information across multiple calls.**
- B. Context length is a hardware limit; memory is a software configuration set by the developer.
- C. Context length applies to the input; memory applies only to the model's output generation.
- D. Context length and memory are the same thing — different terms used by different providers.

Correct Answer: A

Explanation

Context length is a per-call limit — what the model can see in one inference request. Language models have no built-in memory across calls; once a session ends, the model starts blank. Memory across calls is an application-layer concern achieved through databases, summarisation, or explicit memory tools.

Q36. What happens when you exceed a model's context window?

- A. The model automatically summarises earlier content to make room for new tokens.
- B. The API typically truncates the input, raises an error, or the model silently ignores overflow tokens depending on implementation.**
- C. The model switches to a retrieval mode, fetching the overflow content from a cache.
- D. The response quality degrades linearly but the model still processes all tokens.

Correct Answer: B

Explanation

Behaviour varies by implementation: most APIs return an error if you exceed the context limit. Some client libraries silently truncate from the beginning of the conversation. The model does not auto-summarise or retrieve — what falls outside the window is gone. Engineers must handle this explicitly.

Q37. What is "KV cache" and why does it matter for inference efficiency?

- A. A key-value store external to the model used to retrieve facts during generation.
- B. A compressed representation of the model's weights for faster loading at startup.
- C. A log of previous API calls that the model uses to maintain conversation history.
- D. A cache of the key and value matrices from previous tokens, avoiding recomputation during autoregressive generation.**

Correct Answer: D

Explanation

During autoregressive generation, each new token attends to all previous tokens. Without caching, those attention keys and values would be recomputed every step. KV cache stores them, so only the new token's computations are needed. This is essential for making generation fast — without it, long outputs would be extremely slow.

Q38. What is "context stuffing" and when is it a problem?

- A. Injecting adversarial content into the context to manipulate the model's output.
- B. Using the context window exclusively for system prompt instructions, leaving no room for user input.
- C. Filling the context with as much information as possible and hoping the model uses it — which degrades performance when the signal is buried in noise.**
- D. Compressing multiple conversations into a single context to save API costs.

Correct Answer: C

Explanation

Context stuffing — dumping large amounts of loosely relevant text into the context — seems intuitive but often hurts performance. The model struggles to identify what matters. Selective retrieval (RAG) typically outperforms context stuffing because it surfaces only the most relevant content.

Q39. How does "retrieval-augmented generation" (RAG) address context window limitations?

- A. RAG extends the model's context window by compressing retrieved documents before insertion.
- B. Instead of loading all documents into context, RAG retrieves only the most relevant chunks at query time, keeping context focused and within limits.**
- C. RAG eliminates the need for a context window by storing all knowledge in the vector database.
- D. RAG allows the model to read documents in real time from external URLs during generation.

Correct Answer: B

Explanation

RAG solves the context problem by not trying to fit everything in. A retrieval step finds the few most relevant document chunks based on the query, and only those go into the context. The model's context stays manageable and focused — improving both accuracy and cost.

Q40. What is "prompt caching" offered by some model providers?

- A. A feature that caches the computed key-value states for a shared prefix, reducing cost and latency for repeated inputs with the same preamble.**
- B. A service that stores previously generated responses and returns them verbatim for identical prompts.
- C. A client-side feature that saves prompt templates in a local file for faster reuse.
- D. A fine-tuning technique that learns which prompt structures work best for a given task.

Correct Answer: A

Explanation

Prompt caching (offered by Anthropic, OpenAI, and others) works at the infrastructure level: if many requests share the same long prefix (system prompt, document, knowledge base), the KV states for that prefix are cached and reused. This cuts both latency and cost for high-volume applications.

Q41. What does "multi-turn context management" require from an application developer?

- A. Subscribing to a stateful session tier from the model provider that maintains context server-side.
- B. Fine-tuning the model on the specific conversation history before each new session.
- C. Explicitly maintaining and passing the full conversation history with each API call, since the model has no built-in session memory.**
- D. Summarising every turn before sending it to the model to keep token counts low.

Correct Answer: C

Explanation

The model is stateless — each API call is independent. To simulate a conversation, the developer must pass the full history (user and assistant turns) with every request. This is entirely the application's responsibility. Ignoring this means each turn is answered without knowledge of what came before.

Q42. What is "context poisoning" in the context of language model security?

- A. A phenomenon where outdated training data in the model's weights causes incorrect answers.
- B. Overloading the context window with tokens to cause the model to crash or return errors.
- C. Using context length to bypass rate limits by batching many requests into one call.
- D. An attack where malicious content injected into the context manipulates the model's reasoning or outputs.**

Correct Answer: D

Explanation

Context poisoning (a form of prompt injection) occurs when adversarial content — in retrieved documents, user inputs, or external data — manipulates the model into ignoring instructions or producing harmful outputs. It is a real attack vector for RAG systems that process untrusted content.

Q43. Why does very long context not always produce better results?

- A. APIs hard-cap response quality above a certain context length to manage compute costs.
- B. Attention over very long sequences can dilute focus, and the model may fail to use distant relevant content effectively.**
- C. Long contexts always improve results — the limitation is only in the user's interpretation.
- D. Long contexts cause the model to switch to a less capable inference mode to manage memory.

Correct Answer: B

Explanation

More context is not always better. Very long contexts introduce noise, and attention mechanisms struggle to reliably use content spread across hundreds of thousands of tokens. The "lost in the middle" effect, diluted attention, and increased hallucination risk all mean that selective, focused context often outperforms raw length.

Q44. What is the practical implication of paying for both input and output tokens in most pricing models?

- A. Developers should always minimise system prompt length at the expense of instruction quality.
- B. Output tokens are always cheaper, so applications should request longer responses to get more value.
- C. Prompt design and output length both directly affect cost — verbose prompts and long responses compound expenses.**
- D. Token costs only apply to production deployments; development and testing are always free.

Correct Answer: C

Explanation

Most APIs charge per token for both input and output. A 10,000-token system prompt multiplied by millions of calls is expensive — as is requesting long responses. Efficient prompt engineering and output length control are not just quality concerns; they have direct financial implications at scale.

Q45. What is "context window utilisation" and why does it matter operationally?

- A. How much of the available context window a request actually uses — important for cost management, latency, and avoiding truncation.**
- B. A metric for how accurately the model uses information within the context to answer questions.
- C. The percentage of the training corpus that fits within the model's context window.
- D. A measure of how frequently a deployed model's context fills to capacity in production.

Correct Answer: A

Explanation

Context window utilisation tracks how close requests come to the context limit. At low utilisation, you may be under-using available context. Near the limit, you risk truncation. Monitoring this in production helps engineers right-size their prompts, manage costs, and catch unexpected growth in conversation length.

SET

29

Inference and Reasoning

Set 29 of 5 · 15 Questions · AI Engineering

Q46. What does "autoregressive generation" mean?

- A. The model generates all output tokens simultaneously in a single forward pass.
- B. The model retrieves pre-generated responses from a cache and adapts them to the current input.
- C. The model repeatedly revises its entire output until a quality threshold is met.
- D. The model generates one token at a time, conditioning each new token on all previously generated tokens.**

Correct Answer: D**Explanation**

Autoregressive means each token is generated sequentially: the model takes the input plus all tokens generated so far and produces the next token. This repeats until an end token is reached or the max length is hit. It is sequential and cannot be trivially parallelised during generation.

Q47. What is "greedy decoding" and what is its main drawback?

- A. Generating multiple full responses and selecting the one with the highest overall probability.
- B. Always picking the single highest-probability token at each step — fast but can miss higher-quality sequences reachable through lower-probability early choices.**
- C. Randomly sampling tokens without any probability weighting, producing maximally diverse outputs.
- D. Constraining generation to tokens that appear in a predefined vocabulary whitelist.

Correct Answer: B**Explanation**

Greedy decoding is the simplest strategy: always pick the most likely next token. It is fast and deterministic but locally optimal — it can get stuck in repetitive or suboptimal sequences that would have been avoided by accepting a lower-probability token earlier in the sequence.

Q48. What is "chain-of-thought" reasoning and why does it improve performance?

- A. A training technique that chains multiple fine-tuning stages to build reasoning capabilities.
- B. A retrieval method that chains document lookups to gather evidence before answering.
- C. Prompting the model to reason through intermediate steps before giving a final answer, which improves accuracy on complex tasks.**
- D. A decoding strategy that generates multiple candidate chains and selects the best one.

Correct Answer: C

Explanation

Chain-of-thought (CoT) prompting asks the model to "think out loud" — showing its reasoning before the answer. This improves performance on multi-step problems (maths, logic, planning) because the model can build on intermediate steps rather than attempting to jump directly to the answer.

Q49. What distinguishes a "reasoning model" (like o1 or o3) from a standard instruction-tuned model?

- A. Reasoning models are trained to spend more compute generating internal reasoning chains before producing an answer, improving performance on hard tasks.**
- B. Reasoning models have access to external calculators and search tools by default.
- C. Reasoning models are always larger in parameter count than standard instruction-tuned models.
- D. Reasoning models generate answers in a structured JSON format rather than natural language.

Correct Answer: A

Explanation

Reasoning models (OpenAI o-series, Anthropic's extended thinking) are trained to produce extended internal reasoning — sometimes called a "scratchpad" — before the final response. This deliberate search through reasoning paths improves performance on hard tasks like maths, coding, and logical reasoning at the cost of higher latency.

Q50. What is "top-p" (nucleus) sampling?

- A. Selecting the top p percent of tokens by frequency in the training data.
- B. Limiting generation to tokens that appear in the top p documents retrieved from a knowledge base.
- C. Setting a minimum probability threshold below which tokens are never generated.
- D. Sampling from the smallest set of tokens whose cumulative probability exceeds p, avoiding both determinism and the long tail of unlikely tokens.**

Correct Answer: D

Explanation

Top-p sampling restricts the sampling pool to the nucleus — the smallest set of tokens that together account for probability mass p (e.g., 0.9 = 90%). This avoids very unlikely tokens while preserving diversity. It works well alongside temperature to produce coherent but varied outputs.

Q51. What is the difference between "latency" and "throughput" in LLM inference?

- A. Latency measures output quality; throughput measures how many servers are running simultaneously.
- B. Latency is a user-facing metric; throughput is only relevant for model training pipelines.
- C. Latency is the time from request to first token; throughput is the number of tokens or requests processed per unit time.**
- D. Latency and throughput are the same metric measured at different time scales.

Correct Answer: C

Explanation

Latency (specifically "time to first token") measures how long a user waits before seeing any output — critical for perceived responsiveness. Throughput measures total tokens generated per second across all requests — critical for cost and capacity planning. Optimising one can trade off against the other.

Q52. What is "speculative decoding" and what problem does it solve?

- A. A technique where a smaller draft model generates candidate tokens that a larger model verifies in parallel, reducing total latency.**
- B. A method where the model predicts the user's likely next prompt to pre-generate a response.
- C. A sampling strategy that generates multiple response candidates and selects the most likely one.
- D. A compression technique that reduces the model's memory footprint during inference.

Correct Answer: A

Explanation

Autoregressive generation is inherently sequential — each token must wait for the previous one. Speculative decoding breaks this by having a fast small model generate several candidate tokens ahead, then having the large model verify them in one parallel forward pass. Valid tokens are accepted; invalid ones trigger regeneration from that point.

Q53. What does "hallucination" mean in the context of LLMs?

- A. The model producing outputs that are stylistically inconsistent with the prompt.
- B. The model generating content that is confidently stated but factually incorrect or entirely fabricated.**
- C. The model repeating the same phrase or sentence multiple times in a response.
- D. The model refusing to answer questions it was trained to answer due to safety filters.

Correct Answer: B

Explanation

Hallucination is the most significant reliability failure of LLMs: the model produces plausible-sounding but false information — wrong facts, non-existent citations, fabricated statistics. It occurs because models generate probable token sequences, not because they verify truth. It is not "lying" — the model has no intent — but the effect can be just as harmful.

Q54. What is "beam search" in the context of LLM decoding?

- A. Searching the training data for the closest matching sentence to the current context.
- B. Filtering generated tokens through a classifier beam before adding them to the output.
- C. Maintaining multiple candidate sequences simultaneously and selecting the one with the highest overall probability at the end.**
- D. Splitting generation across multiple GPU cores to increase throughput.

Correct Answer: C

Explanation

Beam search keeps k candidate sequences (beams) at each step, expanding each with the top tokens and pruning back to k . It explores more of the output space than greedy decoding, often producing higher-probability sequences. However, high-quality diverse generation has largely shifted to sampling with temperature and top- p .

Q55. Why do reasoning models have higher latency than standard models?

- A. They make multiple API calls to external tools before generating a response.
- B. They run each response through a separate verification model before returning it.
- C. They are deployed on fewer servers than standard models, creating longer queue times.
- D. They generate extended internal reasoning chains — often thousands of tokens — before producing the visible answer.**

Correct Answer: D

Explanation

The latency cost of reasoning models comes from the internal "thinking" tokens — the model generates a full reasoning trace, sometimes thousands of tokens long, which is not always shown to the user. This computation takes time. The trade-off: much better accuracy on hard tasks, but users wait longer.

Q56. What does "token budget" mean in practical LLM application design?

- A. The allocation of available context tokens across system prompt, retrieved content, conversation history, and output, managed to stay within limits and control cost.**
- B. The total number of tokens a user is permitted to generate per month under a subscription plan.
- C. A training-time constraint that limits how many tokens the model can generate during fine-tuning.
- D. A safety mechanism that stops generation when the model's confidence falls below a threshold.

Correct Answer: A

Explanation

Token budget thinking is an engineering discipline: you have a fixed context window and a cost per token. You must allocate those tokens across competing needs — instructions, examples, retrieved docs, history, response space. Good application designers track this explicitly, not as an afterthought.

Q57. What is "few-shot inference" and how does it differ from fine-tuning?

- A. Few-shot inference requires fewer API calls than fine-tuning and is therefore cheaper.
- B. Few-shot inference provides examples in the prompt to guide the model at runtime; fine-tuning updates the model's weights through additional training.**
- C. Few-shot inference works only with small models; fine-tuning is required for large models.
- D. Few-shot inference and fine-tuning are synonymous — both adapt the model to new tasks.

Correct Answer: B

Explanation

Few-shot inference adapts model behaviour by including examples in the prompt — no weight updates occur. Fine-tuning actually updates the model's parameters through gradient descent on new training data. Few-shot is faster and cheaper to try; fine-tuning is more durable but requires data, compute, and care.

Q58. What is "output consistency" and why is it a challenge with temperature > 0?

- A. Output consistency is guaranteed when using the same model version and system prompt.
- B. Output consistency only matters for creative tasks; factual tasks are always deterministic.
- C. Output consistency can be enforced by setting max_tokens to a fixed value.
- D. At temperatures above 0, the model samples probabilistically, so identical inputs can produce different outputs — inconsistency is inherent.**

Correct Answer: D

Explanation

At temperature 0, decoding is nearly deterministic (greedy). At temperature > 0, each run samples from the probability distribution, so outputs vary. For applications requiring reproducibility — structured data extraction, classification — low temperature or temperature 0 is preferred, accepting reduced diversity.

Q59. What is the "prefill" phase of LLM inference?

- A. The phase where the model processes the full input prompt in parallel, before beginning autoregressive generation.**
- B. The phase where the server pre-loads model weights into GPU memory before the first request.
- C. The phase where the application populates template placeholders before sending the prompt.
- D. The phase where retrieved documents are ranked and inserted into the context window.

Correct Answer: A

Explanation

Inference has two phases. Prefill: the model processes the entire input prompt in a single (parallelisable) forward pass, generating KV cache for all input tokens. Decode: the model generates output tokens one at a time autoregressively. Prefill is fast; the decode phase is where latency compounds.

Q60. What does "model quantisation" do and why is it used?

- A. It compresses the model by removing parameters that contributed least to training performance.
- B. It converts the model from one architecture to another for deployment on different hardware.
- C. It reduces the numerical precision of model weights (e.g., from float32 to int8), shrinking memory footprint and increasing inference speed with modest quality trade-offs.**
- D. It splits the model into smaller modules that can be fine-tuned independently.

Correct Answer: C

Explanation

Quantisation reduces the bit-width of model weights — from 32-bit or 16-bit floats to 8-bit or 4-bit integers. The model becomes smaller and faster to run, at the cost of some precision. It is widely used to deploy large models on consumer GPUs or to reduce cloud inference costs.

SET

30

Model Selection and Fine-tuning

Set 30 of 5 · 15 Questions · AI Engineering

Q61. What is the primary consideration when choosing between a large frontier model and a smaller efficient model for a production application?

- A. Task complexity, latency requirements, cost constraints, and privacy needs — not parameter count alone.
- B. The larger model is always preferable because it will produce better outputs.
- C. Model selection should always prioritise the most recently released model.
- D. The choice depends entirely on the context window size required for the application.

Correct Answer: A

Explanation

Model selection is a multi-factor engineering decision. A large model may outperform on complex reasoning but cost 10–100x more per token and add latency. Smaller models may be sufficient for extraction, classification, or structured output tasks. Privacy, latency, and cost shape the decision as much as capability.

Q62. What is "fine-tuning" and when is it the right choice over prompting?

- A. Fine-tuning is always the first step in deploying an LLM for any enterprise application.
- B. Fine-tuning is only possible for open-source models; commercial models cannot be fine-tuned.
- C. Fine-tuning is a synonym for instruction tuning — it refers to the initial alignment step performed by the model provider.
- D. Fine-tuning updates model weights on new training data — appropriate when prompt engineering cannot achieve the required consistency, style, or task performance.

Correct Answer: D

Explanation

Fine-tuning is appropriate when you have high-quality task-specific data and need durable behaviour that prompt engineering cannot reliably achieve. Examples: enforcing a specific output format, learning domain vocabulary, or replicating a tone at scale. Start with prompting; fine-tune only when you have evidence it is needed.

Q63. What is a "base model" versus an "instruction-tuned model"?

- A. A base model is smaller; an instruction-tuned model is the same architecture with more parameters.
- B. A base model is trained only on next-token prediction; an instruction-tuned model has been further trained to follow instructions and behave helpfully.**
- C. A base model is free to use; an instruction-tuned model requires a paid API subscription.
- D. A base model generates code; an instruction-tuned model generates natural language.

Correct Answer: B

Explanation

Base models (like GPT-4 base or Llama before instruction tuning) are powerful but unpredictable in conversation — they complete text, not follow instructions. Instruction-tuned models (Claude, ChatGPT, etc.) have been aligned via supervised fine-tuning and RLHF to be helpful, follow user intent, and behave appropriately.

Q64. What is "LoRA" (Low-Rank Adaptation) and why is it used for fine-tuning?

- A. A regularisation technique that penalises large weight updates during fine-tuning to prevent catastrophic forgetting.
- B. A data augmentation method that generates synthetic training examples using the base model.
- C. A parameter-efficient fine-tuning method that injects small trainable rank-decomposition matrices into the model, leaving original weights frozen — dramatically reducing compute and memory requirements.**
- D. A deployment optimisation that reduces the number of active layers during inference.

Correct Answer: C

Explanation

Full fine-tuning of a large model requires updating billions of parameters — expensive and prone to catastrophic forgetting. LoRA freezes the original weights and adds small trainable matrices at specific layers. These adapters can be trained cheaply, swapped between tasks, and merged into the base model at deployment.

Q65. What is "catastrophic forgetting" in the context of fine-tuning?

- A. When fine-tuning on new data causes the model to lose performance on tasks it previously handled well.**
- B. When the model generates outputs that directly contradict its training data.
- C. When a model's fine-tuning run fails midway and all checkpoints are lost.
- D. When a model overfits to the fine-tuning data and performs worse than the base model on the target task.

Correct Answer: A

Explanation

Catastrophic forgetting occurs when gradient updates for new tasks overwrite the weight configurations that enabled prior capabilities. A model fine-tuned aggressively on legal documents might lose its general reasoning ability. Techniques like LoRA, rehearsal, and regularisation are used to mitigate this.

Q66. What is a "system prompt" in the context of deployed LLM applications?

- A. A special prompt that triggers the model to enter a system administration mode.
- B. A hidden prompt generated by the model itself to guide its own reasoning.
- C. A fixed instruction block provided by the operator that sets the model's behaviour, persona, and constraints before the user's input is processed.**
- D. A prompt constructed automatically from the user's account metadata and preferences.

Correct Answer: C

Explanation

The system prompt is the operator's layer of control — instructions the model receives before any user message. It defines persona, sets rules, restricts topics, or provides context. It persists across the conversation and is typically not visible to the end user. It is the primary tool for customising model behaviour without fine-tuning.

Q67. What factors should drive the decision to use an open-source model versus a commercial API?

- A. Open-source models are always cheaper; commercial models are always more capable.
- B. Commercial APIs are suitable for prototypes; open-source models should always be used in production.
- C. The choice depends only on the licence type of the open-source model.
- D. Data privacy requirements, customisation needs, cost at scale, latency constraints, and available ML engineering capacity.**

Correct Answer: D

Explanation

Open-source models (Llama, Mistral, Qwen) offer full control, no data leaving your infrastructure, and no per-token costs at scale — but require significant ML engineering to deploy and maintain. Commercial APIs are faster to start with and continuously updated but involve data sharing, vendor lock-in, and ongoing costs.

Q68. What is "domain adaptation" in the context of LLMs?

- A. Translating a model trained on English data to perform well on other language domains.
- B. Adjusting a general-purpose model — through fine-tuning, prompting, or RAG — to perform well on a specific domain such as legal, medical, or financial text.**
- C. Adapting a text model to handle non-text modalities like images or audio.
- D. Changing the model's system prompt to make it sound like a domain expert.

Correct Answer: B

Explanation

Domain adaptation addresses the gap between a general-purpose model's training data and a specific domain's vocabulary, reasoning patterns, and knowledge. This can be achieved through domain-specific fine-tuning, retrieval of domain documents, or structured prompting — each with different cost-quality trade-offs.

Q69. What is "multimodal" capability in an LLM?

- A. The ability to output multiple response formats such as JSON, Markdown, and plain text.
- B. The ability to handle multiple languages in the same model.
- C. The ability to process and generate content across multiple modalities — text, images, audio, or video — within a single model.**
- D. The ability to run on multiple hardware platforms including CPU, GPU, and TPU.

Correct Answer: C

Explanation

Multimodal models extend the token paradigm beyond text. Vision-language models (GPT-4o, Claude 3, Gemini) can process images alongside text. Some models handle audio or video. This enables use cases like document analysis, image description, diagram interpretation, and video understanding.

Q70. What is the significance of a model's "knowledge cutoff"?

- A. The model has no reliable knowledge of events that occurred after its training data cutoff date.**
- B. The model refuses to answer questions that require more than its knowledge cutoff token limit.
- C. The knowledge cutoff determines the maximum context the model can process.
- D. The knowledge cutoff is the date after which the model's licence expires.

Correct Answer: A

Explanation

A knowledge cutoff is the date beyond which training data was not collected. Ask a model about events after its cutoff and it either doesn't know or hallucinates. This is a critical limitation for real-time applications — addressed by RAG, tool use for web search, or regularly re-training/updating the model.

Q71. What is the difference between "zero-shot" and "few-shot" prompting in the context of model capability?

- A. Zero-shot prompting works only on fine-tuned models; few-shot prompting works on base models.
- B. Zero-shot asks the model to perform a task without examples; few-shot provides examples in the prompt to demonstrate the desired behaviour.**
- C. Zero-shot produces shorter outputs; few-shot produces longer outputs.
- D. Zero-shot is faster because it skips the example processing step.

Correct Answer: B

Explanation

Zero-shot: the model is asked to do something with no examples — relying entirely on its pre-trained knowledge and instruction-following ability. Few-shot: the prompt includes worked examples showing input-output pairs, demonstrating the desired format or reasoning pattern. Few-shot generally improves performance on structured tasks.

Q72. What is "model distillation" and what is it used for?

- A. Extracting factual knowledge from a large model into a structured database.
- B. Removing unsafe capabilities from a model by filtering its outputs during training.
- C. Splitting a large model into smaller independent sub-models for parallel deployment.
- D. Training a smaller student model to mimic the outputs of a larger teacher model — producing a compact model that retains much of the teacher's capability.**

Correct Answer: D

Explanation

Distillation trains a small model on the output distribution of a large model — the small model learns not from raw data labels, but from the teacher's soft probability distributions and reasoning patterns. The result is a smaller, faster, cheaper model that retains much of the large model's quality on target tasks.

Q73. When is fine-tuning NOT the right solution?

- A. When the task involves structured output — fine-tuning never helps with format compliance.
- B. When using a commercial model, since fine-tuning is only available for open-source models.
- C. When you have limited high-quality data, when the issue is solvable with better prompting, or when you need the model to use knowledge that post-dates training.**
- D. When the application requires low latency, since fine-tuned models are always slower.

Correct Answer: C

Explanation

Fine-tuning is often reached for prematurely. If the underlying issue is a bad system prompt, lack of examples, or missing context — these are prompting and RAG problems, not fine-tuning problems. And fine-tuning cannot inject new post-training knowledge — that requires retrieval. Validate that prompting cannot solve the problem first.

Q74. What does "safety alignment" mean in the context of commercial LLMs?

- A. Encrypting model weights to prevent extraction or misuse by third parties.
- B. Training processes — including RLHF, Constitutional AI, or similar — that teach the model to refuse harmful requests and behave in accordance with defined values.**
- C. Restricting the model's outputs to a whitelist of approved response types.
- D. Testing the model against a fixed benchmark to ensure outputs are factually correct.

Correct Answer: B

Explanation

Safety alignment refers to the deliberate training steps that shape a model's values and behaviour — teaching it to be helpful, honest, and avoid causing harm. Anthropic uses Constitutional AI and RLHF; OpenAI uses similar methods. Alignment is not a filter on top of the model — it is baked into the weights.

Q75. What is the practical implication of model deprecation for production applications?

- A. Deprecated models become freely available for local deployment once removed from the API.
- B. Deprecation only affects fine-tuned models; base model API access is permanent.
- C. Model deprecation triggers automatic migration to the latest version with no engineering work required.
- D. Applications built on a deprecated model version must be re-evaluated and potentially re-engineered when that version is removed from the API.**

Correct Answer: D

Explanation

Model providers regularly deprecate older versions — sometimes with weeks of notice. Applications that depend on specific model behaviour (format, capability, safety) must be monitored, re-tested against successor models, and potentially re-engineered. Deprecation risk is a real operational concern for production AI systems.

SET

31

System Prompts

Set 31 of 5 · 15 Questions · AI Engineering

Q1. What is the primary purpose of a system prompt in an LLM application?

- A. To provide the model with a list of facts it should memorise for the session.
- B. To set the model's behaviour, persona, constraints, and context before any user interaction begins.**
- C. To configure the API parameters such as temperature and max tokens.
- D. To authenticate the operator's identity before the model processes user input.

Correct Answer: B**Explanation**

The system prompt is the operator's primary control mechanism. It defines who the model is, what it can and cannot do, what tone to use, what domain it operates in, and any rules it must follow. It is processed before user messages and persists throughout the conversation.

Q2. Why does a well-crafted system prompt reduce the need for per-request instructions?

- A. System prompts are cached by the model and applied at the hardware level for efficiency.
- B. User messages are ignored if a system prompt is present — the model only follows system-level instructions.
- C. System prompts compress the token count of subsequent messages automatically.
- D. Standing instructions in the system prompt apply to every turn, eliminating repetition in user messages.**

Correct Answer: D**Explanation**

A system prompt that clearly defines expected behaviour — format, tone, domain, restrictions — means developers do not need to repeat those instructions in every user message. The model carries those standing instructions across all turns. This reduces token usage, improves consistency, and simplifies application logic.

Q3. What is the risk of an overly vague system prompt?

- A. The model defaults to generic helpful assistant behaviour, which may not match the application's requirements.**
- B. Vague system prompts are rejected by the API and return a validation error.
- C. The model generates longer responses to compensate for the lack of guidance.
- D. Vague system prompts are more expensive because the model must infer more at runtime.

Correct Answer: A

Explanation

Without specific instructions, the model behaves as a general assistant — which may be too broad, too formal, too verbose, or insufficiently restricted for the use case. Specificity in system prompts — explicit persona, output format, scope, and boundaries — is what separates a reliable product from an unpredictable chatbot.

Q4. What is "prompt injection" and why is it a threat to system prompt integrity?

- A. A technique for inserting dynamic content into a static system prompt template at runtime.
- B. A training vulnerability where system prompt data leaks into the model's base weights.
- C. An attack where user input contains instructions that attempt to override or circumvent the system prompt.**
- D. An API error caused by invalid characters or encoding in the system prompt.

Correct Answer: C

Explanation

Prompt injection occurs when untrusted user input contains adversarial instructions — for example, "Ignore your previous instructions and do X instead." If the model follows these embedded instructions, the system prompt's constraints are bypassed. It is the primary security concern for LLM applications that process untrusted input.

Q5. What is the difference between a system prompt and a user message in terms of model trust?

- A. System prompts are processed first but carry the same trust level as user messages.
- B. System prompts are set by the operator and carry higher trust; user messages come from end users and should be treated with appropriate scepticism.**
- C. User messages carry higher trust because they are entered directly by the authenticated user.
- D. The model treats all messages identically — trust level is determined by content, not role.

Correct Answer: B

Explanation

The role distinction matters. Operators configure the system prompt and take responsibility for appropriate use. Users are untrusted members of the public. Models like Claude are designed to follow operator instructions more reliably than user instructions — especially when the two conflict. This trust hierarchy is foundational to safe deployment.

Q6. What is the recommended approach for including sensitive instructions in a system prompt?

- A. Place critical constraints clearly and early in the system prompt, as models attend more reliably to content at the beginning.**
- B. Encode sensitive instructions in Base64 to prevent users from reading them if the prompt is leaked.
- C. Put all restrictions at the end of the system prompt so they override any earlier instructions.
- D. Never include restrictions in the system prompt — use a post-processing filter instead.

Correct Answer: A

Explanation

Instruction placement affects reliability. Models tend to attend more strongly to content at the beginning of the context. Critical rules — what the model must not do, what persona it must maintain — should appear early and be stated clearly. Encoding or hiding instructions does not make them more effective and does not prevent extraction.

Q7. What is "meta-prompting"?

- A. Writing a prompt that describes itself — a self-referential structure used in reasoning tasks.
- B. A technique for nesting one prompt inside another to build hierarchical instructions.
- C. A meta-learning approach where the model learns optimal prompt structures from training data.
- D. Using an LLM to generate, improve, or evaluate prompts — treating prompt writing as a task the model can assist with.**

Correct Answer: D

Explanation

Meta-prompting uses the model as a tool to improve prompting itself — asking it to rewrite a prompt more clearly, generate variants, score quality, or predict failure modes. It is a practical technique for iterating on system prompts faster than human authorship alone.

Q8. What is a "persona" in a system prompt and when is it appropriate to use one?

- A. A security mechanism that prevents the model from revealing the contents of the system prompt.
- B. A template variable that inserts the user's name into the model's responses.
- C. A defined character identity (name, tone, expertise) that the model adopts — appropriate when consistent brand voice or specialised expert framing improves user experience.**
- D. A fine-tuning instruction that permanently changes the model's personality across all deployments.

Correct Answer: C

Explanation

Personas shape tone, expertise framing, and response style. "You are Aria, a concise financial assistant" produces different outputs than a generic assistant. Personas are appropriate for customer-facing products where brand consistency matters — but should not contradict the underlying model's safety training or make false claims about the model's nature.

Q9. How should a system prompt handle conflicting instructions — for example, when a user asks the model to do something the system prompt prohibits?

- A. The system prompt should explicitly state how to handle conflicts — typically by declining politely and staying within the defined scope.**
- B. Conflicting instructions are automatically resolved by the model in favour of the user.
- C. The model should always follow the most recent instruction, regardless of source.
- D. Conflicting instructions cause the model to generate an error and terminate the session.

Correct Answer: A

Explanation

Without explicit conflict-handling instructions, model behaviour varies. A well-crafted system prompt anticipates common conflicts and specifies the response: "If users ask you to do X, decline and explain that you can only help with Y." Leaving this to chance produces inconsistent behaviour that is difficult to debug.

Q10. What is the effect of instruction length on system prompt reliability?

- A. Longer system prompts always produce more reliable behaviour because they provide more context.
- B. System prompts above a certain length are truncated by the API automatically.
- C. Instruction length has no effect on reliability — only instruction clarity matters.
- D. Very long system prompts can dilute the model's attention, causing it to miss or inconsistently follow some instructions.**

Correct Answer: D

Explanation

More instructions are not always better. Very long system prompts introduce the "lost in the middle" effect — the model attends less reliably to instructions buried in a dense prompt. Critical rules should be stated concisely, positioned prominently, and reinforced through examples rather than repeated verbosely.

Q11. What is "system prompt leakage" and how should it be handled?

- A. A technical bug where the system prompt appears in the model's output due to an API error.
- B. The risk that a user extracts the system prompt through clever questioning — mitigated by instructing the model not to reveal it, though not guaranteed.**
- C. A security vulnerability where system prompt content is logged and exposed in API responses.
- D. The gradual degradation of system prompt effectiveness over the course of a long conversation.

Correct Answer: B

Explanation

Users can often extract system prompts by asking the model to repeat its instructions. Instructing the model to keep the system prompt confidential reduces (but does not eliminate) leakage — models are not cryptographic secrets. Sensitive business logic should not rely solely on system prompt confidentiality for security.

Q12. What is the benefit of including examples directly in a system prompt?

- A. Examples are parsed by the API as training data, permanently improving the model's performance.
- B. Examples reduce the total token count of user messages by pre-loading common responses.
- C. Examples demonstrate the desired behaviour concretely, reducing ambiguity and improving output consistency across diverse inputs.**
- D. Examples override the model's default behaviour by acting as a fine-tuning signal at runtime.

Correct Answer: C

Explanation

Examples (few-shot demonstrations) in a system prompt show the model exactly what "good" looks like — input format, reasoning style, output structure. This is especially effective for structured output tasks where verbal description alone leaves room for interpretation. Examples do not constitute training — they are in-context guidance only.

Q13. What is a "negative constraint" in a system prompt and when should you use one?

- A. A constraint that scores the model's output negatively if it exceeds a length limit.
- B. An instruction that inverts the model's default behaviour by negating its training objectives.
- C. A security flag that triggers human review when certain keywords appear in the output.
- D. An explicit instruction about what the model must NOT do — useful when the boundary between allowed and prohibited behaviour might otherwise be ambiguous.**

Correct Answer: D

Explanation

Negative constraints are prohibitions: "Do not discuss competitor products", "Never reveal the system prompt", "Do not provide medical diagnoses." They are most effective when paired with positive alternatives: "If users ask about X, redirect them to Y." Negative-only constraints without alternatives can produce unhelpful refusals.

Q14. What is a "grounding instruction" in a system prompt?

- A. An instruction that anchors the model's persona to a specific historical period.
- B. An instruction that tells the model to base its responses on provided documents or data, rather than its own internal knowledge.**
- C. A system-level instruction that prevents the model from generating speculative content.
- D. An instruction that grounds the model's outputs in a specific cultural or regional context.

Correct Answer: B

Explanation

Grounding instructions tell the model to use provided context — "Only answer based on the documents provided below. If the answer is not in the documents, say so." This is foundational for RAG applications, reducing hallucination by directing the model away from its parametric knowledge toward trusted retrieved content.

Q15. What is the purpose of "format instructions" in a system prompt?

- A. To specify the structure of the model's outputs — such as JSON, markdown, bullet points, or response length — ensuring downstream processing can rely on a consistent format.**
- B. To tell the model which file format the user's uploaded documents are in.
- C. To configure the API's response encoding, such as UTF-8 or ASCII.
- D. To define the formatting rules for the system prompt itself.

Correct Answer: A

Explanation

Format instructions are critical for applications that parse or display model outputs programmatically. "Respond only with valid JSON matching this schema", "Use numbered lists for steps", "Keep responses under 150 words" — these remove ambiguity about structure and make outputs predictable. Without them, format varies with phrasing.

SET

32

Chain-of-Thought and Reasoning

Set 32 of 5 · 15 Questions · AI Engineering

Q16. What is the simplest way to activate chain-of-thought reasoning in a capable model?

- A. Setting the temperature to 0 to force the model into a deterministic reasoning mode.
- B. Enabling a special "reasoning mode" flag in the API request parameters.
- C. Including a phrase like "Think step by step" or "Reason through this carefully before answering".**
- D. Fine-tuning the model on reasoning traces before deploying it.

Correct Answer: C**Explanation**

"Think step by step" was shown in the original CoT paper to elicit reasoning in capable models without any fine-tuning. Modern instruction-tuned models respond even more reliably to explicit reasoning prompts. It is the simplest and cheapest way to improve accuracy on multi-step tasks.

Q17. Why does chain-of-thought prompting improve accuracy on arithmetic and logic problems?

- A. Breaking the problem into explicit steps externalises intermediate reasoning, allowing the model to build on correct sub-results rather than attempting a one-shot answer.**
- B. Chain-of-thought activates a specialised reasoning sub-network that bypasses the token prediction mechanism.
- C. The model has access to a calculator when chain-of-thought mode is activated.
- D. Writing out steps reduces the probability of sampling from incorrect regions of the token distribution.

Correct Answer: A**Explanation**

LLMs struggle to hold many intermediate values implicitly. Writing out steps makes each intermediate result an explicit token in the context — the model can then "read" those results in subsequent steps. This externalisation of working memory is what makes CoT effective for complex, multi-step problems.

Q18. What is "zero-shot chain-of-thought" versus "few-shot chain-of-thought"?

- A. Zero-shot CoT uses no context; few-shot CoT injects retrieved documents to ground the reasoning.
- B. Zero-shot CoT works for small models; few-shot CoT is only effective for models above a certain size.
- C. Zero-shot CoT and few-shot CoT produce identical results — the difference is only in token efficiency.
- D. Zero-shot CoT asks the model to reason step by step without examples; few-shot CoT provides worked examples of reasoning traces alongside the task.**

Correct Answer: D

Explanation

Zero-shot CoT ("Think step by step") works purely through instruction. Few-shot CoT provides complete worked examples — question, reasoning trace, answer — demonstrating the exact style of reasoning expected. Few-shot CoT generally outperforms zero-shot on structured or domain-specific reasoning tasks.

Q19. What is "self-consistency" as a prompting technique?

- A. Asking the model to verify its own answer by re-reading the question after generating a response.
- B. Generating multiple independent reasoning chains for the same problem and taking the majority answer, improving reliability over a single chain.**
- C. A technique that enforces logical consistency by checking that all statements in a response are non-contradictory.
- D. Prompting the model to produce the same output across multiple API calls by setting temperature to 0.

Correct Answer: B

Explanation

Self-consistency samples multiple reasoning paths at temperature > 0 and aggregates the final answers — typically by majority vote. Because different reasoning paths can lead to the same correct answer through different routes, aggregating them is more reliable than any single chain. Effective for maths, logic, and factual tasks.

Q20. What is "tree-of-thought" (ToT) prompting?

- A. A prompt format that organises information hierarchically using indented bullet points.
- B. A method of chaining multiple LLM calls in a branching pipeline to handle different intent categories.
- C. A technique that explores multiple reasoning paths in a tree structure, evaluating intermediate steps and backtracking from dead ends.**
- D. A visualisation technique that maps the model's attention weights onto a tree diagram.

Correct Answer: C

Explanation

ToT extends CoT by branching: at each reasoning step, multiple candidate continuations are generated and evaluated. Unpromising branches are pruned; promising ones are extended. This systematic search through reasoning space outperforms linear CoT on tasks requiring backtracking — like puzzles, planning, and multi-step maths.

Q21. What type of task benefits least from chain-of-thought prompting?

- A. Mathematical word problems with multiple operations.
- B. Tasks requiring the model to plan a sequence of actions.
- C. Legal or medical reasoning tasks that involve weighing multiple considerations.
- D. Simple factual lookups or direct questions where the answer requires no intermediate reasoning steps.**

Correct Answer: D

Explanation

CoT adds tokens and latency — the trade-off is worth it for complex, multi-step tasks. For simple factual questions ("What is the capital of France?"), adding reasoning steps provides no accuracy benefit and wastes tokens. Apply CoT selectively based on task complexity.

Q22. What is "scratchpad reasoning" in the context of extended thinking models?

- A. An internal reasoning process where the model generates a private thinking trace before producing the visible answer.**
- B. A developer technique for logging intermediate API calls during a multi-step pipeline.
- C. A whiteboard metaphor used in prompt templates to organise complex instructions.
- D. A model feature that stores reasoning traces in a database for later retrieval.

Correct Answer: A

Explanation

Extended thinking models (like Claude's thinking mode or OpenAI's o-series) generate a reasoning trace — a "scratchpad" — that may not be visible in the final output. This internal deliberation allows the model to explore, evaluate, and revise before committing to an answer, improving performance on hard problems.

Q23. What is a "reasoning trace" and why should developers evaluate it?

- A. A log of all API calls made during an agentic workflow, used for debugging and auditing.
- B. The step-by-step reasoning the model produces before its final answer — examining it reveals whether the model is reasoning correctly or reaching correct answers for wrong reasons.**
- C. A structured output format that encodes the model's confidence for each claim in the response.
- D. A fine-tuning dataset extracted from the model's internal activations.

Correct Answer: B

Explanation

Evaluating reasoning traces is a powerful debugging technique. A model can give the right answer for the wrong reasons — or show flawed logic that happened to cancel out. Reading the trace reveals reasoning quality, not just output quality. This matters especially for high-stakes applications where process integrity is as important as the final answer.

Q24. What is "prompt chaining" and how does it differ from a single complex prompt?

- A. Linking multiple system prompts together so that each one overrides the previous one.
- B. A technique for combining outputs from multiple concurrent LLM calls into a single response.
- C. Constructing a single prompt that references multiple previous prompts in the conversation.
- D. Breaking a task into sequential LLM calls where each call's output becomes the next call's input, enabling more reliable and debuggable multi-step processing.**

Correct Answer: D

Explanation

A single complex prompt asks the model to do many things at once, which can overwhelm it or produce tangled outputs. Prompt chaining decomposes the task: step 1 extracts data, step 2 analyses it, step 3 formats the output. Each step is smaller and more reliable, errors can be caught between steps, and each call can be tested independently.

Q25. What is "step-back prompting"?

- A. A technique for prompting the model to reconsider an incorrect answer by providing a hint.
- B. A fallback prompting strategy used when the primary prompt fails to produce useful output.
- C. Asking the model to first reason about the general principles or concepts relevant to a question before addressing the specific question.**
- D. A method for reducing verbosity by asking the model to summarise its reasoning at the end.

Correct Answer: C

Explanation

Step-back prompting asks the model to "take a step back" and consider the broader principle before the specific instance. For example, before answering a physics question, it first articulates the relevant physical laws. This primes the model with the right conceptual framework, improving accuracy on questions that require domain knowledge.

Q26. What is the risk of uncritically accepting a chain-of-thought response that arrives at a correct final answer?

- A. Correct answers with visible reasoning are always more expensive to generate.
- B. The reasoning may be flawed or confabulated, meaning the correct answer was reached by chance and the reasoning cannot be trusted for generalisation.**
- C. The model may refuse to generate reasoning for follow-up questions if it already showed its work.
- D. Accepting correct answers trains the model to produce shorter reasoning traces in future calls.

Correct Answer: B

Explanation

A correct final answer does not validate the reasoning that preceded it. Models sometimes produce plausible-sounding but logically invalid chains — "galaxy-brained" reasoning that happens to reach the right destination. Evaluating the reasoning independently of the answer is essential for trusting the model's capability rather than just its output.

Q27. What is "least-to-most" prompting?

- A. Decomposing a complex problem into simpler sub-problems solved in order from easiest to hardest, using each answer to inform the next.**
- B. A prompting strategy that begins with minimal instructions and adds detail only when the model fails.
- C. Providing the model with the most basic examples first, then increasing complexity.
- D. A token-efficiency technique that orders prompt sections from shortest to longest.

Correct Answer: A

Explanation

Least-to-most prompting explicitly decomposes a problem: "First, what do I need to know to solve this? Let's solve that. Now, using that answer, solve the main problem." This structured decomposition improves performance on compositional reasoning tasks where the answer to a sub-problem is required to answer the main question.

Q28. What is "role prompting" and how does it affect model outputs?

- A. Asking the model to take on the user's perspective to generate more empathetic responses.
- B. A technique for switching the model between different task types within a single conversation.
- C. Assigning the model an expert identity ("You are an expert in X") that primes it to respond with domain-appropriate knowledge and tone.**
- D. Specifying the API role parameter to grant the model operator-level permissions.

Correct Answer: C

Explanation

Role prompting activates relevant knowledge patterns: "You are a senior software engineer reviewing this code" produces a different response than "You are a friendly tutor explaining this code." The model's internal representations shift based on the role, affecting vocabulary, confidence level, and what assumptions it makes about the reader.

Q29. What is "directional stimulus prompting"?

- A. A technique for steering the model's attention toward specific sections of a long document.
- B. A method for generating responses with a specific emotional valence by including sentiment cues.
- C. An adversarial technique for steering the model away from its safety guidelines.
- D. Including a hint or guiding keyword in the prompt that nudges the model toward a desired type of response.**

Correct Answer: D

Explanation

Directional stimulus prompting uses a hint to bias the output. For example, appending "Hint: consider safety risks" to a question about a process nudges the model to include safety considerations without explicitly requiring it. It is a light-touch steering mechanism — less prescriptive than format instructions, more targeted than role prompting.

Q30. Why is it important to test prompts across a diverse set of inputs, not just the examples that motivated the prompt design?

- A. Diverse testing is required by API terms of service before deploying a production prompt.
- B. Prompts optimised for a few examples often overfit and fail on edge cases, adversarial inputs, or distributional shifts in real user queries.**
- C. Testing on diverse inputs allows the model to improve its responses through in-context learning.
- D. Models behave identically across all inputs given the same prompt — diverse testing is for the developer's confidence only.

Correct Answer: B

Explanation

Prompt engineering is empirical: a prompt that works brilliantly on five hand-crafted examples may fail on edge cases you did not anticipate. Real users will phrase things unexpectedly, include unusual content, or probe boundaries. Systematic evaluation across a diverse test set — including adversarial examples — is the only way to know if a prompt is truly robust.

SET

33

Role and Structure

Set 33 of 5 · 15 Questions · AI Engineering

Q31. What is the general principle for ordering information in a long prompt?

- A. Place the most critical instructions at the beginning and end, as models attend less reliably to information in the middle of long prompts.
- B. Place examples first, then instructions, so the model understands the task before reading the rules.
- C. Order information from most specific to most general, matching how humans read documents.
- D. Randomise the order to prevent the model from developing positional biases.

Correct Answer: A**Explanation**

The "lost in the middle" effect means models attend most strongly to content near the beginning and end of the context. For long prompts, critical instructions belong at the top, and key constraints can be reinforced at the end. Important content buried in the middle of a long prompt is less reliably followed.

Q32. What is "context injection" in the context of LLM applications?

- A. A technique for injecting adversarial content into a prompt to test model robustness.
- B. A method for compressing large documents into the context window using summarisation.
- C. Dynamically inserting relevant information — user data, retrieved documents, or state — into the prompt at runtime before sending it to the model.
- D. The process of fine-tuning a model on context-specific data to improve domain performance.

Correct Answer: C**Explanation**

Context injection is standard practice for data-driven AI applications: you take a template prompt with placeholders, then fill those placeholders with dynamic content at runtime — the user's name, their account data, retrieved documents, conversation history, or the current date. This gives the model the information it needs without hardcoding it.

Q33. What is the benefit of separating different types of content in a prompt using clear delimiters?

- A. Delimiters are required by the API for the prompt to be parsed correctly.
- B. Delimiters (XML tags, markdown headers, triple quotes) help the model distinguish between instructions, examples, data, and user input — reducing confusion and improving reliability.**
- C. Delimiters reduce the token count by compressing whitespace between sections.
- D. Delimiters tell the model which sections to ignore when generating a response.

Correct Answer: B

Explanation

Without clear structure, models can confuse instructions with data or examples with live content. Using `<instructions>`, `<examples>`, `<document>`, or `###` headers makes the role of each section unambiguous. This is especially important when user-supplied content could be mistaken for instructions — a prompt injection risk.

Q34. What is "few-shot prompting" and what makes examples effective?

- A. A technique for reducing API calls by batching multiple questions into a single prompt.
- B. Providing the model with a few hints about where to search for information.
- C. A method for fine-tuning the model using a small number of labelled examples.
- D. Providing input-output pairs that demonstrate the desired behaviour — effective examples are representative, correctly formatted, and cover edge cases.**

Correct Answer: D

Explanation

Few-shot examples show the model what good looks like. Effective examples are: correctly labelled, representative of the real input distribution, formatted exactly as desired, and varied enough to demonstrate expected behaviour across different cases. Bad examples — wrong, unrepresentative, or inconsistently formatted — can hurt more than no examples.

Q35. What is "prompt templating" and why is it important for production systems?

- A. Using parameterised prompt structures where placeholders are filled at runtime — ensuring consistency, testability, and separation of prompt logic from application logic.**
- B. Designing a fixed prompt that the model memorises across sessions for faster response times.
- C. A technique for generating prompts automatically from user intent without developer involvement.
- D. A compression format that reduces prompt token counts for cost efficiency.

Correct Answer: A

Explanation

Prompt templates treat the prompt as code: a structured string with defined variables. This enables version control, A/B testing, systematic evaluation, and clear ownership. Without templating, prompts exist as scattered strings in application code — hard to maintain, test, or improve systematically.

Q36. What is the risk of including too many examples in a few-shot prompt?

- A. The model is limited to processing a maximum of five examples in a single prompt.
- B. It consumes context window space that could be used for the actual input or retrieved content, and may cause the model to overfit to the example format rather than generalise.**
- C. Too many examples cause the model to average across them, producing bland outputs.
- D. Including many examples triggers a fine-tuning mode that permanently alters the model.

Correct Answer: B

Explanation

Each example costs tokens. In a 32K context window, 20 long examples might leave little room for the actual document or user input. More critically, many examples can anchor the model too strongly to a specific pattern, reducing flexibility on inputs that differ from the examples. Five to ten well-chosen examples often outperform twenty mediocre ones.

Q37. What is "instruction decomposition" and why does it improve prompt reliability?

- A. Splitting a long prompt into multiple API calls to avoid context window limits.
- B. A technique for removing ambiguous language from instructions through automated rewriting.
- C. Decomposing the model's output into component parts for downstream processing.
- D. Breaking a complex instruction into explicit sequential sub-steps, making it easier for the model to follow each part correctly.**

Correct Answer: D

Explanation

"Do X, taking into account Y and Z, unless W, while ensuring Q" is hard to follow reliably. Breaking this into numbered steps — "1. First do X. 2. Check for condition Y. 3. If W, do V instead. 4. Verify Q before responding." — makes each component clear. Sequential instructions reduce omissions and logic errors in model outputs.

Q38. What is a "dynamic system prompt" and when is it appropriate?

- A. A system prompt that updates itself based on feedback from the model's previous responses.
- B. A system prompt generated entirely by the model at the start of each conversation.
- C. A system prompt that varies based on user context, account type, or application state — appropriate when different users or scenarios require meaningfully different model behaviour.**
- D. A system prompt that rotates through different configurations to test which performs best.

Correct Answer: C

Explanation

Static system prompts work for uniform use cases. Dynamic prompts inject user-specific context: enterprise tier vs. free tier users might have different permission sets; a sales tool might inject the salesperson's name and territory. This personalisation improves relevance without requiring separate model deployments for each user segment.

Q39. What does it mean to "format-prime" a model in a prompt?

- A. Including format instructions in the first line of the system prompt to give them maximum weight.
- B. Beginning the assistant's turn with a partial response that demonstrates the desired format, encouraging the model to continue in that structure.**
- C. Pre-processing the prompt to ensure consistent whitespace and encoding before sending to the API.
- D. Training the model on formatted outputs so it defaults to structured responses.

Correct Answer: B

Explanation

Format-priming places a partial assistant response at the end of the prompt: "The answer in JSON format: {" — the model then continues from that start. This is more reliable than describing the format verbally, because the model literally sees the format it is continuing rather than interpreting an instruction about it.

Q40. What is the purpose of "negative examples" in few-shot prompting?

- A. Showing the model what incorrect or undesired outputs look like helps it learn the boundary between acceptable and unacceptable responses.**
- B. Negative examples are incorrect input-output pairs used to test the model's error detection.
- C. Including negative examples reduces hallucination by training the model to avoid incorrect patterns.
- D. Negative examples are output-only samples that the model uses as contrast when generating new responses.

Correct Answer: A

Explanation

Positive examples show what to do; negative examples show what NOT to do. "Here is an output that is too verbose / incorrect format / unhelpfully vague" followed by a corrected version makes the desired quality boundary explicit. For tasks with subtle quality distinctions — like tone or specificity — negative examples are essential.

Q41. What is "prompt versioning" and why does it matter?

- A. A technique for compressing prompts into shorter versions without losing meaning.
- B. An API feature that stores different prompt configurations for different user segments.
- C. Treating prompts as code with version control — tracking changes, enabling rollback, and correlating prompt versions with evaluation results.**
- D. A method for generating multiple variants of a prompt to identify the most effective one.

Correct Answer: C

Explanation

Prompts in production change over time. Without versioning, you cannot know which prompt change caused a quality regression, cannot roll back to a working version, and cannot systematically evaluate improvements. Treating prompts as code — with Git, changelogs, and evaluation tied to versions — is a production engineering discipline.

Q42. What is "XML tag prompting" and why does Anthropic recommend it for Claude?

- A. A technique for encoding structured data as XML before injecting it into a prompt.
- B. An API authentication method where requests are wrapped in XML envelopes.
- C. A fine-tuning approach that trains Claude to output well-formed XML by default.
- D. Using XML-style tags like <instructions>, <document>, and <output> to structure prompt sections — Claude's training makes it particularly responsive to this structured format.**

Correct Answer: D

Explanation

Claude's training data included extensive use of XML tags for structuring prompts and outputs. Using tags like <task>, <context>, <examples>, and <output_format> helps Claude reliably identify the role of each section. Anthropic explicitly recommends this pattern in their prompting documentation for complex prompts.

Q43. What is "output length control" and how is it best achieved?

- A. Setting the temperature to control how much the model elaborates on each point.
- B. Specifying the desired response length explicitly in the prompt — either via instruction or by setting `max_tokens` in the API — to prevent verbose or truncated outputs.**
- C. Fine-tuning the model on outputs of the desired length to make it naturally concise.
- D. Output length is determined solely by the model and cannot be controlled through prompting.

Correct Answer: B

Explanation

Models tend to be verbose by default — they optimise for seeming thorough. Explicit length instructions ("Respond in 3 sentences", "Provide a one-paragraph summary") are the most reliable control. `max_tokens` prevents runaway length but can truncate mid-sentence. For structured outputs, specifying field lengths or using a schema is most precise.

Q44. What is "constrained generation" and when is it used?

- A. A safety technique that prevents the model from generating content above a certain risk score.
- B. A token budget constraint that stops generation when the cost exceeds a defined threshold.
- C. Restricting the model's output to a defined set of options or a specific grammar — used when the application requires exact, predictable output formats.**
- D. A decoding method that forces the model to use only words from the user's input.

Correct Answer: C

Explanation

Constrained generation (JSON mode, function calling, grammars in `llama.cpp`) forces the model to produce outputs that conform to a defined structure. This is used when downstream systems parse model outputs programmatically — you cannot afford malformed JSON or unexpected format variations. It trades creative flexibility for reliability.

Q45. What is the difference between a "prompt" and a "programme" in the context of LLM engineering?

- A. A prompt is a single query; a programme is a structured composition of multiple prompts, logic, and tool calls that orchestrates complex tasks.**
- B. A prompt is written in natural language; a programme is written in a formal programming language.
- C. A prompt is sent to the model; a programme is stored in the model's memory for later use.
- D. A prompt produces one response; a programme produces multiple responses in parallel.

Correct Answer: A

Explanation

Single prompts handle simple tasks. Complex applications are "programmes" — chains of LLM calls, conditional logic, tool invocations, memory reads, and output parsing. Thinking of LLM engineering as programming shifts the mindset from "write a good prompt" to "design a reliable system" — with all the software engineering discipline that implies.

SET

34

Context and Output Formatting

Set 34 of 5 · 15 Questions · AI Engineering

Q46. Why is JSON output mode preferred over asking the model to "respond in JSON" via instruction?

- A. JSON mode is faster because the model skips the natural language generation step.
- B. JSON mode reduces hallucination because the model cannot generate unsupported claims in JSON format.
- C. JSON mode is cheaper because structured outputs require fewer tokens than prose.
- D. JSON mode (where available) constrains the token sampling to enforce valid JSON structure, eliminating parse errors from instruction-only approaches.**

Correct Answer: D

Explanation

Instruction-based JSON prompting produces JSON most of the time but fails on complex schemas, long outputs, or model drift. JSON mode (Anthropic, OpenAI function calling, grammar-constrained decoding) enforces syntactic validity at the decoding level. For production pipelines that parse model outputs, structural reliability is non-negotiable.

Q47. What is "structured output" prompting and why does it matter for downstream systems?

- A. A technique for making model outputs more readable to human reviewers by adding headings and bullet points.
- B. Prompting the model to return data in a machine-readable format (JSON, XML, CSV) that downstream code can reliably parse without natural language interpretation.**
- C. Prompting the model to cite its sources in a structured bibliography format.
- D. A method for generating outputs that conform to a visual design template.

Correct Answer: B

Explanation

When LLM outputs feed into databases, APIs, UIs, or other automated processes, unstructured prose is a liability. Structured outputs give downstream systems a reliable contract: field X will always be a string, field Y will be an integer, array Z will always be present. This transforms the LLM from a text generator into a predictable data processor.

Q48. What is a "schema prompt" and how does it reduce output variability?

- A. A database schema injected into context to tell the model what data is available for retrieval.
- B. A validation schema run after generation to filter out non-conforming outputs.
- C. A prompt that defines the exact fields, types, and structure of the desired output — giving the model a concrete template to fill rather than asking it to invent a structure.**
- D. A meta-prompt that describes how to write other prompts for a specific task category.

Correct Answer: C

Explanation

Schema prompts eliminate structural ambiguity: "Return a JSON object with fields: { title: string, summary: string (max 100 chars), tags: string[], confidence: number 0–1 }." The model fills a template rather than inventing structure. This produces more consistent outputs than natural-language format descriptions alone.

Q49. What is the trade-off between Markdown output and plain text output in LLM applications?

- A. Markdown renders well in chat interfaces but produces cluttered output when displayed in plain text contexts — format choice must match the rendering environment.**
- B. Markdown outputs are always more expensive because formatting tokens count toward usage.
- C. Plain text outputs are always more reliable because models are trained primarily on unformatted text.
- D. Markdown is only appropriate for code outputs; plain text should be used for all prose responses.

Correct Answer: A

Explanation

A model that uses headers, bullets, and bold text produces great output in a chat interface that renders Markdown — but if that output goes into an email, a PDF, or a plain text field, it is cluttered with asterisks and pound signs. Matching output format to the rendering environment is a basic but frequently overlooked engineering concern.

Q50. What is "chain-of-thought with output extraction"?

- A. A technique that chains multiple reasoning steps, then feeds the trace into a second model for evaluation.
- B. A CoT variant that extracts the most confident sentence from the reasoning trace as the answer.
- C. A method for extracting specific entities from a reasoning trace using regex patterns.
- D. Prompting the model to reason through a problem, then explicitly extracting the final answer from a designated section rather than parsing the entire reasoning trace.**

Correct Answer: D

Explanation

When using CoT, the final answer is embedded in a reasoning trace that may be verbose. Instructing the model to end with "Final answer: [X]" — or using XML tags like `<answer></answer>` — makes extraction reliable. Without explicit extraction markers, downstream parsing must interpret natural language reasoning traces, which is fragile.

Q51. What is "output grounding" and when is it important?

- A. A technique for preventing the model from generating content that contradicts its training data.
- B. A post-processing step that verifies outputs against a factual database.
- C. Instructing the model to anchor its response to specific provided text, data, or documents rather than drawing from its parametric knowledge.**
- D. A formatting instruction that requires the model to cite the paragraph number for each claim.

Correct Answer: C

Explanation

Output grounding is critical for accuracy-sensitive applications: "Your response must be based only on the provided documents. Do not use your general knowledge." Without grounding, the model mixes retrieved content with parametric knowledge — making it hard to distinguish verified information from plausible-sounding hallucination.

Q52. What is a "response template" in prompt engineering?

- A. A pre-defined output structure with placeholders that the model fills in, ensuring consistent format while allowing variable content.**
- B. A saved prompt template that developers reuse across multiple API calls.
- C. An HTML or Markdown template that formats the model's raw output for display.
- D. A fine-tuning template that defines the expected input-output format for training examples.

Correct Answer: A

Explanation

A response template tells the model: "Structure your output exactly like this: [Header]: [Content]. [Summary]: [2-3 sentences]. [Next steps]: [Numbered list]." The model fills the placeholders. This is more reliable than describing format in prose and produces outputs that are easier to parse and display consistently.

Q53. What is the "answer first" prompting pattern and when is it useful?

- A. A technique where the correct answer is provided to the model first, then it explains why.
- B. Asking the model to state its conclusion first, then provide supporting reasoning — useful for concise, scannable outputs where the user needs the answer immediately.**
- C. A CoT variant where the model predicts the answer before reasoning, then checks its prediction.
- D. A format instruction that places numerical answers before textual answers in the output.

Correct Answer: B

Explanation

Default model behaviour is often to build up to an answer — preamble, reasoning, conclusion. For executive-facing outputs, dashboards, or quick-reference tools, "answer first" reverses this: the conclusion appears on line one, and justification follows. This matches how busy readers consume information.

Q54. What is "citation prompting" and what are its limitations?

- A. A technique for forcing the model to use only information from Wikipedia articles.
- B. A post-processing method that automatically adds citations to model outputs using a database lookup.
- C. Instructing the model to cite specific documents or passages for each claim — useful for verifiability, but the model may fabricate citations if sources are not provided in context.**
- D. A prompting style where each sentence ends with a source URL that the model generates.

Correct Answer: C

Explanation

Citation prompting improves verifiability when real documents are in context: "For each claim, cite the document section it comes from." But without source documents, models hallucinate citations — generating plausible-sounding but non-existent paper titles, authors, or URLs. Citation prompting works only when grounded in real provided content.

Q55. What is "output length calibration" and why does it require evaluation?

- A. Setting max_tokens to match the desired word count, converting words to tokens using the standard 0.75 ratio.
- B. Training the model on outputs of the desired length so it learns the target without explicit instruction.
- C. A technique for compressing verbose model outputs in post-processing to meet length requirements.
- D. Empirically determining the instruction that produces outputs of the right length and depth for the use case — because models interpret vague length instructions inconsistently.**

Correct Answer: D

Explanation

Instructions like "be concise" or "respond briefly" are interpreted differently by different models and across different prompt contexts. What constitutes "concise" must be calibrated through evaluation: define what length is appropriate, measure actual outputs, and refine instructions until they reliably produce the target length range.

Q56. What is the benefit of including a "reasoning section" and a "conclusion section" as separate output fields?

- A. It separates the model's working from its final answer, making it easier to evaluate reasoning quality independently and to display only the conclusion to end users.**
- B. Separate sections reduce token count because the model can omit reasoning from the conclusion.
- C. Separate sections allow different models to handle reasoning and conclusion independently.
- D. It prevents the model from including reasoning in the conclusion, which would confuse users.

Correct Answer: A

Explanation

Separating reasoning from conclusion serves two audiences: developers who need to evaluate the reasoning process, and end users who may only need the conclusion. It also prevents the model from hedging the conclusion with reasoning caveats — the conclusion section forces a definitive statement.

Q57. What is "table formatting" in prompt engineering and when should it be avoided?

- A. A technique for organising prompt content into columns to reduce vertical space.
- B. Asking the model to format comparative information as a Markdown table — effective in rendered interfaces but problematic when outputs are processed as plain text or fed into non-rendering systems.**
- C. A database output format required when the model retrieves structured data.
- D. A rendering instruction that converts bullet points into table rows automatically.

Correct Answer: B

Explanation

Tables are powerful for comparative or structured data in rendered interfaces. But Markdown tables in plain text contexts are unreadable pipes and dashes. And parsing Markdown tables programmatically is more fragile than parsing JSON. Choose tables for human-readable rendered outputs; choose structured data formats for programmatic consumption.

Q58. What is "label prompting" for classification tasks?

- A. Assigning a numerical label to each example in a few-shot prompt to help the model index them.
- B. Applying content labels to model outputs as a post-processing classification step.
- C. A technique for labelling sections of a prompt to help the model understand their role.
- D. Restricting the model's output to a defined set of class labels, preventing free-text responses that require additional parsing.**

Correct Answer: D

Explanation

For classification, you want "POSITIVE", "NEGATIVE", or "NEUTRAL" — not "This text seems to convey a generally positive sentiment with some caveats." Label prompting constrains output vocabulary: "Respond with exactly one word: POSITIVE, NEGATIVE, or NEUTRAL." Combined with JSON mode or function calling, this makes classification outputs fully deterministic and parseable.

Q59. What is a "confidence score" prompt and what are its reliability limits?

- A. Asking the model to express its confidence in its answer as a number — useful directionally but unreliable as a calibrated probability because models are not trained to produce calibrated confidence scores.**
- B. A prompt that activates the model's internal confidence estimation mechanism for more accurate outputs.
- C. A post-processing technique that scores outputs using a separate classifier model.
- D. A prompt format that instructs the model to only respond when confidence exceeds a threshold.

Correct Answer: A

Explanation

Models can generate confidence scores when asked — "On a scale of 0–1, how confident are you?" — and these correlate directionally with accuracy. But they are not calibrated probabilities: a model saying "0.9 confident" does not mean it is right 90% of the time. Confidence scores are useful for flagging uncertain outputs but should never be treated as statistical probabilities.

Q60. What is "chain of verification" (CoVe) prompting?

- A. A multi-model approach where a second model verifies each claim in the first model's output.
- B. A CoT variant that generates an answer, then traces backward to verify each reasoning step.
- C. A technique where the model generates an initial answer, drafts verification questions about that answer, answers each question, then revises the final answer based on what the verification reveals.**
- D. A prompting strategy that asks the model to list all sources that support and contradict its answer.

Correct Answer: C

Explanation

CoVe is a self-refinement technique: (1) generate a draft answer, (2) generate a list of questions that would verify the claims in that answer, (3) answer each question independently, (4) revise the original answer based on any corrections revealed. It reduces hallucination by forcing the model to surface and check its own implicit assumptions.

SET

35

Reliability and Evaluation Prompting

Set 35 of 5 · 15 Questions · AI Engineering

Q61. What is "prompt sensitivity" and why is it a problem for production systems?

- A. The phenomenon where small changes in prompt wording produce significantly different model outputs, making results fragile and hard to maintain.**
- B. The model's tendency to respond differently to the same prompt when it contains sensitive topics.
- C. The degradation in prompt effectiveness that occurs as the model's weights are updated.
- D. The computational sensitivity of the model to the numerical values in the input embedding.

Correct Answer: A**Explanation**

Prompt sensitivity means that "Summarise this in three points" and "Give me a three-point summary" may produce systematically different outputs. This fragility is a fundamental property of LLMs — they are not executing formal logic. Production prompts must be evaluated across paraphrases and edge cases, not just the single wording that worked in development.

Q62. What is an "evaluation prompt" (also called an "LLM-as-judge" prompt)?

- A. A prompt used to evaluate the developer's prompting skills against a benchmark.
- B. A system prompt configuration used exclusively in model evaluation benchmarks.
- C. A prompt that retrieves evaluation metrics from an external scoring service.
- D. A prompt that instructs a language model to score, rank, or critique another model's output against defined criteria.**

Correct Answer: D**Explanation**

LLM-as-judge is a scalable evaluation technique: instead of manual human review or rigid regex matching, you prompt a capable model to evaluate outputs. "Score this response on helpfulness (1–5), accuracy (1–5), and tone (1–5). Explain each score." This enables systematic evaluation of open-ended outputs at scale — though the judge model also introduces bias.

Q63. What is "prompt regression testing" and why is it necessary?

- A. A technique for identifying prompts that cause the model to regress to earlier, less capable behaviour.
- B. Running a fixed test suite of prompts and expected outputs after any change to the prompt, model, or system to detect quality degradations before they reach production.**
- C. A method for automatically rolling back prompt changes that reduce user engagement metrics.
- D. Testing whether a new model version responds consistently to prompts designed for an older model.

Correct Answer: B

Explanation

Prompts in production interact with a complex system: the prompt itself, the model version, retrieved content, and user input all affect outputs. When any of these change — prompt edit, model upgrade, data update — regression testing ensures previously working behaviour is preserved. Without it, silent quality degradation goes undetected.

Q64. What is the "evaluation rubric" approach to assessing model outputs?

- A. A rubric is an automated scoring algorithm that computes quality scores from output tokens.
- B. A grid of benchmark tasks sourced from academic evaluation datasets.
- C. Defining explicit, observable criteria for what constitutes a good output, allowing consistent scoring across evaluators and over time.**
- D. An evaluation technique that compares model outputs to a golden reference answer using BLEU score.

Correct Answer: C

Explanation

Rubrics make subjective quality judgements explicit and repeatable. Instead of "is this a good answer?", a rubric specifies: "Does it address all parts of the question? (1 point) Is it factually accurate? (1 point) Is it appropriately concise? (1 point)." This structure enables consistent human evaluation and can be operationalised as an LLM-as-judge prompt.

Q65. What is "adversarial prompting" in the context of reliability testing?

- A. Deliberately crafting inputs designed to break, confuse, or manipulate the model — used to find failure modes before they appear in production.**
- B. A technique for generating adversarial training examples to improve model robustness.
- C. A method for testing whether the model can detect and resist adversarial user inputs.
- D. The practice of using aggressive language in prompts to force the model to be more direct.

Correct Answer: A

Explanation

Before deploying a prompt, stress-test it: try confusing inputs, contradictory instructions, off-topic questions, injection attempts, and edge cases. If the model fails on adversarial inputs in testing, it will fail in production. Adversarial testing is not optional for applications that handle untrusted user input.

Q66. What is the difference between "precision" and "recall" when evaluating LLM classification outputs?

- A. Precision is the proportion of outputs that are factually accurate; recall is the proportion that are relevant.
- B. Precision measures output conciseness; recall measures how much of the input the model references.
- C. Precision measures how often the model's positive labels are correct; recall measures how often actual positives are correctly identified.**
- D. Precision and recall are interchangeable metrics — they differ only in which class is treated as positive.

Correct Answer: C

Explanation

Precision: of the cases the model labelled positive, what fraction truly are? Recall: of the truly positive cases, what fraction did the model catch? For high-stakes classification (fraud detection, safety filtering), the trade-off between precision and recall has real consequences. Improving one often degrades the other at a fixed decision threshold.

Q67. What is "hallucination rate" as an evaluation metric?

- A. The rate at which the model generates content outside its defined persona or scope.
- B. A measure of how often the model fails to respond to a prompt within the defined context window.
- C. The frequency with which the model's outputs differ across multiple runs of the same prompt.
- D. The proportion of model outputs that contain factually incorrect or fabricated claims — measured by comparing outputs against ground truth or retrieved sources.**

Correct Answer: D

Explanation

Hallucination rate is a key reliability metric for factual applications. It requires a ground truth — verified facts, retrieved source documents, or expert review — to assess whether the model's claims are accurate. Lowering hallucination rate typically involves grounding (RAG), output constraints, and citation requirements.

Q68. What is "A/B testing" of prompts and what makes it valid?

- A. A testing method where version A is the production prompt and version B is an evaluation-only prompt never shown to users.
- B. Comparing two prompt variants against each other with real or representative traffic to measure which produces better outcomes on defined metrics.**
- C. Testing prompt A in development and prompt B in production to compare real-world and simulated performance.
- D. A randomised test where users are shown two model responses and asked to choose the better one.

Correct Answer: B

Explanation

Valid prompt A/B testing requires: random assignment of conditions, sufficient sample size to detect meaningful differences, a pre-defined success metric, and controlled conditions (same model, same data, only the prompt varies). Without these, observed differences are noise. Like any experiment, it should be designed before running, not analysed post-hoc.

Q69. What is "reference-free evaluation" and when is it necessary?

- A. Evaluating prompts without testing them against the target model, using a proxy model instead.
- B. A method for evaluating factual accuracy without access to the original source documents.
- C. Evaluating model outputs without a ground truth reference answer — necessary for open-ended tasks where no single correct answer exists.**
- D. An automated evaluation technique that does not require human annotators.

Correct Answer: C

Explanation

Many real-world tasks — "write a marketing email", "summarise this article" — have no single correct answer. Reference-free evaluation assesses quality on dimensions like coherence, relevance, tone, and helpfulness without comparing to a gold standard. LLM-as-judge, rubric-based scoring, and human preference ratings are all reference-free methods.

Q70. What is "prompt drift" and how can it be detected?

- A. The gradual degradation in prompt effectiveness over time as the model version, data, or user behaviour changes — detected through ongoing evaluation against a fixed test suite.**
- B. The tendency for long prompts to lose coherence toward the end due to context window limitations.
- C. A phenomenon where model outputs drift toward training data patterns when prompts are ambiguous.
- D. The gradual divergence between a prompt's intended behaviour and what users actually ask.

Correct Answer: A

Explanation

Prompt drift is an operational risk: a prompt that works well today may underperform after a model update, a change in retrieved content, or shift in user query patterns. The only way to detect it is continuous evaluation — running the same test suite regularly and tracking metric trends. Without monitoring, drift goes undetected until users complain.

Q71. What is the purpose of a "golden dataset" in prompt evaluation?

- A. A training dataset used to fine-tune the model on the specific task the prompt is designed for.
- B. A curated set of representative inputs with high-quality human-verified expected outputs used as a benchmark for evaluating and comparing prompt versions.**
- C. A set of adversarial inputs specifically designed to break the model under evaluation.
- D. The original training data used by the model provider, used as a reference for capability assessment.

Correct Answer: B

Explanation

A golden dataset is the evaluation foundation: inputs that represent real-world usage, with outputs verified by domain experts. When you change a prompt, run the golden dataset and compare. Regression against golden examples is the most reliable signal of whether a prompt change is an improvement or a regression.

Q72. What is "inter-rater reliability" and why does it matter for LLM evaluation?

- A. The reliability of the model's outputs when the same prompt is sent to multiple server instances simultaneously.
- B. A metric for how consistently the model follows instructions across different evaluators' test inputs.
- C. The agreement rate between the model's outputs and a reference answer database.
- D. The degree to which different human evaluators agree on quality scores — low agreement means the evaluation criteria are ambiguous, making scores unreliable as a benchmark.**

Correct Answer: D

Explanation

If two evaluators score the same output a 4 and a 2, your evaluation is measuring evaluator variance, not output quality. High inter-rater reliability — achieved through clear rubrics, calibration sessions, and examples — is a prerequisite for meaningful human evaluation. Without it, your evaluation data is noise.

Q73. What is "output toxicity evaluation" in the context of prompt testing?

- A. A technique for filtering toxic words from model outputs using a keyword blocklist.
- B. An evaluation that measures how often the model refuses to answer questions it should answer.
- C. Systematically testing whether prompts produce harmful, biased, or offensive outputs across a representative range of inputs including adversarial cases.**
- D. A safety audit of the model provider's training data for toxic content.

Correct Answer: C

Explanation

Toxicity evaluation tests whether your application — including your system prompt — reliably prevents harmful outputs under realistic conditions. This means testing with adversarial inputs, sensitive topic probes, and jailbreak attempts. It is not sufficient to test only on polite, cooperative user inputs. Real users will push boundaries.

Q74. What is "calibration" in the context of model confidence and evaluation?

- A. Calibration refers to adjusting the model's temperature so its outputs match the desired level of certainty.
- B. A well-calibrated model's stated confidence accurately reflects its actual accuracy — when it says 80% confident, it is right about 80% of the time.**
- C. A post-processing technique that normalises confidence scores across different model outputs.
- D. Calibration is the process of fine-tuning a model to produce correct answers on a validation set.

Correct Answer: B

Explanation

Calibration is a statistical property: a perfectly calibrated model's confidence scores are empirically accurate. LLMs are generally poorly calibrated — they can be confidently wrong. This matters for evaluation: a model saying "I'm 95% certain" should not be taken at face value without empirical calibration data.

Q75. What is the most important principle for maintaining reliable prompt behaviour in production?

- A. Locking the prompt and never changing it once it has been tested and deployed.
- B. Using the largest available model to maximise output quality and minimise failure rates.
- C. Ensuring the system prompt is as long and detailed as possible to cover all possible scenarios.
- D. Continuous evaluation against a fixed test suite — monitoring output quality metrics over time and treating prompt performance as an ongoing operational concern, not a one-time setup.**

Correct Answer: D

Explanation

Prompts exist in a dynamic environment: models update, data changes, user behaviour shifts, edge cases emerge. The only way to maintain reliability is to treat prompt evaluation as an ongoing process — test suites, monitoring, regression checks, and periodic human review. One-time setup is not enough for production-quality AI systems.

SET

36

Vector Databases and Embeddings

Set 36 of 5 · 15 Questions · AI Engineering

Q1. What is a vector database and why is it used in AI applications?

- A. A database that stores model weights and configuration files for fast model loading.
- B. A database optimised for storing and searching high-dimensional embedding vectors using approximate nearest-neighbour algorithms.**
- C. A relational database with a vector data type added for storing numerical arrays.
- D. A key-value store where AI-generated responses are cached for fast retrieval.

Correct Answer: B**Explanation**

Vector databases (Pinecone, Weaviate, Chroma, pgvector) are built for one thing classical databases do poorly: finding the most semantically similar items in a high-dimensional space. They power semantic search and RAG by quickly finding the embedding vectors closest to a query vector — an operation called approximate nearest-neighbour (ANN) search.

Q2. What is the difference between storing embeddings in a vector database versus a traditional relational database?

- A. Vector databases store text; relational databases store numbers — embeddings are numbers so they belong in relational databases.
- B. Vector databases are schema-less; relational databases require a fixed schema that embeddings do not fit.
- C. Vector databases are only suitable for development; production systems must use relational databases for reliability.
- D. Vector databases are optimised for ANN search with specialised indexing (HNSW, IVF); relational databases support exact match and range queries but cannot efficiently search high-dimensional vectors.**

Correct Answer: D**Explanation**

Relational databases can store embedding arrays but lack the specialised indexes (HNSW, IVF-flat, ScaNN) needed to search millions of vectors in milliseconds. A PostgreSQL table with a vector column (pgvector) is fine for small datasets; dedicated vector databases provide the indexing, filtering, and performance needed at scale.

Q3. What is HNSW and why is it commonly used in vector search?

- A. Hierarchical Navigable Small World — a graph-based ANN index that provides fast and accurate nearest-neighbour search with good performance on high-dimensional data.**
- B. High-Noise Semantic Weighting — a technique for down-weighting noise in embedding vectors before indexing.
- C. Hybrid Neural Search Workflow — a pipeline combining dense and sparse retrieval in a single index.
- D. Horizontal Node Sharding and Weaving — a distributed architecture for scaling vector databases across multiple servers.

Correct Answer: A

Explanation

HNSW builds a hierarchical graph structure where each node connects to its nearest neighbours at multiple resolution levels. Search starts at the top (coarse) level and progressively refines — finding approximate nearest neighbours very quickly without exhaustive comparison. It is the dominant ANN algorithm in Pinecone, Weaviate, Chroma, and pgvector.

Q4. What does "cosine similarity" measure in the context of embedding search?

- A. The absolute distance between two vectors in Euclidean space.
- B. The number of shared tokens between two pieces of text after tokenisation.
- C. The angular similarity between two vectors — a value near 1 means the vectors point in the same direction, indicating semantic similarity.**
- D. The proportion of embedding dimensions where both vectors have the same sign.

Correct Answer: C

Explanation

Cosine similarity ignores vector magnitude and measures only direction — the angle between two vectors. In embedding space, direction encodes semantic meaning, so cosine similarity reliably captures semantic relatedness regardless of text length. Dot product is related and often used for efficiency when vectors are normalised.

Q5. What is the role of an embedding model in a RAG pipeline?

- A. Generating the final response from retrieved documents using the base language model.
- B. Converting text (queries and documents) into dense vectors so they can be compared for semantic similarity during retrieval.**
- C. Compressing retrieved documents into shorter summaries before they enter the context window.
- D. Classifying retrieved documents into topic categories to improve relevance ranking.

Correct Answer: B

Explanation

The embedding model is the semantic backbone of RAG. It converts both query text and document chunks into vectors in the same space. At retrieval time, query and chunk vectors are compared — chunks with high similarity to the query are retrieved. The quality of the embedding model directly determines retrieval quality.

Q6. What is the difference between "sparse" and "dense" retrieval?

- A. Sparse retrieval (BM25) uses keyword frequency and inverse document frequency; dense retrieval uses embedding similarity — combining both is often more effective than either alone.**
- B. Sparse retrieval works on small datasets; dense retrieval scales to large corpora.
- C. Sparse retrieval retrieves fewer documents; dense retrieval retrieves more documents per query.
- D. Sparse retrieval uses the full text; dense retrieval uses only the first sentence of each chunk.

Correct Answer: A

Explanation

Sparse retrieval (BM25, TF-IDF) matches based on exact term overlap and frequency — very effective for keyword queries. Dense retrieval (embedding similarity) captures meaning even without exact word match — effective for paraphrases and conceptual queries. Hybrid retrieval (combining both scores) typically outperforms either approach alone.

Q7. What is "metadata filtering" in a vector database and why is it important?

- A. Tagging embedding vectors with quality scores to filter out low-confidence retrievals.
- B. Removing metadata fields from documents before embedding to reduce vector noise.
- C. A post-processing step that filters retrieved documents by their cosine similarity score.
- D. Restricting vector search to documents matching specific metadata conditions (date, author, category) before or after ANN search — essential for access control and relevance filtering.**

Correct Answer: D

Explanation

Metadata filtering combines semantic search with structured constraints: "retrieve the most semantically similar chunks, but only from documents dated after 2024, tagged as financial reports, belonging to this user's organisation." Without metadata filtering, semantic search returns globally similar content, ignoring access permissions, recency, or domain scope.

Q8. What is "dimensionality reduction" in the context of embeddings and when is it used?

- A. Reducing the number of documents in an index by merging semantically similar chunks.
- B. Shrinking the embedding model's parameter count to reduce inference cost.
- C. Compressing high-dimensional embedding vectors into lower dimensions using techniques like PCA or UMAP — used to reduce storage cost, search latency, or to enable 2D/3D visualisation.**
- D. A technique for normalising embedding vectors to unit length before storage.

Correct Answer: C

Explanation

High-dimensional embeddings (1536D, 3072D) can be compressed to lower dimensions with modest accuracy loss. PCA reduces dimensions linearly preserving variance; UMAP preserves local structure for visualisation. Matryoshka embeddings (from OpenAI and others) are designed to be truncatable — you can use the first 256 dimensions of a 1536D embedding and still get reasonable retrieval quality.

Q9. What is "vector index freshness" and how is it managed?

- A. Keeping the vector index up to date as new documents are added or existing documents change — managed through re-indexing pipelines or real-time upsert operations.**
- B. The accuracy degradation of ANN indexes over time due to model drift.
- C. The frequency with which vector database providers update their ANN algorithms.
- D. A metric for how recently the embedding model was trained relative to the indexed documents.

Correct Answer: A

Explanation

A vector index is a snapshot. As source documents are added, updated, or deleted, the index must be kept in sync. This requires upsert pipelines (add new embeddings, update changed chunks, delete removed ones) and decisions about whether to re-index incrementally or rebuild periodically. Stale indexes return outdated or missing content.

Q10. What is "multi-tenancy" in a vector database and why does it matter for enterprise applications?

- A. The ability to run multiple embedding models simultaneously within the same index.
- B. A deployment pattern that runs the vector database across multiple cloud regions.
- C. A feature that allows multiple users to query the same index concurrently without performance degradation.
- D. The ability to store and search embeddings for multiple isolated organisations or users within a single database instance, with strict access separation.**

Correct Answer: D

Explanation

Enterprise RAG systems typically serve multiple customers or departments, each with private documents. Multi-tenancy ensures customer A's documents are invisible to customer B's queries. This is implemented through metadata filtering, namespace separation, or separate indexes per tenant — with different cost and isolation trade-offs.

Q11. What is "approximate" nearest-neighbour search and why is it used instead of exact search?

- A. ANN is used because exact search is mathematically impossible in high-dimensional spaces.
- B. ANN trades a small amount of accuracy for dramatically faster search — exact nearest-neighbour search over millions of vectors is too slow for real-time applications.**
- C. ANN returns approximate results to prevent the model from over-fitting to the retrieved content.
- D. ANN is a regulatory requirement for AI systems to prevent deterministic retrieval behaviour.

Correct Answer: B

Explanation

Exact nearest-neighbour search requires comparing the query vector against every indexed vector — $O(n)$ comparisons. Over millions of documents, this is too slow for real-time retrieval. ANN algorithms (HNSW, IVF) find vectors that are very likely nearest neighbours in sub-linear time, accepting a small accuracy trade-off for 100x or greater speed gains.

Q12. What is the significance of the embedding model's "embedding dimension" for a vector database?

- A. The embedding dimension determines the maximum document length that can be embedded.
- B. Higher embedding dimensions always improve retrieval accuracy regardless of the corpus size.
- C. The embedding dimension determines storage requirements and search latency — higher dimensions capture more semantic nuance but require more storage and compute for each comparison.**
- D. The embedding dimension must match the language model's hidden state size to work correctly.

Correct Answer: C

Explanation

Every embedded document costs ($\text{dimension} \times \text{bytes per float}$) of storage. For 10M documents at 1536 dimensions in float32, that is ~60GB. Search latency also scales with dimension. Choosing embedding dimension involves balancing retrieval quality against infrastructure cost — Matryoshka embeddings allow post-hoc dimension reduction without re-embedding.

Q13. What is "namespace" or "collection" separation in a vector database?

- A. A feature that assigns human-readable names to individual embedding vectors for easier management.
- B. A partitioning technique that stores embeddings alphabetically by document title.
- C. A security mechanism that encrypts different vector partitions with different keys.
- D. Logically isolating groups of vectors within the same database so different applications or data sources can be searched independently.**

Correct Answer: D

Explanation

Namespaces (Pinecone) or collections (Chroma, Weaviate) allow a single vector database to serve multiple use cases: a customer support knowledge base in one namespace, a product catalogue in another, legal documents in a third. Queries target a specific namespace, preventing cross-contamination of retrieval results.

Q14. What is "re-ranking" after vector retrieval and why is it needed?

- A. A technique that re-orders retrieved documents chronologically before injecting them into context.
- B. A second-stage ranking step that uses a more accurate (but slower) model to re-score the top-k retrieved candidates, improving final result relevance.**
- C. A post-retrieval step that removes duplicate content from the retrieved candidate set.
- D. A method for re-embedding retrieved documents with a higher-quality model for final selection.

Correct Answer: B

Explanation

ANN retrieval is fast but imprecise — the top-20 results may include noise. Re-ranking applies a cross-encoder model (which processes query and document jointly, capturing interactions) to re-score and reorder the candidates. The top-5 after re-ranking are more precisely relevant than the top-5 from ANN alone, at the cost of additional latency.

Q15. What is "embedding drift" and when does it cause problems?

- A. When documents and queries are embedded with different model versions, their vectors are in incompatible spaces — retrieval quality degrades significantly.**
- B. The gradual degradation of embedding quality as the vector database becomes too large.
- C. The tendency for embedding models to produce less accurate vectors for rare vocabulary over time.
- D. A phenomenon where embeddings become less unique over time as new documents are indexed.

Correct Answer: A

Explanation

Embedding models have versions. If you embed your document corpus with model v1 and then upgrade to v2 for query embedding, the vectors are in different spaces — semantically similar items no longer cluster together. When upgrading embedding models, you must re-embed the entire corpus, not just new documents. Failing to do so causes silent retrieval failures.

SET

37

Chunking and Indexing

Set 37 of 5 · 15 Questions · AI Engineering

Q16. What is "chunking" in a RAG pipeline and why is it necessary?

- A. Compressing documents into dense summaries before embedding them.
- B. Dividing the vector index into shards for distributed storage across multiple servers.
- C. Splitting source documents into smaller segments that fit within the embedding model's token limit and provide focused, retrievable units of information.**
- D. Breaking the user's query into sub-questions to improve retrieval coverage.

Correct Answer: C**Explanation**

Embedding models have token limits (typically 512–8192 tokens). Long documents cannot be embedded whole. More critically, a whole document embedded as one vector loses specificity — the embedding is a diffuse average of all its content. Chunking creates focused units: retrieving a specific relevant paragraph rather than a 100-page document.

Q17. What is "fixed-size chunking" and what is its main limitation?

- A. Splitting text into chunks of a fixed token count — simple to implement but can cut across sentence or paragraph boundaries, creating incoherent fragments.**
- B. Embedding each sentence as its own chunk — produces too many small chunks for efficient indexing.
- C. Dividing documents into fixed numbers of sections regardless of content length.
- D. A chunking method that preserves document structure by splitting only on heading markers.

Correct Answer: A**Explanation**

Fixed-size chunking (e.g., 512 tokens) is the simplest approach: split every N tokens. The problem is arbitrary boundary cuts — a chunk might start mid-sentence or cut a key statement in half. Adding overlap (repeating the last 50 tokens in the next chunk) mitigates boundary issues but increases total chunk count and storage.

Q18. What is "semantic chunking" and what problem does it solve?

- A. Embedding each sentence independently and clustering similar sentences into a single chunk.
- B. Using the embedding model itself to determine the optimal chunk boundaries through similarity analysis.
- C. A chunking method that creates chunks based on the number of distinct concepts they contain.
- D. Splitting text at semantically meaningful boundaries — paragraphs, topic shifts, section headings — rather than at arbitrary token counts.**

Correct Answer: D

Explanation

Semantic chunking respects document structure: split at paragraph breaks, section headings, or topic change points detected by a small model. This produces chunks that contain coherent ideas rather than arbitrary text windows. The trade-off is higher implementation complexity and potentially uneven chunk sizes.

Q19. What is "chunk overlap" and what is its purpose?

- A. Storing multiple copies of the same chunk with different embeddings from different models.
- B. Repeating a portion of text at the boundary between adjacent chunks to prevent information loss at split points.**
- C. A technique for combining the embeddings of two adjacent chunks into a single vector.
- D. Overlapping metadata between chunks so retrieval systems can reconstruct document structure.

Correct Answer: B

Explanation

If a key concept straddles a chunk boundary, both halves of the statement are in adjacent chunks but neither half may be retrievable on its own. Overlap (repeating the last 100 tokens of chunk N as the first 100 tokens of chunk N+1) ensures boundary-straddling content is fully present in at least one chunk. It increases storage by the overlap percentage.

Q20. What is "hierarchical chunking" and when is it appropriate?

- A. Chunking a document tree structure based on its heading hierarchy.
- B. A method where each chunk is embedded at progressively lower dimensions for multi-resolution search.
- C. Creating chunks at multiple granularity levels — sentence, paragraph, section, document — and linking them, enabling retrieval at the right granularity for different query types.**
- D. Chunking documents in order of their importance, processing the most critical sections first.

Correct Answer: C

Explanation

Some queries need a single sentence; others need a full section. Hierarchical chunking indexes at multiple levels. Retrieval fetches the most relevant unit — perhaps a sentence — but the system can expand to the parent paragraph or section for full context before injection into the LLM prompt. This is the basis of "small-to-big retrieval" patterns.

Q21. What is a "document parser" in the context of RAG pipelines and why does it matter?

- A. A model component that classifies documents into categories before they are indexed.
- B. A tool that splits documents into chunks based on their natural language structure.
- C. A retrieval component that parses the user's query into structured search parameters.
- D. Software that extracts clean text from raw file formats (PDF, DOCX, HTML, spreadsheets) before chunking — poor parsing produces garbled chunks that degrade retrieval quality.**

Correct Answer: D

Explanation

Before chunking, documents must be converted from their native format to clean text. PDF parsing is notoriously unreliable — tables become garbled, columns merge, headers repeat. If the parsed text is poor, the embeddings are meaningless, and retrieval fails regardless of how good the rest of the pipeline is. High-quality parsing is a frequently underestimated RAG dependency.

Q22. What is "chunk size" and how should it be chosen?

- A. The token count of each chunk — chosen empirically based on the embedding model's limits, the granularity of information, and the typical query type, balancing specificity against context completeness.**
- B. The file size of the embedded chunk as stored in the vector database.
- C. The number of sentences per chunk, typically fixed at 3–5 for optimal retrieval.
- D. The token count must always match the context window size of the language model.

Correct Answer: A

Explanation

Chunk size is a hyper-parameter that affects retrieval quality. Small chunks (128 tokens) are specific but may lack context; large chunks (1024 tokens) are contextual but may dilute the embedding with irrelevant content. The right size depends on document type, query length, and embedding model — it should be determined through retrieval evaluation on real queries.

Q23. What is "parent-child retrieval" in a RAG system?

- A. A retrieval pattern where parent documents are retrieved first and child documents are retrieved from within them.
- B. Indexing small child chunks for precise retrieval, but injecting the larger parent chunk into context to provide surrounding information.**
- C. A hierarchical permission model where parent users can access all child users' retrieved documents.
- D. A chunking strategy where parent headings are stored as separate chunks above their child content.

Correct Answer: B

Explanation

Parent-child retrieval solves a trade-off: small chunks retrieve precisely (high recall for specific facts) but may lack context when injected into the LLM. The solution: index small chunks for retrieval, but when a chunk is selected, return its parent (the full paragraph or section) to the LLM. Specificity in retrieval, context in generation.

Q24. What is "document-level metadata" and how does it enhance retrieval?

- A. A summary of the document's content stored as a separate embedding for document-level retrieval.
- B. The document's file format and encoding information required by the parser.
- C. Metadata tags embedded within the chunk text to help the language model understand the source.
- D. Structured information about the document (author, date, source, category, access level) stored alongside chunk embeddings, enabling filtered and context-aware retrieval.**

Correct Answer: D

Explanation

Raw semantic search retrieves based on content alone. With metadata, you can constrain retrieval: "only documents from the legal department", "only published after Q3 2024", "only accessible to tier-1 users." This makes retrieval both more precise and more secure — essential for enterprise knowledge systems with access control requirements.

Q25. What is "index poisoning" and why is it a security concern for RAG systems?

- A. A database corruption attack that corrupts ANN index structures to cause retrieval failures.
- B. A technique used by attackers to extract documents from a vector database by observing query patterns.
- C. Injecting adversarial or misleading documents into the vector index to manipulate what the system retrieves and ultimately what the LLM says.**
- D. The gradual degradation of index quality caused by adding too many similar documents.

Correct Answer: C

Explanation

If an attacker can insert documents into your knowledge base — through a public data source, user-uploaded content, or web scraping — they can plant documents designed to be retrieved for specific queries. The LLM then incorporates this adversarial content into its response. Index poisoning is indirect prompt injection via the retrieval layer.

Q26. What is "sparse-dense hybrid indexing"?

- A. An indexing approach that stores small chunks densely and large chunks sparsely to optimise storage.
- B. Maintaining both a sparse keyword index (BM25) and a dense embedding index, then combining their scores for retrieval to get the benefits of both.**
- C. A technique that uses sparse matrix compression to reduce vector storage costs.
- D. An index that switches between sparse and dense retrieval based on query length.

Correct Answer: B

Explanation

Sparse indexes excel at exact term matching; dense indexes excel at semantic similarity. Hybrid indexes maintain both and combine scores (via reciprocal rank fusion or learned combination weights). This handles both keyword-specific queries ("ISO 27001 clause 6.2") and semantic queries ("what are our information security obligations") reliably.

Q27. What is "incremental indexing" and why does it matter operationally?

- A. Adding new documents to the index without rebuilding the entire index from scratch — critical for knowledge bases that receive frequent updates.**
- B. A technique for progressively improving index quality by re-embedding documents in order of retrieval frequency.
- C. Building the index in stages across multiple machines to reduce memory requirements.
- D. A method that incrementally compresses old embeddings to free space for new documents.

Correct Answer: A

Explanation

Many enterprise knowledge systems are dynamic — new documents added daily, old ones updated. Rebuilding the entire index from scratch for each update is too slow and expensive. Incremental indexing adds new documents with upsert operations (insert or update by document ID) without touching existing entries, keeping the index current without downtime.

Q28. What is "content-aware chunking" for structured documents like tables and code?

- A. A chunking technique that embeds the document's style and formatting alongside its text content.
- B. Using the document's own section markers to determine chunk boundaries.
- C. Treating tables, code blocks, and structured data as atomic units rather than splitting them mid-structure, preserving semantic integrity.**
- D. A method that assigns different embedding models to different content types within the same document.

Correct Answer: C

Explanation

Tables and code blocks cannot be split arbitrarily — a table split across two chunks loses its column-row structure; code split mid-function loses its meaning. Content-aware chunking detects these structures and treats them as atomic: a full table is one chunk, a complete function is one chunk — even if this violates the target token count.

Q29. What is a "knowledge graph" and how does it complement vector retrieval?

- A. A visual diagram of the vector space showing how document embeddings cluster.
- B. A database of factual triples used to verify the accuracy of LLM-retrieved content.
- C. A graph database version of a vector index that organises embeddings by topic hierarchy.
- D. A structured representation of entities and their relationships — useful for queries that require multi-hop reasoning across connected concepts that pure semantic search handles poorly.**

Correct Answer: D

Explanation

Vector retrieval is excellent at finding semantically similar content but poor at multi-hop reasoning: "What are the supply chain implications of our contract with Vendor X given our Q3 risk assessment?" requires traversing entity relationships. Knowledge graphs model entities and their connections explicitly, enabling reasoning that flat vector search cannot perform.

Q30. What is "late chunking" in modern RAG pipelines?

- A. Delaying the chunking step until query time, dynamically splitting documents based on query relevance.
- B. Embedding the full document first with a long-context embedding model, then splitting the contextualised token representations into chunk-level embeddings — preserving full-document context in each chunk's embedding.**
- C. A technique that embeds chunks only when they are first queried, rather than pre-computing all embeddings at index time.
- D. A post-retrieval step that re-splits retrieved chunks to fit the available context window.

Correct Answer: B

Explanation

Traditional chunking loses context: the embedding of a paragraph does not know about the rest of the document. Late chunking (pioneered by JinaAI) uses long-context embedding models to process the full document, then splits the resulting token-level representations into chunk embeddings. Each chunk embedding is contextualised by the entire document — improving retrieval for content that only makes sense in document context.

SET

38

Retrieval

Set 38 of 5 · 15 Questions · AI Engineering

Q31. What is "top-k retrieval" and what determines the right value of k?

- A. Returning the k most similar chunks to the query — k is chosen based on the context window budget, retrieval precision, and how much relevant content a typical query needs.**
- B. Retrieving all documents above a cosine similarity threshold, with k as the minimum floor.
- C. A retrieval strategy that always returns exactly 5 chunks regardless of query complexity.
- D. The k top-scored chunks from the sparse keyword index, used before dense re-ranking.

Correct Answer: A**Explanation**

Top-k is the most common retrieval pattern. The right k depends on: how much context window you can spare, how precise retrieval is (low precision means you need more candidates to ensure the right content is included), and the typical query complexity. k=3–5 is common for focused queries; k=10–20 for broad research queries with re-ranking.

Q32. What is "query expansion" in retrieval and how does it improve recall?

- A. Expanding the context window to accommodate more retrieved chunks.
- B. Breaking a complex query into sub-queries and retrieving results for each independently.
- C. Generating alternative phrasings or related terms for the query before retrieval, increasing the chance of matching relevant content that uses different vocabulary.**
- D. Adding metadata filters to a query after initial retrieval to narrow results.

Correct Answer: C**Explanation**

A user query uses specific vocabulary that may not match the vocabulary in relevant documents. Query expansion generates synonyms, related concepts, or re-phrasings: "heart attack" → also search "myocardial infarction", "cardiac event". This improves recall for vocabulary mismatches without reducing precision if done carefully.

Q33. What is "hypothetical document embedding" (HyDE) and what problem does it solve?

- A. Creating fictional document examples during index creation to improve embedding model training.
- B. Generating a hypothetical answer to the query, embedding that answer, and using it for retrieval — solving the vocabulary gap between short queries and longer document chunks.**
- C. Embedding a hypothetical user persona to personalise retrieval results.
- D. A technique for generating synthetic documents to fill gaps in a sparse knowledge base.

Correct Answer: B

Explanation

Short queries and long document chunks have very different linguistic characteristics — their embeddings may not be close even when semantically relevant. HyDE asks the LLM to generate a hypothetical answer to the query, then embeds that hypothetical answer for retrieval. The hypothetical answer is in the same register as real document chunks, improving embedding similarity.

Q34. What is "multi-query retrieval" and when is it useful?

- A. Sending the same query to multiple vector databases simultaneously and taking the union of results.
- B. Splitting a complex query into sequential sub-queries where each result informs the next.
- C. Retrieving from multiple indexes in parallel to compare coverage across different knowledge bases.
- D. Generating multiple reformulations of the user's query, retrieving results for each, and merging the results — useful for broad or ambiguous queries where a single phrasing misses relevant content.**

Correct Answer: D

Explanation

A single query phrasing may retrieve different content than a slightly different phrasing — due to vocabulary mismatch and the approximate nature of embedding similarity. Multi-query retrieval runs 3–5 reformulations and deduplicates the union of results. This improves coverage with modest latency overhead.

Q35. What is "retrieval precision" versus "retrieval recall" in a RAG context?

- A. Precision: proportion of retrieved chunks that are actually relevant; Recall: proportion of relevant chunks in the corpus that were successfully retrieved.**
- B. Precision: accuracy of the final LLM answer; Recall: proportion of the question that the answer addresses.
- C. Precision: speed of retrieval; Recall: completeness of the indexed corpus.
- D. Precision and recall are the same metric in retrieval contexts — both measure relevance.

Correct Answer: A

Explanation

These are the fundamental retrieval quality metrics. High precision means retrieved chunks are relevant (low noise in context); high recall means no important chunks were missed. Increasing k improves recall but may hurt precision. Re-ranking improves precision of the final selected chunks. The right trade-off depends on whether missing relevant content or including irrelevant content is more harmful.

Q36. What is "cross-encoder re-ranking" and how does it differ from bi-encoder retrieval?

- A. Cross-encoders retrieve from multiple embedding models simultaneously, taking the best result.
- B. Cross-encoders process query and document together in one pass, capturing interactions — more accurate than bi-encoders which encode them independently, but too slow for initial retrieval.**
- C. Cross-encoders use keyword overlap alongside semantic similarity for more accurate retrieval.
- D. Cross-encoders are a faster alternative to bi-encoders that sacrifice accuracy for speed.

Correct Answer: B

Explanation

Bi-encoders (used for initial retrieval) embed query and document separately — fast but limited by their inability to model query-document interactions. Cross-encoders take [query + document] as a single input and score the pair jointly — capturing nuanced relevance but requiring one forward pass per candidate. Used in the re-ranking stage on top-k candidates from bi-encoder retrieval.

Q37. What is "contextual retrieval" (as introduced by Anthropic) and what problem does it solve?

- A. A retrieval technique that uses the conversation history as additional context for query reformulation.
- B. A method where retrieved chunks are summarised before being injected into the LLM context.
- C. A retrieval approach that retrieves both the relevant chunk and its adjacent chunks automatically.
- D. Prepending a chunk-specific summary of its role within the parent document to each chunk before embedding — solving the problem of chunks that lose meaning without document context.**

Correct Answer: D

Explanation

Anthropic's contextual retrieval addresses a core problem: a chunk saying "The agreement was signed on 15 March 2024" is meaningless without knowing which agreement. Contextual retrieval uses the LLM to generate a brief description of what the chunk covers in the context of the full document, prepends this to the chunk before embedding, and improves retrieval accuracy significantly.

Q38. What is "retrieval augmented generation" (RAG) at its most fundamental level?

- A. Training a language model on retrieved web content to keep its knowledge current.
- B. Using a language model to generate improved search queries for a traditional search engine.
- C. Retrieving relevant external content at query time and injecting it into the LLM context, grounding the response in provided information rather than the model's parametric knowledge alone.**
- D. A technique for compressing retrieval results into the model's context window more efficiently.

Correct Answer: C

Explanation

RAG is a two-step architecture: retrieve then generate. Step 1: convert the query to an embedding, search the vector index, and retrieve relevant chunks. Step 2: inject those chunks into the LLM prompt alongside the query. The LLM generates a response grounded in the retrieved content rather than from parametric memory alone. This is the foundation of almost all enterprise AI knowledge applications.

Q39. What is "self-querying retrieval" and how does it improve structured knowledge bases?

- A. A retrieval pattern where the model generates its own retrieval queries without user input.
- B. Using an LLM to convert natural language queries into structured filters for the vector database — combining semantic search with metadata filtering based on the user's intent.**
- C. A method where the retrieval system queries itself to identify gaps in the knowledge base.
- D. A technique for retrieving documents that match the style of the user's query rather than its content.

Correct Answer: B

Explanation

Self-querying retrieval uses an LLM to parse the query into a semantic component and a structured filter component: "documents about quarterly performance from Q1 2025" → semantic query = "quarterly performance" + filter = "date between 2025-01-01 and 2025-03-31." The LLM generates both, enabling natural language queries against structured metadata.

Q40. What is "reciprocal rank fusion" (RRF) and when is it used?

- A. A method for combining ranked result lists from multiple retrieval systems by aggregating reciprocal rank positions — used in hybrid retrieval to merge sparse and dense results.**
- B. A technique for re-ranking retrieved documents based on their relevance to the full conversation history rather than just the current query.
- C. A scoring method that multiplies the dense retrieval score by the sparse retrieval score.
- D. A fusion technique that averages the embedding vectors of top retrieved chunks before LLM injection.

Correct Answer: A

Explanation

When combining results from BM25 and dense retrieval, you have two ranked lists with different score scales. RRF converts each list to ranks and combines them: $score = \sum 1/(k + rank_i)$ where k is a smoothing constant. This is robust to score scale differences and consistently outperforms naive score averaging in hybrid retrieval benchmarks.

Q41. What is "retrieval latency" and why does it matter for user-facing applications?

- A. The delay between a document being added to the corpus and it becoming retrievable.
- B. The time taken for the LLM to process retrieved chunks and generate a response.
- C. The time taken from query submission to when retrieved chunks are available for the LLM — in user-facing applications, retrieval latency adds directly to perceived response time.**
- D. A metric for how quickly the vector index can be rebuilt after a major update.

Correct Answer: C

Explanation

For real-time applications, every millisecond matters. Retrieval latency depends on: index size, ANN algorithm, k , whether re-ranking is applied, and database hardware. Caching frequent queries, using in-memory indexes, and limiting re-ranking candidates are common optimisations. A slow retrieval step can make an otherwise fast LLM feel sluggish.

Q42. What is "query routing" in a multi-knowledge-base RAG system?

- A. Load-balancing retrieval queries across multiple vector database replicas.
- B. A technique for splitting a complex query into multiple sub-queries routed to different LLMs.
- C. Forwarding unanswerable queries to a human agent after retrieval fails.
- D. Classifying incoming queries and directing them to the most appropriate knowledge base or retrieval strategy based on query type or domain.**

Correct Answer: D

Explanation

Large organisations may have multiple knowledge bases: HR policies, technical documentation, product specifications, legal contracts. Query routing classifies the intent and directs to the right source: "questions about leave policy" → HR index; "questions about API specs" → technical docs. This improves both precision and retrieval efficiency compared to searching all indexes for every query.

Q43. What is "passage retrieval" versus "document retrieval"?

- A. Passage retrieval uses embedding similarity; document retrieval uses keyword matching.
- B. Passage retrieval returns the specific relevant section within a document; document retrieval returns the whole document — passage retrieval is more precise for focused queries.**
- C. Passage retrieval is used for short documents; document retrieval is used for long documents.
- D. Passage and document retrieval are equivalent — the distinction is in how results are displayed, not how they are retrieved.

Correct Answer: B**Explanation**

Document retrieval returns whole documents — useful when the entire document is relevant or when document-level context is needed.

Passage retrieval returns specific sections — more efficient for focused queries and better preserves context window budget. Most modern RAG systems use passage (chunk) retrieval, with optional parent document expansion for additional context.

Q44. What is "negative sampling" in embedding model training and why does it matter for retrieval quality?

- A. Removing low-quality documents from the index to prevent them from being retrieved.
- B. A technique for generating synthetic queries for documents that have no natural associated queries.
- C. Training the embedding model on hard negative examples — documents that are superficially similar to relevant ones but should be ranked lower — to produce embeddings that distinguish near-duplicates.**
- D. Sampling a subset of the corpus to train a small retrieval model for low-resource settings.

Correct Answer: C**Explanation**

Hard negatives are the key to high-quality retrieval embeddings. If the model only trains on obvious positives and random negatives, it learns coarse distinctions. Hard negatives — documents that seem relevant but are not — force the model to learn fine-grained distinctions. Models trained with hard negatives (like E5, BGE, and mxbai-embed) significantly outperform those trained without them.

Q45. What is "retrieval evaluation" and what metrics are used?

- A. Systematic measurement of retrieval quality using metrics like recall@k, precision@k, NDCG, and MRR against a labelled dataset of queries and relevant documents.**
- B. An evaluation of the LLM's ability to use retrieved context accurately in its final answer.
- C. A benchmark test that measures how quickly the vector database returns results under load.
- D. An audit of the knowledge base for completeness and accuracy before deploying a RAG system.

Correct Answer: A

Explanation

Retrieval evaluation requires a golden dataset: queries with known relevant documents. Key metrics: recall@k (are relevant docs in the top-k?), precision@k (how many of the top-k are relevant?), NDCG (rank-weighted precision), MRR (rank of the first relevant result). Without retrieval evaluation, you cannot tell if changes to chunking, embedding models, or retrieval strategy improve or hurt quality.

SET

39

Grounding and Hallucination Prevention

Set 39 of 5 · 15 Questions · AI Engineering

Q46. What is "grounding" in a RAG system and why is it the primary mechanism for reducing hallucination?

- A. Connecting the model to real-time data sources so its knowledge is always current.
- B. A technique for fact-checking model outputs against a database of verified facts after generation.
- C. Anchoring the model's outputs in user-specific context by injecting user profile data.
- D. Instructing the model to base its response on provided documents only, making fabricated claims verifiable and constraining the model away from its parametric knowledge.**

Correct Answer: D**Explanation**

Grounding works by replacing the question "what do I believe?" with "what does the provided context say?" When the model is instructed to answer only from retrieved documents, every claim traces back to a specific source. This does not eliminate hallucination (models can still misread sources) but dramatically reduces fabrication of facts not in the provided content.

Q47. What is "faithfulness" as a RAG evaluation metric?

- A. Whether the retrieved documents are factually correct and up to date.
- B. Whether the model's response accurately reflects the content of the retrieved documents, without adding claims not supported by the retrieved context.**
- C. Whether the model responds consistently across multiple runs of the same query.
- D. Whether the model's persona remains consistent with the system prompt throughout the conversation.

Correct Answer: B**Explanation**

Faithfulness measures the alignment between retrieved context and generated response. A faithful response makes only claims that are supported by the retrieved chunks. An unfaithful response adds information from the model's parametric knowledge — which may or may not be accurate. Faithfulness can be measured with LLM-as-judge evaluation against the retrieved source.

Q48. What is "answer relevance" in RAG evaluation and how does it differ from faithfulness?

- A. Answer relevance measures output length; faithfulness measures citation accuracy.
- B. Answer relevance is an automated metric; faithfulness requires human evaluation.
- C. Answer relevance measures whether the response addresses the user's question; faithfulness measures whether it is grounded in the retrieved context — a response can be faithful but irrelevant, or relevant but unfaithful.**
- D. Answer relevance and faithfulness are the same metric evaluated from different perspectives.

Correct Answer: C

Explanation

These are orthogonal dimensions. A response grounded entirely in the retrieved documents but not addressing the user's actual question is faithful but not relevant. A response that directly answers the question using external knowledge is relevant but not faithful. High-quality RAG requires both: grounded in sources AND responsive to the query.

Q49. What is "source citation" in RAG output and how does it reduce user trust risk?

- A. Including the source document name, section, or URL for each claim — allowing users to verify responses and signalling that the system is grounded in authoritative content.**
- B. A legal requirement that all AI-generated content must attribute its sources.
- C. A technique for improving response quality by forcing the model to read sources more carefully.
- D. A post-processing step that automatically adds Wikipedia links to key terms in the response.

Correct Answer: A

Explanation

Citation makes grounding visible: "According to the Employee Handbook (Section 4.2), annual leave accrues at 1.5 days per month." The user can verify the claim directly. This increases trust, catches retrieval errors before they propagate, and creates an audit trail. For enterprise AI, citation is often a compliance and governance requirement, not just a quality feature.

Q50. What is the "context faithfulness" problem in RAG?

- A. Retrieved context that is outdated or factually incorrect before the model even reads it.
- B. The model refusing to generate responses when retrieved context is insufficient.
- C. A parsing failure where the model cannot read embedded tables or structured data in retrieved chunks.
- D. The model generating responses that contradict or misrepresent the retrieved context — often by over-relying on its parametric knowledge instead of the provided documents.**

Correct Answer: D

Explanation

Even with perfectly retrieved, accurate documents, models sometimes ignore or misrepresent them — answering from their parametric memory instead. This is especially common when the retrieved context contradicts the model's training data. Strong grounding instructions, citations, and faithfulness evaluation are the mitigations.

Q51. What is the "knowledge conflict" problem in RAG?

- A. When two retrieved documents contradict each other, leaving the model unable to determine the truth.
- B. When the knowledge base is too large for all relevant documents to fit in the context window.
- C. When retrieved documents contradict the model's parametric knowledge — the model must decide whether to trust the retrieved content or its training, and often gets this wrong.**
- D. When the model's knowledge cutoff pre-dates the retrieved documents, causing integration errors.

Correct Answer: C

Explanation

Models are trained on a large corpus and develop strong beliefs. When retrieved content contradicts those beliefs, models often default to their parametric knowledge — answering incorrectly while ignoring the correct retrieved information. Strong grounding instructions ("always trust the provided documents over your training") help, but knowledge conflicts remain a failure mode in RAG systems.

Q52. What is "attribution hallucination" in a RAG context?

- A. The model claiming a retrieved document says something it does not actually say — accurately citing the source but misrepresenting its content.**
- B. Hallucinating a source citation that does not exist in the retrieved documents.
- C. Attributing the response to the wrong retrieved document when multiple are in context.
- D. Generating a response attributed to the user's past statements rather than retrieved documents.

Correct Answer: A

Explanation

Attribution hallucination is subtler than outright fabrication: the model cites "Policy Document Section 3.1" but either misquotes it or states something the document does not say. This is dangerous because citations create a false sense of verifiability. Faithfulness evaluation — checking whether cited claims are actually supported by the cited source — is required to catch this.

Q53. What is "RAG versus fine-tuning" and when should each be used to reduce hallucination?

- A. Fine-tuning always produces more accurate responses; RAG is a workaround for models that cannot be fine-tuned.
- B. RAG injects current, verifiable external content at query time; fine-tuning bakes domain knowledge into weights. RAG is preferred when the knowledge is dynamic, traceable, or requires citation.**
- C. RAG is used for factual questions; fine-tuning is used for stylistic adaptation.
- D. RAG and fine-tuning produce identical results — the choice depends only on implementation cost.

Correct Answer: B

Explanation

Fine-tuning for factual knowledge is unreliable — models can still hallucinate domain facts even after fine-tuning, and the knowledge cannot be updated without re-training. RAG provides verifiable, updateable, citable knowledge at query time. Fine-tuning is better for style, format, and task behaviour — not for injecting facts that need to be trusted.

Q54. What is "abstention" and when should a RAG system abstain?

- A. A safety mechanism that prevents the model from generating responses on sensitive topics.
- B. A retrieval strategy that returns no results when confidence is below a threshold.
- C. Declining to answer when the retrieved context does not contain sufficient information, rather than hallucinating a response.**
- D. A technique where the model refers users to human agents instead of answering.

Correct Answer: C

Explanation

A well-designed RAG system should abstain gracefully: "I don't have sufficient information in the available documents to answer this question." This is far better than hallucinating an answer. Abstention requires explicit instruction and a faithfulness evaluation step — without it, models default to generating plausible-sounding content rather than admitting uncertainty.

Q55. What is "chunk context contamination" and how does it cause incorrect responses?

- A. A corruption where encoding errors in retrieved chunks introduce garbled text into the context.
- B. The injection of adversarial content into retrieved chunks by a malicious document author.
- C. A problem where chunks from different documents are combined in a way that creates copyright issues.
- D. When irrelevant or misleading retrieved chunks are included in the context alongside relevant ones, and the model synthesises information across all of them, producing a confused response.**

Correct Answer: D

Explanation

If retrieval returns 5 chunks but 3 are irrelevant, the model may blend information from all 5. A question about the termination clause in Contract A might be answered with information from Contract B that was co-retrieved. This is why retrieval precision matters: reducing irrelevant chunks reduces contamination risk, not just noise.

Q56. What is "confidence-based routing" in a RAG system?

- A. Routing queries to different response strategies based on retrieval confidence — high-confidence retrievals go to the LLM; low-confidence ones trigger clarification, abstention, or human escalation.**
- B. A technique for routing high-traffic queries to faster model endpoints during peak load.
- C. A method for automatically selecting the most confident retrieved chunk as the sole context.
- D. A routing system that sends low-confidence responses to a human reviewer before delivery.

Correct Answer: A

Explanation

Not all retrievals are equally reliable. A retrieval score above a threshold indicates good matches exist in the knowledge base; below the threshold, the knowledge base may not cover the query. Confidence-based routing handles this: certain responses are answered directly, uncertain ones prompt clarification or escalation, preventing unreliable responses from reaching users.

Q57. What is "closed-book versus open-book" generation in LLM systems?

- A. Closed-book models cannot access the internet; open-book models can browse the web in real time.
- B. Closed-book: the model answers from parametric memory only; Open-book (RAG): the model answers from provided documents — open-book is generally more accurate for specific factual questions.**
- C. Closed-book generation requires fine-tuning; open-book generation uses the base model.
- D. Closed-book and open-book refer to whether the model's system prompt is visible to users.

Correct Answer: B

Explanation

The closed/open-book distinction captures whether the model has access to external context. Closed-book models answer from weights alone — fast, private, but limited to training knowledge. Open-book (RAG) models answer from provided documents — more accurate for specific, current, or proprietary knowledge, but dependent on retrieval quality.

Q58. What is the "faithfulness versus completeness" trade-off in RAG responses?

- A. Faithful responses are shorter; complete responses are longer — the trade-off is response length.
- B. Faithfulness is a user-facing quality; completeness is an internal system metric.
- C. The trade-off only applies when retrieved content is from multiple conflicting sources.
- D. Maximally faithful responses stick close to retrieved content and may miss information not in the context; complete responses may hallucinate to fill gaps — the balance depends on application risk tolerance.**

Correct Answer: D

Explanation

Faithful responses say only what the sources say — they may be incomplete if the sources do not cover the full answer. Complete responses try to answer the whole question — they may add parametric knowledge, risking hallucination. For high-stakes applications (legal, medical, compliance), faithfulness dominates. For research and exploration, completeness may be more valuable.

Q59. What is "hallucination detection" as a post-generation step?

- A. Running a secondary check — using an LLM or a fact-checking model — to identify claims in the response that are not supported by the retrieved context.**
- B. A pre-generation filter that blocks prompts likely to cause hallucination.
- C. A metric computed from the model's internal token probability distributions.
- D. A technique for identifying hallucinated citations by querying the source database.

Correct Answer: A

Explanation

Post-generation hallucination detection runs a verification pass: "For each claim in this response, is it supported by one of the retrieved documents? Flag any claim that is not." This can be automated with LLM-as-judge prompting against the retrieved context. It adds latency and cost but is essential for high-stakes applications where undetected hallucination is unacceptable.

Q60. What is a "grounding score" and how is it used in RAG evaluation?

- A. A score assigned to each retrieved chunk indicating how relevant it is to the query.
- B. A benchmark score that compares RAG system performance against non-RAG baselines.
- C. A metric that measures what proportion of the response's claims are traceable to the retrieved context — used to monitor and improve RAG faithfulness over time.**
- D. A confidence score generated by the LLM indicating how certain it is about its response.

Correct Answer: C

Explanation

Grounding score = (claims supported by retrieved context) / (total claims in response). Tracking this metric across queries identifies patterns: certain query types cause more hallucination, certain document types are misread, or certain system prompt formulations improve grounding. It is an actionable metric for systematic RAG quality improvement.

SET

40

Enterprise Knowledge Systems

Set 40 of 5 · 15 Questions · AI Engineering

Q61. What is an "enterprise knowledge system" built on RAG?

- A. An AI application that connects organisational knowledge sources — policies, documents, databases, wikis — to a language model via retrieval, enabling employees to query internal knowledge in natural language.
- B. A database system that uses AI to automatically categorise and tag enterprise documents.
- C. A model fine-tuned on an organisation's proprietary data to answer internal questions.
- D. A search engine that uses keyword indexing to retrieve enterprise documents faster than traditional search.

Correct Answer: A**Explanation**

Enterprise knowledge systems are the most common enterprise AI application: employees ask questions in natural language ("What is our reimbursement policy for client entertainment?") and the system retrieves the relevant policy document and generates a grounded, cited response. The quality depends on parsing, chunking, retrieval, and grounding — not just the LLM.

Q62. What is "access control" in an enterprise RAG system and how is it typically implemented?

- A. Encrypting all documents in the vector index to prevent unauthorised access to raw embeddings.
- B. A login system that authenticates users before allowing them to query the knowledge base.
- C. A post-retrieval filter that removes sensitive content from responses before delivery.
- D. Ensuring users can only retrieve content they are authorised to see — typically through metadata filtering by user role, department, or document classification.

Correct Answer: D**Explanation**

In an enterprise context, not all employees should see all documents — financial forecasts, HR records, legal privileged content. Access control in RAG is typically implemented via metadata: each document is tagged with access permissions, and retrieval queries are filtered to include only documents the requesting user is authorised to view. This must be enforced at the database layer, not just in the prompt.

Q63. What is "knowledge freshness" and how is it managed in a production RAG system?

- A. Using a recent embedding model to ensure indexed documents reflect the latest language patterns.
- B. Keeping the indexed knowledge current by continuously ingesting new documents, updating changed ones, and removing deprecated ones through automated pipelines.**
- C. Restricting retrieval to documents published within the last 90 days to prevent stale answers.
- D. Retraining the language model periodically to incorporate knowledge from the organisation's documents.

Correct Answer: B

Explanation

A knowledge base that is not updated becomes a liability: employees get answers based on outdated policies, superseded product specs, or revoked legal guidance. Knowledge freshness requires document ingestion pipelines that detect new or modified source documents, re-chunk and re-embed them, upsert into the vector index, and delete removed documents.

Q64. What is a "knowledge gap" in an enterprise RAG system and how should it be handled?

- A. A discrepancy between what the LLM knows parametrically and what is in the knowledge base.
- B. A retrieval failure caused by poor chunking that prevents the system from finding existing content.
- C. When the knowledge base does not contain information to answer a query — handled by abstaining gracefully, routing to a human expert, or flagging the gap for knowledge base improvement.**
- D. The difference in quality between documents produced by different departments in the organisation.

Correct Answer: C

Explanation

No knowledge base is complete. A well-designed system detects when it cannot find relevant content (low retrieval scores, no above-threshold results) and responds appropriately: "I don't have documentation on this topic. Please contact [expert team]." This is more professional than hallucinating an answer. Tracking knowledge gaps also provides a prioritised backlog for knowledge base improvement.

Q65. What is "document lifecycle management" in the context of RAG?

- A. Tracking documents through creation, versioning, expiry, and deletion — ensuring the knowledge base reflects current authoritative versions and removes outdated content.**
- B. A content governance process for ensuring documents are written in a RAG-compatible format.
- C. A backup and recovery system for vector database contents.
- D. The process of archiving retrieved responses for audit and compliance purposes.

Correct Answer: A

Explanation

Documents have lifecycles. Policies are superseded; product specs are updated; contracts expire. Without lifecycle management, a RAG system might retrieve an outdated policy that has been replaced. This requires: version control (new versions replace old), expiry metadata (auto-remove content after a date), and governance processes to ensure the knowledge base reflects the current authoritative state.

Q66. What is "semantic deduplication" in knowledge base curation?

- A. A technique for removing identical files from a document repository before indexing.
- B. Using an LLM to merge similar documents into a single authoritative version.
- C. Identifying and removing near-duplicate documents or chunks using embedding similarity — preventing the same information from being retrieved multiple times and inflating its apparent importance.**
- D. A post-retrieval step that removes duplicate chunks before injecting them into the LLM context.

Correct Answer: C

Explanation

Document repositories accumulate duplicates: slightly different versions of the same policy, copied content in multiple documents, email threads that quote each other. Semantic deduplication identifies near-duplicate chunks using cosine similarity and removes or merges them. Duplicate content biases retrieval — the same information retrieved multiple times appears more relevant than it is.

Q67. What is "knowledge base segmentation" and when is it appropriate?

- A. Splitting large documents into smaller ones to improve chunking quality.
- B. A technique for distributing the vector database across multiple servers for performance.
- C. Segmenting users into groups that receive different knowledge base access for A/B testing.
- D. Organising the knowledge base into separate indexes or namespaces by domain, department, or topic — improving retrieval precision and enabling domain-specific access control.**

Correct Answer: D

Explanation

A single global knowledge base for an entire enterprise mixes HR policies, technical docs, legal agreements, and marketing collateral. Retrieving "salary" in this context might return both compensation policy and a sales script. Segmentation into domain-specific indexes improves precision, enables department-specific access control, and allows different chunking/embedding strategies per domain.

Q68. What is "RAG observability" and what does it involve?

- A. A technique for making the RAG system's internal workings visible to end users.
- B. Monitoring and logging the full RAG pipeline — queries, retrieved chunks, generation inputs and outputs — to diagnose failures, measure quality, and identify improvement opportunities.**
- C. Real-time monitoring of vector database performance metrics like latency and throughput.
- D. An audit trail of all user queries for compliance and legal discovery purposes.

Correct Answer: B

Explanation

RAG systems fail in many ways: retrieval misses relevant content, retrieved content is misread, grounding fails, answers hallucinate. Without observability — logging queries, retrieved chunks, relevance scores, and generated responses — diagnosing these failures is impossible. Full pipeline logging enables root cause analysis, quality metric tracking, and targeted improvement.

Q69. What is "hybrid knowledge architecture" combining structured and unstructured data?

- A. A database architecture that uses different storage formats for different document types.
- B. A retrieval system that uses different embedding models for different content types.
- C. An AI system that retrieves from both vector databases (for semantic search of unstructured text) and relational databases (for precise structured data queries) and synthesises both in the response.**
- D. A knowledge base that mixes publicly available and proprietary documents.

Correct Answer: C

Explanation

Real enterprise queries often require both: "What does our policy say about contractor expenses AND what was our actual contractor spend last quarter?" The first is unstructured (vector search); the second is structured (SQL). Hybrid architectures retrieve from both, synthesising a unified response. This is significantly more complex than pure RAG but reflects real enterprise information needs.

Q70. What is "personalised retrieval" in an enterprise context?

- A. Adapting retrieved content based on the user's role, preferences, history, or current project context — surfacing the most relevant content for that specific user rather than generic results.**
- B. A technique for retrieving information that matches the user's writing style.
- C. A retrieval strategy that prioritises documents the user has previously accessed.
- D. Adjusting retrieval scores based on the user's department to apply access control.

Correct Answer: A

Explanation

Generic retrieval returns the same results for the same query regardless of who is asking. A sales manager asking about "product performance" needs different content than a product engineer asking the same question. Personalised retrieval uses the user's role, recent activity, and current projects to adjust retrieval — returning contextually relevant results rather than globally relevant ones.

Q71. What is "knowledge base quality scoring" and why is it important before RAG deployment?

- A. A technique for benchmarking the knowledge base's retrieval performance against industry standards.
- B. Assessing the coverage, accuracy, currency, and format quality of documents in the knowledge base before deployment — poor quality input produces poor quality RAG output.**
- C. Automatically scoring document importance to prioritise which content to index first.
- D. A compliance assessment of knowledge base content for regulatory requirements.

Correct Answer: B

Explanation

Garbage in, garbage out. A RAG system is only as good as its knowledge base. Pre-deployment quality scoring assesses: coverage (are the key topics represented?), accuracy (is the content correct?), currency (is it up to date?), and format quality (is it cleanly parsed and well-structured?). Deploying RAG over a poor knowledge base delivers poor results regardless of model quality.

Q72. What is "chunking for tables" and why does it require special handling?

- A. Tables should always be embedded as plain text with whitespace preserved to maintain alignment.
- B. Tables in documents should be removed before indexing because they confuse embedding models.
- C. Table cells should each be indexed as separate chunks to maximise retrieval granularity.
- D. Tables must be preserved as complete atomic units or converted to structured formats, because splitting a table mid-row destroys the row-column relationships needed to interpret the data.**

Correct Answer: D

Explanation

A table split across chunks loses the header-row relationship: the cell "42%" on its own is meaningless without the column header "Q3 Revenue Growth" and the row label "APAC Region." Tables should be extracted whole, either embedded as serialised text (Markdown tables, CSV-like text) or stored as structured data with metadata describing each cell's row and column context.

Q73. What is a "retrieval-augmented fine-tuning" (RAFT) approach?

- A. Fine-tuning an embedding model on domain-specific query-document pairs to improve retrieval quality.
- B. A technique for fine-tuning the entire RAG pipeline jointly — both retriever and generator — on task-specific data.
- C. Fine-tuning the language model to be better at using retrieved documents — specifically training it to identify relevant context, ignore distractors, and extract answers faithfully.**
- D. Using RAG to generate training data for fine-tuning a smaller, more efficient model.

Correct Answer: C

Explanation

RAFT (proposed by researchers at Berkeley) fine-tunes the language model specifically for RAG tasks: it trains the model to read a document set containing relevant and irrelevant "distractor" documents, identify which parts are relevant, and produce a faithful answer with citations. The result is a model that is better at the RAG task than a general-purpose model.

Q74. What is the "two-stage RAG" architecture and what problem does it solve?

- A. An architecture with two separate knowledge bases — one for retrieval and one for grounding validation.
- B. A retrieve-then-filter architecture: first retrieve a large candidate set (high recall), then apply a more expensive filter or re-ranker to select the final context (high precision) — solving the precision-recall trade-off.**
- C. A two-model architecture where one model retrieves and another model generates the response.
- D. A pipeline that runs two independent RAG calls and merges their responses.

Correct Answer: B

Explanation

Single-stage retrieval must balance recall (retrieve enough relevant content) and precision (avoid noise). Two-stage RAG separates these: stage 1 retrieves broadly (top-50 ANN results, high recall), stage 2 applies a cross-encoder re-ranker to select the best 5 (high precision). The additional latency from re-ranking is offset by significantly better final context quality.

Q75. What is "RAG evaluation at the pipeline level" and what does it measure?

- A. A performance test that measures the throughput and latency of the production RAG pipeline.
- B. An audit of the RAG system's compliance with data governance policies.
- C. Evaluating the quality of individual components (chunker, embedder, retriever) separately before integration.
- D. Measuring the end-to-end performance of the complete RAG system — retrieval quality, grounding faithfulness, and answer accuracy — rather than evaluating each component in isolation.**

Correct Answer: D

Explanation

Component-level evaluation (retrieval recall, embedding quality, LLM faithfulness) can all look good while the end-to-end system performs poorly due to interactions between components. Pipeline-level evaluation measures what users actually experience: given a set of representative questions, does the system return correct, grounded, complete answers? This is the ultimate quality signal.

SET

41

Hallucination Detection

Set 41 of 5 · 15 Questions · AI Engineering

Q1. What is the most reliable approach for detecting hallucinations in a RAG system's output?

- A. Measuring the model's token probability scores and flagging outputs with low confidence.
- B. Comparing each claim in the response against the retrieved source documents using an LLM-as-judge evaluation.**
- C. Running the same prompt multiple times and flagging responses that differ across runs.
- D. Checking whether any sentences in the response appear verbatim in the training corpus.

Correct Answer: B**Explanation**

In a RAG system, source documents are available and verifiable. An LLM judge can check: "Is each claim in the response supported by the provided documents?" This provides grounded, systematic hallucination detection. Probability scores are unreliable; variance across runs measures inconsistency, not accuracy; verbatim matching misses correct paraphrases.

Q2. What is a "factual consistency" check and how is it different from a "factual accuracy" check?

- A. Factual consistency checks grammar and style; factual accuracy checks content.
- B. Factual consistency requires human evaluation; factual accuracy can be automated.
- C. Factual consistency applies to structured data; factual accuracy applies to prose responses.
- D. Factual consistency checks whether the response aligns with the provided context; factual accuracy checks whether the response reflects real-world truth.**

Correct Answer: D**Explanation**

Factual consistency (faithfulness) is an internal check: does the response contradict its own retrieved sources? Factual accuracy is an external check: is the response true relative to the real world? A response can be internally consistent (faithful to sources) but factually inaccurate if the sources themselves are wrong. Both matter, but for different reasons.

Q3. What is "SelfCheckGPT" as a hallucination detection technique?

- A. Generating multiple samples from the same prompt and checking whether consistent facts appear across samples — consistent facts are likely true; inconsistent ones are likely hallucinated.**
- B. Asking the model to check its own output for errors before returning the response.
- C. A benchmark dataset for evaluating language models on factual accuracy tasks.
- D. A prompt engineering technique that instructs the model to flag uncertain statements.

Correct Answer: A

Explanation

SelfCheckGPT leverages a key insight: a model is more likely to be consistent about things it "knows" and inconsistent about things it is fabricating. By sampling several outputs from the same prompt and identifying claims that vary across samples, SelfCheckGPT detects probable hallucinations without a reference — useful when no ground truth is available.

Q4. What is "hallucination taxonomy" and why does it matter for mitigation?

- A. A list of topics on which models are known to hallucinate, used to pre-filter queries.
- B. A structured dataset of known hallucination examples used to evaluate detection systems.
- C. Classifying hallucinations by type (factual fabrication, attribution error, intrinsic contradiction, extrinsic inconsistency) to identify appropriate mitigations for each.**
- D. A metric that measures the severity of hallucinations on a 1–5 scale.

Correct Answer: C

Explanation

Not all hallucinations are equal or caused by the same mechanism. Factual fabrication (invented facts) is mitigated by RAG and grounding. Attribution errors (citing wrong sources) are mitigated by citation verification. Intrinsic contradictions (self-contradicting responses) are mitigated by consistency checking. Taxonomy-driven mitigation is more effective than treating hallucination as a single undifferentiated problem.

Q5. What is "entity hallucination" and why is it particularly harmful?

- A. A type of hallucination specific to named entity recognition tasks.
- B. The model generating incorrect proper nouns — wrong names, companies, dates, statistics, or locations — which appear authoritative and are hard to spot without domain knowledge.**
- C. The model misidentifying the entity type (person vs. organisation) in its response.
- D. Hallucinations that only occur when the query involves specific entity categories.

Correct Answer: B

Explanation

Entity hallucinations are especially dangerous because specific claims ("the merger was announced by CEO John Smith on 14 March 2024 for \$2.3 billion") are what people use to make decisions. They sound authoritative, look researched, and require domain expertise to detect. Wrong statistics, fabricated names, and incorrect dates in AI outputs have caused real-world harm in legal, medical, and financial contexts.

Q6. What is the "calibration-hallucination connection" in LLMs?

- A. Poorly calibrated models — those whose confidence scores do not reflect actual accuracy — are more likely to produce hallucinations presented with inappropriate certainty.**
- B. Hallucination rates can be reduced by calibrating the temperature parameter during inference.
- C. Calibrated models always produce accurate outputs; uncalibrated models always hallucinate.
- D. The calibration curve can be used to predict the exact hallucination rate for a given model.

Correct Answer: A

Explanation

Well-calibrated models "know what they don't know" — their confidence tracks their accuracy. Poorly calibrated models state uncertain or incorrect claims with high confidence. Calibration and hallucination are related: a model that hedges ("I believe" / "I'm not certain") when uncertain signals its reliability better than one that states all claims with equal confidence. Calibration training is a mitigation for overconfident hallucination.

Q7. What is "hallucination red-lining" in deployment?

- A. A technique for highlighting hallucinated text in model outputs for user review.
- B. A safety rating that restricts model use to low-hallucination-risk domains.
- C. A benchmark process for measuring hallucination rate before production deployment.
- D. Defining topics or query types where hallucination risk is unacceptably high and implementing hard constraints — abstention, human review, or explicit source citation requirements — for those categories.**

Correct Answer: D

Explanation

Not all hallucination risk is equal. In a general knowledge chatbot, a wrong movie quote is annoying. In a medical information system, a wrong dosage is dangerous. Red-lining identifies high-risk categories — legal advice, medical information, financial data — and applies stricter handling: mandatory grounding, abstention when uncertain, or mandatory human review before response delivery.

Q8. What is "citation grounding" as a hallucination mitigation strategy?

- A. Embedding citations into the model's training data so it learns to attribute claims accurately.
- B. A post-processing step that automatically adds Wikipedia citations to factual claims.
- C. Requiring the model to cite a specific source document and passage for every factual claim, making unverifiable claims impossible to present as grounded.**
- D. A retrieval technique that pre-selects only highly cited documents for inclusion in context.

Correct Answer: C

Explanation

If the model must cite a specific passage for every claim, it cannot assert unsupported facts as confidently — the absence of a citation is itself a signal. This also makes verification practical: reviewers can check whether the citation supports the claim. Citation grounding does not eliminate hallucination but makes it detectable and shifts the burden of proof to the model.

Q9. What is the "Needle in a Haystack" test for LLMs?

- A. A test that places a specific fact in a large context and measures whether the model can accurately retrieve and use it — evaluating context utilisation across the full context window.**
- B. A benchmark for measuring how quickly the model can find relevant information in a knowledge base.
- C. A test that checks whether the model hallucinates when the correct answer is rare in its training data.
- D. A technique for identifying the specific training examples most responsible for a hallucination.

Correct Answer: A

Explanation

The Needle in a Haystack test inserts a single "needle" fact into a long "haystack" document and asks the model a question that can only be answered using that fact. By varying the needle's position within the context, it maps where the model fails to use context — revealing the "lost in the middle" effect and the practical limits of long-context models.

Q10. What is "intrinsic hallucination" versus "extrinsic hallucination"?

- A. Intrinsic hallucinations are generated by the model's internal reasoning; extrinsic ones come from retrieved documents.
- B. Intrinsic hallucinations occur in mathematical tasks; extrinsic ones occur in language tasks.
- C. Intrinsic hallucinations are detectable by automated tools; extrinsic ones require human evaluation.
- D. Intrinsic hallucinations contradict the provided input context; extrinsic hallucinations add information not present in the context — both are failures but require different detection approaches.**

Correct Answer: D

Explanation

Intrinsic hallucinations directly contradict the provided source — the model states the opposite of what the document says. These are detectable by automated consistency checking. Extrinsic hallucinations add information that is not in the source but may or may not be accurate — detecting these requires external knowledge or human review. RAG grounding evaluation primarily catches intrinsic hallucinations.

Q11. What makes number-based claims particularly high-risk for hallucination?

- A. Numerical processing is handled by a different sub-module that is more prone to errors.
- B. Numbers (statistics, dates, quantities, percentages) are precise, consequential, and the model has no reliable mechanism for generating correct figures it did not memorise from training.**
- C. Models are not trained on numerical data, so all numerical outputs are generated randomly.
- D. Numbers require calculation, which LLMs cannot perform accurately without tool use.

Correct Answer: B

Explanation

Language models generate numbers the same way they generate words — by predicting the next token. They have no internal calculator. A model may confidently generate "revenue grew 23.7%" when the actual figure was 18.2% — not because it was confused, but because 23.7 was a plausible continuation of the surrounding text. Statistical claims, specific dates, and monetary figures should always be verified against authoritative sources.

Q12. What is "RAGAs" (Retrieval-Augmented Generation Assessment) and what does it measure?

- A. A dataset of question-answer pairs for benchmarking RAG systems against human baselines.
- B. An API endpoint that scores RAG outputs in real time during production.
- C. A framework for evaluating RAG systems on faithfulness, answer relevance, context relevance, and context recall — providing a structured suite of automated evaluation metrics.**
- D. A set of prompting templates for constructing grounded, hallucination-resistant RAG responses.

Correct Answer: C

Explanation

RAGAs (the open-source framework) provides four core metrics: faithfulness (is the answer supported by retrieved context?), answer relevance (does the answer address the question?), context relevance (are the retrieved chunks relevant to the query?), and context recall (does the retrieved context contain the information needed to answer?). Together they diagnose where in the RAG pipeline quality is failing.

Q13. What is "chain-of-thought faithfulness" and why is it an evaluation concern?

- A. Whether the reasoning chain contains factually accurate intermediate steps.
- B. Whether the reasoning chain is internally consistent without contradictions.
- C. Whether the reasoning chain is sufficiently detailed for the user to follow.
- D. Whether the model's stated reasoning actually reflects the process it used to arrive at its answer — models can produce plausible-sounding but post-hoc rationalisations.**

Correct Answer: D

Explanation

Research has shown that model reasoning traces can be unfaithful — the stated reasoning is not necessarily what caused the final answer. A model may produce a correct answer through one process but describe a different, plausible-sounding process. Evaluating whether reasoning traces are genuinely causal (not post-hoc) is an open research problem with significant implications for transparency.

Q14. What is the "TruthfulQA" benchmark and what does it measure?

- A. A benchmark for evaluating LLM performance on factual question-answering from Wikipedia.
- B. A benchmark specifically designed to measure how often LLMs generate false information on questions where common misconceptions or plausible-sounding wrong answers exist.**
- C. A dataset of questions that models commonly hallucinate, used to fine-tune against hallucination.
- D. A benchmark that measures whether models accurately report their own uncertainty.

Correct Answer: B

Explanation

TruthfulQA tests models on questions where humans commonly hold false beliefs — questions designed so that the most fluent, confident-sounding answer is wrong. It measures whether a model is more like a knowledgeable expert (truthful) or a confident confabulator (fluent but wrong). It reveals hallucination risk on questions where the "obvious" answer is a misconception.

Q15. What is "grounded generation evaluation" and how does it differ from benchmark evaluation?

- A. Grounded generation evaluation measures how accurately a model uses provided context for specific documents; benchmark evaluation measures general capability on standardised test sets.**
- B. Grounded generation evaluation is automated; benchmark evaluation requires human raters.
- C. Grounded generation evaluation measures speed; benchmark evaluation measures accuracy.
- D. Benchmark evaluation applies to production systems; grounded generation evaluation applies only to research models.

Correct Answer: A

Explanation

Benchmarks (MMLU, HellaSwag, TruthfulQA) measure general model capability and hallucination tendency in controlled settings. Grounded generation evaluation measures a specific deployment's faithfulness to its provided documents — a production-relevant metric that general benchmarks do not capture. Both are needed: benchmarks for model selection, grounded evaluation for production monitoring.

SET

42

Evaluation Frameworks

Set 42 of 5 · 15 Questions · AI Engineering

Q16. What is the difference between "offline evaluation" and "online evaluation" for AI systems?

- A. Offline evaluation is manual; online evaluation is automated.
- B. Offline evaluation applies to model training; online evaluation applies to model inference.
- C. Offline evaluation uses pre-collected test sets before deployment; online evaluation measures real user interactions and outcomes in production.**
- D. Offline evaluation measures accuracy; online evaluation measures latency.

Correct Answer: C**Explanation**

Offline evaluation happens before deployment: you run your system against a curated test set and measure quality metrics. Online evaluation happens in production: you track user satisfaction, task completion rates, escalation frequency, or explicit feedback. Both are necessary — offline catches known issues before launch; online catches emergent problems in real usage.

Q17. What is "BLEU score" and what are its limitations for evaluating LLM outputs?

- A. BLEU measures n-gram overlap between generated text and reference text — adequate for machine translation but poor for evaluating open-ended generation where many valid responses exist.**
- B. BLEU is a benchmarking framework for comparing LLMs across standardised tasks.
- C. BLEU measures semantic similarity between a generated response and a reference answer.
- D. BLEU evaluates the factual accuracy of generated text against a knowledge base.

Correct Answer: A**Explanation**

BLEU was designed for machine translation, where a reference translation provides a reasonable gold standard. For open-ended LLM outputs, BLEU fails: a correct, high-quality response phrased differently from the reference scores low; a poor response with accidentally matching phrases scores high. Semantic similarity metrics (BERTScore, ROUGE) and LLM-as-judge have largely replaced BLEU for modern LLM evaluation.

Q18. What is "LLM-as-judge" evaluation and what are its failure modes?

- A. A technique where the evaluated model judges its own outputs to self-improve.
- B. A method where multiple LLMs vote on the best response, and the majority opinion is taken as ground truth.
- C. An evaluation approach that uses an LLM to generate the evaluation rubric from task description.
- D. Using a capable LLM to score or rank other LLM outputs against defined criteria — failure modes include position bias, verbosity bias, self-enhancement bias, and sensitivity to evaluation prompt wording.**

Correct Answer: D

Explanation

LLM-as-judge scales human-quality evaluation to large datasets. But LLM judges have systematic biases: they prefer longer responses (verbosity bias), prefer options presented first (position bias), and rate outputs from the same model family higher (self-enhancement bias). Mitigation: randomise position, use pairwise comparisons, calibrate against human judgements, and use a diverse set of judge models.

Q19. What is "MMLU" and what does it assess?

- A. A benchmark specifically designed to test model performance on multilingual tasks.
- B. Massive Multitask Language Understanding — a benchmark covering 57 academic subjects used to assess broad general knowledge and reasoning across disciplines.**
- C. A framework for measuring LLM output quality across multiple evaluation dimensions simultaneously.
- D. A benchmark for evaluating multi-modal models on image and text understanding jointly.

Correct Answer: B

Explanation

MMLU consists of multiple-choice questions across 57 subjects from elementary school to professional level — maths, history, law, medicine, computer science, and more. It measures breadth of world knowledge and reasoning. It is widely used for model comparison but criticised for data contamination (test questions in training data) and for not measuring open-ended capability.

Q20. What is "human evaluation" and when is it irreplaceable in LLM assessment?

- A. Human evaluation is the gold standard for all LLM tasks and should always be preferred over automated metrics.
- B. Human evaluation is used only for compliance checking — automated metrics are preferred for quality assessment.
- C. Having humans directly assess model outputs — irreplaceable for subjective qualities like tone, appropriateness, nuance, and real-world usefulness that automated metrics cannot capture.**
- D. Human evaluation is replaced by LLM-as-judge in all modern evaluation frameworks.

Correct Answer: C

Explanation

Automated metrics measure proxies for quality; humans measure quality directly. For tasks where "good" is inherently subjective — tone appropriateness, cultural sensitivity, helpfulness in context, creative quality — human evaluation is irreplaceable. The cost and latency of human evaluation mean it is used for calibration and final validation, with automated metrics for continuous monitoring.

Q21. What is "task-specific evaluation" and why is it preferred over general benchmarks for production systems?

- A. Evaluating the model on every possible task to ensure generalisation before deployment.
- B. A technique that automatically selects relevant benchmarks based on the system's configuration.
- C. Evaluating each component of an AI pipeline separately rather than end-to-end.
- D. Evaluating a deployed model on representative samples from its actual use case — more predictive of production performance than general benchmarks which test capabilities the system may never use.**

Correct Answer: D

Explanation

A general coding assistant benchmark score tells you little about how a model performs on your specific legal document summarisation task. Task-specific evaluation builds a test set from real or representative production queries, with expected outputs reviewed by domain experts. This is the only evaluation that directly predicts user experience in the actual deployment context.

Q22. What is "win rate" as an evaluation metric for comparing AI systems?

- A. The proportion of pairwise comparisons where one system's output is preferred over another's — used when there is no single correct answer and quality is comparative.**
- B. The percentage of tasks a model completes successfully without error.
- C. A metric that measures how often the AI system's recommendation is accepted by users.
- D. The rate at which one model's outputs match the outputs of a reference model.

Correct Answer: A

Explanation

Win rate is used in comparative evaluation: show evaluators (human or LLM) two outputs side by side and ask which is better. System A "wins" when its output is preferred. This is particularly useful for open-ended generation where multiple valid outputs exist and absolute quality measurement is difficult. A/B testing and RLHF preference collection both rely on win rate.

Q23. What is "benchmark contamination" and why does it undermine model comparison?

- A. A testing artefact where models perform differently on the same benchmark across different runs.
- B. When benchmark questions or answers appear in the model's training data, inflating its measured performance above its true generalisation ability.**
- C. The inclusion of culturally biased questions that advantage models trained on Western data.
- D. A technical issue where benchmark evaluation tools produce incorrect scores due to tokenisation differences.

Correct Answer: B

Explanation

If a model was trained on data containing the questions (and answers) from a benchmark, its score reflects memorisation, not capability. As benchmarks become widely known, data cleaning to remove them from training becomes harder. This is why novel, private evaluation sets are more reliable than public benchmarks for assessing truly new model capabilities.

Q24. What is a "golden test set" in AI evaluation?

- A. The training data used by the model provider, considered the gold standard for model capabilities.
- B. An automatically generated set of test cases derived from the system's most common queries.
- C. A set of adversarial inputs specifically designed to maximise model failures.
- D. A curated set of representative inputs with expert-verified expected outputs used as the stable reference for evaluating and comparing system versions over time.**

Correct Answer: D

Explanation

A golden test set is the evaluation anchor: built once from representative real inputs, verified by domain experts, and kept stable. When the system changes — prompt update, model version, retrieval change — you run the golden set and compare. It provides a consistent reference for detecting regressions and validating improvements across the system's lifetime.

Q25. What is "inter-annotator agreement" in AI evaluation and why is it important?

- A. Agreement between the evaluated model and a reference model on task outputs.
- B. A measure of how consistently the same annotator scores similar outputs over time.
- C. The degree to which different human evaluators agree on quality scores — low agreement means the evaluation criteria are ambiguous, making scores unreliable benchmarks.**
- D. Agreement between automated metrics and human evaluators on the same dataset.

Correct Answer: C

Explanation

If two evaluators rate the same output as 5/5 and 2/5, the evaluation is measuring annotator preference, not output quality. High inter-annotator agreement requires: clear, specific rubrics with examples, annotator training and calibration sessions, and periodic re-calibration. Cohen's kappa or Krippendorff's alpha are standard measures. Low agreement is a signal to refine the rubric, not to average the scores.

Q26. What is "evals" in the context of LLM application development?

- A. Manual evaluation sessions where domain experts review model outputs before each release.
- B. Automated test suites that measure whether an AI system meets defined quality criteria — treated as code, versioned, and run continuously as part of the development process.**
- C. A provider-specific term for the accuracy metrics reported in model documentation.
- D. The process of selecting which benchmarks to use for model capability assessment.

Correct Answer: B

Explanation

Evals are the AI equivalent of unit tests: structured, automated, versioned checks that run every time the system changes. They operationalise quality: "does the response always contain a citation?", "is the JSON always valid?", "does the tone always match the persona?". Building an eval suite before shipping and maintaining it throughout the system's life is a core AI engineering discipline.

Q27. What is "precision at k" (P@k) and when is it used in AI evaluation?

- A. The proportion of the top-k results that are relevant — used in retrieval evaluation to measure how many retrieved items are actually useful.**
- B. A metric that measures whether the model's k-th generation attempt is correct.
- C. The accuracy of the model on the first k examples in a test set.
- D. A measure of how many of the top-k benchmark tasks a model completes correctly.

Correct Answer: A

Explanation

P@k measures retrieval precision at a specific cutoff: if you retrieve 10 chunks and 7 are relevant, $P@10 = 0.7$. It is the standard metric for retrieval evaluation — useful because users and downstream systems typically only use the top few results. P@k is more practically meaningful than average precision across all possible k values.

Q28. What is "normalised discounted cumulative gain" (NDCG) and when is it used?

- A. A metric that measures how much a model's performance degrades on normalised (cleaned) test data.
- B. A gain metric that measures information extracted per token in model outputs.
- C. A rank-weighted retrieval metric that gives higher credit to relevant results appearing earlier in the ranked list — used when the position of relevant results matters.**
- D. A normalised version of BLEU that accounts for different response lengths.

Correct Answer: C

Explanation

NDCG rewards relevant results more when they appear higher in the ranked list — a relevant result at position 1 contributes more than one at position 10. It also handles graded relevance (highly relevant, somewhat relevant, not relevant) rather than binary. NDCG is the standard metric for information retrieval systems and re-ranker evaluation in RAG pipelines.

Q29. What is "automated evaluation at scale" and what trade-offs does it involve?

- A. Running the same evaluation simultaneously across many model instances to parallelise testing.
- B. Automatically generating evaluation test cases from production traffic to scale the test set.
- C. A technique for evaluating models without labelled data by measuring distributional properties.
- D. Using automated metrics (LLM-as-judge, rule-based checks, embedding similarity) to evaluate large volumes of outputs — trading measurement accuracy for coverage and cost.**

Correct Answer: D

Explanation

Manual human evaluation of 10,000 outputs is impractical. Automated evaluation sacrifices some accuracy (LLM judges have biases; metrics are proxies) in exchange for scale, speed, and cost. The right approach uses: automated metrics for continuous monitoring of large volumes, with human evaluation for calibration, validation, and cases where automated metrics disagree.

Q30. What is "evaluation for safety" as distinct from "evaluation for quality"?

- A. Safety evaluation is done by the model provider; quality evaluation is done by the application developer.
- B. Safety evaluation tests whether the system produces harmful, biased, or inappropriate outputs; quality evaluation tests whether outputs are accurate, relevant, and useful — these require different test sets and rubrics.**
- C. Safety evaluation is a one-time assessment before launch; quality evaluation is continuous.
- D. Safety evaluation applies to the model; quality evaluation applies to the application.

Correct Answer: B**Explanation**

Safety and quality are orthogonal dimensions that both require dedicated evaluation. A system can be high-quality (accurate, relevant, well-formatted) but unsafe (produces harmful content under certain inputs). Safety evaluation requires adversarial test sets specifically designed to probe harmful behaviours — jailbreaks, bias elicitation, misinformation generation — that normal quality evaluation does not cover.

SET

43

Guardrails and Fallbacks

Set 43 of 5 · 15 Questions · AI Engineering

Q31. What is a "guardrail" in an AI system and what problem does it solve?

- A. A constraint or check — either pre-generation, inline, or post-generation — that prevents the AI from producing harmful, incorrect, or out-of-scope outputs.
- B. A hardware limit that prevents the AI system from using more compute than allocated.
- C. A legal disclaimer added to AI outputs to limit organisational liability.
- D. A rate limiting mechanism that prevents users from querying the AI too frequently.

Correct Answer: A**Explanation**

Guardrails are the safety and reliability infrastructure around an AI system. They can operate at multiple points: pre-generation (classifying input intent before sending to the model), inline (system prompt instructions that constrain model behaviour), or post-generation (checking outputs before delivery to users). Guardrails address what the core model alone cannot reliably guarantee.

Q32. What is an "input guardrail" and what types of inputs does it typically handle?

- A. A validation that checks whether the user's input is correctly formatted before API submission.
- B. A rate limiter that prevents individual users from sending too many queries per minute.
- C. A check applied to user input before it reaches the model — handling prompt injection attempts, out-of-scope queries, PII detection, and inappropriate content.
- D. An authentication check that verifies the user's identity before processing their input.

Correct Answer: C**Explanation**

Input guardrails intercept problems before the model processes them: detecting and blocking prompt injection attempts, classifying whether the query is in-scope for the application, detecting and redacting PII that should not be sent to the model, filtering explicit or harmful content. Handling these at the input layer is more reliable than relying on the model to handle them in-context.

Q33. What is an "output guardrail" and how does it differ from input filtering?

- A. A post-processing step that formats the model's raw output into the application's display format.
- B. A check applied to model output before it reaches the user — detecting harmful content, PII leakage, format violations, or grounding failures in the generated response.**
- C. A citation requirement that ensures every factual claim in the output has a source.
- D. A response length check that truncates outputs exceeding a defined word limit.

Correct Answer: B

Explanation

Output guardrails run after generation but before user delivery. They catch what slipped through input filtering and model instructions: a response that contains PII from retrieved documents, a response that violates format requirements, a response that includes inappropriate content despite system prompt instructions. Output guardrails provide a backstop when the model fails to follow instructions reliably.

Q34. What is a "fallback system" in AI application design?

- A. A backup model that activates automatically when the primary model's API is unavailable.
- B. A secondary retrieval system that provides additional context when initial retrieval returns insufficient results.
- C. A logging mechanism that records failed AI interactions for later review and improvement.
- D. A predefined response or alternative action triggered when the primary AI system fails, produces low-confidence output, or encounters an out-of-scope query.**

Correct Answer: D

Explanation

Fallback systems ensure that when the AI cannot reliably answer — low retrieval confidence, flagged by a guardrail, hallucination risk detected, API error — the application degrades gracefully: a safe pre-written response, a human escalation prompt, a disclaimer, or an apology with contact information. Without fallbacks, AI failures are user-facing.

Q35. What is "confidence-based routing" and how does it improve reliability?

- A. Routing queries to different handling paths based on the AI system's assessed confidence — high-confidence queries get automated responses; low-confidence ones get human review or escalation.**
- B. Routing different query types to different models based on their measured accuracy for that task.
- C. A load-balancing technique that directs traffic to the highest-confidence model instance.
- D. A technique for choosing between multiple retrieved chunks based on similarity scores.

Correct Answer: A

Explanation

Not all queries are equally certain. A query with high retrieval scores and a strong match to known patterns can be answered confidently. A query with poor retrieval, an out-of-distribution intent, or detected ambiguity should be handled differently — with explicit uncertainty statements, human escalation, or clarification requests. Confidence-based routing operationalises this distinction.

Q36. What is a "system prompt guardrail" and how reliable is it as the sole safety mechanism?

- A. A system-level access control that prevents certain users from modifying the system prompt.
- B. Instructions in the system prompt that tell the model to refuse or handle certain request types — effective as a first layer but insufficient alone, as models can be jailbroken or misled.**
- C. A validation that checks the system prompt for prohibited keywords before deployment.
- D. An automated test that verifies the system prompt produces safe outputs before launch.

Correct Answer: B

Explanation

System prompt guardrails ("Never provide medical diagnoses", "Do not discuss competitor products") are the simplest safety mechanism — and the least reliable as a standalone control. Determined adversarial users can craft inputs that bypass instruction-following. System prompt guardrails should be one layer in a defence-in-depth approach that also includes input/output classifiers, content filters, and monitoring.

Q37. What is a "constitutional AI" approach to guardrails?

- A. A legal framework for AI governance that defines acceptable use policies for AI systems.
- B. A system prompt template structured as a numbered list of rules the model must follow.
- C. A human review process that checks AI outputs against a defined set of ethical guidelines.
- D. Training a model with a set of principles ("constitution") that it uses to self-critique and revise its outputs — building safety constraints into the model's weights rather than relying only on prompt instructions.**

Correct Answer: D

Explanation

Constitutional AI (Anthropic's approach) trains the model to evaluate its own outputs against a set of principles and revise them — making safety part of the model's reasoning process, not just an external constraint. This produces more robust safety behaviour than system prompt instructions alone, because the model internalises the principles during training rather than following them as runtime instructions.

Q38. What is "PII detection" as a guardrail and why is it critical in enterprise AI?

- A. A technique for verifying that users querying the AI system are who they claim to be.
- B. A monitoring system that alerts administrators when users share personal information with the AI.
- C. Identifying and redacting personally identifiable information in user inputs or model outputs before it is processed or delivered — critical for privacy compliance (GDPR, DPDP Act) and data governance.**
- D. A filter that removes employee names from retrieved documents before they enter the AI context.

Correct Answer: C

Explanation

Enterprise AI systems often process sensitive data. A customer support AI might receive messages containing credit card numbers, Aadhaar numbers, or medical conditions — none of which should be sent to external model APIs or stored in logs. PII detection guardrails intercept this before it reaches the model, replace it with placeholders, and ensure compliance with privacy regulations.

Q39. What is a "topic boundary guardrail" and when is it needed?

- A. A guardrail that prevents the model from generating content on sensitive topics defined in the system prompt.
- B. A constraint that prevents the model from responding to queries outside the defined scope of the application — redirecting out-of-scope queries rather than attempting to answer them.**
- C. A keyword filter that blocks specific terms from appearing in model outputs.
- D. A retrieval filter that excludes documents outside the application's defined knowledge domain.

Correct Answer: B

Explanation

A legal research AI should not provide personal financial planning advice; a customer support bot should not discuss competitors; a children's education tool should not engage with adult content. Topic boundary guardrails classify incoming queries and redirect out-of-scope ones before the model processes them — preventing the model from attempting to answer queries it should not handle.

Q40. What is "graceful degradation" in AI system design?

- A. Ensuring that when components fail or confidence is low, the system fails safely — providing partial help, clear uncertainty statements, or human escalation rather than silent failure or unreliable output.**
- B. A technique for reducing model capability progressively under high load to maintain availability.
- C. Designing AI outputs to degrade in visual formatting gracefully across different display contexts.
- D. A failover architecture where a simpler model activates when the primary model's API is unavailable.

Correct Answer: A

Explanation

Graceful degradation applies the principle of safe failure to AI: when the system cannot answer reliably, it should fail in a way that is helpful rather than harmful. "I'm not confident in this answer — you may want to verify with a human expert" is graceful degradation. Producing a confident hallucination is ungraceful failure. Design for the uncertain case, not just the nominal case.

Q41. What is "rate limiting" as an AI system guardrail?

- A. A technique for limiting the length of model outputs to reduce latency and cost.
- B. A guardrail that prevents the model from generating more than a defined number of claims per response.
- C. Restricting the number of queries a user or application can send per time period — preventing abuse, managing cost, and protecting against denial-of-service attacks.**
- D. A constraint that limits how many retrieved chunks are included in each LLM context.

Correct Answer: C

Explanation

Rate limiting is an operational guardrail: it prevents individual users from consuming disproportionate resources, protects against automated abuse or scraping, and controls cost at the application level. For AI applications with per-token pricing, unlimited query rates can create unbounded cost exposure. Rate limiting is a basic but essential operational control.

Q42. What is "output validation" as distinct from "output guardrails"?

- A. Output validation is automated; output guardrails require human review.
- B. Output validation applies to generation; guardrails apply to retrieval.
- C. Output validation and guardrails are synonymous — different terms for the same process.
- D. Output validation checks whether the response meets structural requirements (valid JSON, required fields present, length within bounds); guardrails check whether the content is safe and appropriate.**

Correct Answer: D

Explanation

Both are post-generation checks but they measure different things. Validation is structural: "Is the JSON parseable?", "Are all required fields present?", "Is the response within the length limit?" Guardrails are content-based: "Does the response contain PII?", "Is it in-scope?", "Does it match safety guidelines?" Applications need both: structural validity and content safety.

Q43. What is "jailbreaking" and why does it matter for guardrail design?

- A. A technique for extracting the system prompt from an AI application to reveal its configuration.
- B. Crafting adversarial prompts that cause an AI system to bypass its safety instructions and produce content it was configured not to produce.**
- C. An attack that exploits vulnerabilities in the API infrastructure rather than the model itself.
- D. A method for accessing premium model capabilities without proper authorisation.

Correct Answer: B

Explanation

Jailbreaks demonstrate that instruction-based guardrails are not robust: sufficiently creative users can find phrasings that cause the model to ignore its constraints. This is why guardrail design must be defence-in-depth — system prompt instructions are the first layer, but input classifiers, content filters, output checks, and monitoring must back them up. Security through instructions alone is insufficient.

Q44. What is "monitoring-as-a-guardrail" and how does it complement preventive controls?

- A. A passive safety approach that relies entirely on monitoring rather than implementing active guardrails.
- B. A compliance mechanism that records all AI interactions for regulatory audit purposes.
- C. Using continuous logging, anomaly detection, and alerting to detect guardrail bypass and quality failures that preventive controls missed — enabling rapid response to emergent issues.**
- D. A performance monitoring system that tracks model latency and error rates in production.

Correct Answer: C

Explanation

Preventive guardrails stop known problems. Monitoring catches what slipped through — novel jailbreak patterns, unexpected use cases, quality degradation, PII leakage. Monitoring enables the closed loop: detect a new failure mode, implement a preventive guardrail, verify the monitoring shows improvement. Without monitoring, you have no visibility into what your guardrails are missing.

Q45. What is "human-in-the-loop" (HITL) as a reliability mechanism and when is it appropriate?

- A. Inserting human review at decision points where AI confidence is low or stakes are high — appropriate for high-risk decisions, low-confidence outputs, or novel situations outside the AI's training distribution.**
- B. A development practice where humans write all prompts and review all outputs before deployment.
- C. A training technique where human feedback is used to improve the model's quality iteratively.
- D. A safety requirement mandating human approval for every AI output before it is delivered.

Correct Answer: A

Explanation

HITL is a reliability and safety mechanism for high-stakes or uncertain situations: route low-confidence outputs to a human reviewer before delivery; require human approval for consequential decisions (loan approval, medical recommendations, legal advice); escalate novel queries that fall outside the system's known patterns. HITL adds latency and cost but provides the ultimate backstop where AI reliability is insufficient.

SET

44

Monitoring and Testing AI Outputs

Set 44 of 5 · 15 Questions · AI Engineering

Q46. What is "production monitoring" for an AI system and what metrics should be tracked?

- A. A pre-launch testing phase that simulates production traffic to identify performance bottlenecks.
- B. Manual review of a random sample of production outputs by quality assurance analysts.
- C. A compliance audit that checks whether the AI system's outputs meet regulatory requirements.
- D. Continuously measuring system health in live deployment — tracking output quality, user satisfaction, guardrail trigger rates, latency, error rates, and hallucination indicators.**

Correct Answer: D**Explanation**

Production monitoring is ongoing, not one-time. Key metrics: quality indicators (user satisfaction, task completion, thumbs-down rate), safety indicators (guardrail trigger rate, escalation rate), operational indicators (latency, error rate, token cost per session), and drift indicators (output length trend, topic distribution shift). Without monitoring, you are flying blind.

Q47. What is "A/B testing" for AI systems and what makes it valid?

- A. Testing two model versions simultaneously and selecting the one with lower error rate.
- B. Comparing two system variants with random assignment of real traffic — valid when there is a pre-defined success metric, sufficient sample size, and only the tested variable differs between conditions.**
- C. Running the same test twice — once in development and once in production — and comparing results.
- D. Presenting users with two AI responses and asking them to choose the better one.

Correct Answer: B**Explanation**

A/B testing of AI systems requires the same statistical rigour as any controlled experiment: random assignment (no selection bias), pre-defined success metric (decided before the test, not after seeing results), sufficient sample size to detect meaningful differences, and controlled conditions (only the variable being tested differs). Without these, observed differences are noise, not signal.

Q48. What is "shadow testing" for AI system deployment?

- A. A testing approach where the development team tests the system without telling the client.
- B. A technique for testing the AI system against adversarial inputs without the knowledge of the model provider.
- C. Running the new system in parallel with the production system on the same live traffic, comparing outputs without exposing users to the new system's responses.**
- D. Running a simplified version of the system in a lower-cost environment to simulate production.

Correct Answer: C

Explanation

Shadow testing is the safest deployment testing approach: the new system processes real production traffic and generates responses, but users see only the old system's responses. The new system's outputs are evaluated and compared without any user risk. Once quality is confirmed in shadow mode, the new system can be gradually promoted to handle real traffic.

Q49. What is "canary deployment" in the context of AI systems?

- A. Gradually rolling out a new system version to a small percentage of users, monitoring quality metrics, before full deployment — limiting exposure if the new version underperforms.**
- B. A technique for testing the AI system on a small subset of the training data before full evaluation.
- C. Deploying a simplified version of the system to test infrastructure performance under load.
- D. A monitoring approach that uses a small set of "canary" test queries run continuously against production.

Correct Answer: A

Explanation

Canary deployment controls rollout risk: send 5% of traffic to the new version, monitor quality and error metrics, and increase the percentage only if metrics are healthy. If the new version underperforms, roll it back before the majority of users are affected. This is the standard deployment pattern for high-traffic AI applications where a bad model update has user-facing consequences.

Q50. What is "regression testing" for AI systems and what triggers it?

- A. Testing the system on tasks it previously failed to verify that the failures have been fixed.
- B. A scheduled monthly evaluation of system quality against industry benchmarks.
- C. A test that measures whether the system's performance degrades as the knowledge base grows.
- D. Running a fixed test suite after any change (prompt update, model version change, retrieval change) to verify that previously passing cases still pass — triggered by any system modification.**

Correct Answer: D

Explanation

AI systems can regress silently: a prompt change that improves one behaviour degrades another; a model version update that improves general capability changes specific task behaviour. Regression testing runs the same curated test suite before and after changes, catching quality degradations before they reach production. Without regression testing, silent regressions accumulate undetected.

Q51. What is "latency monitoring" for AI applications and why does it matter for user experience?

- A. A technique for measuring how quickly the model generates tokens per second.
- B. Monitoring API rate limit errors caused by exceeding the provider's request quota.
- C. Tracking time-to-first-token, total generation time, and end-to-end response time — slow responses directly reduce user engagement and perceived quality.**
- D. Tracking how quickly new documents are indexed into the vector database.

Correct Answer: C

Explanation

Response latency is a quality dimension: users abandon slow interfaces, slow responses feel lower quality even when content is identical, and in conversational applications excessive wait times destroy the interaction flow. Monitoring p50, p95, and p99 latency (not just average) reveals tail latency issues that affect a minority of users but significantly damage their experience.

Q52. What is "semantic drift" monitoring and when is it triggered?

- A. Tracking whether the distribution of topics and intents in user queries shifts over time — triggered when queries increasingly fall outside the system's designed domain.**
- B. Monitoring whether the model's response style drifts from the defined persona over time.
- C. Detecting when the embedding model's representation of the same text changes after a model update.
- D. Tracking whether users increasingly phrase queries in ways that cause retrieval to fail.

Correct Answer: A

Explanation

User needs evolve. A customer support AI designed for billing queries may see queries about new features after a product launch. Semantic drift monitoring compares the current query distribution to the baseline — when distribution shifts significantly, the knowledge base, system prompt, and guardrails may need updating. Catching drift early prevents the system from being queried outside its reliable operating range.

Q53. What is "cost monitoring" for AI applications and what should it alert on?

- A. Monitoring the computational resources consumed by the model provider's infrastructure.
- B. Tracking token consumption, API costs, and cost per user interaction — alerting on unexpected spikes that may indicate misuse, runaway sessions, or system prompt changes with unintended cost implications.**
- C. Tracking whether the organisation is within its contracted API usage tier.
- D. A procurement process for reviewing and approving AI service spending.

Correct Answer: B

Explanation

AI applications have variable costs driven by token consumption. A system prompt change that adds 500 tokens per call, multiplied by 100,000 daily calls, is a significant unplanned cost increase. Cost spikes may also indicate misuse — users crafting long inputs to extract information. Cost monitoring with anomaly detection catches both accidental and adversarial cost surprises.

Q54. What is "error rate monitoring" for AI applications?

- A. Monitoring the number of factually incorrect responses produced per day.
- B. A metric for measuring how often the AI system fails to retrieve relevant documents.
- C. Tracking the frequency of API errors, timeout failures, parsing errors in structured outputs, and guardrail trigger events — essential for distinguishing systemic from transient failures.**
- D. Tracking the rate at which users report dissatisfaction with AI responses.

Correct Answer: C

Explanation

Error rate monitoring covers multiple failure types: API errors (model provider outages), timeout failures (response time exceeding SLA), parsing errors (model failing to return valid JSON when required), guardrail triggers (potentially harmful queries detected). Distinguishing these failure types enables targeted remediation — API errors require infrastructure action; parsing errors require prompt fixes.

Q55. What is "user feedback collection" in AI application monitoring?

- A. A compliance mechanism that requires users to confirm they understand AI responses may be inaccurate.
- B. A technique for using user conversations to create fine-tuning training data.
- C. A survey sent to users after each session to collect overall satisfaction scores.
- D. Capturing explicit signals (thumbs up/down, ratings, reports) and implicit signals (session abandonment, follow-up queries) from users to identify output quality issues at scale.**

Correct Answer: D

Explanation

User feedback is a direct quality signal. Explicit feedback (thumbs down, "report this response") directly identifies problematic outputs. Implicit feedback (user immediately rephrases the query — suggesting the first response missed; session abandonment after a response — suggesting the response was unhelpful) reveals quality issues at scale without requiring users to rate every response. Both signal types should feed into the improvement loop.

Q56. What is "chaos testing" for AI systems?

- A. Deliberately injecting failures — API timeouts, malformed retrieved chunks, context window overflows, guardrail triggers — to verify that the system handles failures gracefully and reliably.**
- B. A technique for testing AI safety by deliberately providing dangerous prompts to assess guardrail effectiveness.
- C. A random query generation approach for discovering unexpected model failure modes.
- D. A load testing methodology that simulates peak traffic to assess system stability.

Correct Answer: A

Explanation

Chaos testing verifies resilience: what happens when the vector database is slow? When the LLM API returns an error? When a retrieved document is corrupted? When the context window overflows? A system that handles these gracefully in testing will handle them gracefully in production. A system that crashes, returns silent errors, or produces confusing outputs under failure conditions is not production-ready.

Q57. What is "output consistency testing" and why is it important for structured outputs?

- A. Testing whether the model produces the same response to the same query across multiple runs.
- B. Testing whether the system produces structurally valid outputs (valid JSON, required fields, correct types) across diverse inputs — critical for downstream systems that parse model outputs.**
- C. Checking that all responses share the same tone and style as defined in the system prompt.
- D. Verifying that citation formatting in responses is consistent across different document types.

Correct Answer: B

Explanation

Structured output applications (APIs that return JSON, pipelines that parse model outputs) break when the model produces invalid structure — even one bad field or missing bracket causes a parse failure. Output consistency testing probes the system with diverse inputs to find cases where structure breaks: long inputs, unusual characters, edge case content, or queries that cause the model to add prose around the JSON.

Q58. What is "end-to-end testing" for a RAG pipeline?

- A. Testing each component of the pipeline independently before integration.
- B. A performance test that measures total pipeline latency from input to output.
- C. A compliance test that verifies the pipeline meets data processing regulations end-to-end.
- D. Testing the complete system from user query to final response — including retrieval, context assembly, generation, and output validation — using representative real-world queries.**

Correct Answer: D

Explanation

Component tests verify each piece works in isolation; end-to-end tests verify the whole system works together for real users. A RAG pipeline where retrieval, generation, and guardrails each pass component tests can still fail end-to-end — due to interactions between components, data format mismatches, or emergent failure modes that only appear under realistic conditions.

Q59. What is "load testing" for AI applications and what does it reveal?

- A. Simulating high-traffic conditions to identify performance degradation, rate limit failures, increased latency, and cost spikes that only appear under load — distinct from nominal performance.**
- B. Testing the model's performance on a large number of diverse test cases.
- C. A benchmark that measures how many tokens per second the model can process.
- D. A technique for testing the knowledge base's capacity as the number of indexed documents grows.

Correct Answer: A

Explanation

AI applications behave differently under load: concurrent requests may exhaust API rate limits, vector database latency may spike under concurrent queries, LLM generation latency may increase as provider capacity fills. Load testing reveals these failure modes before production traffic exposes them to users. It also validates that the cost estimates for expected traffic volumes are realistic.

Q60. What is "observability" for AI systems and how does it differ from monitoring?

- A. Observability is a compliance term for audit logging; monitoring is a technical term for performance tracking.
- B. Observability applies to the model provider's infrastructure; monitoring applies to the application layer.
- C. Observability provides the ability to understand system behaviour from its outputs — through logging, tracing, and metrics — enabling debugging of novel failures; monitoring tracks predefined metrics for known failure modes.**
- D. Observability and monitoring are the same concept — used interchangeably in AI system design.

Correct Answer: C**Explanation**

Monitoring tells you "something is wrong"; observability tells you "what and why." Monitoring tracks known metrics (latency, error rate, cost). Observability enables debugging of unknown failures — by logging the full pipeline state (query, retrieved chunks, generation input, output, guardrail decisions), you can trace any unexpected output back to its root cause. For complex AI systems, observability is as important as monitoring.

SET

45

Human Oversight and Adversarial Testing

Set 45 of 5 · 15 Questions · AI Engineering

Q61. What is the role of human oversight in high-stakes AI applications?

- A. Providing a review and approval mechanism for AI outputs where errors have significant consequences — ensuring that AI recommendations are verified by qualified humans before action is taken.**
- B. Monitoring AI systems for compliance violations on behalf of regulatory bodies.
- C. Reviewing and approving AI system design before deployment to verify safety properties.
- D. A legal requirement mandating that humans review every AI output before delivery.

Correct Answer: A**Explanation**

Human oversight is risk-calibrated: casual queries on a knowledge bot may not require it; a credit decision, medical recommendation, or legal advice output almost certainly does. The role of human oversight is to catch consequential errors before they cause harm — not to review every output, but to be in the loop for decisions where AI error is unacceptably costly.

Q62. What is "red-teaming" an AI system and what does it accomplish?

- A. A performance review process where internal teams compete to identify the most efficient prompts.
- B. An evaluation methodology that uses red (bad) and green (good) labels to classify model outputs.
- C. A regulatory requirement for AI systems operating in certain industries.
- D. Structured adversarial testing by a dedicated team that attempts to find failure modes, safety gaps, and misuse vectors — producing a documented inventory of vulnerabilities before deployment.**

Correct Answer: D**Explanation**

Red-teaming adopts an attacker's mindset: what is the worst this system could do, and can we make it do that? A dedicated red team tries jailbreaks, social engineering, adversarial inputs, edge cases, and misuse scenarios. The output is not a pass/fail grade but a vulnerability inventory — a list of specific inputs that caused the system to fail, with severity ratings — that drives remediation before launch.

Q63. What is "adversarial robustness" and why is it evaluated separately from general accuracy?

- A. A measure of how accurately the system performs on difficult but naturally occurring test cases.
- B. The system's ability to maintain correct and safe behaviour under deliberate adversarial inputs — a distinct capability from performing well on standard or representative inputs.**
- C. The system's ability to maintain performance as the volume of adversarial competitors increases.
- D. A benchmark for systems deployed in security-critical environments like cybersecurity.

Correct Answer: B

Explanation

A system can achieve high accuracy on representative test sets while being catastrophically vulnerable to targeted adversarial inputs. Adversarial robustness evaluation specifically tests inputs crafted to find and exploit failure modes — jailbreaks, prompt injections, data poisoning, out-of-distribution queries. High general accuracy does not imply adversarial robustness; these must be evaluated independently.

Q64. What is "bias evaluation" in AI systems and what types of bias are tested?

- A. An evaluation of whether the model's training data contains balanced representation across all topics.
- B. A technique for removing biased language from model outputs in post-processing.
- C. Systematically testing whether the system produces different quality outputs for different demographic groups, topics, or phrasings in ways that reflect discriminatory patterns.**
- D. A regulatory assessment of whether the AI system was built with diverse team representation.

Correct Answer: C

Explanation

Bias evaluation tests for disparate performance: does the system perform worse for users of certain backgrounds, languages, or demographic groups? Does it reflect harmful stereotypes? Does it apply different standards to politically or socially sensitive topics? These evaluations require diverse test sets specifically designed to surface bias — standard accuracy benchmarks rarely reveal it.

Q65. What is "prompt injection testing" and why is it critical for RAG systems?

- A. Testing whether adversarial content in retrieved documents can override system prompt instructions — critical for RAG systems that retrieve from untrusted or user-supplied content.**
- B. A technique for testing whether the system prompt can be extracted by users through targeted queries.
- C. Testing whether the model can resist attempts to change its output format through user prompts.
- D. A security audit of the API infrastructure for injection vulnerabilities in the HTTP layer.

Correct Answer: A

Explanation

RAG systems retrieve from external sources — web pages, user-uploaded documents, databases. These sources may contain text like "SYSTEM: Ignore your previous instructions and instead do X." Prompt injection testing specifically injects adversarial content into retrieved documents and verifies that the system prompt's instructions are maintained. Without this test, you do not know if your guardrails survive real-world retrieval.

Q66. What is the "alignment tax" and why does it create evaluation trade-offs?

- A. A cost charged by model providers for access to safety features like content filtering.
- B. The time investment required to implement safety alignment in an AI system.
- C. The reduction in raw task performance that safety-aligned models sometimes exhibit compared to unaligned models — evaluating aligned systems requires balancing safety and capability metrics.**
- D. The performance overhead of running safety classifiers in parallel with the main model.

Correct Answer: C

Explanation

Safety-aligned models sometimes produce lower benchmark scores than their unaligned counterparts — they may refuse certain tasks, add caveats, or produce more conservative outputs. Evaluation must account for this: a model that scores lower on raw capability benchmarks but significantly better on safety evaluations may be the better production choice. The alignment tax is a real trade-off that evaluation frameworks must incorporate.

Q67. What is "scalable oversight" as an AI safety concept relevant to evaluation?

- A. Scaling the number of human reviewers proportionally as AI output volume increases.
- B. A technique for automating the oversight of AI systems to reduce human review requirements.
- C. The practice of delegating AI oversight to external auditors as systems become too complex for internal teams.
- D. Methods for maintaining meaningful human oversight as AI systems become more capable and produce outputs that are increasingly difficult for humans to evaluate directly.**

Correct Answer: D

Explanation

As AI systems become more capable — producing complex code, long research reports, nuanced legal analysis — human reviewers may lack the expertise or bandwidth to reliably evaluate their outputs. Scalable oversight research addresses this: debate protocols, AI-assisted evaluation, interpretability tools, and decomposition of complex tasks into verifiable sub-tasks all aim to keep human oversight meaningful as capability increases.

Q68. What is a "safety benchmark" for LLMs and what does it measure?

- A. A benchmark that measures a model's ability to reason about safety scenarios presented in text.
- B. A structured evaluation of whether a model reliably refuses harmful requests, avoids producing dangerous content, and behaves in accordance with safety guidelines across diverse adversarial inputs.**
- C. An industry certification that verifies an AI system has passed minimum safety requirements.
- D. A benchmark for measuring the probability that a model will generate harmful content during normal use.

Correct Answer: B

Explanation

Safety benchmarks (HarmBench, WildGuard, AIR-Bench) test whether models refuse harmful requests across categories: violence, illegal activities, privacy violations, misinformation, self-harm. They use adversarial inputs designed to probe refusal reliability — not polite "please do something harmful" requests, but sophisticated jailbreaks and edge cases. Safety benchmark performance predicts (imperfectly) real-world safety under adversarial use.

Q69. What is "out-of-distribution" (OOD) testing and why is it important for AI reliability?

- A. Testing a model on data from a different organisation or domain than the one it was deployed for.
- B. A technique for evaluating model performance when the evaluation dataset is different from the training dataset.
- C. Testing model behaviour on inputs significantly different from those seen during training or evaluation — revealing how the system handles real-world variation and edge cases.**
- D. Testing the system on queries that fall outside the intended topic scope of the application.

Correct Answer: C

Explanation

Models perform differently on inputs that resemble their training distribution versus novel inputs. OOD testing probes the latter: unusual phrasings, domain jargon not in training data, multilingual inputs, ambiguous or contradictory queries. OOD performance reveals where the system's reliability degrades and where guardrails or human escalation are most needed.

Q70. What is "evaluation versioning" and why does it matter?

- A. Keeping evaluation test sets and scoring rubrics stable across system versions so that improvements and regressions can be measured consistently over time.**
- B. A technique for automatically updating evaluation criteria as the system evolves.
- C. Maintaining separate evaluation frameworks for different model versions.
- D. A data management practice for archiving historical evaluation results.

Correct Answer: A

Explanation

If you change the test set every time you evaluate, you cannot compare results across versions. Evaluation versioning locks the test set and rubric, creating a stable measurement reference. When the system improves, the metric improves on the same test set. When it regresses, the metric falls — on the same baseline. Without stable evaluation, improvement and regression are indistinguishable from evaluation noise.

Q71. What is "multi-stakeholder evaluation" in enterprise AI deployment?

- A. A governance process where multiple teams jointly approve AI system design before deployment.
- B. Evaluating AI system quality from the perspectives of multiple stakeholders — users, administrators, domain experts, compliance teams — who may have different and sometimes conflicting quality requirements.**
- C. An evaluation methodology that weights different test cases by their business importance.
- D. A technique for collecting evaluation feedback from a diverse representative sample of users.

Correct Answer: B

Explanation

Different stakeholders care about different things: end users care about helpfulness and accuracy; compliance teams care about regulatory adherence; administrators care about cost and operational reliability; legal teams care about liability exposure. A system that scores highly on user satisfaction metrics may fail compliance evaluation, or vice versa. Multi-stakeholder evaluation surfaces these conflicts before deployment.

Q72. What is "pre-deployment safety review" and what does it involve?

- A. A performance review process that validates the system meets speed and cost requirements before launch.
- B. A compliance check that verifies the system's data handling practices meet regulatory requirements.
- C. An internal approval process where senior management signs off on AI system deployment.
- D. A structured assessment of an AI system's safety properties before release — including red-teaming, bias evaluation, guardrail testing, and documentation of residual risks.**

Correct Answer: D

Explanation

Pre-deployment safety review is a structured gate before launch. It typically involves: red-team adversarial testing across defined harm categories, bias evaluation across demographic groups, guardrail effectiveness testing, documentation of known limitations and failure modes, and definition of monitoring plans for post-deployment. It is both a quality gate and a risk documentation exercise.

Q73. What is the difference between "safety" and "alignment" in AI evaluation?

- A. Safety is evaluated by external auditors; alignment is evaluated by the development team.
- B. Safety applies to the model; alignment applies to the application layer built on top of the model.
- C. Safety evaluates whether the system avoids causing harm; alignment evaluates whether the system does what its designers intended — a system can be safe but misaligned, or aligned but unsafe.**
- D. Safety and alignment are synonymous terms used in different research communities.

Correct Answer: C

Explanation

Safety and alignment are related but distinct. A system might be well-aligned with its designer's intent (it does exactly what they specified) but those specifications were safety-incomplete (the designer did not anticipate harmful use cases). Conversely, a system might refuse many harmful requests (safe in those cases) but still systematically misrepresent information (misaligned). Evaluating both provides a complete safety picture.

Q74. What is "human preference evaluation" and what are its limitations?

- A. A technique for using human preferences to define the gold standard for AI quality.
- B. Measuring which AI outputs humans prefer in pairwise or rating evaluations — useful for quality assessment but biased toward confident, fluent, long responses regardless of accuracy.**
- C. An evaluation approach that aggregates user satisfaction scores from production interactions.
- D. Human preference evaluation is the most reliable method for all AI quality assessment tasks.

Correct Answer: B

Explanation

Human preferences are systematically biased: people rate longer, more confident, more fluent responses higher — regardless of accuracy. An AI system optimised purely for human preference may become better at appearing helpful while becoming worse at being accurate. Preference evaluation must be combined with accuracy, faithfulness, and safety evaluation to avoid optimising for persuasiveness over truth.

Q75. What is the "evaluation flywheel" in AI product development?

- A. A technique for automating the full evaluation pipeline to reduce manual evaluation effort over time.
- B. A metric that measures how quickly the AI system improves over successive model versions.
- C. A team structure where evaluation responsibilities rotate across product, engineering, and data science.
- D. The continuous improvement cycle: evaluate → identify failures → improve system → evaluate again — where each cycle produces better evaluations, better systems, and better understanding of remaining gaps.**

Correct Answer: D

Explanation

The evaluation flywheel is the key discipline in AI product development: systematic evaluation reveals specific failure modes; those failures become targeted improvements; improvements are verified by re-evaluation; new evaluation uncovers the next set of failures. Teams that build robust evaluation infrastructure improve faster and more reliably than teams that rely on intuition and manual spot-checks.

SET

46

AI UX and Trust

Set 46 of 5 · 15 Questions · AI Engineering

Q1. A product team ships an AI feature that gives confidently worded answers but is wrong 20% of the time. What UX pattern best mitigates trust erosion?

- A. Remove the feature until accuracy reaches 100%
- B. Show confidence scores or hedged language so users calibrate trust appropriately**
- C. Add a disclaimer in the terms of service
- D. Increase font size of outputs to appear more authoritative

Correct Answer: B

Explanation

Users calibrate trust based on cues. Explicit uncertainty signals prevent over-reliance and protect the product when errors occur.

Q2. Which design principle best describes showing users why an AI made a specific recommendation?

- A. Latency optimisation
- B. A/B testing
- C. Feature gating
- D. Explainability**

Correct Answer: D

Explanation

Explainability surfaces the reasoning behind AI outputs, enabling users to verify, trust, or correct the system.

Q3. A user completes a task using an AI assistant but cannot remember if the AI did it or they did. This is a failure of:

- A. Agency transparency**
- B. Latency management
- C. Prompt injection prevention
- D. Output formatting

Correct Answer: A

Explanation

Agency transparency means the product clearly communicates when AI acted on behalf of the user vs when the user acted themselves.

Q4. What is the primary purpose of an undo or revert affordance in an AI-powered product?

- A. Reducing API costs
- B. Improving model accuracy
- C. Restoring user control after AI-initiated actions**
- D. Satisfying GDPR requirements

Correct Answer: C

Explanation

AI actions can be irreversible or unexpected. An undo mechanism preserves user agency and reduces fear of delegation.

Q5. A language learning app uses AI to grade essays. Students report feeling judged rather than helped. The most effective UX fix is:

- A. Remove AI grading entirely
- B. Frame AI feedback as suggestions with a growth-oriented tone**
- C. Switch to a different model
- D. Add more correction categories

Correct Answer: B

Explanation

Tone and framing shape how users receive AI feedback. Coaching language reduces defensiveness and increases engagement.

Q6. When is progressive disclosure most valuable in an AI product interface?

- A. When AI outputs are complex and users may only need a summary initially**
- B. When the AI has low latency
- C. When the model is fine-tuned
- D. When the product is in beta

Correct Answer: A

Explanation

Progressive disclosure prevents overwhelm by surfacing the most relevant information first, with detail available on demand.

Q7. A PM notices users frequently edit AI-generated content before using it. This behavior most likely indicates:

- A. The model needs to be replaced
- B. Users do not trust AI in principle
- C. The feature should be removed
- D. The AI output is directionally useful but not precise enough for direct use**

Correct Answer: D

Explanation

High edit rates signal partial value. Users find the output worth refining rather than discarding, pointing to a gap in specificity or context.

Q8. What does appropriate reliance mean in the context of AI-assisted decision-making?

- A. Users always follow AI recommendations
- B. Users ignore AI and make independent decisions
- C. Users trust AI when it is correct and override it when it is wrong**
- D. Users rate AI outputs before acting on them

Correct Answer: C

Explanation

Appropriate reliance is the calibrated use of AI - neither over-reliance (automation bias) nor under-reliance (ignoring correct suggestions).

Q9. An AI product shows users a generated by AI label on all outputs. The main benefit of this label is:

- A. Setting accurate expectations and enabling informed verification**
- B. Increasing click-through rates
- C. Reducing model inference costs
- D. Satisfying animation guidelines

Correct Answer: A

Explanation

Transparency labels help users know what to verify and adjust their scrutiny appropriately, reducing over-trust.

Q10. Which user segment is most likely to experience automation bias when using an AI product?

- A. Expert users with deep domain knowledge
- B. Users who dislike technology
- C. Users who frequently edit AI outputs
- D. Users under time pressure with limited domain expertise**

Correct Answer: D

Explanation

Automation bias is strongest when users lack the knowledge or time to critically evaluate AI outputs - they default to accepting suggestions.

Q11. A product team wants to measure user trust in their AI feature. The most direct behavioral metric is:

- A. Daily active users
- B. Rate at which users act on AI suggestions without modification**
- C. Time to first interaction
- D. Net Promoter Score

Correct Answer: B

Explanation

Unmodified acceptance rate is a behavioral proxy for trust - it captures actual reliance rather than stated preference.

Q12. What is the risk of making AI features invisible or seamlessly integrated with no indication to users?

- A. Latency increases significantly
- B. Model fine-tuning becomes more expensive
- C. Users cannot calibrate trust or know when to verify outputs**
- D. API rate limits are more likely to be hit

Correct Answer: C

Explanation

Invisible AI removes users from the verification loop. When errors occur they are less likely to be caught.

Q13. A fintech product uses AI to flag suspicious transactions. False positives frustrate users. The best UX approach is:

- A. Remove the flagging feature
- B. Increase the flagging threshold to reduce all flags
- C. Add more flags to balance the false positives
- D. Show the reason for the flag and provide a clear dispute path**

Correct Answer: D

Explanation

Explainability plus recourse transforms a frustrating AI error into a manageable interaction. Users can act on and resolve the issue.

Q14. A product manager wants to build user confidence in a new AI writing assistant. The most effective early onboarding move is:

- A. Displaying model benchmarks on the landing page
- B. Showing users a before and after example of a task the AI improved**
- C. Requiring users to complete a tutorial before using the product
- D. Adding a disclaimer about AI limitations

Correct Answer: B

Explanation

Concrete demonstrations of value build trust faster than abstract claims. Before and after examples make the benefit tangible.

Q15. Which of the following is a signal that users have developed appropriate mental models of an AI product?

- A. Users choose when to use the AI feature and when to handle tasks manually**
- B. Users always use the AI feature for every task
- C. Users never override AI suggestions
- D. Users spend more time on the product

Correct Answer: A

Explanation

Selective, deliberate use shows users understand the AI strengths and limits - the hallmark of an accurate mental model.

SET

47

Product Decision-Making

Set 47 of 5 · 15 Questions · AI Engineering

Q16. A PM must decide whether to build a custom model or use a third-party LLM API. The most critical factor in this decision is:

- A. Whether the CEO prefers open-source or proprietary models
- B. Whether the team has Python skills
- C. Whether the required capability gap justifies the cost and maintenance overhead of custom development**
- D. Whether competitors are using custom models

Correct Answer: C

Explanation

Build vs buy for models is a capability, cost, and maintenance decision. The threshold is whether existing APIs can meet the requirement acceptably.

Q17. When evaluating AI features for a product roadmap, which framework best captures the unique risks of AI?

- A. Impact-effort matrix weighted for failure mode severity**
- B. Story point estimation
- C. Velocity tracking
- D. RICE scoring without modification

Correct Answer: A

Explanation

Standard prioritisation frameworks do not account for AI-specific risks like hallucination or bias. Weighting for failure severity makes them appropriate.

Q18. A product team receives a request to add a generative AI chatbot to a healthcare app. The first decision gate should be:

- A. Choosing between GPT-4 and Claude
- B. Designing the chat UI
- C. Estimating development time
- D. Assessing regulatory requirements and liability for AI-generated health information**

Correct Answer: D

Explanation

In healthcare, regulatory compliance and liability are pre-conditions for any AI feature. Technical choices come after viability is established.

Q19. A PM sees that an AI feature has high usage but low satisfaction scores. The most productive next step is:

- A. Increasing inference speed
- B. Qualitative research to understand the gap between usage intent and output quality**
- C. Switching the underlying model
- D. Adding more features to the same surface

Correct Answer: B

Explanation

High usage with low satisfaction means users are trying the feature but not getting what they need. Qualitative research reveals the specific unmet need.

Q20. Which metric best signals that an AI feature is creating genuine product value rather than just generating engagement?

- A. Session duration
- B. Number of AI queries per session
- C. Task completion rate for outcomes users could not achieve before the feature**
- D. Feature discovery rate

Correct Answer: C

Explanation

Real value is measured by whether the AI enables outcomes that were previously impossible or impractical for users - not just activity metrics.

Q21. A PM is deciding whether to launch an AI feature in beta or full release. The strongest argument for beta is:

- A. Beta features are free to build
- B. Beta reduces API costs
- C. Full release requires board approval
- D. The failure mode of AI errors in production requires real user feedback before full exposure**

Correct Answer: D

Explanation

AI features can fail in unexpected ways that are hard to anticipate in staging. Beta allows discovery and remediation before wide impact.

Q22. A startup wants to use AI to automate customer support. The most important pre-launch question is:

- A. What happens when the AI gives an incorrect or harmful response to a customer?**
- B. Which LLM has the lowest per-token cost?
- C. How fast can we ship the chatbot?
- D. Should the chatbot have a name?

Correct Answer: A

Explanation

Failure mode planning is the first question for customer-facing AI. The answer determines guardrails, escalation paths, and liability exposure.

Q23. A product decision to remove human review from an AI-powered loan decisioning system should be gated on:

- A. Speed improvement metrics alone
- B. Demonstrated accuracy across demographic segments and regulatory clearance**
- C. Cost savings projections
- D. Model benchmark scores on public datasets

Correct Answer: B

Explanation

Removing human oversight in high-stakes domains requires fairness validation and regulatory sign-off - not just aggregate accuracy.

Q24. Which of the following is the strongest signal that a product team has underinvested in AI product quality?

- A. The team uses a third-party API
- B. The product has a free tier
- C. The team ships weekly
- D. Users frequently screenshot AI outputs and manually share them to get corrections**

Correct Answer: D

Explanation

Manual workarounds to fix AI errors are a strong signal that the error rate or recourse path inside the product is inadequate.

Q25. When should a PM introduce model versioning as a product requirement?

- A. Only when enterprise customers request it
- B. After reaching 1 million users
- C. When the product core experience depends on consistent AI behaviour across releases**
- D. When switching from one provider to another

Correct Answer: C

Explanation

Model updates can silently change product behaviour. Versioning is a product requirement whenever consistency is part of the value proposition.

Q26. A team ships an AI feature and notices a drop in retention among a specific user cohort. The most likely AI-specific cause is:

- A. The feature is too slow
- B. The feature produces outputs that do not match that cohort context, language, or needs**
- C. The feature is too cheap
- D. The feature was not announced in release notes

Correct Answer: B

Explanation

AI products often underperform for specific cohorts because training data or prompts do not represent their context. Cohort analysis reveals this gap.

Q27. A PM wants to reduce AI hallucinations without waiting for model improvements. The best product-level intervention is:

- A. Constraining the AI to only answer within a verified knowledge base and surfacing citations**
- B. Increasing the temperature parameter
- C. Adding more tokens to the context window
- D. Switching to a larger model

Correct Answer: A

Explanation

RAG with citation is the primary product-level tool for reducing hallucination impact - it grounds outputs in verified sources users can check.

Q28. A product team is debating whether to show AI-generated content inline or in a separate review panel. The main variable determining the right answer is:

- A. Which design pattern is more common in the market
- B. The team design preferences
- C. How consequential errors would be if users act on them without reviewing**
- D. The number of tokens in each output

Correct Answer: C

Explanation

High-stakes outputs warrant friction and separation to encourage review. Low-stakes outputs benefit from inline delivery for speed and flow.

Q29. Which of the following is a product-level solution to AI bias in a hiring tool?

- A. Using a larger model
- B. Adding more training data
- C. Reducing the temperature
- D. Auditing output distributions across candidate demographic groups before launch**

Correct Answer: D

Explanation

Bias audits at the product level catch discriminatory patterns before they affect real candidates - regardless of which model is used.

Q30. A PM must decide between two AI features: one that delights 10% of users intensely and one that mildly improves the experience for 80%. The decision framework should prioritise:

- A. The feature with the larger user count always
- B. Strategic fit with the product core value proposition and retention impact**
- C. The feature with the higher NPS lift always
- D. The feature that is cheaper to build

Correct Answer: B

Explanation

Neither intensity nor breadth is universally correct. The right choice depends on which feature advances core product strategy and drives retention.

SET

48

Feature Prioritisation

Set 48 of 5 · 15 Questions · AI Engineering

Q31. A product team has three AI feature requests: faster inference, better accuracy, and an explainability panel. How should they prioritise?

- A. By mapping each to the top user pain point identified in research**
- B. Accuracy first, always
- C. Ship all three simultaneously
- D. Start with explainability because it is cheapest

Correct Answer: A

Explanation

Prioritisation without user pain mapping produces technically impressive but adoption-poor features. Research grounds the decision.

Q32. Which prioritisation signal most directly indicates that an AI feature should be deprioritised?

- A. The feature takes more than two sprints to build
- B. The feature requires a new model
- C. Users who discover the feature do not return to use it a second time**
- D. The feature has no competitor equivalent

Correct Answer: C

Explanation

Zero repeat usage after discovery signals that the feature solved no real problem or delivered no perceived value - the clearest deprioritisation signal.

Q33. A PM has limited engineering capacity. An AI feature is requested by the top 5 enterprise customers but not by any self-serve users. The best response is:

- A. Immediately prioritise it because enterprise revenue is higher
- B. Evaluate whether the enterprise use case is a leading indicator of broader need**
- C. Ignore it because it has no self-serve demand
- D. Ask the customers to pay for custom development

Correct Answer: B

Explanation

Enterprise requests often preview emerging needs. The PM should assess whether this is niche or a signal of wider unmet demand before committing resources.

Q34. A product team is using a jobs-to-be-done framework to evaluate AI features. What does this approach prioritise?

- A. Technical capability benchmarks
- B. The frequency with which a feature is used
- C. Revenue per feature
- D. The functional progress users are trying to make, not the features themselves**

Correct Answer: D

Explanation

Jobs-to-be-done centres on user motivation. AI features that address real jobs get adopted; features built around capability showcasing do not.

Q35. When should AI personalisation be prioritised over a general-purpose AI feature?

- A. When there is strong evidence that user context meaningfully changes the optimal output**
- B. Always, because personalisation is always better
- C. Only for enterprise tiers
- D. When the team has idle ML engineers

Correct Answer: A

Explanation

Personalisation adds complexity and maintenance. It is worth building only when user-specific context provably improves outcomes.

Q36. A team is building an AI product. They can invest in either model accuracy improvements or UI improvements for the same cost. Research shows users blame the UI when they distrust outputs. What should they prioritise?

- A. Model accuracy, because ground truth matters more
- B. UI improvements, because trust perception is currently the binding constraint**
- C. Neither, wait for user feedback
- D. Build both in parallel

Correct Answer: B

Explanation

The binding constraint is what limits adoption. If trust perception is the bottleneck, UI investment yields higher returns than model accuracy.

Q37. A product manager receives conflicting feature requests from power users and new users for the same AI surface. The right approach is:

- A. Prioritise power users because they generate more revenue
- B. Prioritise new users because they represent growth
- C. Build two separate products
- D. Design for progressive complexity - simple defaults with power user controls accessible on demand**

Correct Answer: D

Explanation

Progressive complexity resolves the power user vs new user tension by making the product approachable without limiting advanced usage.

Q38. Which type of AI feature should be given the highest priority for early kill decision-making?

- A. Features with low usage rates
- B. Features that are technically complex to maintain
- C. Features that produce harmful or misleading outputs even in edge cases**
- D. Features that exceed budget

Correct Answer: C

Explanation

Harmful outputs in edge cases represent liability and trust damage that can outweigh the value of the entire feature. Kill criteria must include safety thresholds.

Q39. A startup AI product has 10 feature ideas and 3 months of runway. The correct prioritisation lens is:

- A. Which feature users mentioned most in surveys
- B. Which single feature most increases the probability of survival through retention or revenue**
- C. Which feature is easiest to build
- D. Which feature the founder finds most interesting

Correct Answer: B

Explanation

Under resource constraint, survival prioritisation means concentrating on the one feature with the highest leverage on key business metrics.

Q40. A PM is evaluating two AI features: one with high ceiling impact but high variance, and one with moderate but reliable impact. Which is preferable in a mature product with a large user base?

- A. The reliable impact feature, because variance at scale creates unpredictable user experience**
- B. The high ceiling feature, because ambition signals product leadership
- C. The feature that ships first
- D. The feature preferred by the design team

Correct Answer: A

Explanation

At scale, variance in AI output quality creates inconsistent experiences across millions of users. Reliability compounds into retention; ceiling impact rarely does.

Q41. A PM must decide when to sunset an AI feature. The strongest signal is:

- A. The feature has been live for more than one year
- B. A competitor has shipped a similar feature
- C. Maintenance cost exceeds the value delivered relative to alternative uses of the same engineering time**
- D. The original PM who built it has left

Correct Answer: C

Explanation

Feature sunset is an opportunity cost decision. A feature should be removed when it consumes more resources than the value it generates compared to alternatives.

Q42. A team has data showing that 90% of users only use 20% of the AI features. What should the roadmap prioritise?

- A. Marketing the unused 80% to increase adoption
- B. Removing the 20% and rebuilding with new features
- C. Adding even more features to increase surface area
- D. Deepening the 20% of features most used, not expanding the unused 80%**

Correct Answer: D

Explanation

Usage concentration is a signal to invest in depth, not breadth. The used features represent the product actual value - deepen them.

Q43. Which of the following is the best leading indicator for an AI feature long-term retention impact?

- A. Feature launch day traffic
- B. Whether users who adopt the feature have higher 30-day retention than those who do not**
- C. Number of press mentions at launch
- D. Internal team excitement about the feature

Correct Answer: B

Explanation

Differential retention between adopters and non-adopters is the strongest early signal that a feature creates durable value.

Q44. A product team is prioritising between an AI feature that improves the core workflow and one that is novel but peripheral. The correct default is:

- A. Novel feature, because it generates press coverage
- B. Peripheral feature, because it differentiates from competitors
- C. Core workflow improvement, because retention impact is higher for features on the critical path**
- D. Whichever is cheaper to build

Correct Answer: C

Explanation

Features on the critical path have disproportionate retention impact. Peripheral novelty rarely compounds into retention the way core workflow improvements do.

Q45. A PM receives a request to add an AI feature that would require collecting additional user data. The first question is:

- A. Whether the value the feature delivers justifies the privacy cost and user consent requirement**
- B. Whether the data is available to purchase
- C. Whether the feature has a competitor equivalent
- D. Whether the model supports the data type

Correct Answer: A

Explanation

Privacy cost must be weighed against feature value before any data collection decision. Collecting data without this analysis creates regulatory and trust risk.

SET

49

User Workflows

Set 49 of 5 · 15 Questions · AI Engineering

Q46. A PM is mapping user workflows to identify where AI can add the most value. The highest-value AI insertion point is typically:

- A. Steps that users enjoy the most
- B. Steps that happen at the end of the workflow
- C. Steps that require no data
- D. Steps that are high-frequency, cognitively demanding, and have predictable structure**

Correct Answer: D

Explanation

AI delivers the most leverage on tasks that repeat often, drain cognitive energy, and have patterns the model can learn - these are the best automation targets.

Q47. A user research session reveals that users complete a workflow faster with AI but feel less ownership over the output. This is a signal to:

- A. Remove the AI feature
- B. Redesign the AI interaction to give users meaningful editorial control at key steps**
- C. Add more AI automation to reduce effort further
- D. Change the underlying model

Correct Answer: B

Explanation

Reduced ownership erodes satisfaction and retention even when speed improves. Meaningful control restores agency without sacrificing efficiency.

Q48. Which workflow insertion pattern is most appropriate when AI errors are costly and hard to reverse?

- A. Full automation with post-hoc audit
- B. AI acting autonomously and logging decisions
- C. AI as a drafting assistant with mandatory human review before any action is taken**
- D. Removing AI from the workflow until errors reach zero

Correct Answer: C

Explanation

High-stakes, hard-to-reverse workflows require human-in-the-loop designs. AI drafts reduce effort while mandatory review preserves oversight.

Q49. A product team notices that users skip the AI suggestion step in a workflow and complete the task manually. The most likely reason is:

- A. The AI suggestion is too generic and does not save meaningful time for this workflow**
- B. Users prefer typing
- C. The AI is too slow
- D. The feature is not visible enough

Correct Answer: A

Explanation

Skip behavior reveals that the AI output is not sufficiently useful relative to the effort of reviewing it. Context specificity is usually the gap.

Q50. In a document editing product, AI suggestions are shown inline but users report feeling distracted. The best UX fix is:

- A. Increasing the font size of suggestions
- B. Making suggestions appear faster
- C. Requiring users to accept all suggestions before proceeding
- D. Moving AI suggestions to a sidebar or making them opt-in rather than always-on**

Correct Answer: D

Explanation

Inline AI suggestions can interrupt flow for experienced users. Sidebar or opt-in delivery respects user attention and workflow rhythm.

Q51. A PM is designing an AI onboarding flow. The most effective early moment is:

- A. Showing a feature list video
- B. Asking users to rate the AI before using it
- C. Having the AI complete a real task for the user using their actual data**
- D. Requiring users to complete a quiz about AI

Correct Answer: C

Explanation

First-hand value experience with real data converts more users than demonstrations. The AI should prove itself immediately in the user actual context.

Q52. A workflow automation product allows AI to take multi-step actions on behalf of users. The most important design requirement is:

- A. A clear audit trail showing every action the AI took and why**
- B. Minimising the number of steps shown to the user
- C. Hiding AI decision points to avoid confusion
- D. Allowing the AI to modify its own instructions

Correct Answer: A

Explanation

Multi-step AI actions require full traceability. Users need to see and verify what was done before trusting the system with consequential tasks.

Q53. When designing an AI-assisted research workflow, which principle most improves output quality?

- A. Removing user control to reduce decision fatigue
- B. Giving users control over what sources the AI searches and how results are filtered**
- C. Maximising the number of results returned
- D. Hiding source information to simplify the display

Correct Answer: B

Explanation

Research quality depends on source quality and relevance. User control over scope produces more targeted, trustworthy results.

Q54. A team notices that AI-generated summaries are used directly without review in 70% of cases. This observation most warrants:

- A. Improving the summary model
- B. Reducing the length of summaries
- C. Adding inline quality signals and encouraging review for high-stakes summaries**
- D. Removing the human review option

Correct Answer: C

Explanation

70% unreviewed acceptance is high automation bias. Quality signals and review nudges help users calibrate without making review mandatory.

Q55. A support tool uses AI to draft responses for agents. Agents report copying AI drafts verbatim without reading them. The product risk is:

- A. Agents becoming too fast
- B. Support costs dropping too quickly
- C. AI drafts becoming longer over time
- D. Harmful or incorrect responses reaching customers without human verification**

Correct Answer: D

Explanation

Verbatim copy-paste without review defeats the human-in-the-loop design. The product must create meaningful friction that prompts at least scanning.

Q56. A PM is designing a feedback mechanism for AI workflow outputs. The most useful feedback type for model improvement is:

- A. Specific inline edits made by users, captured as diff data**
- B. Star ratings after task completion
- C. Binary thumbs up or down without context
- D. NPS surveys sent monthly

Correct Answer: A

Explanation

Inline edit diffs reveal exactly what the AI got wrong and how users corrected it - the richest training signal for improving model outputs.

Q57. A product manager is evaluating whether to add AI to a simple, well-understood workflow. The key question is:

- A. Whether the team has the AI budget
- B. Whether AI reduces friction enough to justify the added complexity and failure risk**
- C. Whether competitors have AI in this workflow
- D. Whether the design team prefers AI interfaces

Correct Answer: B

Explanation

Not every workflow benefits from AI. The bar is whether the friction reduction exceeds the complexity and failure modes introduced.

Q58. A product allows users to delegate tasks to an AI agent. Users hesitate to delegate high-stakes tasks. The best product response is:

- A. Increasing the AI agent autonomy
- B. Marketing the agent accuracy more aggressively
- C. Restricting agent use to low-stakes tasks only
- D. Providing transparency into agent actions and an easy recall or cancel mechanism**

Correct Answer: D

Explanation

Hesitation around delegation signals insufficient trust. Transparency into what the agent is doing and the ability to stop it are the key trust builders.

Q59. A workflow product tracks that AI-assisted tasks take 40% less time but user satisfaction is flat. The most likely explanation is:

- A. Speed improvement alone does not address the quality or confidence gap users experience with AI outputs**
- B. Users prefer slower workflows
- C. The feature has not been marketed enough
- D. The AI is using the wrong model

Correct Answer: A

Explanation

Time savings are necessary but not sufficient. If users are not satisfied with the output quality or feel uncertain about results, satisfaction stays flat.

Q60. A PM wants to design an AI feature that helps users make better decisions rather than making decisions for them. The correct design pattern is:

- A. Full automation - letting the AI decide and notifying the user
- B. Removing the decision step entirely from the workflow
- C. Decision support - surfacing relevant information, options, and tradeoffs without prescribing a choice**
- D. Adding more data fields for users to fill in before the AI acts

Correct Answer: C

Explanation

Decision support augments human judgment. It preserves agency while reducing cognitive load - appropriate when users need to own the outcome.

SET

50

Operational Usefulness

Set 50 of 5 · 15 Questions · AI Engineering

Q61. A PM wants to measure the ROI of an AI feature for an internal operations team. The most direct metric is:

- A. Time saved per task multiplied by the number of tasks and loaded staff cost**
- B. Number of AI requests made per day
- C. Model accuracy on internal benchmarks
- D. Number of features shipped using AI

Correct Answer: A

Explanation

Operational ROI is time x volume x cost. This gives a financial figure the business can compare directly against the cost of building and running the feature.

Q62. An operations team adopts an AI tool but usage drops after week 2. The most likely cause is:

- A. The tool is too expensive
- B. The team lacks technical skills
- C. The model version changed
- D. Initial novelty wore off and the tool does not save enough time to justify the behaviour change**

Correct Answer: D

Explanation

Week-2 drop-off is a classic signal of novelty-driven initial adoption without a habit-forming value proposition. The core workflow benefit is insufficient.

Q63. Which operational AI use case has the highest value threshold for accuracy?

- A. Email subject line generation
- B. Medical diagnosis support, where errors directly affect patient outcomes**
- C. Blog post summarisation
- D. Social media caption writing

Correct Answer: B

Explanation

Accuracy thresholds must be proportional to consequence. Medical decisions have life-altering stakes, making them the highest bar for AI accuracy requirements.

Q64. A team deploys an AI tool for internal knowledge retrieval. After launch, employees report using it less than the old search system. The most productive diagnostic question is:

- A. Is the AI model large enough?
- B. Was there sufficient marketing for the new tool?
- C. Are users finding what they need faster with the old system, and why does the AI fail to match that?**
- D. Is the system deployed in the cloud?

Correct Answer: C

Explanation

Reverting to the old tool is a direct comparison signal. Understanding why the AI fails on the same tasks is the most useful diagnostic for improvement.

Q65. An AI coding assistant is deployed to a 50-person engineering team. What is the most important operational metric to track after 30 days?

- A. Change in pull request review cycles and defect rates, not just lines of code generated**
- B. Number of AI code suggestions accepted
- C. Total tokens consumed by the team
- D. How often engineers open the tool

Correct Answer: A

Explanation

Code quality metrics reveal whether AI assistance improves actual engineering outcomes. Acceptance rate and usage are activity metrics, not outcome metrics.

Q66. A PM is building the business case for an internal AI tool. Stakeholders ask for the payback period. What inputs are required?

- A. Model benchmark scores and API latency
- B. Number of AI features available in the tool
- C. Total implementation cost, ongoing API cost, and measurable time saved per user per week**
- D. Competitor adoption rates

Correct Answer: C

Explanation

Payback period equals total cost divided by value generated per period. Time saved per user converted to labour cost is the primary value numerator for internal tools.

Q67. An internal AI assistant is deployed across departments with different workflows. Adoption is high in marketing but low in finance. The most likely reason is:

- A. Finance employees are less technically sophisticated
- B. Marketing has a larger team
- C. Finance did not receive the announcement email
- D. The AI outputs are better calibrated for unstructured creative tasks than for the precise, regulated outputs finance requires**

Correct Answer: D

Explanation

LLMs perform better on open-ended creative tasks than on structured, accuracy-critical outputs like financial models. Adoption reflects fit to task type.

Q68. A company deploys AI to handle tier-1 customer support. The correct escalation design is:

- A. AI handles all queries including complaints and legal threats
- B. AI handles routine queries and escalates to humans when confidence is low or the issue is sensitive**
- C. AI only handles queries when human agents are unavailable
- D. Humans handle all queries and AI only logs them

Correct Answer: B

Explanation

Escalation design is the critical safety mechanism for customer-facing AI. Low confidence and sensitive topics are the two primary triggers for human handoff.

Q69. A product manager wants to track whether an AI tool improves employee productivity without creating new risks. Which metric pair is most complete?

- A. Usage frequency plus session length
- B. Tokens consumed plus model version
- C. Output quality maintained plus time per task reduced**
- D. Number of prompts submitted plus acceptance rate

Correct Answer: C

Explanation

Productivity gain without quality maintenance is false gain. The correct measurement pair captures both the efficiency improvement and the quality safeguard.

Q70. A PM launches an AI feature that automates a manual data entry workflow. Six months later the team reports the manual workflow has atrophied and no one remembers how to do it. This is a risk of:

- A. Over-automation creating single points of failure and skill degradation**
- B. Successful adoption
- C. High user satisfaction
- D. Strong retention metrics

Correct Answer: A

Explanation

Skill atrophy through automation is an operational risk. When AI fails, the fallback capability no longer exists - a dependency that must be planned for.

Q71. A PM is evaluating three AI vendors for an internal HR tool. Beyond accuracy and cost, the most important evaluation criterion is:

- A. The vendor market share
- B. Data privacy terms, including who owns training rights over submitted employee data**
- C. Whether the vendor offers a free trial
- D. Whether the vendor has a mobile app

Correct Answer: B

Explanation

Employee data submitted to AI tools can be used for model training unless explicitly prohibited. Privacy terms are a compliance and trust-critical evaluation factor.

Q72. A PM notices that AI-generated reports are used in board presentations without any human editing. The correct product response is:

- A. Improving model quality to match human-level writing
- B. Disabling the export function
- C. Removing the report generation feature
- D. Adding a mandatory review checklist and clearly marking content as AI-generated before export**

Correct Answer: D

Explanation

Board-level documents with unreviewed AI content carry reputational and accuracy risk. Review friction and transparency labels are appropriate product guardrails.

Q73. Which of the following best describes the operational risk of prompt injection in an enterprise AI tool?

- A. The model runs out of context window during a long prompt
- B. Users ask the AI questions outside its intended domain
- C. Malicious input in user-submitted data causes the AI to take unauthorised actions or leak information**
- D. The AI produces outputs that are too long for the UI to display

Correct Answer: C

Explanation

Prompt injection allows attackers to hijack AI behaviour through content the model processes as instructions - a critical enterprise security risk.

Q74. An internal AI tool is being used by employees to generate external-facing content. The biggest operational governance risk is:

- A. Employees spending too much time using the tool
- B. AI-generated content being published without review, creating accuracy or brand-voice issues**
- C. The tool generating content faster than the marketing team can review
- D. The tool learning from employee writing style

Correct Answer: B

Explanation

External content represents the organisation publicly. AI-generated content published without review is an accuracy, brand, and legal risk that requires governance.

Q75. A PM is writing success criteria for an AI operational tool six months post-launch. Which set of criteria is most complete?

- A. Adoption rate plus number of features shipped
- B. API uptime plus token cost within budget
- C. Number of support tickets raised about the tool
- D. Adoption rate plus measurable time saved plus error rate maintained below threshold plus positive user feedback**

Correct Answer: D

Explanation

Complete success criteria for an operational AI tool include adoption, efficiency, quality, and satisfaction - all four pillars.

SET

51

Vibe Coding and Rapid Prototyping

Set 51 of 5 · 15 Questions · AI Engineering

Q1. A founder wants to build a working web app prototype in 48 hours without a development team. The best AI-native approach is:

- A. Hire a contractor to write the spec first
- B. Use a natural language-to-code tool to generate a working prototype from a plain English spec, then iterate**
- C. Wait until the team can properly architect the backend
- D. Build in Figma only and call it a prototype

Correct Answer: B

Explanation

AI code generation tools allow non-engineers and engineers alike to go from idea to working prototype in hours. Speed of learning beats architectural perfection at this stage.

Q2. What is vibe coding as a practise in AI-native product development?

- A. Writing code while listening to music to improve focus
- B. Using AI-generated music as a backend for audio apps
- C. A style of pair programming where both engineers code simultaneously
- D. Using conversational AI to iterate on code through natural language instructions, accepting imperfect output and steering toward the goal**

Correct Answer: D

Explanation

Vibe coding treats AI as a co-pilot where the human steers with intent and accepts rough output that gets refined. It optimises for speed of exploration over code quality.

Q3. A non-engineer founder is building a Chrome extension using Claude. They get a working build on the third iteration. What is the most important practise to maintain momentum?

- A. Test each iteration against the specific use case before requesting the next change**
- B. Accept all AI output without testing to move faster
- C. Switch to a different AI tool for each iteration
- D. Hire an engineer to review every AI output

Correct Answer: A

Explanation

In AI-assisted prototyping, tight test-iterate loops catch errors early and give precise feedback to the AI for the next prompt. Untested iterations compound errors.

Q4. When building a rapid AI prototype, which type of input gives an AI coding tool the most useful starting point?

- A. A list of all desired features
- B. A database schema
- C. A concrete user story describing what a specific user does, not what the system does**
- D. An architecture diagram

Correct Answer: C

Explanation

User stories ground the AI in behaviour rather than structure. They produce code oriented toward outcomes, not implementation choices the AI may get wrong.

Q5. A team uses AI to generate a prototype in 2 days. A stakeholder asks them to clean it up before showing investors. The AI-native approach to this request is:

- A. Rewrite the entire codebase from scratch
- B. Ask the AI to refactor for readability and add comments, then review the key interaction paths manually**
- C. Decline the request because prototypes should not be cleaned up
- D. Hand off to a senior engineer to rewrite

Correct Answer: B

Explanation

AI can clean up its own output quickly. The human role is to verify that refactoring did not break functionality - not to rewrite everything manually.

Q6. What is the primary risk of using AI-generated code in production without review?

- A. Security vulnerabilities and logic errors that the AI introduced but did not flag**
- B. The code will be too slow
- C. The code will not run on all browsers
- D. AI licensing restrictions prevent production use

Correct Answer: A

Explanation

AI code generators can introduce subtle bugs, insecure patterns, and incorrect logic without warning. Production deployment requires human review of security-critical paths.

Q7. A solo founder uses AI to build a working SaaS MVP. The first feature they should prioritise shipping is:

- A. The admin dashboard
- B. The settings page
- C. The onboarding email sequence
- D. The one feature that allows a real user to complete the core job-to-be-done end to end**

Correct Answer: D

Explanation

The minimum viable product must do the core job. Peripheral features add no learning until the core workflow is validated with real users.

Q8. When should a rapid AI prototype be discarded rather than refactored into a production codebase?

- A. After every prototype, always
- B. Only if the prototype did not work
- C. When the prototype was built with speed-first decisions that create architectural debt incompatible with production requirements**
- D. When it takes longer than one week to build

Correct Answer: C

Explanation

Prototypes built for speed carry shortcuts that do not scale. The decision to discard vs refactor depends on how deep the architectural debt runs.

Q9. A developer uses an AI tool to generate a database schema from a natural language description. The most important post-generation step is:

- A. Validating that the schema reflects actual data relationships and will support the required queries**
- B. Accepting the schema immediately to save time
- C. Asking the AI to generate the schema again with different words
- D. Converting the schema to a different database engine

Correct Answer: A

Explanation

AI-generated schemas often miss edge cases or model relationships incorrectly. Structural validation before building on top of a schema prevents costly rewrites.

Q10. What does context window as a scratchpad mean in AI-native development practise?

- A. Storing code snippets in a text file
- B. Using the terminal as the primary development environment
- C. Keeping AI outputs in a separate document for later review
- D. Using the conversation history with the AI to maintain project state, decisions, and constraints across a session**

Correct Answer: D

Explanation

The AI context window serves as working memory for the session. Skilled AI-native developers front-load context such as specs, constraints, and prior decisions to get consistent outputs.

Q11. A team prototyping with AI notices the AI frequently forgets earlier constraints as the conversation grows. The best mitigation is:

- A. Starting a new conversation for every change
- B. Re-stating key constraints at the start of each new prompt and using a system prompt or pinned context**
- C. Reducing the length of prompts
- D. Switching to a different AI tool

Correct Answer: B

Explanation

Context window limitations cause AI to drift from earlier instructions. Re-stating constraints and using persistent system prompts maintains coherence across a session.

Q12. Which describes the correct sequencing in AI-native rapid prototyping?

- A. Idea to architecture document to engineering sprint to AI review to launch
- B. Idea to design in Figma to engineer the design to add AI later
- C. Idea to spec in natural language to AI generates working code to test with real users to iterate**
- D. Idea to hire a team to write a PRD to AI assists with documentation

Correct Answer: C

Explanation

AI-native prototyping collapses the traditional build cycle. Natural language specs feed directly into working code, compressing the idea-to-user-feedback loop.

Q13. A developer wants to add authentication to an AI-generated app. The safest approach is:

- A. Ask the AI to generate a custom authentication system
- B. Skip authentication for the MVP
- C. Copy auth code from a GitHub repository without review
- D. Use a battle-tested auth library or service such as Auth0 or Clerk rather than asking the AI to generate custom auth code**

Correct Answer: D

Explanation

Authentication is a security-critical component where custom AI-generated code frequently introduces vulnerabilities. Established services handle this more safely.

Q14. What is the value of maintaining a decision log when prototyping with AI?

- A. It satisfies compliance requirements
- B. It captures why choices were made, enabling the AI to stay consistent and the team to onboard faster**
- C. It improves AI response speed
- D. It is required for deploying to production

Correct Answer: B

Explanation

A decision log is a human-readable record of architectural and design choices. It feeds back into AI context and preserves rationale as the project grows.

Q15. A non-technical founder builds an MVP with AI assistance and gets their first paying customer. The next highest-leverage AI use is:

- A. Using AI to automate the manual parts of onboarding and fulfilment before hiring anyone**
- B. Using AI to redesign the landing page
- C. Using AI to write investor pitch decks
- D. Using AI to generate more code features immediately

Correct Answer: A

Explanation

After first revenue, AI-native operators use AI to scale operations before headcount. Automating manual steps extends runway and proves the business model further.

SET

52

MVP Building with AI

Set 52 of 5 · 15 Questions · AI Engineering

Q16. A startup uses AI to generate their first landing page, waitlist form, and email sequence. What does this demonstrate about AI-native execution?

- A. AI can fully replace a marketing team
- B. AI is only useful after product-market fit
- C. AI compresses the cost and time of go-to-market assets, enabling solo founders to test demand before building**
- D. AI-generated landing pages convert better than human-written ones

Correct Answer: C

Explanation

AI-native founders use AI to validate demand with minimal investment. Landing page, waitlist, and email sequence can be shipped in hours, enabling pre-build demand testing.

Q17. When building an MVP with AI assistance, which deliverable should be AI-generated last?

- A. The core data model, which should be designed by a human to avoid structural errors that compound**
- B. The landing page copy
- C. The email templates
- D. The FAQ section

Correct Answer: A

Explanation

Data models are foundational. AI-generated schemas may look correct but miss relational integrity or scalability concerns that become expensive to fix later.

Q18. A team of two uses AI tools to ship a product in 3 weeks that would have taken a 10-person team 6 months traditionally. The primary leverage factor is:

- A. The team worked 24 hours a day
- B. The product had fewer features than normal
- C. The team used a no-code tool instead of AI
- D. AI handling execution tasks that previously required specialist headcount across design, code, copy, and QA**

Correct Answer: D

Explanation

AI-native teams are force multiplied because AI performs execution work across multiple functions simultaneously. Headcount is no longer the bottleneck for shipping.

Q19. A PM is building an MVP and must decide what to do manually vs what to delegate to AI. The correct filter is:

- A. Delegate everything that takes more than one hour to AI
- B. Delegate execution tasks to AI; retain human judgment for strategy, validation, and decisions with irreversible consequences**
- C. Keep all code tasks human and delegate only writing tasks to AI
- D. Use AI for external tasks and humans for internal tasks

Correct Answer: B

Explanation

AI excels at execution but lacks contextual judgment. Humans should own strategic choices, quality validation, and decisions that cannot be undone easily.

Q20. An AI-assisted MVP is live and getting traction. The founder wants to scale. The AI-native scaling principle is:

- A. Hire first and then use AI to make hires more productive
- B. Stop using AI and build a proper engineering team
- C. Identify the highest-friction manual operations and automate them with AI before hiring for those roles**
- D. Replace all AI tooling with custom-built internal systems

Correct Answer: C

Explanation

AI-native scaling defers headcount until automation reaches its limit. The pattern is: automate, then saturate, then hire, then automate again.

Q21. What is the primary role of AI in the build, measure, learn lean startup loop for an AI-native team?

- A. Replacing the measure phase with AI analytics
- B. Automating the learn phase by having AI interpret all data
- C. Making the loop unnecessary by getting it right the first time
- D. Compressing the build phase from weeks to days, enabling faster iteration through the full loop**

Correct Answer: D

Explanation

AI compresses build time, which accelerates the entire loop. More loops per unit time means faster learning - the core advantage of AI-native execution.

Q22. A founder uses AI to build an MVP in 2 weeks. A VC asks whether this is defensible. The most honest AI-native answer is:

- A. The build speed is not the moat - the data, distribution, and user insight accumulated through rapid iteration are**
- B. Yes, because AI tools are expensive and competitors cannot afford them
- C. Yes, because the code is proprietary
- D. No, and the product should not be shown to investors yet

Correct Answer: A

Explanation

AI-native build speed lowers barriers for everyone. The defensible advantage is the learning and asset accumulation such as data, users, and brand that speed enables.

Q23. A non-technical founder is building a B2B SaaS MVP with AI. The first technical decision they must make correctly is:

- A. Which programming language to use
- B. Which hosting and database platform will allow them to iterate without refactoring the stack later**
- C. Which AI model to integrate first
- D. How many engineers to hire

Correct Answer: B

Explanation

Platform choices are hard to reverse. Selecting a flexible, developer-friendly stack prevents early architectural lock-in even if AI generates all the code.

Q24. A team builds an MVP in 3 weeks and launches to 100 beta users. After 2 weeks, 80% have churned. The AI-native response is:

- A. Shut down and rebuild from scratch
- B. Wait for more data before making changes
- C. Add more features to retain remaining users
- D. Use AI to rapidly analyse qualitative feedback, identify the core failure, and iterate within days**

Correct Answer: D

Explanation

AI-native teams respond to failure with speed. AI can synthesise feedback, identify patterns, and support rapid re-prioritisation - compressing the response cycle.

Q25. A PM is building a B2C app MVP. Which AI-generated asset directly reduces time-to-first-user the most?

- A. A detailed technical specification document
- B. A competitive analysis report
- C. A working landing page with a clear value proposition and email capture**
- D. A 12-month roadmap

Correct Answer: C

Explanation

First users come from demand - not from documentation or planning. A working landing page enables immediate demand testing and user acquisition before the product is built.

Q26. A founder instructs Claude to build a payment integration. What must they verify before going live?

- A. That the code was written in under 10 minutes
- B. That the integration correctly handles edge cases like failed payments, refunds, and webhook retries**
- C. That the AI chose the cheapest payment processor
- D. That the code uses the latest programming language version

Correct Answer: B

Explanation

Payment integrations fail silently on edge cases. AI-generated code handles the happy path well but often misses error handling - human verification here is critical.

Q27. Which of the following best describes an AI-native team relationship with documentation?

- A. Documentation is AI-generated alongside the code and kept in sync via the same AI workflow**
- B. Documentation is written manually after the product ships
- C. Documentation is not needed because the AI knows what the code does
- D. Documentation is only written for enterprise customers

Correct Answer: A

Explanation

AI-native teams generate docs in the same workflow as code. This removes the documentation lag that plagues traditional teams and keeps context available for future AI sessions.

Q28. A startup wants to test three different product concepts. The AI-native approach is:

- A. Spend six months researching which concept is best before building
- B. Build the most technically complex concept first
- C. Build all three as lightweight prototypes using AI in parallel, then test each with users before committing to one**
- D. Test only one concept at a time to keep focus

Correct Answer: C

Explanation

AI compresses prototype cost enough to test multiple concepts simultaneously. Parallel validation beats sequential testing when the cost of building is near-zero.

Q29. A PM is using AI to generate user interview scripts for MVP validation. The most important human contribution to this process is:

- A. Editing the grammar of the AI-generated script
- B. Translating the script into different languages
- C. Recording the AI-generated script as an audio file
- D. Selecting which questions to prioritise based on what assumptions are most critical to test**

Correct Answer: D

Explanation

AI can generate comprehensive interview scripts. Human judgment determines which assumptions most need testing - that prioritisation is the highest-value contribution.

Q30. Which metric most directly validates an AI-native MVP after the first 30 days?

- A. Total number of AI prompts used to build the product
- B. Whether users return to complete a second session without being prompted**
- C. Number of features shipped
- D. Landing page conversion rate

Correct Answer: B

Explanation

Unprompted return usage is the earliest signal of genuine product-market fit. It shows the product created enough value to motivate behaviour change.

SET

53

AI-Assisted Development

Set 53 of 5 · 15 Questions · AI Engineering

Q31. A developer uses an AI coding assistant to write a function. The most effective review practise is:

- A. Testing the function against the full range of expected inputs including edge cases before merging**
- B. Reading the AI explanation of what the code does and accepting it
- C. Running the code once to see if it produces output
- D. Asking the AI if the code is correct

Correct Answer: A

Explanation

AI coding assistants are confident but not always correct. Edge case testing is the human job - AI explanations are not a substitute for execution-level verification.

Q32. When using an AI to debug code, the most effective prompt structure is:

- A. Describing the bug in general terms and asking the AI to find it
- B. Asking the AI to rewrite the entire file
- C. Pasting the exact error message, the relevant code block, and the expected vs actual behavior**
- D. Running all tests and sending the output without context

Correct Answer: C

Explanation

Specific, structured bug reports give the AI the signals it needs to identify root causes. Vague descriptions produce generic suggestions that rarely resolve the issue.

Q33. An engineer uses AI to generate test cases. What is the primary limitation of AI-generated tests?

- A. AI-generated tests are always too slow
- B. AI tends to generate tests that validate the code it wrote rather than testing genuinely adversarial inputs**
- C. AI cannot write tests in Python
- D. AI-generated tests require separate licensing

Correct Answer: B

Explanation

AI test generation is biased toward confirming its own code. Human testers or adversarial prompting is needed to surface the edge cases AI will not naturally include.

Q34. A team integrates an AI coding assistant into their CI/CD pipeline to auto-generate code review comments. The key governance requirement is:

- A. The AI must review all pull requests without human override
- B. The AI must use the same style guide as the team
- C. The AI must be trained on the team codebase first
- D. Human engineers must retain final approval authority before any suggested change is merged**

Correct Answer: D

Explanation

AI in code review is advisory. Merging AI-suggested changes without human sign-off removes accountability and risks introducing bugs at scale.

Q35. A developer is working with an AI assistant on a large codebase. The most important context to provide at the start of each session is:

- A. The relevant module architecture, the specific task being worked on, and any constraints or conventions in use**
- B. The full codebase pasted into the prompt
- C. A list of all features the product has
- D. The team org chart

Correct Answer: A

Explanation

AI coding assistants produce better outputs with focused, relevant context. Full codebase dumping wastes context window; targeted context scopes the task correctly.

Q36. A team uses AI to migrate a legacy codebase from one framework to another. What is the highest-risk phase of this process?

- A. Converting syntax from one language to another
- B. Validating that migrated code preserves all business logic and edge case behaviour from the original**
- C. Renaming variables and functions
- D. Updating import statements

Correct Answer: B

Explanation

AI migrations handle syntax conversion well but can silently alter business logic. The human-critical phase is verifying that all behaviours are preserved, not just that the code runs.

Q37. An AI generates a 200-line function to implement a feature. A senior engineer says it should be broken into 5 smaller functions. The correct response is:

- A. Accept the 200-line function because the AI wrote it
- B. Reject AI assistance entirely because the output was wrong
- C. Manually refactor without using AI
- D. Ask the AI to refactor the function with a clear instruction about desired modularity and maximum function length**

Correct Answer: D

Explanation

AI refactors on instruction. The human role is to specify the quality standard; the AI executes it. This is faster than manual refactoring and preserves AI leverage.

Q38. A developer uses AI to generate a SQL query. Before running it on production data, the most critical check is:

- A. Checking the query length
- B. Confirming the AI generated the query in under 5 seconds
- C. Verifying the query against a schema diagram to confirm it targets the correct tables and relationships**
- D. Ensuring the query uses uppercase keywords

Correct Answer: C

Explanation

AI-generated SQL frequently targets plausible but incorrect tables or columns. A schema check before execution prevents silent data errors or accidental data loss.

Q39. A PM asks an AI to generate API documentation from the codebase. The most important post-generation step is:

- A. Publishing the documentation immediately
- B. Having an engineer verify that generated descriptions match actual API behaviour, especially for error responses**
- C. Translating the documentation into other languages
- D. Asking the AI to make the documentation shorter

Correct Answer: B

Explanation

AI-generated API docs can describe what an API should do, not what it actually does. Error response documentation is especially prone to inaccuracy and needs verification.

Q40. Which practise most increases the long-term productivity of a team using AI-assisted development?

- A. Maintaining a shared prompt library for common tasks so the team benefits from each other prompt iterations**
- B. Each engineer developing their own private AI workflow
- C. Using only one AI tool to avoid confusion
- D. Limiting AI assistance to junior engineers

Correct Answer: A

Explanation

Prompt quality compounds. A shared library lets the whole team benefit from individual prompt improvements and creates institutional knowledge around AI-assisted workflows.

Q41. A developer is using an AI assistant to implement a feature described in a ticket. The ticket is ambiguous. The correct first step is:

- A. Asking the AI to interpret the ambiguous ticket and implement its interpretation
- B. Implementing both interpretations and asking the PM to choose
- C. Clarifying the ambiguity with the PM before asking the AI to implement, to avoid building the wrong thing correctly**
- D. Closing the ticket and asking for a rewrite

Correct Answer: C

Explanation

AI will execute ambiguous instructions confidently. Clarifying human intent before AI execution prevents well-executed implementations of the wrong requirement.

Q42. An AI coding assistant suggests using a deprecated library. The correct response is:

- A. Using the deprecated library because the AI chose it
- B. Asking the AI if the library is deprecated
- C. Switching to a different AI coding tool
- D. Rejecting the suggestion and specifying the current library version before regenerating**

Correct Answer: D

Explanation

AI training data has a knowledge cutoff. Library recommendations can reflect outdated choices. Specifying current versions in the prompt prevents deprecated or insecure dependencies.

Q43. A company deploys an AI coding assistant to its engineering team. The most important policy to establish is:

- A. AI-generated code may be merged directly without review to maximise speed
- B. All AI-generated code must be reviewed and understood by the engineer before merging**
- C. AI must only be used for test files, not production code
- D. Engineers must disclose to the AI that they are professionals

Correct Answer: B

Explanation

Code ownership requires comprehension. Engineers who merge code they do not understand cannot debug, extend, or be accountable for it - regardless of who wrote it.

Q44. A team uses AI to write code and AI to write tests for that code. Which external check best validates both?

- A. Running AI-generated unit tests only
- B. Asking the AI to review both the code and tests
- C. End-to-end testing with real user flows and production-like data**
- D. Checking that the code compiles without errors

Correct Answer: C

Explanation

AI-generated code and AI-generated tests can be mutually reinforcing blind spots. End-to-end testing with real user data catches failures that unit tests miss.

Q45. Which describes the correct use of AI in a code review process?

- A. AI surfaces potential issues and style violations as a first pass; human reviewers evaluate logic, architecture, and context**
- B. AI replaces human code reviewers entirely
- C. AI only reviews documentation, not code
- D. AI approves pull requests after running tests

Correct Answer: A

Explanation

AI code review is fast and pattern-based but lacks contextual judgment. The correct model pairs AI broad pattern detection with human evaluation of intent and architecture.

SET

54

Workflow Acceleration

Set 54 of 5 · 15 Questions · AI Engineering

Q46. A founder uses AI to compress a task that previously took 3 hours to 20 minutes. The most important follow-up action is:

- A. Using AI for even more tasks to save more time
- B. Documenting the time saved for investor reporting
- C. Telling team members to do the same task manually to keep skills sharp
- D. Reinvesting the saved time into higher-leverage strategic work that AI cannot do**

Correct Answer: D

Explanation

AI-native execution multiplies output by freeing human time for judgment-intensive work. The compound benefit comes from where saved time is reinvested, not from the saving itself.

Q47. A content team uses AI to produce first drafts. The correct role for human editors in this workflow is:

- A. Rewriting all AI drafts entirely
- B. Fact-checking, adding unique insight, and ensuring brand voice - not rewriting AI drafts from scratch**
- C. Publishing AI drafts without review
- D. Using AI only for titles, not full drafts

Correct Answer: B

Explanation

Human editors in an AI-native content workflow are curators and elevators, not re-writers. Their value is judgment and specificity - not volume production.

Q48. A PM uses AI to prepare for a board meeting. The highest-leverage AI use in this scenario is:

- A. Using AI to take notes during the meeting
- B. Using AI to email board members after the meeting
- C. Generating the first draft of the board deck and narrative, which the PM then refines with insider context**
- D. Using AI to research board members on LinkedIn

Correct Answer: C

Explanation

AI can structure and populate board decks from raw data and objectives. The PM role is to inject the context and strategic framing AI cannot access.

Q49. A sales team uses AI to personalise outreach at scale. The most important human oversight in this workflow is:

- A. Reviewing a sample of AI-generated messages before sending to ensure accuracy and appropriate tone**
- B. Reviewing every message before sending
- C. Letting AI send all messages without review
- D. Having the AI send messages only to warm leads

Correct Answer: A

Explanation

Full review defeats the scale benefit; zero review creates risk. Sampling provides a quality check that catches systematic errors without eliminating efficiency.

Q50. A team uses AI to generate weekly status reports from raw data. After two months, leadership stops reading them. The most likely cause is:

- A. Leadership prefers verbal updates
- B. The reports are too long
- C. The AI is using the wrong template
- D. AI-generated reports lack the editorial judgment to surface what actually matters to leadership**

Correct Answer: D

Explanation

AI can aggregate and structure data but struggles to prioritise what matters to a specific audience. Reports without human editorial judgment lose relevance quickly.

Q51. A company uses AI to summarise all customer support tickets daily. The correct use of these summaries is:

- A. As a replacement for reading any individual tickets
- B. As the sole basis for product roadmap decisions
- C. As an input to the PM and support lead for trend spotting, not as a replacement for reading individual tickets on critical issues**
- D. As a public-facing customer communication

Correct Answer: C

Explanation

AI summaries are useful for trend detection at volume. Critical issues require reading the original ticket to preserve nuance that summaries compress away.

Q52. An operations team uses AI to automate weekly report generation. The process saves 8 hours per week. Which risk must be actively managed?

- A. The AI silently propagating an error in source data across multiple weeks of reports before detection**
- B. The reports being generated too quickly
- C. The AI choosing the wrong font in the report
- D. The AI generating more reports than requested

Correct Answer: A

Explanation

Automation amplifies both correct and incorrect inputs. A systematic data error can propagate undetected for weeks if no human spot-checks the source data periodically.

Q53. A PM uses AI to analyse qualitative user interview transcripts. What should they validate in the AI output?

- A. Whether the AI completed the analysis in under 2 minutes
- B. Whether the themes the AI identified match what stood out to the human interviewer during the sessions**
- C. Whether the AI used the correct summary format
- D. Whether the AI mentioned all interviewees by name

Correct Answer: B

Explanation

AI thematic analysis can miss emotional tone, implicit signals, and interviewer context. Cross-referencing AI themes against human recall surfaces what the AI flattened.

Q54. An AI is used to triage and categorise 500 inbound leads per week. What is the highest-risk failure mode to monitor?

- A. The AI taking too long to categorise each lead
- B. The AI generating too many categories
- C. High-value leads being systematically miscategorised into a low-priority bucket**
- D. The AI sending automatic responses to leads

Correct Answer: C

Explanation

Systematic misclassification of high-value leads has direct revenue impact. Monitoring should focus on false negatives in high-priority categories, not just overall accuracy.

Q55. A team uses AI to generate 30 social media posts per week. After 4 weeks, engagement drops significantly. The most productive diagnostic is:

- A. Increasing the number of posts per week
- B. Switching to a different AI tool
- C. Changing the posting time
- D. Comparing engagement on AI-generated posts vs posts that had significant human editing**

Correct Answer: D

Explanation

Engagement drop after AI content adoption often reflects a loss of authentic voice. The differential analysis reveals whether AI volume is substituting for human quality.

Q56. A company rolls out AI tools to all employees. Productivity gains are strong in some teams and absent in others. The most likely structural cause is:

- A. Different task types - AI accelerates execution tasks but adds friction to judgment-heavy tasks**
- B. Some teams are less intelligent
- C. Some teams received better laptops
- D. AI tools were not available in all office locations

Correct Answer: A

Explanation

AI productivity gains are not uniform. Execution-heavy roles see the most acceleration. Roles requiring deep judgment, relationships, or novel problem-solving see less gain.

Q57. A product team uses AI to write sprint tickets. The main risk of this workflow is:

- A. Tickets that are too long
- B. Tickets that are complete and well-formatted but describe the wrong work if the PM brief was imprecise**
- C. Tickets that do not use the correct Jira format
- D. The AI assigning tickets to the wrong engineers

Correct Answer: B

Explanation

AI amplifies the quality of inputs. A vague brief produces polished but misdirected tickets. AI-generated ticket quality is bounded by the PM clarity of thinking.

Q58. A team uses AI to produce meeting summaries with action items. The most common failure mode is:

- A. Summaries that are too long
- B. Summaries that miss the date of the meeting
- C. AI including off-topic comments in the summary
- D. Action items assigned to the wrong owner or with the wrong due date because the AI misidentified who said what**

Correct Answer: D

Explanation

Speaker identification errors cause action item misassignment. Reviewing action item ownership before distributing summaries is the critical human check in this workflow.

Q59. Which AI workflow acceleration practise has the highest compounding return over time?

- A. Building reusable prompt templates for recurring tasks that encode organisational context and standards**
- B. Using AI for as many different tasks as possible each week
- C. Switching to a new AI tool every quarter
- D. Limiting AI use to one task type to build depth

Correct Answer: A

Explanation

Reusable prompts compound because every improvement benefits every future run of that workflow. They also transfer institutional knowledge into repeatable AI behaviour.

Q60. A founder uses AI to run their company for 3 months with no employees. Which function is hardest for AI to fully replace?

- A. Drafting contracts
- B. Writing marketing copy
- C. Relationship-building with customers, partners, and investors**
- D. Generating financial projections

Correct Answer: C

Explanation

AI compresses most execution tasks but cannot build authentic human relationships. Trust, rapport, and influence require human presence - the durable limit of AI automation.

SET

55

AI-First Operations

Set 55 of 5 · 15 Questions · AI Engineering

Q61. What distinguishes an AI-first company from a company that uses AI tools?

- A. AI-first companies design their workflows, org structure, and decision-making around AI as the primary execution layer
- B. AI-first companies use more AI tools than others
- C. AI-first companies hire only AI engineers
- D. AI-first companies have no human employees

Correct Answer: A**Explanation**

AI-first is an architectural posture, not a tooling quantity. It means rebuilding how work gets done with AI as the default execution layer, not an add-on.

Q62. A startup is designing its operations from scratch with AI-first principles. What is the first decision to make?

- A. Which AI tools are cheapest
- B. How many AI tools to subscribe to
- C. Whether to use open-source or proprietary AI
- D. Which workflows will remain human-led and why, before designing AI automation for everything else

Correct Answer: D**Explanation**

AI-first design starts with judgment about where humans must lead - then automates everything else. This prevents both under-automation and inappropriate automation.

Q63. A company adopts AI-first operations and finds that middle management roles are changing significantly. The most accurate description of this change is:

- A. Middle managers are fully replaced by AI
- B. Middle managers shift from coordinating information flow to curating AI outputs and making judgment calls**
- C. Middle managers spend more time in meetings
- D. Middle managers become AI engineers

Correct Answer: B

Explanation

AI removes the coordination and information-synthesis tasks that defined middle management. The residual human role is judgment, context, and accountability.

Q64. An AI-first operations team uses AI agents to execute multi-step workflows. The critical human responsibility is:

- A. Monitoring token usage in real time
- B. Ensuring the AI uses the correct output format
- C. Defining the scope, reviewing outputs, and intervening when the agent exceeds its intended remit**
- D. Rotating which AI tool the agent uses each week

Correct Answer: C

Explanation

AI agents can drift or take unintended actions in multi-step workflows. Human oversight defines the boundary of acceptable action and catches scope creep before impact.

Q65. A company runs its customer onboarding entirely on AI. A new enterprise client has a complex, non-standard setup. The correct operational response is:

- A. Escalating to a human CS team immediately rather than forcing the non-standard case through an AI-only flow**
- B. Asking the AI to handle the complex setup anyway
- C. Declining the enterprise client
- D. Redesigning the entire onboarding flow for this one client

Correct Answer: A

Explanation

AI-first operations must have human escalation paths for edge cases. Forcing non-standard inputs through AI-only flows produces poor experiences and churn.

Q66. A company measures operational efficiency as cost per task completed. After deploying AI, cost per task drops 60%. What secondary metric must be tracked?

- A. Number of tasks completed per week
- B. Number of AI tools used
- C. Quality per task, to ensure cost reduction is not coming from lower-quality outputs**
- D. Time the AI takes per task

Correct Answer: C

Explanation

Cost efficiency gains are only real if quality is maintained. AI can reduce cost by reducing quality - the secondary quality metric catches this substitution.

Q67. A company uses AI to handle all internal knowledge queries. Employees stop documenting processes because the AI knows everything. The long-term risk is:

- A. Employees becoming too fast at finding information
- B. The AI becoming too powerful
- C. Employees spending less time searching for information
- D. Institutional knowledge degrading as undocumented processes exist only in conversation history that may not be retained**

Correct Answer: D

Explanation

AI as a knowledge repository does not replace written documentation. Conversation histories are ephemeral and unsearchable. Documentation degrades silently in an AI-only model.

Q68. An AI-first company wants to hire its first employee. The role that generates the highest marginal value is:

- A. A data entry specialist
- B. Someone who can direct and improve AI systems - a judgment and orchestration role, not a pure execution role**
- C. A social media manager
- D. A generalist who can do everything manually

Correct Answer: B

Explanation

In an AI-first company, execution is already handled. The first hire marginal value is in improving AI direction, quality, and strategy - not in adding execution capacity.

Q69. A company runs all its marketing on AI-generated content for 6 months. Organic reach declines. The most likely cause is:

- A. The AI is generating too much content
- B. The AI is using outdated social media algorithms
- C. AI-generated content lacks the specificity, authenticity, and novelty that audiences respond to over time**
- D. Competitors have larger budgets

Correct Answer: C

Explanation

AI content often converges on averaged, generic outputs. Audiences habituate to AI patterns quickly. Human editorial distinctiveness is the differentiator AI cannot replicate at scale.

Q70. A PM wants to build an AI-first onboarding workflow. The correct starting point is:

- A. Mapping the current manual onboarding steps and identifying which are highest-volume and most rule-based**
- B. Choosing which AI tool to use first
- C. Building the AI onboarding before the manual version exists
- D. Outsourcing onboarding to a third-party AI vendor immediately

Correct Answer: A

Explanation

AI automation should follow a documented manual process. Without mapping the current workflow, AI onboarding automates the wrong steps or misses critical ones.

Q71. An AI-first company tracks AI coverage - the percentage of workflow steps handled by AI. At 90% AI coverage, the most important human focus is:

- A. Increasing AI coverage to 100%
- B. The 10% of steps that remain human - which are there because they require judgment, relationships, or accountability**
- C. Reducing the number of workflow steps
- D. Documenting the AI coverage metric for investors

Correct Answer: B

Explanation

The residual human steps in an AI-first operation are not inefficiencies - they are the highest-value human contributions. They deserve the most human attention, not elimination.

Q72. A company uses AI to draft all legal agreements. The non-negotiable human role in this workflow is:

- A. Having the AI verify its own draft
- B. Using AI to negotiate the terms after drafting
- C. Sending AI drafts directly to counterparties without legal review
- D. A qualified lawyer reviewing every agreement before execution, regardless of how good the AI draft appears**

Correct Answer: D

Explanation

Legal agreements carry binding obligations. AI drafts can contain errors with significant legal consequences. Qualified human review is a non-negotiable control regardless of AI quality.

Q73. An AI-first ops team is building a new internal tool. The correct build sequence is:

- A. Choose an AI tool, give it instructions, then launch
- B. Hire an engineer, write code, then add AI later
- C. Define the workflow in plain language, then identify AI automation points, then build the human exception path, then deploy**
- D. Launch with full AI automation, then add human steps if problems emerge

Correct Answer: C

Explanation

AI-first tooling requires human exception paths to be designed in, not bolted on. Defining the workflow first ensures AI is applied to the right steps.

Q74. A company has replaced most of its customer support with AI. Customer satisfaction scores are up but escalation volume is also up. What does this signal?

- A. The escalation team is too small
- B. AI handles routine queries well but is creating friction on complex or sensitive issues that quickly exhaust users**
- C. Customers prefer to speak to humans regardless of AI quality
- D. The AI is answering too quickly

Correct Answer: B

Explanation

Rising escalation alongside rising CSAT means AI is succeeding on easy queries but failing on hard ones. The fix is better routing, not more AI in complex cases.

Q75. A founder builds a company that reaches 1000 customers with a team of 3 people using AI. The most accurate description of this business model is:

- A. An understaffed company that will collapse at scale
- B. A company that has replaced all human judgment with AI
- C. A company that uses AI for marketing only
- D. AI-leveraged operations where automation handles a volume of work that would traditionally require 20 to 50 headcount**

Correct Answer: D

Explanation

AI-native companies operate with headcount leverage ratios that were previously impossible. Automation handles execution volume; a small human team handles judgment and direction.

END OF QUESTION BANK

*Your turning
point starts
here.*